Potential Energy Surface and State-to-State Dynamics of Chemical Reactions

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by

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Certificate

This is to certify that the work contained in this thesis, titled '**Potential energy surface and state-to-state dynamics of chemical reactions**' by **Ajay Mohan Singh Rawat (Reg. No. 17CHPH62)**, has been carried out under my supervision and is not submitted elsewhere for a degree.

This thesis is free from plagiarism and has not been submitted previously in part or in whole to this or any other University or Institute for the award of any degree or diploma.

Part of this thesis has been published in the following publications:

1. Jayakrushna Sahoo, **Ajay Mohan Singh Rawat**, and S. Mahapatra*, Quantum interference in the mechanism of H + $LiH^+ \rightarrow H_2 + Li^+$ reaction dynamics. Phys. Chem. Chem. Phys. **23**, 27327 (2021).

2. Jayakrushna Sahoo, **Ajay Mohan Singh Rawat**, and S. Mahapatra*, Theoretical Study of the Energy Disposal Mechanism and the State-Resolved Quantum Dynamics of the H + $LiH^+ \rightarrow H_2 + Li^+$ Reaction. J. Phys. Chem. A **125**, 3387 (2021).

3. **Ajay Mohan Singh Rawat**, Jayakrushna Sahoo, and S. Mahapatra*, Combined Quantum Mechanical and Quasi-Classical State-to-State Dynamical Study on the Isotopic Effect in H/D + $LiH^+/LiD^+ \rightarrow H_2/HD/D_2 + Li^+$ Reactions. J. Phys. Chem. A **127**, 10733 (2023).

4. **Ajay Mohan Singh Rawat**, Mohammed Alamgir, Sugata Goswami, and S. Mahapatra*, A new ground electronic state potential energy surface of $HeLiH^+$: analytical representation and investigation of the dynamics of He + LiH^+ (v=0, j=0) $\rightarrow LiHe^+$ + H reaction. J. Chem. Phys. **161**, 124308 (2024).

Part of the thesis is also planned to be communicated in the following papers:

1. **Ajay Mohan Singh Rawat**, Jayakrushna Sahoo, and S. Mahapatra*, State-to-state dynamics of $^{18}O + {}^{16}O^{16}O$ ($v = 0, j = 1$) $\rightarrow {}^{18}O^{16}O + {}^{16}O$ isotopic exchange reaction.

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4. Poster presentation in 'Structure and Dynamics: Spectroscopy and Scattering (SDSS-2023)'. October 2023 at IACS Kolkata, India.
5. Poster presentation in '18th Theoretical Chemistry Symposium (TCS-2023)'. November 2023 at IIT Madras, India.

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3	CY-801	Research Proposal	4	Pass
4	CY-806	Instrumental Methods-B	4	Pass

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Declaration

I, **Ajay Mohan Singh Rawat**, hereby declare that this thesis entitled '**Potential energy surface and state-to-state dynamics of chemical reactions**' submitted by me under the guidance and supervision of **Prof. Susanta Mahapatra** is a bonafide research work. I also declare that it has not been submitted previously in part or in full to this University or any other University or Institution for the award of any degree or diploma.

Date: Oct 3rd 2024

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a small effort to push the boundaries of knowledge.

dedicated to my

grandparents ***and*** **parents**

सा विद्या या विमुक्तये।

Knowledge sets us free.

Acknowledgement

I joined PhD in January 2018. On the first day of my joining, I met my supervisor, and his first question was why I had joined the PhD program in theoretical chemistry. My answer was a cumulative inspiration of the stories of famous scientists and the awe factor associated with scientific miracles rather than the laboratory rigor of theoretical science. Now, when I am finally writing my thesis six years later with the experience of 'laboratory rigor,' I feel that my journey as a PhD student has brought me closer to my answer. I came as a young boy in awe of science, and I am going out as a PhD with the same awe and the realization of what it takes to do real science. To survive 'the science,' as Carl Sagan termed it, I consider it a great feat. However, I am afraid that I am not the one to be credited for it. Here is the acknowledgment to the people and experiences where the real credit lies.

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Ajay M. Rawat
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List of Abbreviations

SE	Schrödinger equation
QCT	Quasi-classical trajectory
PES	Potential energy surface
ML	Machine learning
QM	Quantum mechanical
AP	Aguado Paniagua
NN	Neural network
ICS	Integral cross section
DCS	Differential cross section
FCI	Full configuration interaction
CBS	Complete basis set
RMSE	Root mean square error
TDQM	Time-dependent quantum mechanical
CID	Collision induced dissociation
ZPE	Zero-point energy
TDSE	Time-dependent Schrödinger equation
TISE	Time-independent Schrödinger equation
KE	Kinetic energy
BO	Born-Oppenheimer
PEC	Potential energy curve
NACT	Non-adiabatic coupling term
PIP	Permutationally invariant polynomial
SO	Split operator
TDSE	Time-dependent Schrödinger equation
FT	Fourier transformation
FFT	Fast Fourier transformation
DVR	Discrete variable representation
FBR	Fixed basis representation
FD	Fermi Dirac
TD	Time-dependent
TI	Time-independent
FWHM	Full width half maximum
HB	Histogram binning
GB	Gaussian binning

MRCI	Multi-reference configuration interaction
BSSE	Basis set superposition error
MEP	Minimum energy path
CASSF	Complete active space self-consistent field
QSM	Quasi-statistical method
CBS	Complete basis set
MSE	Mean square error
MIF	Mass-independent fractionation

Contents

Chapter 1

Introduction

A chemical reaction is a change in the chemical nature of participating species, and the study of these chemical transformations is at the heart of chemistry. Different branches of chemistry deal with different aspects of these chemical reactions. Thermodynamics describes the changes related to the energy and temperature of a reaction. These quantities provide information about the energetic feasibility or, in other words, the comparative stability of the reactant and product fragments. The reaction is exothermic (exoergic) when the product is energetically lower than the reactant, and if the reactants are energetically lower than the products, the reaction is endothermic (endoergic). An exothermic reaction results in a net release of energy, and an endothermic reaction requires energy. However, this energy favourability does not necessarily translate into a swift conversion of reactants into products for all exothermic reactions. The answers to such discrepancies are found in chemical kinetics, a branch of chemistry that tells us about the rate of reactions or, in other words, the swiftness of the transformation of one chemical species into another. It introduces the concept of activation energy, the height of the barrier preventing the swift conversion of reagents to products in all exothermic reactions without providing a certain amount of energy. Chemical kinetics connects the concentrations of reactants to the reaction rate and explains the effect of temperature on the reaction rate. However, figuring out the correlation of the rate of the reaction with the concentration of reactants is not a trivial task because, whereas a reaction shows the net changes in the chemical species involved, the reaction rate is about the mechanism of the reactions, which can only be determined with the knowledge of all the elementary reactions involved in a reaction. Even if the reaction under consideration is elementary, Chemical Kinetics studies have limitations originating from the fact that these studies are done at the bulk level, meaning that a group of molecules at thermal equilibrium are involved. So, the answers obtained from these studies are averaged, and what goes into a reaction at the molecular level remains a mystery.

The answer to these mysteries is molecular reaction dynamics. From its perspective, a chemical reaction is the movement of atoms from the reactant to the product states. In other words, a reaction is a dynamic event, and molecular reaction dynamics aims to understand the reactions by following these events from one point to another in time and space at the molecular level. In molecular reaction dynamics, the elementary reactions are studied at the microscopic level, enabling us to understand the effect of the initial quantum states, orientation, energy distributions, and other molecular factors of reactants on the reaction, providing increased control over the reactions.

The dynamics of the reactions at such molecular magnification can be studied both experimentally and theoretically. However, the current state of the art in both experimental and theoretical fields limits us from fully investigating the reactions involving more than a few atoms. Experimental techniques in molecular reaction dynamics are difficult due to the sophistication needed to control the reagent's initial energy and orientation and detect products in its nascent stage. As for the theoretical methods, it is possible to handle complex systems; however, approximations become necessary as the complexities of reactions increase. Thus, theoretical and experimental studies complement each other. A theoretical study helps understand the experimental behavior of the reaction, providing insight into the reactions where experimental studies are not feasible.

To implement theoretical methods to follow the motion of atoms in a reaction, classical or quantum mechanical equations of motion are solved. In the classical framework, the equations of motion are either Newton's or Hamilton's equations of motion, while the quantum framework uses the Schrödinger equation (SE). In the quantum mechanical approach, time-dependent or time-independent methods are used, whereas the Quasi-classical trajectory (QCT) is used in the classical approach. The method is called quasi-classical because the initial state of reactants (diatoms) at the beginning of the trajectory is approximated to the quantum state of the reactant. In both approaches, the differential equations are initial value problems solved at a time (t) to get information about the system at a future time. Classical methods may not be very accurate for reactions where quantum effects, i.e., quantum tunneling and resonances features, are prominent, but at higher collision energies, classical methods approach the accuracy of quantum methods. In addition, classical methods provide mechanistic insights, such as trajectory calculations, which are otherwise impossible with quantum methods.

Since the motion of atoms is also controlled by the system's potential at different configurations, the potential energy surface (PES) becomes the central part of molecular reaction dynamics. PES, composed of electronic potential points in the reaction space, decides the past, present, and future of a reaction. The effect of the initial rotational and vibrational states, initial translational energy, and reactant orientation can be elucidated from the knowledge of the underlying potential. This makes the potential information at different nuclear configurations critical. This potential information can be calculated at the required configurations during dynamical calculations (on-the-fly dynamics) or from the potential subroutine fitted with the precalculated potential energy points. Since thousands of points are required during the dynamics, calculating the potential on the fly proves to be daunting, especially when the system has many atoms or the required potential is to be calculated at a highly sophisticated level of theory. This makes fitted potential subroutines a go-to practice in the molecular reaction dynamics community. This rigorous process requires sampling the entire reaction space with the data points, using an accurate functional form, and performing a highly accurate fit. Various methods with different analytical forms have been proposed to perform the fitting. However, most of these analytical forms are limited by the size of the system they can fit, but with the advent of machine learning (ML) methods, a highly accurate PES for systems with many atoms is possible.

1.1 Thesis objective

As pointed out, dynamics calculations provide detailed insight into a reaction, the precision of which depends on the underlying PES. Keeping this in mind, the objective of this thesis is two-fold: first, to understand the various phenomena involved in the atom-diatom reactive collisions under different reaction conditions through theoretical means and to add to the understanding of the specific reactions during individual dynamical studies, and second, to construct the accurate PES.

Now, when it comes to reactions involving atom + diatom collisions, they can be of different types. Exothermic, endothermic, and thermoneutral when the energetics of the reactions are considered. When the topography of the PES is considered, reactions are either barrierless, early-barrier, late-barrier, reaction with deep wells, etc. Each of these different reactions has distinct mechanisms to form products, and the unique signature of their nature can be observed in various reaction observables. Among the reactions reported in this thesis, H + LiH^+ $\rightarrow$ H_2 + Li^+ is barrierless exothermic with shallow well on the product side, He + LiH^+ $\rightarrow$ $LiHe^+$ + H is endothermic and also has a shallow well, and ^{18}O + $^{16}O^{16}O$ $\rightarrow$ $^{18}O^{16}O$ + ^{16}O is thermoneutral in nature. The PESs of these three reactions exhibit different topographic features. The H + LiH^+ and He + LiH^+ reactions occurs on the ground electronic state of LiH_2^+ and $HeLiH^+$, respectively, have much simpler potential surface than ^{18}O + $^{16}O^{16}O$ reaction. The ^{18}O + $^{16}O^{16}O$ reaction occurs on the ground state of the O_3 system and exhibits a complex potential surface with deep wells. So, one can understand their effect on the reaction observables by spanning the whole spectrum of reactions with different energetics and surface topographies. The findings from these three distinct types of reactions cumulatively provide a relatively complete picture of reaction dynamics.

As far as the individual reactions are concerned, the first two reactions, H + LiH^+ and He

+ LiH^+ are the destruction pathways of LiH^+ molecular ion. This molecular ion is an important member of the Li reaction network, and shedding light on it can result in the clarity of Li chemistry. The H + LiH^+ reaction, along with its isotopic counterparts, is studied using QCT and QM methods, and a one-to-one comparison of the two methods is made on several reaction observables, evaluating the feasibility of using computationally cheaper QCT method than quantum mechanical (QM) methods. This insight is important because for meaningful conclusions to be made about the reactions in various reaction conditions, dynamical calculations at various initial reagent states are required, and such a significant number of calculations are computationally expensive. About the reaction, He + LiH^+, until our surface, only one global PES is available in the literature, and only one dynamical study has been conducted. So, there is not much information available about this reaction. The construction of the global PES at a much higher level of theory than the earlier surface is performed and reported in this thesis, which is the first step toward studying the reaction. In addition, the surface facilitates a second reaction H + $LiHe^+ \rightarrow LiH^+$ + He, a formation channel of LiH^+ molecular ion. The new surface also opens up the possibility of studying these reactions in greater detail.

1.2 Thesis overview

The theoretical methodology used in investigating the different research problems is discussed in Chapter 2 of the thesis. Since the work presented in this thesis starts from the fitting of the PES and goes all the way to studying the dynamics of the triatomic (A + BC) reaction systems using both the classical and quantum methods of fitting, the theoretical idea behind fitting and dynamics is discussed in detail. The discussion on potential fitting starts with exploring the reasons that inspired a generation of scientists to work and perfect the concept of analytical fitting. Two different fitting approaches, ML and non-machine learning, which include using the Aguado Paniagua (AP) form, are discussed. The ML method used in the fitting reported in this thesis is the neural network (NN) method. A complete description of the different aspects of NN, such as the construction of the network, choices of training data, optimization algorithm, and fine-tuning of the network, is presented. The chapter also discusses the procedure to overcome the overfitting and underfitting issues in the fitting. The conventional fitting procedure, which uses the AP functional form, is also discussed, and an effort is made to compare the two fitting approaches.

After the discussion on fitting, the nuclear dynamics of the triatomic (A + BC) system are presented. The dynamical calculations reported in this thesis are adiabatic, meaning that the potential surface of only one electronic state is involved in the reaction, or, in other words, the reaction is taking place in only one electronic state. Since solving the system dynamics is finding a solution to the equation of motion, the chapter also discusses both the classical and QM solutions of the dynamics. Procedures to solve Hamilton's equation as a classical method and SE as a quantum mechanical method are discussed in detail. Further, the mathematical expressions to calculate the different reaction attributes, e.g., reaction probability, integral cross section (ICS), differential cross section (DCS), product energy disposal, etc., are discussed for both the classical and QM methods.

At the core of the chapter 3 is $H + LiH^+$ ($v = 0, j = 0$) $\rightarrow H_2$ (v', j') $+ Li^+$ reaction. This is an important reaction channel for the destruction of LiH^+ molecular ions. The work reported in the chapter deals with two different research problems: one, the effect of isotopic substitution on the different reaction observables, and two, the comparison of classical and quantum mechanical methods. The four different isotopic variants of the above reaction in the same initial rovibrational level ($v = 0, j = 0$) are used in the study.

$$H + LiH^+ \rightarrow H_2 + Li^+ \qquad R1$$

$$D + LiH^+ \rightarrow HD + Li^+ \qquad R2$$

$$H + LiD^+ \rightarrow HD + Li^+ \qquad R3$$

$$D + LiD^+ \rightarrow D_2 + Li^+ \qquad R4$$

For R1 and R2, the quantum mechanical results from our previous publication are used, whereas the QCT calculations are performed for four reactions. The isotopic effect on different reaction observables is studied based on these QCT calculations. Since substituting the isotopes doesn't change the electronic potential surface but the nuclear masses involved in the dynamics, the concept of mass-independent coordinates is also explored. The QM and QCT dynamical attributes are compared for the R1 and R2, and a one-to-one comparison is made for several different reaction observables. The comparison study between the QM and QCT results is important because a fair match provides confidence in using the QCT calculations for the exothermic reactions system where the QM results are quantitatively and computationally demanding. The different mechanisms by which the reactants transform into the products are also discussed.

In Chapter 4, the construction of the ground electronic PES for the $HeLiH^+$ system is performed. The analytical fitting is performed on full configuration interaction (FCI) level ab initio energy points, which are extrapolated to the complete basis set (CBS) limit. Both the NN and AP functional forms are used in combination to generate the analytical fit for the surface. The potential energy of the chemical species is expressed as the many-body expansion, which breaks down the overall potential of a triatomic system into one, two, and three-body parts. Out of three 'two-body' potential parts, two diatomic parts are fitted using the NN, and one diatomic part is fitted using the AP functional. The three-body potential part is fitted using the AP. The overall root mean square error (RMSE) of the fitted PES is 14.21 cm^{-1}, which decreases to 7.48 cm^{-1} when the points of high energy configurations (> 2 eV) are discarded. Further, the fitted potential surface is used to perform the QCT and time-dependent quantum mechanical (TDQM) calculations for He + LiH^+ (v = 0, j = 0) $\rightarrow$ $LiHe^+$ (v', j') + H reaction in the collision energy range of 0.0 eV - 1.0 eV. The quantum dynamical investigations are performed using the DIFFREALWAVE code. Different reaction attributes, e.g., total ICS, reaction probability, DCS, etc., were calculated. Since the above endothermic reaction has a collision-induced dissociation (CID) opening at very low collision energy, its effect on the reaction mechanism is also discussed. The CID channel (He + LiH^+ (v = 0, j = 0) $\rightarrow$ Li^+ + He + H) is also studied separately, and the QCT-HB CID ICSs are reported up to 1.0 eV.

In chapter 5, the full Coriolis coupled TDQM method is implemented to investigate the isotopic exchange reaction ^{18}O + $^{16}O^{16}O$ (v = 0, j = 1) $\rightarrow$ $^{18}O^{16}O$ (v', j') + ^{16}O. The same DIFFREALWAVE code as used in Chapter 4 is also used in this work. The reaction is thermoneutral, excluding the zero-point energy (ZPE) of reactant and product diatoms, which becomes exothermic ($\sim$ 0.002 eV) when ZPEs are included. This reaction poses a different set of complexities than the other reactions discussed in this thesis due to the deep wells on the PES. When studied theoretically, the presence of the deep wells increases the computational cost because of the requirement for a longer propagation time and denser grid. The chapter presents the reaction probability and state-specific and state-to-state ICSs for the isotopic exchange reaction.

At last, the summary of the thesis and outlook is presented in Chapter 6. This chapter summarises the works reported in the thesis and discusses the motivation behind them. The limitations of the different theoretical approaches used in dynamics and potential construction are also discussed, and future direction is presented.

Chapter 2

Theory and Methodology

2.1 Introduction

In the realm of quantum mechanics, everything from the smallest particle to the vast galaxy has a functional representation called wave function. The wave function doesn't carry any physical meaning by itself. However, all the properties of the system, such as position, energy, momentum, etc., can be calculated just by applying a mathematical expression called an operator to the wave function. When it comes to the time evolution or energy of the system, one equation which is at the heart of quantum mechanics is the Schrödinger equation (SE). For a molecular system, the time-dependent form of Schrödinger equation (TDSE) is written as

$$i\hbar\frac{\partial\psi(r_e, R_N, t)}{\partial t} = \hat{H}\psi(r_e, R_N, t). \tag{2.1}$$

Another form of the SE that is independent of the time and which can be derived from the TDSE (*cf.* Appendix A) is written as follows.

$$\hat{H}\psi(r_e, R_N) = E\psi(r_e, R_N) \tag{2.2}$$

In both the Eqs. 2.1 and 2.2, $\hat{H}$ is the total Hamiltonian of the system, $\hbar = \frac{h}{2\pi}$ where h is Planck constant and ψ is the total wave function of the system. In the case of TDSE ψ depends on the position of electrons (r_e) and nucleus (R_N), and time (t), whereas in the case of time-independent Schrödinger equation (TISE) ψ is independent of time. The total Hamiltonian $\hat{H}$ that is the quantum mechanical operator of the total energy of the system can be written as

$$\hat{H} = -\sum_A \frac{\hbar^2}{2M_A}\nabla_A^2 - \sum_i \frac{\hbar^2}{2m_e}\nabla_i^2 - \sum_i\sum_A \frac{Z_A e^2}{4\pi\epsilon_0 r_{iA}} + \sum_i\sum_{j>i}\frac{e^2}{4\pi\epsilon_0 r_{ij}} + \sum_A\sum_{B>A}\frac{Z_A Z_B e^2}{4\pi\epsilon_0 r_{AB}}, \tag{2.3}$$

where, i and A are the designations for electron and nucleus, respectively, m_e and M_A are mass of electron and mass of nucleus, e is electron charge, and Z_A is atomic number of A^{th} nucleus. The other parameters r_{iA}, r_{ij}, and r_{AB} are the distances between electron-nucleus, electron-electron, and nucleus-nucleus, respectively. In Eq. 2.3, the first term on the RHS of the equation is the nuclear kinetic energy (KE) operator, the second term is the electronic

KE operator, and the third, fourth, and fifth terms are potential energies between the pair of electron-nucleus, electron-electron, and nucleus-nucleus, respectively. Eq. 2.3 can be made to look a little less intimidating by writing it in the form of the atomic unit. To convert Eq. 2.3 in atomic unit form, the values of $\hbar$, m_e, and e are kept as unit (one). The equation is as follows.

$$\hat{H} = -\sum_{A}\frac{1}{2M_A}\nabla_A^2 - \sum_{i}\frac{1}{2}\nabla_i^2 - \sum_{i}\sum_{A}\frac{Z_A}{r_{iA}} + \sum_{i}\sum_{j>i}\frac{1}{r_{ij}} + \sum_{A}\sum_{B>A}\frac{Z_A Z_B}{r_{AB}} \tag{2.4}$$

To find the energy or information about the dynamical system in time, one can simply solve Eq. 2.1, which is an initial value problem for a first-order differential equation, and 2.2, which is an eigenvalue equation. However, solving these differential equations is not straightforward. So, the Born-Oppenheimer approximation (BO) is invoked to make the process of solving the differential equation a little simpler [1].

2.2 Born-Oppenheimer Approximation

BO approximation is the separation of nuclear and electronic motions, given that electrons are a lot lighter but quicker than nuclei ($m_{\mathrm{p}} \gtrsim 1836.15 m_e$) [1]. On providing equal KE to them, the electron is $\sim$ 42 times faster than the nucleus, a difference which increases with increasing mass of the nucleus. This makes the nucleus appear stationary with respect to the electronic motion. In other words, this is described as electronic motion in the field of stationary nuclei. This approximation mathematically translates into separating the total molecular Hamiltonian in a sum of nuclear KE terms, a constant nuclear-nuclear interaction, and electronic terms parametrically dependent on fixed nuclear positions.

$$\hat{H} = \hat{H}_N + \hat{H}_e + \sum_{A}\sum_{B>A}\frac{Z_{\mathrm{A}} Z_B}{r_{AB}}$$

where,

$$\hat{H}_N = -\sum_{\mathrm{A}}\frac{1}{2M_{\mathrm{A}}}\nabla_{\mathrm{A}}^2$$

$$\hat{H}_e = -\sum_{i}\frac{1}{2}\nabla_i^2 - \sum_{i}\sum_{\mathrm{A}}\frac{Z_{\mathrm{A}}}{r_{i\mathrm{A}}} + \sum_{i}\sum_{j>i}\frac{1}{r_{ij}}$$

Now, neglecting $\hat{T}_N$, and solving the SE.

$$\hat{H}_e \Phi_i(r_e; R_N) = V_i^{el}(R_N)\Phi_i(r_e; R_N) \tag{2.5}$$

where, $\Phi_e(r_e; R_N)$ is the electronic wave function for i^{th} electronic state and $V_i^{el}(R_N)$ is electronic energy at fixed nuclear configuration R_N. Now, one extra term, $\hat{V}_{\mathrm{NN}}$, left in the

Hamiltonian is a constant and (thus) has the same wave function as $\hat{H}_e$. It can simply be added to the electronic energy to get the total energy of stationary nuclei.

$$V_i^{tot}(R_N) = V_i^{el}(R_N) + \hat{V}_{\mathrm{NN}}(R_N) \tag{2.6}$$

The calculation of these energies, $V_i^{tot}(R_N)$, is termed 'electronic structure calculation.' When energies calculated at different configuration spaces **R** are clubbed together, they form the potential energy curve (PEC)for diatoms and PES for polyatomic systems. However, calculating these electronic energy terms, even at a single point in configuration space, let alone the whole reaction space is complex and computationally tedious. Bypassing this bottleneck requires fitting PES with analytical functions. **The task of fitting PES with an analytical form makes up the first part of this thesis.**

Now, as discussed, electronic wave functions are parametrically dependent on R_N, meaning that at each (different) value of nuclear positions (R_N), there exists a different set of equations $\Phi_i(r_e; R_N)$. This set of electronic wave functions $\Phi_i(r_e; R_N)$ form the complete basis set, and the total molecular wave function can be written as a sum of these electronic basis sets as follows.

$$\psi(r_e, R_N) = \sum_i \Phi_i(r_e; R_N)\mathcal{X}_{\mathrm{i}}(R_N) \tag{2.7}$$

Here, $\mathcal{X}_i(R_N)$ is the coefficient of i^{th} electronic state, which is also called the nuclear wave functions.

The above formulation can be used to solve the molecular SE (Eq. 2.2) by putting $\psi(r_e, R_N)$ from Eq. 2.7 to Eq. 2.2 as follows.

$$\sum_i (\hat{H}_{\mathrm{N}} + \hat{H}_{\mathrm{e}} + \hat{V}_{\mathrm{NN}})\Phi_i(r_e; R_N)\mathcal{X}_i(R_N) = \sum_i E\Phi_i(r_e; R_N)\mathcal{X}_i(R_N)$$

Left multiplying the above equation by $\Phi_j(r_e; R_N)$ and integrating in electronic space gives the following form. A detailed discussion of it can be found in Appendix B.

$$(\hat{T}_{\mathrm{N}} + V_j^{tot} - E)\mathcal{X}_j(R_N) - \sum_i \hat{\Lambda}_{ji}\mathcal{X}_i(R_N) = 0 \tag{2.8}$$

Here, $\hat{\Lambda}_{ji}$ in non-adiabatic coupling between j^{th} and i^{th} electronic states.

$$\hat{\Lambda}_{ji} = \frac{1}{2m}(2\hat{F}_{ji} \cdot \nabla + \hat{G}_{ji})$$

Here,

$$\hat{F}_{ji} = \langle \Phi_j(r_e; R_N)|\nabla|\Phi_i(r_e; R_N)\rangle$$

$$\hat{G}_{ji} = \langle \Phi_j(r_e; R_N)|\nabla^2|\Phi_i(r_e; R_N)\rangle$$

The two operators, $\hat{F}_{ji}$ and $\hat{G}_{ji}$, are the coupling between the two electronic states through the nuclear motion. The $\hat{F}_{ji}$ is derivative coupling, which is vector in nature, whereas $\hat{G}_{ji}$ coupling is scalar. The derivative coupling $\hat{F}_{ji}(R_N)$ is calculated using the Hellmann-Feynman theorem [2, 3].

$$F_{ji}(R_N) = \frac{< \Phi_j(r_e; R_N)|\nabla \hat{H}_e(r_e; R_N)|\Phi_i(r_e; R_N) >}{V_j(R_N) - V_i(R_N)} \tag{2.9}$$

The Eq. 2.8 can be written in a matrix form as follows.

$$\hat{\mathbf{T}}_N \mathcal{X} + (\mathbf{V}^{tot} - \mathbf{E})\mathcal{X} - \hat{\mathbf{\Lambda}}\mathcal{X} = 0 \tag{2.10}$$

For a n electronic state system, the $\mathcal{X}$ is $(n{\times}1)$ column matrix, $\mathbf{V}^{tot}$ is $(n{\times}n)$ matrix with only diagonal elements, $\hat{\mathbf{\Lambda}}$ is also a $(n{\times}n)$ matrix where diagonal elements are coupling between the same electronic state and non-diagonal terms which are also called non-adiabatic coupling terms (NACT), are coupling between two different electronic states. Now, based on different approximations, the Eqs. 2.10 takes different forms as follows.

1. **Born-Oppenheimer adiabatic approximation or adiabatic approximation:** This is the case when $\hat{\mathbf{\Lambda}} = 0$. It means that nuclear coupling between two same and different electronic states is zero.
$$\hat{\mathbf{T}}_N \mathcal{X} + (\mathbf{V}^{tot} - \mathbf{E})\mathcal{X} = 0 \tag{2.11}$$
2. $\hat{\mathbf{\Lambda}}_{ji} = 0;\ \forall\ j{\neq}i$, meaning that all NACTs are zero or nuclear coupling between two different states is zero.

Eq. 2.11 gives us the total nuclear energy E at point R_N and nuclear wave function $\mathcal{X}_{\mathrm{i}}(R_N)$ in R_N coordinate space for i^{th} electronic state. In reality, the total molecular or nuclear wave function of a system evolves over time. The evolution of the molecular system in time is one of the important fields of study, viz. Molecular dynamics. The time evolution can be investigated using classical and quantum mechanical methods. **These dynamical investigations for a triatomic system involving atom and diatom reactions make up the second part of this thesis.**

2.3 Fitting of Potential Energy Surface

As discussed, representing potential surfaces using analytical, functional forms saves computational efforts. This analytical function for the PES or PES can be obtained using different fitting methods. These methods are unique based on the choice of analytical function and optimization algorithms. These functions are fitted to the *ab initio* data, which are generated either for the complete molecule or different smaller fragments of it, a choice that depends on the representation of the PES. This representation of PES is a way to represent the total potential of the system analytically. There are different representations of the potential energy functions, such as Global representation, Many-body expansion (MBE), Double many-body expansion (DMBE), etc. In the 'global representation' approach, the potential of a system is represented by one term, the full molecule (and molecular potential) itself. Here, the analytical form of the total potential is obtained by fitting the *ab initio* points representing the full molecular potential. The MBE approach of Murrel and Sorrier [4] and the method of Varandas [5], DMBE, both represent the total molecular potential as a sum of smaller fragments.

Once the method to represent the PES is decided, the next step is to calculate the *ab initio* points. These potential values are calculated at different grid points. The choice of coordinates is critical because the functional form of the PES has a few requirements that depend largely on the choice of coordinates. The potential (V) should follow the invariant properties of physics. The invariant properties of the potential mean that the potential should not change for the configurations that appear different but are essentially the same under certain symmetry operations. This invariant nature of the potential function is introduced by using the coordinates with inbuilt invariant properties, which are as follows.

1. **Translational and rotational invariance**: The potential of the system (V) shouldn't change with the rotation and translation of the whole system. In other words, the coordinates should be invariant to the translation and rotation of the molecular system.
2. **Invariance with the permutation of like atoms:** The potential (V) of the system should remain invariant (unchanged) as the identical atoms in the molecule are interchanged [6].

When it comes to the invariance with respect to the permutation of like atoms, there are successful attempts to fit the potential surface by employing the coordinates that were not permutationally invariant [7, 8] and including configurations with similar atoms permuted among each other. The only drawback is that the fitting algorithm takes these similar configurations (in reality) as different data points having the same energies, and thus, these otherwise reluctant data points crowd the fitting space. This is not a major issue if the molecular system is small, but the process soon turns complex with the increasing number of atoms. Another issue with the above approach is that each point and region of space is fitted independently with their own local errors, meaning that two similar but similar atom permuted geometries fit differently. Thus, it is preferred to use the permutationally invariant coordinate system [6].

Different coordinate systems behave differently under the operations given above. The invariant properties of these coordinate systems are given in Table 2.1. To incorporate this

Table 2.1: Cartesian coordinates and invariant properties

Coordinate system	**Translation**	**Rotation**	**Permutation of same atoms**
Cartesian coordinates	No	No	No
Internal coordinates	Yes	Yes	No
Jacobi coordinates	Yes	Yes	No

property of permutation invariance in coordinates, permutationally invariant polynomials (PIPs) of Bowman et al. [6] are used.

Once the coordinate system is decided and *ab initio* energies are calculated, the next step is to fit the *ab initio* data with an accurate function analytically. This is known as the potential fitting, a curve fitting or a regression problem (refer to Appendix C). This curve-fitting problem has two parts: one is the choice of mathematical functions to represent the analytical potential, and the second is the optimization algorithm. Several mathematical functions and optimization algorithms are available in the literature, and the categorization of these fitting methods can be made in several ways. One such way is the interpolative or non-interpolative nature of the fitting procedures. A short description is given below.

1. **Interpolative methods**

   These fitting methods fit the data points so that the fitted curve passes exactly or very closely through the input data points, thus making the error of fitting negligibly small at the data points. When using these methods, the quality of the fit depends on the number of data points, and also any outlier in the input data reduces the accuracy of the fit. A few such methods are as follows.

   a) Splines
   b) Interpolating moving least square (IMLS)
   c) Reproducing Kernel Hilbert space (RKHS)
   d) Modified Shepherd interpolation

2. **Non-interpolative methods**

   Non-interpolative methods provide the fit that doesn't necessarily pass through all the data points but chooses the best path by reducing the error at the input data points. The error that is minimized can be root mean square error (RMSE), mean square error (MSE), etc. Examples of such methods are as follows.

a) Least square approach
b) Neural network (NN) approach
c) Permutation invariant polynomials (PIP) method

2.3.1 Machine learned fitting approach

The fitting of PES is a problem of multidimensional curve fitting. Now, when it comes to fitting a curve, it is usually done by choosing a suitable function or set of functions that follow the general behavior of the curve and fine-tuning the parameters associated with the functions. The choice of a function having a general behavior results in an intrinsic bias in the fitting procedure. When the curve is of low dimensionality, it is easy to find a functional form. However, as the dimensionality of the curve increases, finding the function that follows the behavior of the potential surface starts getting difficult. These bottlenecks in constructing PES make using machine learning methods lucrative.

So, **what is Machine learning?** [9]
It is the ability of machines (computers) to learn with negligible or no human intervention. It learns from the labeled or unlabelled data and finds a pattern. When the data is labeled, it is called supervised learning, and when unlabeled data is used, learning is unsupervised learning. There are different ML algorithms, and artificial NN is one of them.

The idea of ANN is inspired by the brain's learning process. Like the brain, it is a network of neurons that are interconnected with each other. Each neuron takes the inputs, weighs them, sums them up, and after passing the sum through a mathematical filter, it outputs the result. An ANN can have multiple layers, each composed of several neurons. The neurons of a layer are connected with the neurons of the previous and next layers but not with each other. A network with only one layer is called a perceptron, and it is a multilayer perceptron (MLP) that is called a NN. A short description of the Perceptron is as follows.

Perceptron

Perceptron is an algorithm used in supervised learning to solve binary classification problems. The concept of the Perceptron was invented by Frank Rosenblatt in 1957 based on the mathematical rules of McCulloch-Pitts (MCP) neurons [10].

It is composed of an input layer, weights associated with them, a bias neuron, and an activation function. These components combined together give the following output.

$$\text{Output} = f(b + \Sigma_i w_i x_i) \tag{2.12}$$

Here, x_i's are the input values, w_i's are weights associated with the input values, b is the

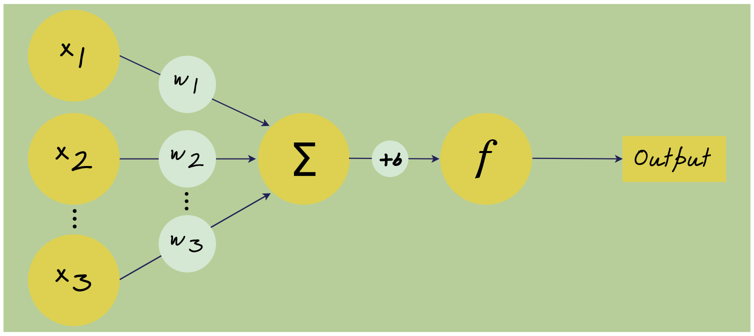


Figure 2.1: Diagram of Perceptron.

value of bias term, and f is the activation function. A short description of the activation function is as follows.

Activation function

It is a mathematical function that accepts the input data as a variable and, depending on the type of function outputs the modified or unchanged data. Eq. 2.12 defines the use of the activation function. It is clear that the term inside the bracket $(b+\Sigma_i w_i x_i)$ is linear in nature, and it is the activation function (f) which provides the non-linear character to the output. This introduction of non-linearity is what makes Perceptron, and as will be discussed later, NN, so good at fitting unknown data of any nature. There are different activation functions, and a brief discussion of some of them is given below.

Table 2.2: Activation Functions.

Activation function	**Mathematical form /**$f(x)$	**Derivative form /**$f'(x)$
Linear function	x	1
Logistic function	$1/(1+e^x)$	$e^{-x}/(1+e^{-x})^2$
Hyperbolic tangent function (tanh)	$(1-e^{-2x})/(1+e^{2x})$	$4e^{-2x}/(1+e^{-2x})^2$
Softplus function	$ln(1+e^x)$	$1/(1+e^{-x})$
Binary step function	$\begin{cases} 0, & x<0 \\ 0, & x \geq 0 \end{cases}$	$\begin{cases} 0, & x<0 \\ \text{DNE}, & x=0 \\ 0, & x>0 \end{cases}$
Rectified linear unit (ReLu)	$\begin{cases} 0, & x<0 \\ x, & x \geq 0 \end{cases}$	$\begin{cases} 0, & x<0 \\ \text{DNE}, & x=0 \\ 1, & x>0 \end{cases}$
Arctan	$\tan^{-1}(x)$	$1/(1+x^2)$

As seen from the Table 2.2 and Fig. 2.2, different activation functions have different behav-

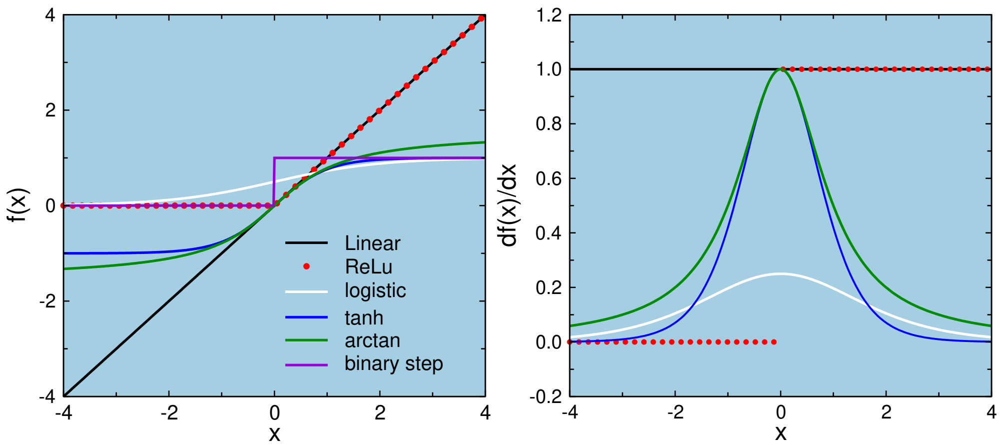


Figure 2.2: Activation function and their derivatives

iors and act as filters. The linear activation function makes no changes in the value passing through it, whereas the binary step function outputs either zero (0) or one (1) based on the value passing through it. Another broad category of functions is called 'sigmoid function'. These functions, such as logistic, tanh, arctan, etc., look like S-curves and follow a continuous curve between two boundaries, e.g., 0 to 1 or -1 to 1. Activation functions are an important part of the learning process, the choice of which depends on many factors, such as the optimization algorithm used in the learning process and the requirement of the problem.

1. A network that is expected to give output both in the -ve and +ve values, such as NN to fit potentials, can't be trained properly by using a sigmoid or binary type of activation function as an activation function of the final layer.

When there is more than one choice of activation function, not all functions will behave the same, and the overall training accuracy depends largely on the activation function, among other parameters. Often, more than one activation function can be used in a problem, but not all activation functions behave the same.

2.3.1.1 Neural network

A NN is made up of more than one Perceptron. It has one input layer, one output layer, and hidden layers. A brief description is as follows.

1. **Input layer**
   It is the first layer of the NN. It takes the input data directly from the neurons. These are special neurons that directly output the data given to them without any modification. An input layer can have multiple neurons, and these neurons are also called features.

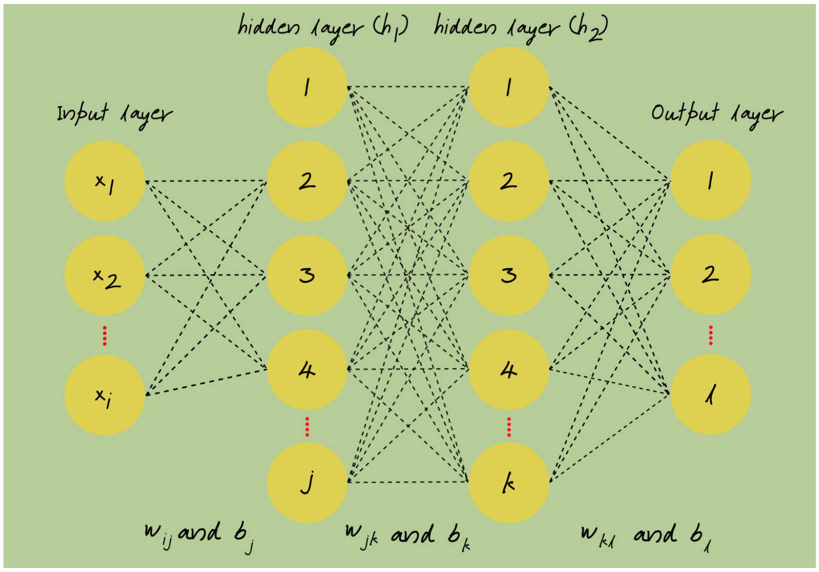


Figure 2.3: Architecture of a NN.

The number of features depends on the input parameter in a data set. *Example:* data for diatomic potential has one feature, that is, the interatomic distance, and data for triatomic potential have three interatomic distances as the features.

2. **Output layer**

   The output layer is the outermost or the final layer of the network. It receives the data from the previous layers of the network and outputs it. The total neurons in this layer depend on the total outputs in a data set; e.g., a binary classification problem and adiabatic potential fitting will have only one output neuron, whereas a diabatic potential fitting problem will have two output neurons.

3. **Hidden layer**

   Hidden layers are sandwiched between the input and output layers. There can be any number of hidden layers in a network. There are no restrictions on the number of neurons in a hidden layer, but the accuracy of the training process indirectly controls their number. As far as their work is concerned, it is by far the most uncertain in the learning of the NN.

Types of neural network

The working of the NN depends on the connection between different layers and the sharing of data among them. Based on these factors, there are several types of NNs, and a few of these are as follows.

1. Feed-forward or artificial NN (FNN/ ANN)
2. Convolution NN (CNN)

3. Recurrent NN (RNN)

In our work, FNNs are used in potential fitting. A detailed discussion is as follows.

In FNN, the neurons of a layer are connected (with some weighting factor) to the neurons of layers before and after, but there is no direct connection between the neurons in the same layer. Here, the information flows in the forward direction so that the neurons of a layer feed the data to the next layer but not to the previous one. As for the number of hidden layers, there is another term associated with the NN called 'deep NN', which is usually a network with more than one hidden layer. It is usually believed that the network becomes deeper with an increasing number of hidden layers. However, the definition of the deep NN is becoming blurry because, nowadays, it is quite common to have NNs with multiple hidden layers stacked.

The working and outcome of the NN depend on all the network's neurons. These neurons combine together and give the final output, which has the following mathematical form. The detailed discussion can be found in Appendix D.

$$\text{Output}_l = f^o \left[b_l^o + \sum_k w_{kl}^o f^{h2} \left(b_k^{h2} + \sum_j w_{jk}^{h2} f^{h1} \left(b_j^{h1} + \sum_i w_{ij}^{h1} x_i \right) \right) \right] \tag{2.13}$$

Here, x_i's are the values of the input neurons, w_{ij}^{h1}, w_{jk}^{h2}, and w_{lk}^{o} are the weights associated with the first hidden layer-input layer, second hidden layer - first hidden layer, and output layer - second hidden layer, respectively. The parameters b_j^{h1}, b_k^{h2}, and b_l^o are biases associated with the first hidden layer, second hidden layer, and the output layer. f^{h1}, f^{h2}, and f^o are the activation functions of the first hidden, second hidden, and output layers, respectively. The variables i, j, k, and l are the number of neurons in the input, first hidden, second hidden, and output layers. Now, in the case of adiabatic potential fitting for a triatomic system, the number of neurons in the input and output layers are 3 and 1, respectively. The activation function used in the final (output) layer is linear, and thus, the final fitted potential looks as follows.

$$\mathrm{V}^{fit} = b_1^O + \sum_k w_{k1}^O f^{h2} \left(b_k^{h2} + \sum_j w_{jk}^{h2} f^{h1} \left(b_j^{h1} + \sum_i^3 w_{ij}^{h1} x_i \right) \right) \tag{2.14}$$

The reason potential fit uses a linear activation or no activation function in the output layer is that the final potential is not scaled, and it has potential values spanning negative and positive numbers. If any activation function of Sigmoid type or any other form that restricts the output in a fixed range is used, it will result in errors.

2.3.1.2 Process of learning

Training in machine learning is a process of how a machine or a model learns. Learning is an optimization process to find the minima or maxima of a function. However, it is the minimization of function that is performed in potential fitting problems. One starts with initial values of weights and biases, and as the training progresses, these parameters are optimized to minimize the error between the training and training output. There are different optimization algorithms or optimizers used in literature for the purpose; some of them are as follows [9].

1. Gradient descent
2. Stochastic gradient descent (SGD)
3. Levenberg Marquardt (LM)
4. Adam optimizer
5. Adagrad
6. Mini-Batch Gradient Descent
7. Newton's Method
8. Particle swarm
9. Random search
10. Genetic algorithm
11. Powell's Method
12. Simulated annealing
13. Differential evolution

The algorithms given above are broadly divided into two categories: 'gradient-based' algorithms (1-7 in the above list) and 'gradient-free' algorithms (7-13 in the above list). Gradient-based algorithms use the derivative or gradient value to perform optimization, whereas gradient-free algorithms don't need gradient information for optimization. These gradient-free optimization methods are used in problems where the derivatives are complex or not defined.

Note: Often, the derivative-based algorithm uses backpropagation to calculate the derivative (partial), and this derivative-calculating algorithm, along with the optimization algorithms, which use the calculated derivative, is collectively called learning. Derivative-free optimizers don't use backpropagation.

2.3.1.3 Hyperparameters in a machine learning models

A machine learning model has different parameters. Some of these parameters, like weights and biases, are converged during the training. However, a model depends on many factors other than just these. The other parameters are called Hyperparameters. These are the parameters that don't get optimized during the single training instance. Hyperparameters are equally important for the model and affect its accuracy. There are two types of hyperparameters.

1. **Model Hyperparameters**
   These parameters are dependent on the machine learning model. E.g., number of hidden layers and neurons, activation function, initial weights and biases, and dropout.
2. **Optimizer Hyperparameters**
   These parameters depend on the learning algorithm. E.g., learning rate, Momentum, number of epochs, batch size, etc.

A brief description of these hyperparameters is given below.

A. **Initial weights and biases**
   Initial weights and biases decide the starting point of the training. In the case of potential fitting, a completely different set of initial weights and biases can affect the overall accuracy of the model. There are various ways weights and biases can be initialized in an NN model, such as zero, one, random values, etc.

B. **Learning rate**
   The optimization or learning of a model is a function minimization problem. The model starts from a random point in space, controlled by the weights and biases, and reaches the point of minimum. The step size used by the optimization algorithm to reach the point of minima is controlled by a factor called learning rate. The choice of learning rate is a complicated decision, and only an educated guess can be made. Neither a low nor a high value of learning rates are suitable for the training of a model because the lower the learning rate, the higher the chances that the model can get trapped in a local minima and take a huge amount of time to converge and higher the learning rate, greater are the chances of jumping over and missing the global minima.

C. **Epoch**
   The number of iterations or cycles of optimizing process during the training is called an epoch. The number of epochs affects the accuracy of the model; an excessively high number of epochs increases the chances of overfitting, and a low number of epochs can result in an underfit model.

Note:

These parameters are not optimized for a single instance of training and have to be optimized manually. However, different machine learning modules or tools have their own hyperparameter optimizers. These optimizers automatically run several instances of training by randomly selecting different combinations of hyperparameter values. The optimized hyperparameters are chosen from the best-trained model.

Bottlenecks in NN training

NN algorithms are excellent when it comes to fitting the data used in the training and predicting the unknown from the training space. However, these trained models are not always as accurate when predicting the outputs for unknown data. There are two terms used for such accuracy fluctuation: Overfitting and Underfitting.

1. **Underfitting**
   When the NN model doesn't learn the relationship between input and output data in the training set accurately.
2. **Overfitting**
   It is the situation when the model captures the relationship accurately for the training set, but when it comes to the unknown data, it doesn't perform as accurately.

A user can make the decision about the state of the model being underfitted or overfitted by comparing the 'accuracy vs. epochs' or 'error vs. epochs' curves (learning curve) of training and validation sets. Ideally, learning curves for the training and validation set converge with each other for a perfectly trained model. So, the model is deemed overfit when the learning curves of training and validation sets diverge from each other, such that the error of the validation set starts increasing as compared to that of the training set. In the case of underfitting, the two learning curves don't converge, and the training stops while the learning curve for the training set is still decreasing. These two instances are shown in Fig. 2.4.

2.3.1.4 How to deal with overfitting?

As already discussed, a complex model can cause overfitting during the training of a network. The process to avoid overfitting is termed regularization. There are many techniques for regularizing or regularizing network training. A brief discussion of some of these approaches is as follows.

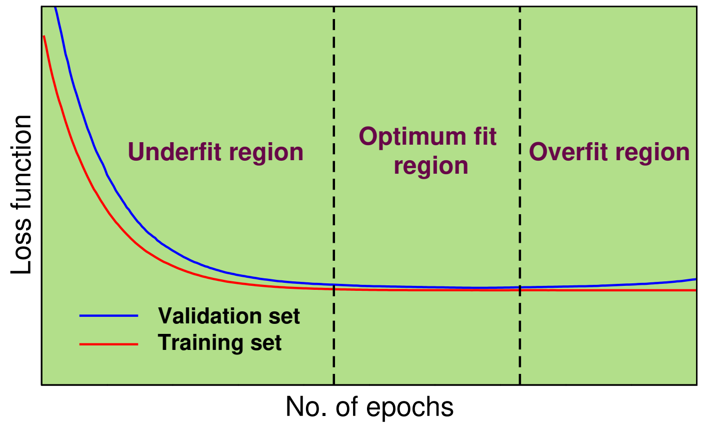


Figure 2.4: Underfitting, optimum fitting, and overfitting during the training.

1. **Early stopping**

   Overfitting is a situation when the model performs better with the training set but not with the unknown data. During the training, this can be confirmed by increasing (decreasing) the error (accuracy) value of the validation set to the training set. One can stop the model from overfitting by stopping the training process as the validation error (accuracy) of an epoch starts increasing (decreasing) compared to the validation error from the previous epoch. However, generally, the training is not stopped immediately after the validation error starts increasing; instead, a patience factor (p) is used, and training is stopped only when the validation error doesn't improve for p epochs after the epoch where error starts increasing.

   E.g., Let's say that in the case of AB diatoms, the first increase of validation error takes place at n^{th} epoch; here, loss increases from 0.0342 unit to 0.0416 unit. The patience value p is 10. Now, the training doesn't stop here, but it checks for loss of next p = 10 epochs after 0.0342 unit, which is 0.0416, 0.0414, 0.0449, 0.1037, 0.0852, 0.0352, 0.0625, 0.0510, 0.0947, and 0.0386 (10^{th} step after n^{th} step). It is found that the loss of $(n+10)^{\text{th}}$ epoch is 0.0386 unit, which is still higher than 0.0342 unit. Finding no improvement, training stops at this point. This process of stopping the training before the specified number of epochs is called early stopping.

   Note:

   Sometimes, the weights and biases are restored to the epoch where the first increase takes place, n^{th} epoch having error 0.0342 unit in the above example. However, this is optional, and the user may choose not to restore the weights and biases and take them the same as they are for $(n+p)^{\text{th}}$ epoch.

2. **Dropout**

   Dropout is the technique of randomly dropping out the connections between neurons while training [11].

2.3.1.5 Steps to fit potentials using NN

NN can be used to fit the electronic potential of diatomic, triatomic, or any polyatomic species. Once the potential data is available, following are the steps.

1. **Scaling of data** (optional)
   Once the fitting data is available, both the input (or features) and output can be scaled either between 0 to 1, -1 to 1, or any other range based on the different scaling approaches, such as data normalization, data standardization, etc. Scaling data is not a compulsory step, and based on learning algorithms, it may drastically affect the learning or not at all. Especially in the case of NN, with gradient-based optimization schemes, such as gradient descent, etc., scaling makes the training a lot faster and more accurate.
   In the case of potential fit, the potential range can be very broad: low potential at high bond distances and high potential at short distances. This huge range can destabilize the network because now the algorithm has to tune the weights and biases while taking care of both the high and low end of potential. Scaling the potential brings the points a lot closer and helps in the convergence of the network.
2. **Splitting of data set**
   NN or any ML algorithm, as already discussed, has to overcome the bottlenecks of overfitting and underfitting. Although this is done by employing different techniques, the litmus test for the success of these techniques is the comparison of loss function for training, validation, and test set. The complete data set is split either in 'training and validation sets' or 'training, validation, and test sets'. A short description of these different split sets is as follows.

   A. **Training set**
      The training set is the part of the data used by the learning algorithm to train. The learning algorithm sees this set, learns the pattern, and minimizes the error or maximizes the accuracy.
   B. **Validation set**
      This is the part of the data the learning algorithm sees but doesn't get trained on.
   C. **Test set**
      The test set is the data that is neither involved in the training nor seen by the model. It is kept aside before constructing the model and used by the user as additional testing for the accuracy of the model.

   Note: It is to be pointed out that the ratio of training, validation, and test sets is not fixed.

Also, one can train the network the network by splitting the full data set only into training and validation sets, leaving the test set empty.

3. **Construction of NN**
The choice of the NN is an important step in learning. It involves choosing the number of layers and number of neurons in each layer, allotting activation functions to all the layers, excluding the input layer, and initialing weights and biases. The neurons in the input layer are decided based on the features in the data, which are the number of independent variables in the input data. In the case of a diatomic curve fitting, interatomic distance is the only feature. In the case of triatomic molecular potential fitting, the number of features will be three interatomic distances or three degrees of freedom.

4. **Choice of hyperparameters**
There are different hyperparameters when it comes to training a NN. Some of these hyperparameters have already been chosen while constructing a network before starting the training. The other hyperparameters, such as optimization algorithm, learning rate, batch size, and number of epochs, are decided.

5. **Training**
Once the data is scaled and the network is ready with all the hyperparameters set, training can be started using any of the optimization algorithms. Training is performed until the algorithm has been trained for the number of cycles equal to epochs. In the case of the early stopping method being used and also invoked during the learning, the training stops before the end of a total number of epochs.

It is often the case that training may not produce the best network right away and may need multiple instances of training. These multiple instances can be completely random or can be repeated if the weight initialization is not randomized.

2.3.2 Aguado Paniagua function for fitting

Aguado Paniagua function was purposed by Aguado and Paniagua [12]. The functional form was proposed for two-body and three-body fragments in MBE representation. The functional form of two body potentials is as follows.

$$V^2_{\mathrm{AB}}(R_{\mathrm{AB}}) = \frac{c_0 e^{-\alpha_{\mathrm{AB}} R_{\mathrm{AB}}}}{R_{\mathrm{AB}}} + \sum_{i=1}^{N} c_i \rho^i_{\mathrm{AB}} \tag{2.15}$$

Here, $\rho_{\mathrm{AB}} = R_{\mathrm{AB}} e^{-\beta^{(2)}_{\mathrm{AB}} R_{\mathrm{AB}}}$, $c_i's$ are linear parameters, α_{AB}, and $\beta^{(2)}_{\mathrm{AB}}$ are nonlinear parameters. In the case of the three-body fragments in the MBE, the function is as follows.

$$V^3_{\mathrm{ABC}}(R_{\mathrm{AB}}, R_{\mathrm{BC}}, R_{\mathrm{AC}}) = \sum_{i,j,k}^{M} d_{ijk} \rho^i_{\mathrm{AB}} \rho^j_{\mathrm{BC}} \rho^k_{\mathrm{AC}} \tag{2.16}$$

Here, $i + j + k \neq i \neq j \neq k$ and $i + j + j \leq$ M. Parameters ρ_{AB}, ρ_{BC}, and ρ_{AC} have the mathematical form: $\rho_{\text{AB}} = R_{\text{AB}} e^{-\beta^{(3)}_{\text{AB}} R_{\text{AB}}}$, where $\beta^{(3)}_{\text{AB}}$ is different than $\beta^{(2)}_{\text{AB}}$.

Both the two-body and three-body functions are non-linear in nature, and various optimization schemes, such as gradient descent, Levenberg Marquardt scheme, etc., can be used to fit the data with analytical form or, in other words, to minimize the loss function.

2.3.2.1 Steps to fit potential using Aguado Paniagua function

The steps to fit potential using the Aguado Paniagua function are almost the same as the steps in the NN method, except that no scaling is needed. Once the data curation is done according to the need of the MBE approach, as discussed already, the fitting can be performed using any non-linear regression optimization algorithm.

2.4 Nuclear Dynamics of Triatomic Molecular System

As discussed earlier, the dynamics of the system is its motion in time. In the context of any reaction, it is the transformation of reactants into products. The problems taken up in this thesis involve three atoms in the form of atom-diatom collisions. A collision between the atom (A) and diatom (BC) can result in different products, such as

$$
\begin{aligned}
\text{A} + \text{BC}(v, j) &\rightarrow \text{BC}(v, j) + \text{A} && \text{elastic scattering,}\\
&\rightarrow \text{BC}(v', j') + \text{A} && \text{inelastic scattering,}\\
&\rightarrow \text{AB}(v', j') + \text{C} && \text{reactive scattering,}\\
&\rightarrow \text{AC}(v', j') + \text{B} && \text{reactive scattering,}\\
&\rightarrow \text{A} + \text{B} + \text{C} && \text{dissociative scattering.}
\end{aligned}
$$

Here, (v, j) and (v', j') depict the initial and final rovibrational level of the reactant and product diatoms, respectively. The reactions shown above are self-explanatory and are the different possible channels of atom-diatom collision. Although these reaction channels are equally important, the atom and diatomic collisions of reactive scattering type are discussed in detail here.

There are two ways to study the dynamics: one involves quantum treatment, and the other involves classical treatment of the propagation.

2.4.1 Quantum mechanical method

In the QM method, Eqs. 2.10 and 2.11 are solved to study the dynamics of reactions using quantum mechanical equations and rules. The Eq. 2.10 solves the equation of motion nonadiabatically by including more than one electronic state in the dynamics. To study the dynamics of the reaction on a single state, the adiabatic equation, Eq. 2.11, is solved.

The TDSE for nucleus is expressed as

$$i\hbar\frac{\chi(\vec{R}_N,t)}{\partial t} = \hat{H}_N\chi(\vec{R}_N,t) \tag{2.17}$$

Starting from the Eq. 2.17, the wave function at the time t is

$$\chi(\vec{R}_N,t) = exp\left(\frac{-i\hat{H}_N t}{\hbar}\right)\chi(\vec{R}_N,0), \tag{2.18}$$

where $\chi(\vec{R}_N,0)$ is the nuclear wave function at initial time $t = 0$ and $exp\left(\frac{-i\hat{H}_N t}{\hbar}\right)$ is the time evolution operator $\hat{U}(t,t_0 = 0)$. This operator is a unitary operator ($UU^\dagger = 1$), which acts on the wave function at the initial time ($t = 0$) and outputs the wave packet at a later time t. When implementing the Eq. 2.17 in computation, total time t is used as grid points of Δt time length, and the propagation is done by iteratively applying $exp\left(\frac{-i\hat{H}_N\Delta t}{\hbar}\right)$ as follows.

$$\hat{U}(t,t_0 = 0) = exp\left(\frac{-i\hat{H}t}{\hbar}\right) \Rightarrow exp\left(\frac{-i\hat{H}N\Delta t}{\hbar}\right)$$

$$= exp\left(\frac{-i\hat{H}}{\hbar}(\Delta t_1 + \Delta t_2 + \Delta t_3 + ...\Delta t_N)\right)$$

$$\hat{U}(t,t_0 = 0) = \Pi_{n=0}^{N-1}\hat{U}(\Delta t(n+1), n\Delta t)$$

2.4.1.1 Choice of Coordinates

The first step in the process of wave packet propagation is choosing the coordinates, which depends on the ease of solving the motion equation and on the analysis of the wave packet at the end of time evolution when the product forms. Among the different types of coordinate systems, the Jacobi coordinates are used in the work presented in this thesis. Jacobi coordinates can transform a system of n non-interacting particles from a n-particle problem to n, one-particle problems, a property which helps in simplifying the action of the Hamiltonian on the wave function [13]. In the case of A + BC → AB + C reaction, there are two different sets of Jacobi coordinates (*cf.* Fig. 2.5). One set (R_a, r_a, γ_a) describe the reactant channel and other (R_c, r_c, γ_c) describes the product channel.

Product coordinates are best suited to represent the product side, and reactant coordinates are best suited to the reactant side. This means that a single coordinate system among the two may not represent the whole reaction space. When only the initial state-selected observables are needed, the reactant Jacobi coordinates are enough, as discussed in Appendix E.

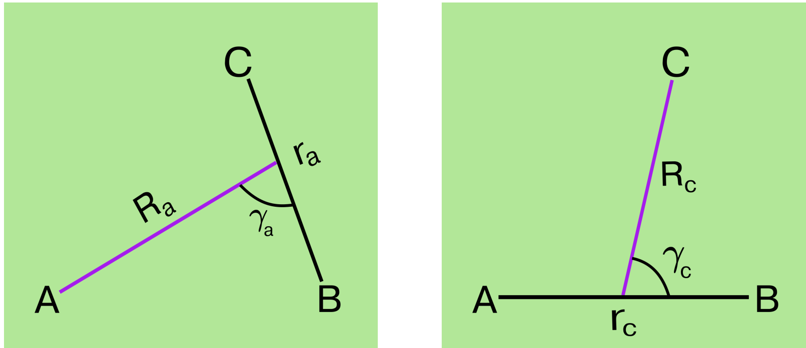


Figure 2.5: Jacobi coordinates for reactant (A + BC) and product (AB + C) channels.

However, regarding the state-to-state properties, product Jacobi coordinates are necessary [14]. This problem of choosing the proper coordinates is termed as 'coordinate-problem' in the reaction dynamics community. This notorious problem attracted immense interest resulting in the development of several methods [14–20]. Among these, the 'reactant-product decoupling' (RPD) method of Peng & Zhang [15] and modified RPD method of Althrope [17] attempt to avoid the coordinate-problem by decoupling the reactant and product space in a reaction. The real wave packet method (DIFFREALWAVE code) of Balint-Kurti [21] constructs the initial wave function using the reactant Jacobi coordinates and propagates the wave function in the product Jacobi. In this case, the coordinate system is transformed from reactant Jacobi to product just before the propagation starts. Some of the methods discussed require the transformation between the two coordinate systems. This mathematical transformation, which included the concept of Wigner rotation [22–24], is the major computational bottleneck.

In the quantum dynamical method used in the current thesis, DIFFREALWAVE is used [21, 25]. Different steps are discussed in detail below.

2.4.1.2 Construction of initial wave function

The coordinate system used in the propagation is body-fixed in nature, meaning that the coordinate system moves as the body moves, making it fixed with the body. Also, the z-axis of Jacobi's and R_a or R_c coincide with each other. This choice of coordinate system has its benefits, which will be discussed later in the text. Reactant Jacobi coordinates are used to construct the initial wave function. These coordinates are then transformed into the product Jacobi. The initial wave function for a (J, Ω) state where J and Ω are total angular momentum and reactant helicity quantum number in reactant Jacobi coordinates is [26]

$$\chi^{J,\Omega}(R_a, r_a, \gamma_a, t = t_0) = G(R_a - R_0)\chi^{\mathrm{BC}}_{v,j}(r_a, \gamma_a), \tag{2.19}$$

where $G(R_a - R_0)$ is a one-dimensional wave packet along the scattering coordinate (R_a), $\phi^{\mathrm{BC}}_{v,j}(r_a, \gamma_a)$ is the initial rovibrational state of the diatom BC, and R_0 is the center of the wave function. The function $\phi^{\mathrm{BC}}_{v,j}(r_a, \gamma_a) = \phi_{v,j}(r_a)P_j(\cos\gamma_a)$, where $\phi_{v,j}(r_a)$ is the vibrational wave function of the reactant diatom at rotational level j and $P_j(\cos\gamma_a)$ is the Legendre polynomial of degree j. $G(R_a - R_0)$ is a modified SINC function and expressed as [27]

$$G(R_a) = \frac{1}{\sqrt{\alpha\pi}} e^{-ik_0(R_a-R_0)} e^{-\beta(R_a-R_0)^2} \frac{sin(\alpha(R_a - R_0))}{R_a - R_0},$$

where α and β control the width and smoothness of the wave function, respectively, and $k_0 = \sqrt{2\mu E_{col}}$ is the initial momentum given to the wave function towards the interaction region. The combination of α, β, and k_0 is crucial for quantum dynamics. A detailed discussion of these parameters, along with their effect on the dynamics calculations, is given in section 4.2.3.1 of chapter 4. This initial wave packet in reactant Jacobi is converted into product Jacobi as [26]

$$\chi^{J,\Omega'}(R_c, r_c, \gamma_c, t = t_0) = \sum_{\Omega} \chi^{J,\Omega}(R_a, r_a, \gamma_a, t = t_0) \frac{R_c r_c}{R_a r_a} d^J_{\Omega\Omega'}(\beta).$$

Since the initial Ω value is fixed for a calculation, the above form of the equation is converted to the following form.

$$\chi^{J,\Omega'}(R_c, r_c, \gamma_c, t = t_0) = \chi^{J,\Omega}(R_a, r_a, \gamma_a, t = t_0) \frac{R_c r_c}{R_a r_a} d^J_{\Omega\Omega'}(\beta), \tag{2.20}$$

where, $d^J_{\Omega\Omega'}(\beta)$ is the reduced Wigner matrix elements and β is the angle between R_a and R_c. The reduced Wigner matrix element $d^J_{\Omega\Omega'}(\beta) = D^J_{\Omega\Omega'}(0\beta0)$, where, $D^J_{\Omega\Omega'}(0\beta0)$ is Wigner rotation matrix. The values of Ω' vary from $-J$ to $+J$, meaning that when a $\psi^{J,\Omega}$ state is converted to product Jacobi, $(2J + 1)$ $\psi^{J,\Omega'}$ states are generated and each of such states is expressed as Eq. 2.20. In the DIFFREALWAVE code, the wave packet propagation uses a wave packet in the form of Eq. 2.20 for propagation.

Grid representation of the wave packet

The wave packet in Eq. 2.20 is a continuous function, but when propagating it numerically, the wave packet is expressed on grids. A uniform grid is used for r and R variables, and a non-uniform grid where the grid points are the Gauss-Legendre quadrature points is used for the angular variable γ. The grid representation of the wave function is as follows.

$$\chi^{J,\Omega'}(R^i_c, r^j_c, \gamma^k_c, t) = \sqrt{\omega_k}\, \chi^{J,\Omega'}(R^i_c, r^j_c, \gamma^k_c, t) \tag{2.21}$$

Here, i, j, and k are the r, R, and γ grid points, respectively, and $\omega_k{}'$s are the weight associated with Gauss-Legendre quadrature grid points. In the DIFFREALWAVE code used in the investigations presented in this thesis, the γ grid is the same for all the K' values.

2.4.1.3 Time propagation

Solving Eq. 2.18 is solving the time propagation for a system. In the time evolution of the system, two essential conditions are as follows.

1. **Conservation of the norm of wave function**

   Let's consider the wave function $\mathcal{X}(R_N, t)$ at time t, that is found after applying the time evolution operator ($\hat{U}$) on the wave function at the initial time t_0. Starting from the norm of the wave function at time t [$\langle \mathcal{X}(R_N, t)|\mathcal{X}(R_N, t)\rangle$], the conservation of norm in the propagation can be proved as follows.

   $$\begin{aligned}\langle \mathcal{X}(R_N, t)|\mathcal{X}(R_N, t)\rangle &= \langle \hat{U}(t, t_0)\mathcal{X}(R_N, t_0)|\hat{U}(t, t_0)\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0)|\hat{U}^{\dagger}(t, t_0)\hat{U}(t, t_0)|\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0)|\hat{U}^{\dagger}(t, t_0)\hat{U}(t, t_0)|\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0)|\mathcal{X}(R_N, t_0)\rangle\end{aligned}$$

   It is clear from the above procedure that the norm of the wave function, $\langle \mathcal{X}(R_N, t)|\mathcal{X}(R_N, t)\rangle = \langle \mathcal{X}(R_N, t_0)|\mathcal{X}(R_N, t_0)\rangle$, is conserved during the time evolution.

2. **Conservation of the total energy of the system**

   Mathematically, the conservation of the energy is,

   $$\langle \mathcal{X}(R_N, t)|\hat{\mathrm{H}}|\mathcal{X}(R_N, t)\rangle = \langle \mathcal{X}(R_N, t_0)|\hat{H}|\mathcal{X}(R_N, t_0)\rangle.$$

   Let's take the left-hand side and proceed.

   $$\begin{aligned}\langle \mathcal{X}(R_N, t)|\hat{H}|\mathcal{X}(R_N, t)\rangle &= \langle \hat{U}(t, t_0)\mathcal{X}(R_N, t_0)|\hat{H}|\hat{U}(t, t_0)\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0|\hat{\mathrm{U}}^{\dagger}(t, t_0)\hat{H}|\hat{\mathrm{U}}(\mathrm{t},\mathrm{t_0})\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0)|\hat{U}^{\dagger}(t, t_0)\hat{U}(t, t_0)\hat{H}|\mathcal{X}(R_N, t_0)\rangle \\ &= \langle \mathcal{X}(R_N, t_0)|\hat{H}|\mathcal{X}(R_N, t_0)\rangle\end{aligned}$$

   Here, $\hat{H}\hat{U}(t, t_0) = \hat{U}(t, t_0)\hat{H}$ because $\left[\hat{H}, e^{-i\hat{H}_N\Delta t/\hbar}\right] = 0$. This relationship can be proved easily by expanding and checking the commutation relation of the time evolution operator $e^{-i\hat{H}_N\Delta t/\hbar}$ with $\hat{H}$.

   It is to be noted that the two conservation properties are a direct consequence of the unitary nature of the time evolution operator.

There are various approaches to approximate the time evolution operator, e.g., Split operator (SO) method, Chebyshev polynomial method, etc. Some of these methods are discussed below.

1. **Split operator method**

   The time evolution operator is an exponential function of $\hat{H}$, where $\hat{H} = \hat{T} + \hat{V}$. The $\hat{T}$ and $\hat{V}$ are kinetic energy and potential energy operators. In the SO scheme, the time evolution operator is split into potential and kinetic energy parts, and then both parts act on the wave function separately.

   $$\chi(R_N, t + \Delta t) = e^{-i\hat{H}\Delta t/\hbar}\chi(R_N, t) \Rightarrow e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}\chi(R_N, t) \tag{2.22}$$

   The above procedure is straight forward except that the $\hat{T}$ and $\hat{V}$ operators are not commutative: $e^{-i\hat{H}\Delta t/\hbar} \neq e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$. This introduces an error of order two in the computation.

   $$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar} + O(\Delta t^2) \tag{2.23}$$

   The term $O(\Delta t^2)$ in Eq. 2.23 is the error. It shows that the two forms of the time evolution operator, $e^{-i\hat{H}\Delta t/\hbar}$: the actual form of the operator and $e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$: operator form that acts on the wave function are not equivalent. This error contains Δt^2 and other higher-order terms. This error can further be reduced by splitting $\hat{T}$ or $\hat{V}$ in two symmetric terms. When $\hat{T}$ is split, it is termed as potential referenced, and when $\hat{V}$ is split, it is termed as kinetic energy referenced split operator method.

   $$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{T}\Delta t/2\hbar}e^{-i\hat{V}\Delta t/\hbar}e^{-i\hat{T}\Delta t/2\hbar} + O(\Delta t^3) \quad \text{Potential energy referenced}$$
   $$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{V}\Delta t/2\hbar}e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/2\hbar} + O(\Delta t^3) \quad \text{Kinetic energy referenced}$$

   This technique is called the second-order SO method. In this method, the error starts from the third order of Δt^3, details of which are discussed in the Appendix F. The results coming out of both the kinetic energy referenced and potential energy referenced methods are equivalent; the only difference is the computational cost, which in the case of the potential energy referenced method is high because the computationally expensive KE term acts twice.

2. **Chebyshev polynomial method**

   As has been discussed already, the idea behind solving the time propagation of a system is to express the time evolution operator in a simple form. The method discussed here uses the Chebyshev polynomials to express the time evolution operator.

   These Chebyshev polynomials are the solutions of the Chebyshev differential equation (see Appendix G for detailed discussion). A series of these polynomials of order n ($T_n(x)$) with a corresponding weight A_n can be used to expand any arbitrary continuous and single-valued function $f(ax)$ $\forall\, x \in$ [-1,1] as follows.

   $$f(ax) = \sum_{n=0}^{\infty} A_n(a)T_n(x) \tag{2.24}$$

$$f(ax) = A_0T_0(x) + A_1T_1(x) + A_3T_3(x) + ...A_nT_n(x).$$

This property of the Chebyshev polynomials is utilized to approximate the time evolution operator, which is a function of $\hat{H}$, as a sum of Chebyshev polynomials, given that $\hat{H}$ ranges from -1 to 1.

Now, before we start using the Chebyshev polynomial approach to expand and solve Eq. 2.18, it is to be pointed out that there are various modified forms of Eq. 2.18, which are used to perform the time propagation of a system. Here, a short discussion on these different approaches of Eq. 2.18 is presented.

A. **Complex wave packet approach**

Let's say that the time evolution operator operates on the wave function at initial time t and outputs the wave function at time $t + \Delta t$ [28]. This can be written as

$$\mathcal{X}(R_N, t + \Delta t) = exp\left(-\frac{i\hat{H}\Delta t}{\hbar}\right)\mathcal{X}(R_N, t). \tag{2.25}$$

Eq. 2.25 can simply be solved iteratively at each time step t to get the wave function information at the next time step $t + \Delta t$. However, while solving the equation, the exponential of the operator is not operated directly, but first, the exponential form is expanded using the Chebyshev polynomial, and that expanded form acts on wave function at time t. The Chebyshev expansion of the above form (*cf.* Appendix G) is written as

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = \sum_0^\infty i^n(2 - \delta_{n0})T_n(-\hat{H})J_n\left(\frac{t}{\hbar}\right) \qquad \text{or} \tag{2.26}$$

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = \sum_0^\infty (2 - \delta_{n0})Q_n(-i\hat{H})J_n\left(\frac{t}{\hbar}\right). \tag{2.27}$$

The Eq. 2.26 is the expansion of time evolution operator in real Chebyshev polynomials, and 2.27 is in complex Chebyshev polynomials. The $J_n\left(\frac{t}{\hbar}\right)$ are the Bessel functions of order n.

B. **Real wave packet approach**

Let's take the wave packet ($\mathcal{X}(R_N, t)$) and backpropagate it by applying the time evolution operator as

$$\mathcal{X}(R_N, t - \Delta t) = exp\left(\frac{i\hat{H}\Delta t}{\hbar}\right)\mathcal{X}(R_N, t). \tag{2.28}$$

Expressing $exp\left(\frac{i\hat{H}\Delta t}{\hbar}\right)$ as $\cos\left(\frac{i\hat{H}\Delta t}{\hbar}\right) + i\sin\left(\frac{i\hat{H}\Delta t}{\hbar}\right)$ in Eqs. 2.25 and 2.28 and adding both equations gives us the following expression.

$$\mathcal{X}(R_N, t + \Delta t) = 2\cos\left(\frac{\hat{H}\Delta t}{\hbar}\right)\mathcal{X}(R_N, t) - \mathcal{X}(R_N, t - \Delta t) \tag{2.29}$$

The wave function is complex in nature and can be written as real and imaginary fragments as $q(R_N, t) + ip(r_N, t)$. Splitting the wave function in Eq. 2.29 in real and imaginary parts and only considering the real part gives us the following form of the equation.

$$q(R_N, t + \Delta t) = 2\cos\left(\frac{\hat{H}\Delta t}{\hbar}\right) q(R_N, t) - q(R_N, t - \Delta t) \tag{2.30}$$

The Eq. 2.30 calculates the wave function at $t + \Delta t$ by using the wave function information at time t and $t - \Delta t$. To get the wave function information at time t, operate the time evolution operator on $q(R_N, t - \Delta t)$, expand the time evolution operator as sin and cosin, and only consider the real part of the expression, as expressed in the following equation.

$$q(t) = \cos\left(\frac{\hat{H}\Delta t}{\hbar}\right) q(t - \Delta t) + \sin\left(\frac{\hat{H}\Delta t}{\hbar}\right) p(t - \Delta t) \tag{2.31}$$

Here, Eqs. 2.30 and 2.31 are used for wave packet propagation.

Note:

In the Chebyshev method [28, 29], $f(\hat{H})$ is approximated according to Eq. 2.24. This is a straightforward process, but the problem is that the $\hat{H}$ is not bound in the [-1,1] range. So, before expanding the $f(\hat{H})$, $\hat{H}$ is scaled as:

$$\hat{H}_s = a_s \hat{H} + b_s, \tag{2.32}$$

where, $a_s = 2/\Delta E$, $b_s = -1 - a_s E_{min}$, and $\Delta E = E_{max} - E_{min}$. E_{max} and E_{min} are the maximum and minimum boundaries of $\hat{H}$. It can easily be seen from Eq. 2.32 and expressions of a_s and b_s that at E_{max} and E_{min} values of $\hat{H}$, $\hat{H}_s$ is equals to -1 and +1, respectively. In the complex wave packet and real wave packet approaches, the $f(\hat{H})$ is first expressed as $f(\hat{H}_s)$ and then expanded using the Chebyshev approximation.

C. **Modified real wave packet approach**

Solving the Eqs. 2.30 and 2.31, which use only the real part of the complex wave function to solve the TDSE, is as difficult as it is to solve the Eq. 2.25 for complex TDSE. This difficulty is tackled using an approach that solves the modified Schrödinger equation. A modified Schrödinger equation is written as [18, 26]

$$i\hbar \frac{\partial \chi_f(R_N, t)}{\partial t} = f(\hat{H}_s)\chi_f(R_N, t), \tag{2.33}$$

where, $f(\hat{H}_s) = -\frac{\hbar}{\Delta t}\cos^{-1}(\hat{H}_s)$ and $\chi_f(R_N, t)$ are the parts of modified SE equation. The $\hat{H}_s$ has the same meaning as specified in Eq. 2.32. The Eqs. 2.30 and 2.31, which are used to solve the time propagation, can be written as

$$q_f(R_N, t + \Delta t) = 2\hat{H}_s q_f(R_N, t) - q_f(R_N, t - \Delta t), \tag{2.34}$$

$$q_f(t) = \hat{H}_s q_f(t - \Delta t) - \sqrt{1 - \hat{H}_s^2}\, p_f(t - \Delta t). \tag{2.35}$$

Here, Eq. 2.35 is solved only once to get the $q_f(r_N, t)$ from $q_f(r_N, t - \Delta t)$ and then Eq. 2.34 is solved iteratively to get wave function at subsequent time steps. The operator form $\sqrt{1 - \hat{H}_s^2}$ in Eq. 2.35 is expanded using Chebyshev polynomials as

$$\sqrt{1 - \hat{H}_s^2} = \frac{2}{\pi}\left(1 - \sum_{n=1} \frac{2T_{2n}(\hat{H}_s)}{4n^2 - 1}\right).$$

The complete mathematical process to get the above expansion is discussed in Appendix G.

Note: In the Chebyshev polynomial-based schemes, the expansion is exactly equal to the time evolution operator only when $n \to \infty$. However, in principle, a relatively accurate approximation can be found with a few hundred terms only.

2.4.1.4 Action of Hamiltonian on the wave packet

In the time propagation, the time evolution operator, which is a function of $\hat{H}$, acts on the wave packet. In all the different approaches adopted to approximate the action of the time evolution operator, the action of the $\hat{H}$ is essential. In the modified Chebyshev approximation approach, the Hamiltonian is scaled, but that doesn't change the way different parts of the Hamiltonian act on the wave packet. For an atom + diatom (A + BC) collision, Hamiltonian has the following form in Jacobi coordinates [30].

$$\hat{H} = -\frac{1}{2\mu_R}\frac{\partial^2}{\partial^2 R} - \frac{1}{2\mu_r}\frac{\partial^2}{\partial^2 r} + \frac{\hat{j}^2}{2I_r} + \frac{\hat{L}^2}{2I_R} + \hat{V}_{elec} \tag{2.36}$$

In Eq. 2.36, the first two terms are the radial parts of the Hamiltonian, one along r and the other along R, $\hat{j}$ is diatomic rotational angular momentum operator, $\hat{L}$ is orbital angular momentum for diatom-atom system, $\hat{V}_{elec}$ is the electronic energy operator, while $I_r(= \mu_r r^2)$ and $I_R(= \mu_R R^2)$ are the moments of inertia along r and R, respectively. The Eq. 2.36 can be written in terms of total angular momentum operator $\hat{J}$ as follows. A detailed derivation is given in Appendix H.

$$\hat{H} = -\frac{\hbar^2}{2\mu_r}\frac{\partial^2}{\partial \mathrm{r}^2} - \frac{\hbar^2}{2\mu_R}\frac{\partial^2}{\partial R^2} + \left(\frac{1}{2I_r} + \frac{1}{2I_R}\right)\hat{j}^2 + \frac{(\hat{J}^2 - 2\hat{J}_z\hat{j}_z)}{2I_R} + \hat{V}_{elec} - \frac{(\hat{J}_+\hat{j}_- + \hat{J}_-\hat{j}_+)}{2I_R} \tag{2.37}$$

The first two terms in the Eq. 2.37 are the radial kinetic energy terms, the third term is the rotational angular momentum term, the fourth term is the total angular momentum and z component of it, and the last term contains the Coriolis couplings.

Different terms of the Hamiltonian act differently on the wave packet ($\psi^{J,\Omega'}(R_c, r_c, \gamma_c, t = t_0)$ or $q^{J,\Omega'}(R_c, r_c, \gamma_c, t)$ or $q_f^{J,\Omega}(R_c, r_c, \gamma_c, t)$) and their form and action on the wave packet depends on the coordinates system. In this case, the propagation is done in a body-fixed coordinate system, which is briefly discussed here and readers are referred to a work of Gabriel G. Balint-Kurti et al. (International Reviews in Physical Chemistry, 11(2):317–344, 1992) for the detailed discussion.

$$
\begin{aligned}
\hat{H}q^{J,\Omega'}(R_c, r_c, \gamma_c, t) = &-\left(\frac{\hbar^2}{2\mu_{r_c}}\frac{\partial^2}{\partial r_c^2} + \frac{\hbar^2}{2\mu_{R_c}}\frac{\partial^2}{\partial R_c^2}\right) q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \\
&+ \left(\frac{1}{2I_{r_c}} + \frac{1}{2I_{R_c}}\right)\left(-\frac{1}{\sin\gamma_c}\frac{\partial}{\partial \gamma_c}\sin\gamma_c\frac{\partial}{\partial \gamma_c} + \frac{\Omega'}{\sin^2\gamma_c}\right) q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \\
&+ \frac{(\hat{J}^2 - 2\hat{J}_z\hat{j}_z)}{2I_{R_c}} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) + \hat{V}_{elec} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \\
&- \frac{(\hat{J}_+\hat{j}_- + \hat{J}_-\hat{j}_+)}{2I_{R_c}} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \quad (2.38)
\end{aligned}
$$

Before we start the discussion about the action of individual terms in the Hamiltonian (Eq. 2.38) on the wave packet, we need to know that the local nature of an operator in a wave packet space makes it easy to calculate the action. In those cases where the operator is local in a wave packet space, the action of the operator on the wave packet is just the multiplication of the physical value of the operator and the wave packet. Whereas, in the case when the operator is not local in wave packet space, different collocation methods are used to transform the wave packet in a space where the operator is local. An example of such a non-local operator and wave packet is the action of radial terms of kinetic energy operator on the wave packet in the position space. Further discussion on the action of individual terms of the Hamiltonian is presented below.

1. **Action of potential**
   The fifth term in Eq. 2.38 is the electronic potential operator. This operator is local in the position coordinates. Thus, its action on the wave function at the grid points in position space is just the multiplication of electronic potential and corresponding wave packet at the grid points.

$$
\hat{V}_{elec} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) = V_{elec}(R_c, r_c, \gamma_c, t) q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \quad (2.39)
$$

2. **Action of radial terms**
   The first two terms in Eq. 2.38 are the radial terms. The radial kinetic energy operator terms are not local in the position space, but the terms can be made local when transformed in the momentum space. In the momentum coordinates, the radial KE operator

is expressed as $\frac{\hat{p}_k^2}{2\mu}$, where $\hat{p}_k$ is the momentum operator. The wave packet in the position space, $q^{J,\Omega'}(R_c, r_c, \gamma_c, t)$, is converted to the momentum space using a type of collocation method. In the present case, this method is the Fourier transformation (FT). Following are the steps to calculate the action of radial kinetic energy on wavepacket [31].

a. Fourier transforms the wave packet from the position to the momentum space. It is important to understand that information of the wave packet at all the grid points in position space is needed to get the information of the wave packet at one grid point in momentum space.

b. Multiply the transformed wave packet at all the momentum grid points with the radial kinetic energy operator term expressed at that momentum grid point. In the momentum space, the radial kinetic energy operator acts on wave packet [31] as follows.

$$\frac{\hat{p}_k^2}{2\mu}|q(k)\rangle = \frac{(\hbar k)^2}{2\mu}|q(k)\rangle$$

Here, $q(k)$'s are the wave packet in momentum space, which are obtained after Fourier transforms the wave packet from position space.

c. Inverse Fourier transform the above terms $\left(\frac{(\hbar k)^2}{2\mu}|q(k)\rangle\right)$ from the momentum to position space. As in step 1, information at all the grid points in momentum space is required to get the wave packet information at one grid point in position space.

These three steps can be combined as

$$-\frac{\hbar^2}{2\mu_r}\frac{\partial^2}{\partial r^2}q^{J,\Omega'}(R_c^i, r_c^j, \gamma_c^k, t) = \mathcal{F}^{-1}\left\{\frac{(\hbar k)^2}{2\mu_r}\mathcal{F}\left\{q^{J,\Omega'}(R_c^i, r_c^j, \gamma_c^k, t)\right\}\right\},$$

$$-\frac{\hbar^2}{2\mu_R}\frac{\partial^2}{\partial R^2}q^{J,\Omega'}(R_c^i, r_c^j, \gamma_c^k, t) = \mathcal{F}^{-1}\left\{\frac{(\hbar k)^2}{2\mu_R}\mathcal{F}\left\{q^{J,\Omega'}(R_c^i, r_c^j, \gamma_c^k, t)\right\}\right\},$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the FT and inverse FT, respectively. The k in $\hbar k$ is momentum on grid points (i, j, k), which may not be the same as the grid point k of angular grid γ. A fast and efficient computer algorithm to perform the tasks of computer radial energies is called Fast Fourier transformation (FFT).

Note: The (wave) function at a grid point in position and momentum space has no one-to-one correlation. To represent a wave function at a grid point in one space in terms of a wave function at another space, the wave function information at all the grid points of another space is needed. Thus, to calculate the action of the radial kinetic energy operator on the wave function at grid point r_i, information about the action of the kinetic energy operator on the wave function at all the grid points in momentum space is required.

3. **Action of angular terms**

The third, fourth, and sixth terms in Eq. 2.38 are the angular terms of the kinetic energy operator acting on the wavepacket on a grid. The action of these operators is not local on the wavepacket in the grid representation, and solving them becomes complicated. To overcome these problems, different methods are introduced [30, 31]. One of the methods uses the property of the angular kinetic energy operator being local in the basis representation. E.g., Legendre polynomials are the eigenfunctions of the angular kinetic energy terms and thus, expressing the wavepacket in term of Legendre polynomials ($P_n(\cos\gamma)$) where $n = j$ (j') for reactant (product) diatom, makes the operation of radial kinetic energy straightforward. This method of transforming the wavepacket from discrete to basis set representation is called 'discrete variable representation - finite basis representation' (DVR-FBR) approach.

In the discrete variable representation (DVR) approach, the wave function is represented on a set of grid points. E.g., a wavefunction $\psi(x)$ is represented as a set of $\psi(x_i)$ where x_i's are the grid points. These grid points are often the roots of the orthogonal polynomials, such as Chebyshev, Legendre, etc. In finite basis representation (FBR), the wave function is written as a linear combination of basis functions. E.g., the wave function ($\psi(x)$) is written as: $\psi(x) = \sum_i c_i\phi_i(x)$, where $\phi_i(x)$'s and c_i's are the basis and corresponding coefficients, respectively. These two representations are connected to each other through a transformation matrix.

The DVR-FBR method used in the DIFFREALWAVE code was proposed by Leforestier [32] and Corey et al. [33], which is derived from the work of Light et al. [34]. In their approach, the same θ grid can be used for the wave function irrespective of the Ω' value. Each Ω' has a different transformation matrix to transform the wavepacket from grid to basis set representation and vice versa. To calculate the operation of the third and fourth terms of Eq. 2.38 on the (J, Ω') state, the DVR-FBR transformation matrix for Ω' is required, but to calculate the action of the sixth term (Coriolis coupling operators), the transformation matrix for $\Omega' - 1$ and $\Omega' + 1$ are also required.

The following steps are followed to calculate the action of the angular kinetic energy operator on the wavepacket.

A. Transform the wavepacket at (R, r, γ) grid point from DVR to the FBR.

B. Multiply the wavepacket in FBR with the eigenvalue of the angular operator part of the Hamiltonian. These eigenvalues for different angular kinetic energy operators are as follows.

a. $\left(-\frac{1}{\sin\gamma_c}\frac{\partial}{\partial\gamma_c}\sin\gamma_c\frac{\partial}{\partial\gamma_c} + \frac{\Omega'}{\sin^2\gamma_c}\right)$ operator: $j(j+1)$.

b. $\hat{J}^2 - 2\hat{J}_z\hat{j}_z$ operator: $J(J+1) - 2\Omega'^2$.

c. $\hat{J}_\pm\hat{j}_\mp$ operator: $\sqrt{[(J(J+1) - \Omega'(\Omega' \pm 1))(j(j+1) - \Omega'(\Omega' \mp 1))]}$

C. Back transform the above term to the DVR.

Three of the above steps can be combined as follows. The expression is given for $\hat{j}$.

$$\frac{\hat{j}^2}{2I_{R_c^i/r_c^l}} q^{J,\Omega'}(R_c^i, r_c^l, \gamma_c^{k'}) = \sum_{k,j'} \left[U^\dagger_{j',k'} \frac{j'(j'+1)}{2I_{R_c^i/r_c^l}} U_{j',k} q^{J,\Omega'}(R_c^i, r_c^l, \gamma_c^k) \right] \tag{2.40}$$

Here, $\frac{1}{I_{R_c^i/r_c^l}} = \left(\frac{1}{I_{r_c^l}} + \frac{1}{I_{R_c^i}}\right)$. The grid points along r_c are shown as r_c^l, not r_c^j as referred earlier, and j' in the above equation are rotation quantum numbers for the product diatom. Also, the action of the angular kinetic energy is shown on the $q^{J,\Omega'}(R_c^i, r_c^l, \gamma_c^{k'})$, where k' has the same meaning as k.

2.4.1.5 Calculation of observables

This section discusses the mathematical expressions to calculate different observables. A detailed mathematical description can be found in refs. 26 and 35. Information about the wave function in the product states is required to calculate the different observables such as reaction probability, integral cross section (ICS), differential cross section (DCS), rate constants, and product energy distribution. This information at each time t is obtained by taking a cut (analyzing) through the wave packet ($q^{J,\Omega'}(R_c, r_c, \gamma_c, t)$ or one implemented in the dynamics approach based on approximation used for time propagation) at the analysis line (R_∞), where R_∞ is the distance between the product diatom and atom fragments which corresponds to the asymptotic potential part of the product fragments. The wave packet at the analysis line is: $q^{J,\Omega'}(R_c = R_\infty, r_c, \gamma_c, t)$ and this information about the wave packet at each time t is obtained by propagating the wave packet using the Eqs. 2.34 and 2.35. At each time t, the contribution of a certain product state $\left(\phi_{v',j'}(r_c, \gamma_c)\right)$ can be calculated as

$$C^J_{v,j,\Omega\to v',j',\Omega'}(t) = \int \phi^*_{v',j'}(r_c, \gamma_c) q^{J,\Omega'}(R_c = R_\infty, r_c, \gamma_c, t) \sin\gamma \, dr \, d\gamma, \tag{2.41}$$

where $C^J_{v,j,\Omega\to v',j',\Omega'}(t)$ is termed as time-dependent expansion coefficient. The Eq. 2.41 can be derived when the wave function at the analysis line R_∞ is expanded as the sum of product eigenstates as:

$$q^{J,\Omega'}(R_c = R_\infty, r_c, \gamma_c, t) = \sum_{v',j'} C^J_{v,j,\Omega\to v',j',\Omega'}(t) \phi_{v',j'}(r_c, \gamma_c). \tag{2.42}$$

These time-dependent expansion coefficients are converted into energy-dependent expansion coefficients by half-Fourier transformation as

$$A^J_{v,j\Omega\rightarrow v'j'\Omega'}(E) = \frac{1}{2\pi}\int_0^\infty e^{iEt/\hbar} C^J_{v,j\Omega\rightarrow v'j'\Omega'}(t)dt. \tag{2.43}$$

These energy-dependent expansion coefficients are in the body-fixed coordinate systems, but for the purpose of including the effects of long-range Coriolis coupling in the body-fixed coordinate system, these energy-dependent coefficients are transformed into the space-fixed coordinate system. This transformation is as follows.

$$A^J_{v,jl\rightarrow v'j'l'}(E) = \sum_{\Omega'\Omega}^{\min(J,j')} T^J_{l\Omega} A^J_{v,j\Omega\rightarrow v'j'\Omega'}(E) T^J_{l'\Omega'} \tag{2.44}$$

These space-fixed energy-dependent expansion coefficients are further used to calculate S-matrix elements [26, 35] as follows.

$$S^J_{v,jl\rightarrow v'j'l'}(E) = \frac{\hbar^2 a_s}{\sqrt{1-E_s^2}}\left(\frac{k_{v'j'}k_{vj}}{\mu_{R_a}\mu_{R_c}}\right)^{1/2} \frac{2A^J_{v,jl\rightarrow v'j'l'}(E)}{\bar{g}(-k_{vj})} e^{-i(k_{v'j'}R_\infty + \delta n_{v'j'l'} + \delta n_{vjl})} \tag{2.45}$$

Here, a_s is the scaling parameter as explained in Eq. 2.32, and E_s is the scaled energy parameter. $\mu_{R_a} = \frac{m_A(m_B+m_c)}{m_A+m_B+m_C}$ and $\mu_{R_c} = \frac{m_C(m_A+m_B)}{m_A+m_B+m_C}$ are the reduced masses of reactant (A+BC) and product (C+AB) scattering coordinates. k_{vj} and $k_{v'j'}$ are the wave vector components (wave numbers) associated with the reactant and product channels.

$$k_{v'j'} = \sqrt{2\mu_p\left(E - \left(E_{v'j'} + \frac{l'(l'+1)}{2\mu_{R_c}R_\infty^2}\right)\right)} \tag{2.46}$$

The mathematical expression of space-fixed S-matrix in Eq. 2.45 is for the real wavepacket approach with a modified form of Chebyshev approximation for the time propagator. The S-matrix expression for the other form of Chebyshev approximation of the time propagator is a little different. The space-fixed S-matrix in Eq. 2.45 is transformed to the body-fixed coordinates as follows.

$$S^J_{v,j\Omega\rightarrow v'j'\Omega'}(E) = \sum_{ll'}^{\min(J,j')} T^J_{\Omega l} S^J_{v,jl\rightarrow v'j'l'}(E) T^J_{\Omega'l'} \tag{2.47}$$

Once the S-matrix elements are calculated, different reaction observables can be calculated. For a reaction A + BC $\rightarrow$ AB + C, a detailed discussion on the calculation of different reaction observables is given below.

1. **Reaction probability**

The reaction probability when the initial rotational level of reactant diatom (j) is equal to zero (0) is calculated using the following equation.

$$P^J_{vj\Omega\rightarrow v'j'\Omega'}(E) = |S^J_{v,j\Omega\rightarrow v'j'\Omega'}(E)|^2 \tag{2.48}$$

The initial rovibrational state selected reaction probability is expressed as a sum of state-to-state reaction probability $P^J_{vj}(E) = \sum_{v'}\sum_{j'}\sum_{\Omega'}|S^J_{v,j\Omega\rightarrow v'j'\Omega'}(E)|^2$. Here, only one initial Ω value (= 0) is possible; thus, no summation in Ω is required. In the case of the initial value of $j > 0$, the state-to-state reaction probability is expressed as [36]

$$P^J_{vj\Omega\rightarrow v'j'\Omega'}(E) = \frac{1+\delta_{\Omega'0}}{2}\sum_{p=0}^{1}|S^{J,p}_{v,j\Omega\rightarrow v'j'\Omega'}(E)|^2. \tag{2.49}$$

Here, $S^{J,p}_{v,j\Omega\rightarrow v'j'\Omega'}(E)$ are state-to-state S-matrix elements for parity p. In the case of initial rotational level $j = 1$, two values of $\Omega = 0, 1$ are possible. Further, for each Ω value, two values of $p = 0$ and 1 exist. In the case of $J = 0$, an initial state with only one parity ($p = 0$) exists, and Eq. 2.49 becomes equal to Eq. 2.48. The initial rovibrational and helicity state-selected probability can also be calculated as follows.

$$P^J_{vj\Omega}(E) = \sum_{v'}\sum_{j'}\sum_{\Omega'}\frac{1+\delta_{\Omega'0}}{2}\sum_{p=0}^{1}|S^{J,p}_{v,j\Omega\rightarrow v'j'\Omega'}(E)|^2 \tag{2.50}$$

The initial rovibrational state-selected reaction probability, which includes all the helicity (Ω) quantum numbers of the reactant diatom, is as follows.

$$P^J_{vj}(E) = \sum_{\Omega=0}^{j}\frac{g_\Omega}{2j+1}\sum_{v'}\sum_{j'}\sum_{\Omega'}\frac{1+\delta_{\Omega'0}}{2}\sum_{p=0}^{1}|S^{J,p}_{v,j\Omega\rightarrow v'j'\Omega'}(E)|^2 \tag{2.51}$$

Here, parameter g_Ω is equals to 1 for $\Omega = 0$ and 2 for $\Omega > 1$.

2. **Integral cross section**

$$\sigma_{vj\rightarrow v'j'}(E) = \frac{\pi}{k^2_{vj}}\sum_{\Omega=0}^{j}\frac{g_\Omega}{2j+1}\sum_{J\geq\Omega}^{J_{\max}}(2J+1)\sum_{\Omega'}^{\min(J,j')}P^J_{vj\Omega\rightarrow v'j'\Omega'}(E) \tag{2.52}$$

In the above Eq. 2.52, $k_{vj} = \frac{\sqrt{2\mu_{\text{A-BC}}E_{\text{col}}}}{\hbar}$ is the modulus of the wave vector for reactants. The units are the atomic units. The term g_Ω is is equals to 1 for $\Omega = 0$ and 2 for $\Omega > 1$. It is to be noted that the Eq. 2.52 is the cross section from the initial (v, j) state to the final product (v', j') state, where reactant and product states are summed over Ω and Ω' states, respectively. The initial state-selected (state-specific) cross sections can be calculated by summing over the product (v', j') states as follows.

$$\sigma_{vj}(E) = \frac{\pi}{k_{vj}^2} \sum_{\Omega=0}^{j} \frac{g_\Omega}{2j+1} \sum_{J \geq \Omega}^{J_{\max}} (2J+1) \sum_{v'} \sum_{j'} \sum_{\Omega'}^{\min(J,j')} P^J_{vj\Omega \to v'j'\Omega'}(E) \tag{2.53}$$

The ICS equations above have slightly different forms when the product diatom is homonuclear. This difference originates because of the role Fermi-Dirac (FD) nuclear spin statistics plays in the reaction process, a detailed discussion of which can be found in literature [36, 37] and thus only the mathematical formula is provided below.

$$\sigma^{\mathrm{FD}}_{vj \to v'j'}(E) = g_s \sigma_{vj \to v'j'}(E) \tag{2.54}$$

Here, g_s is called post antisymmetrization factor with values 3/2 and 1/2 for odd and even j', respectively.

3. **Rate constant**

These state-to-state and state-selected ICSs are further used to calculate the state-to-state and state-selected rate constants. Following is the formula to calculate the initial reactant rovibrational (v, j) state to product rovibrational (v', j') state rate constant.

$$k_{vj \to v'j'}(T) = \sqrt{\frac{8k_BT}{\pi\mu_{\text{A-BC}}}} \frac{1}{(k_BT)^2} \int_0^\infty \sigma_{vj \to v'j'}(E) e^{-E/k_BT} E\, dE \tag{2.55}$$

Here, k_B=1.38×10^{-23} JK^{-1} is Boltzmann constant and $\mu_{\text{A-BC}}$ is reduced mass of reactant atom-diatom pair. The above rate constants are derived using the collision theory, a detailed discussion on which can be found in Ref. 38. Following is the form of Eq. 2.55, which is used in the numerical solution.

$$k_{vj \to v'j'}(T) = \sqrt{\frac{8k_BT}{\pi\mu_{R_a}}} \frac{1}{(k_BT)^2} \sum_{E=0}^{\infty} \sigma_{vj \to v'j'}(E) e^{-E/k_BT} E_i dE \tag{2.56}$$

Here, E goes from 0 to ∞, and grid spacing is dE. The denser the grid, the closer the numerical $k(T)$ to the analytical $k(T)$ value. It can be seen from Eq. 2.56 that to get the rate constant at any temperature T, the ICSs at all the energies from $E = 0$ to $E = \infty$ are required. However, these conditions of dense E grid with corresponding ICSs values and availability of ICSs data up to $E = \infty$ can be relaxed to get the convergence of numerical and analytical values of $k(T)$. A detailed discussion on this relaxation aspect is presented in section 4.3.3 of Chapter 4.

Another thing to be noted is that the rate constants calculated in all the works presented in this thesis are either initial state-to-state or initial state-selected rate constants. Another type of rate constants are Boltzmann or thermal averaged rate constants, which are more important for astrochemical discussion of the reactions. These thermal averaged rate constants consider the Boltzmann averaged contribution of all the initial rovibrational levels of a reactant possible at a specific temperature.

4. **Differential cross section**

Initial reactant rovibrational state to product rovibrational state DCS is expressed as follows.

$$\sigma^{\mathrm{DCS}}_{vj\to v'j'}(E,\theta) = \frac{1}{4k^2_{vj}}\sum_{\Omega=0}^{j}\left(\frac{g_\Omega}{2j+1}\right)\sum_{J\geq\Omega}^{J_{\max}}(2J+1)\sum_{\Omega'}|S^J_{v,j\Omega\to v'j'\Omega'}(E)d^J_{\Omega\Omega'}(\pi-\theta)|^2 \tag{2.57}$$

Here, $d^J_{\Omega\Omega'}(\pi-\theta)$ is the reduced Wigner rotation matrix at scattering angle θ [22, 39, 40], where scattering angle is the angle between the product diatom and attacking atom. Scattering angles of 0° and 180° depict forward and backward scattering, respectively. The initial rovibrational state-selected DCS can be calculated as follows.

$$\sigma^{\mathrm{DCS}}_{vj}(E,\theta) = \sum_{v'}\sum_{j'}\sigma^{\mathrm{DCS}}_{vj\to v'j'}(E,\theta) \tag{2.58}$$

Also, DCS can be used to calculate the ICS at a collision energy by integrating it over all the angles as

$$\sigma_{vj\to v'j'}(E) = \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi}\sigma^{\mathrm{DCS}}_{vj\to v'j'}(E,\theta). \tag{2.59}$$

As is seen above, the S-matrix is an important quantity that is used to calculate several reaction observables. The procedure discussed above to calculate the S-matrix is a time-dependent (TD) approach. However, it can also be calculated using the time-independent (TI) approach. The TD approach solves the time-dependent Schrödinger equation (initial-value problem). In contrast, the TI version of the equation is solved using the TI dynamical method (eigenvalue problem). For total angular momentum J collision, TI method scales as $(J\Pi_i N_i)^3$, which is computationally more expensive than the TD approach, which scales as $J(\Pi_i N_i)(\sum_i N_i)$, where N_i is the number of basis functions in i^{th} degree of freedom [17]. Also, whereas only one-time propagation can calculate the S-matrix for a collision energy range in the TD method, TI calculations are performed for the specific collision energies, and when the collision energy is closer to or greater than the dissociation energy, TI calculations are erroneous [41, 42]. Such situations where collision energy is greater than dissociation energy can easily be handled using the TD method. Since the TD method can give information about the reaction in the time domain, a time-evolving view of the reaction is obtained in this method, which is not possible in the TI approach.

2.4.2 Classical method

In the above discussion, the nuclear dynamics was solved quantum mechanically using the SE. The same event of nuclear motion can also be studied using classical mechanics, which is discussed below in detail.

There are various approaches, such as Newton's mechanics, Lagrangian mechanics, Hamilton's mechanics, etc., to study the time evolution of a system classically. These different mechanics approach the same problem differently while reaching the same solution. Newton's mechanics solve the problem in the physical space, and Lagrangian mechanics solve it in the configuration space. On the other hand, Hamilton's mechanics is solved in the phase space constituted of momentum and position space. Hamilton's mechanics are used in the problems studied here and thus are discussed below in detail.

2.4.2.1 Hamilton's mechanics based classical dynamics

For a n particle system, with momentum $\vec{p}_i$ and position $\vec{q}_i$ (p_i and q_i hereon), the equations of motion are

$$\frac{\partial p_i}{\partial t} = \frac{\partial H}{\partial q_i}, \tag{2.60}$$

$$\frac{\partial q_i}{\partial t} = \frac{\partial H}{\partial p_i}, \tag{2.61}$$

where H is the classical Hamiltonian that is equal to the sum of kinetic energy (T) and potential energy (V) of the dynamical system: $H = T+V$. The equations of motion consider the p_i and q_i) coordinates of each particle and solve the time evolution. The time evolution path in the phase space is called the phase flow of the system.

The equations 2.60 and 2.61 are general equations which can be modified for a specific reaction of interest. Suppose we have a molecular system ABC and a reaction A + BC $\rightarrow$ AB + C is to be studied classically. In this case, the Hamiltonian for the system is written as the sum of the total kinetic energy of three atoms and the potential energy of the system, which is the electronic energy of the ABC molecular configuration.

$$\begin{aligned} H &= T + V \\ &= T_A + T_B + T_C + V_{tot}(R_{AB}, R_{BC}, R_{AC}) \end{aligned}$$

If the dynamics is solved by taking the Cartesian coordinates of position and momentum for each atom, 18 first-order differential equations need to be solved. This required number of equations and computational efforts can be reduced by using the Jacobi coordinates. In Jacobi coordinates, the reactant diatom BC is represented by $\vec{r}$ and the distance of atom A from BC center of the mass point is represented by $\vec{R}$ as shown in Fig. 2.6. The $x, y,$ and z

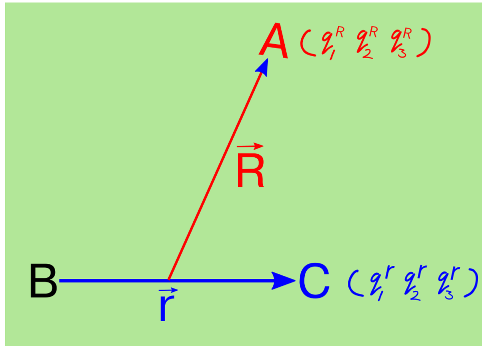


Figure 2.6: Jacobi coordinates for A+BC reaction system.

axial coordinates are represented by r_i and R_i; $i = 1, 2, 3$, respectively. In the same way, the momentum coordinates of diatom BC and A from the center of mass of BC are represented by p_i^r and p_i^R; $i = 1, 2, 3$, respectively. The kinetic energy part of the Hamiltonian will have the following form in this case.

$$\begin{aligned} T &= T_R + T_r + T_{CM} \\ &= \frac{p_r^2}{2\mu_r} + \frac{p_R^2}{2\mu_R} + \frac{p_{CM}^2}{2M} \end{aligned}$$

When the center of mass motion of the whole system is neglected, the Kinetic part reduces as follows.

$$T = \frac{p_r^2}{2\mu_r} + \frac{p_R^2}{2\mu_R}$$

In the above representation of the kinetic part of the Hamiltonian, $\mu_r = \frac{m_B m_C}{(m_B+m_C)}$, $\mu_R = \frac{m_B m_C}{(m_B+m_C)}$, and $M = m_A + m_B + m_C$. Here, μ_r, μ_R, and M are reduced mass of reactant diatom BC, reduced mass of A + BC, and total mass of ABC system, respectively. Now, the equations of motion can be written as follows.

$$\begin{aligned} \frac{\partial q_i^r}{\partial t} &= \frac{\partial H}{\partial p_i^r} = \frac{p_i^r}{\mu_r} \\ \frac{\partial q_i^R}{\partial t} &= \frac{\partial H}{\partial p_i^R} = \frac{p_i^R}{\mu_R} \\ \frac{\partial p_i^r}{\partial t} &= -\frac{\partial H}{\partial q_i^r} = -\sum_{j=1}^{3}\left(\frac{\partial V}{\partial R_j}\right)\left(\frac{\partial R_j}{\partial q_i^r}\right) \\ \frac{\partial p_i^R}{\partial t} &= -\frac{\partial H}{\partial q_i^R} = -\sum_{j=1}^{3}\left(\frac{\partial V}{\partial R_j}\right)\left(\frac{\partial R_j}{\partial q_i^R}\right) \end{aligned} \tag{2.62}$$

In the case of Jacobi coordinates, a total of 12 equations are solved during dynamical studies.

2.4.2.2 Solving the equations of motion

The time propagation of the reaction system is mathematically modeled by solving Hamilton's first-order differential equations. The information about the reactive system at time t is calculated using the information at the initial time t_0. Following are some of the methods used to solve such equations iteratively.

1. Euler's method
2. Fourth order fixed step-size Runge–Kutta method
3. Fourth order adaptive step-size Runge–Kutta method
4. Adams-Moulton predictor-corrector method

The fixed step-size Runge-Kutta method uses the same step size throughout the propagation, whereas the step size gets updated after each propagation step in the adaptive step-size Runge-Kutta method. This continuous updation of step size depends on the topography of the potential surface, which attains smaller and larger values at the asymptotic and interactive regions, respectively.

2.4.2.3 Accuracy of the trajectory propagation

The accuracy of dynamics results is the direct consequence of the level of accuracy used in the integrator scheme. The accuracy can be checked using the following methods [43].

1. **Conservation of total energy and angular momentum**
   The total energy, which is the sum of the rovibrational energy of the reactant diatom and the translational energy of the atom-diatom pair, should be the same at each time step of trajectory propagation. The same is the case with the total angular momentum. The conservation of these two properties can be checked at each time step by comparing the values with the initial values. Usually, the initial values are compared with the values at the end of the trajectories.
2. **Back propagation**
   Non-conservation of total energy and total angular momentum indicates an instability in the numerical integration scheme. However, the conservation of these properties doesn't necessarily mean an accurate propagation. The back-propagation is a much more sophisticated check of accuracy. In back-propagation, the trajectory is back-propagated from the final points. The time step is taken as the negative of the time step used for the forward propagation, and signs of momentum coordinates obtained at the final point are also reversed. The accuracy is checked by comparing the initial points obtained after the back-propagation with the initial points used in the forward propagation.

2.4.2.4 Choice of initial parameters

There are 12 equations of motion and, thus, 12 initial parameters needed to run the dynamics. 6 of these parameters, q_i^r and p_i^r where i = 1 to 3, belong to diatom BC, and the rest of the parameters, q_i^R and p_i^R where i = 1 to 3, belongs to the motion of atom A from the center of mass of BC. The subscripts 1, 2, and 3 denote x, y, and z coordinates, respectively.

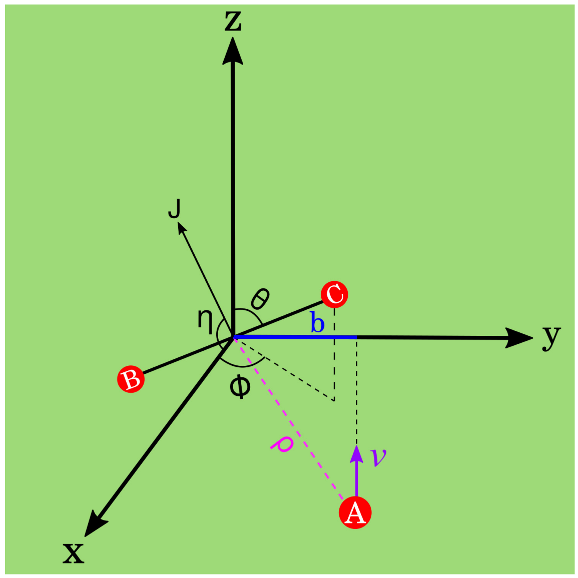


Figure 2.7: Representation of atom + diatom reaction system at an initial point in time.

The initial position of the system is taken such that the atom A from the center of BC diatom is in the yz plane and attacks from -z to +z side. Under these conditions, the initial position and momentum coordinates of atom A from BC center of mass are as follows.

$$q_1^{R0} = 0, \qquad q_2^{R0} = b, \qquad q_3^{R0} = -\sqrt{\rho^2 - b^2} \tag{2.63}$$

$$p_1^{R0} = 0, \qquad p_2^{R0} = 0, \qquad p_3^{R0} = \sqrt{2\mu_{A-BC}E_{col}} \tag{2.64}$$

Here, ρ is the initial distance between atom A and the center of mass of diatom BC, b is the impact parameter value that is the distance between the centers of atom A and BC diatom, and E_{col} is the relative collision energy of A - BC. The position coordinates of diatom BC are as follows.

$$q_1^{r0} = r_0 \sin\theta \cos\phi, \qquad q_2^{r0} = r_0 \sin\theta \sin\phi, \qquad q_3^{r0} = r_0 \cos\theta$$

Here, θ is the angle between $\vec{r}$ and z-axis, ϕ is angle between x-axis and projection of $\vec{r}$ on xy plane, and r_0 is initial diatomic separation. As far as the momentum coordinates of the diatom are concerned, it isn't straightforward, and two main approaches are discussed in literature [44]. The one implemented in the work presented in the current thesis is discussed below.

The initial separation (r_0) of the diatom is taken as equal to the inner (r_-) or outer (r_+) turning point. At these points, the kinetic energy due to the vibrational oscillation is zero, and only the rotational angular momentum ($|\vec{j}_{am}|$) of BC diatom contributes to the initial momentum. The x, y, and z components of momentum can be written as follows [44].

$$\begin{aligned}
p_1^{r0} &= |\vec{j}_{am}| \left(\sin\phi\cos\eta - \cos\theta\cos\phi\sin\eta\right) / r_0 \\
p_2^{r0} &= -|\vec{j}_{am}| \left(\cos\phi\cos\eta + \cos\theta\sin\phi\sin\eta\right) / r_0 \\
p_3^{r0} &= |\vec{j}_{am}| \left(\sin\theta\sin\eta\right) / r_0
\end{aligned} \tag{2.65}$$

After the above condition, the initial position coordinates for the BC diatom are transformed as follows.

$$q_1^{r0} = r_\pm \sin\theta\cos\phi, \qquad q_2^{r0} = r_\pm \sin\theta\sin\phi, \qquad q_3^{r0} = r_\pm \cos\theta \tag{2.66}$$

Where $r_\pm$ is either the inner turning point (r_-) or outer turning point (r_+) of a diatom.

However, it is important to note that the diatom, when vibrating, can have interatomic distance values anywhere between r_- to r_+, not just r_- and r_+. So, when starting the time propagation for a new trajectory for the atom-diatom collision, the actual initial position and momentum coordinates of the diatom are obtained by integrating Hamilton's equations of motions only for the diatom for $\tau\epsilon$ time if $\epsilon < 0.5$ or $\tau(\epsilon - 0.5)$ time, if $\epsilon \geq 0.5$, where ϵ is the 'vibration-phase/2π'. Now, these initial coordinates for diatom, along with the coordinates represented by Eq. 2.63 and 2.64, are finally used for running a trajectory for an atom-diatom collision.

As seen above, the initial values of position and momentum coordinates depend on parameters, such as b, θ, ϕ η, and ϵ. These parameters are called collision parameters, and calculating the values of these is a crucial step, which is discussed in section 2.4.2.10.

2.4.2.5 Separation of product channels

In an atom + diatomic collision of A + BC type, there are four possible product channels.

A + BC → **(1)** AB + C, **(2)** AC + B, **(3)** BC + A, **(4)** A + B + C

The first and second channels are reactive collisions, the third product channel is non-reactive, and the fourth channel is collision-induced dissociation (CID) product. Now, when a trajectory propagates, it doesn't discriminate between any channels by itself, and the onus is on the algorithm to distinguish these channels.

The point is that the trajectory propagation occurs in reactant coordinates throughout the propagation. After each time step, the following conditions are checked, and a decision about the product channels is taken.

1. **A + BC:** If $r < r_{\text{exit}}$ and $br_1 > br_1^{\text{exit}}$
2. **B + AC:** If $r > r_{\text{exit}}$ and $br_2 > br_2^{\text{exit}}$
3. **C + AB:** If $r > r_{\text{exit}}$, $br_3 > br_2$, and $br_3 > br_3^{\text{exit}}$

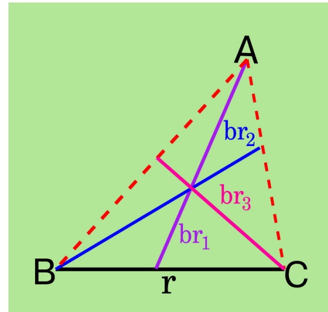


Figure 2.8: Product separation.

Here, r_{exit} is the interatomic distance of BC diatom beyond which diatom BC disintegrates. This value is taken as the outer turning point of the highest possible vibrational level of the diatom. The parameters br_1^{exit}, br_2^{exit}, and br_3^{exit} are the distance between A to the center of mass of BC, B to the center of mass of AC, and C to the center of mass of AB, respectively. These are the distances beyond which the atom and respective diatomic fragments are considered separated. These distances br_1^{exit}, br_2^{exit}, and br_3^{exit} should be chosen in such a way that the interaction between the respective atom-diatom pairs is negligible and the fragment configuration at the point is asymptotic.

Now, as far as the dissociative fragment is considered, the separation is made in each of the three atom-diatom asymptotes after comparing the rovibrational energy of the respective diatom with the dissociation energy of the same diatom. A detailed discussion is presented in section 2.4.2.7.

2.4.2.6 Analysing the energetic of the product fragments

Once the trajectory is separated in a product channel, the energy of the product fragments can be calculated. The initial energy of the reactant atom-diatom pair is the sum of the collision energy given to the reactant atom-diatom pair (E_{col}) and rovibrational energy of the initial state of the reactant diatom (E_{vj}). Since the trajectory run follows the rule of energy conservation, the total energy on the product side is also the same as the initial total energy. This energy, however, may get redistributed in translational, rotational, and vibrational modes of product fragments.

In the case of the A + BC product channel (non-reactive), reactant and product coordinates are the same, and the same reactant coordinates can be used to calculate the energy distribution at the product asymptote. However, in the case of the other two reactive channels, calculating energy distribution straight away is not simple, and the transformation of coordinates from reactant (A + BC) to respective product coordinates is needed. These transformed position coordinates for product channels are shown in Fig. 2.9 and corresponding

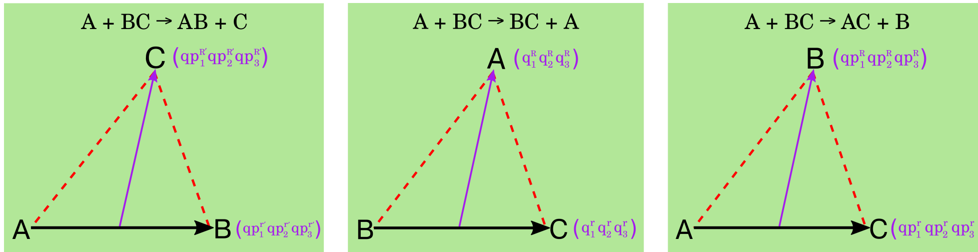


Figure 2.9: Coordinates for reactant and product channels.

momentum coordinates are: yp_i^R, yp_i^r for B + AC and $yp_i^{R'}$, $yp_i^{r'}$ for C + AB channels. Here, i = 1 to 3, and in terms of these new coordinates, the total kinetic energy for AC + B and AB + C can be written as follows.

1. **A + BC channel:**
$$T_{\text{A-BC}} = \left[\left(p_1^R\right)^2 + \left(p_2^R\right)^2 + \left(p_3^R\right)^2 \right] / 2\mu_{\text{A-BC}}$$

2. **B + AC channel:**
$$T_{\text{B-AC}} = \left[\left(yp_1^R\right)^2 + \left(yp_2^R\right)^2 + \left(yp_3^R\right)^2 \right] / 2\mu_{\text{B-AC}}$$

3. **C + AB channel:**
$$T_{\text{C-AB}} = \left[\left(yp_1^{R'}\right)^2 + \left(yp_2^{R'}\right)^2 + \left(yp_3^{R'}\right)^2 \right] / 2\mu_{\text{C-AB}}$$

Once we have the translation energy at the product asymptotes, the rovibrational energy can be calculated. One way to calculate that is to subtract translational energy from the total energy, and the other way is to calculate using the following formula:

$$E_{vj} = \frac{p_x^2}{2\mu} + \frac{p_y^2}{2\mu} + \frac{p_z^2}{2\mu} + V(r). \tag{2.67}$$

Here, p_x, p_y, and p_z are the x, y, and z component of the relative momentum of product diatom, μ is the reduced mass, and $V(r)$ is the potential energy of diatomic configuration at the point of separation of trajectory stoppage. The diatoms are BC, AB, and AC for A + BC, C + AB, and B + AC product channels, respectively.

2.4.2.7 Separation of bound, quasi-bound, and dissociative product states

The procedure to separate the products in different channels is based on the distance between the product fragments. These product states may not necessarily be the bound diatomic states only but quasi-bound and dissociative states, too.

The diatomic **bound state** has the rovibrational energy less than the dissociation energy of diatom (E_{vj}) and is supported by the potential energy well. The **quasi-bound** and **dissocia-**

tive states have the E_{vj} greater than the dissociation energy of the diatom. The quasi-bound states, despite having the E_{vj} energy greater than the dissociation energy, are not in the dissociative state.

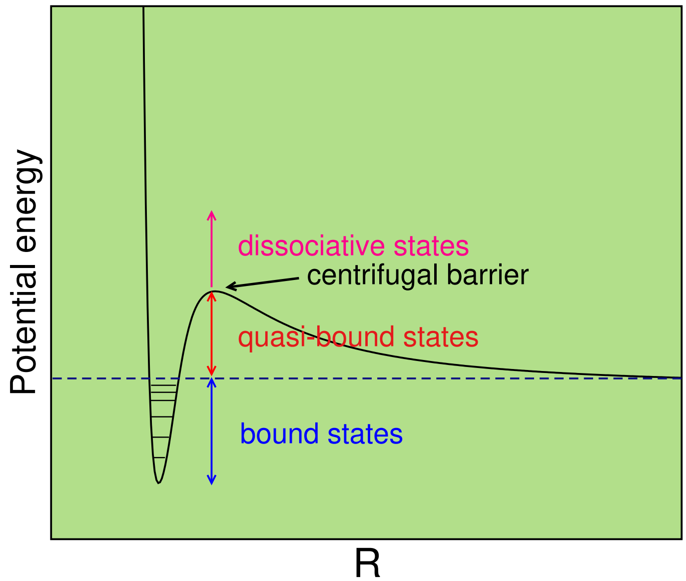


Figure 2.10: Diatomic bound, quasi-bound, and dissociative states.

These quasi-bound states are stabilized by the centrifugal barrier, and only through quantum tunneling do these states slowly leak to the dissociative region. These three states can be understood more clearly from Fig. 2.10. The categorization of trajectories falling in a particular product channel to these three types of diatomic states is performed as follows.

1. The bound states can be separated from quasi-bound and dissociative states by removing the trajectories with E_{vj} greater than the dissociation energy.
2. The separation of quasi-bound and dissociative states from each other needs a little more involvement. As already discussed, due to the centrifugal barrier, the effective potential (electronic potential of diatom + centrifugal barrier) of the diatom has a maximum (V_{eff}^{max}) at some interatomic distance R_{cf}.

$$\text{If '}E_{vj} > V_{eff}^{max}\text{' or '}E_{vj} < V_{eff}^{max} \text{ and } r > R_{cf}\text{' : Dissociative state.}$$
$$\text{If '}E_{vj} < V_{eff}^{max} \text{ and } r < R_{cf}\text{' : Quasi-bound state.}$$

Since, in reality, the quasi-bound state slowly tunnels to the dissociative state, a feature that can not be captured in the classical method, the quasi-bound states can be considered dissociative states when performing classical dynamics.

2.4.2.8 Rotational and vibrational quantum states of product diatom

After calculating the energies of the product diatom, the next step is to find the vibrational and rotational quantum numbers of these states. These rovibrational level resolutions of the product are an important step in finding the state-to-state properties of reactions.

The rotation angular momentum of a diatom ($|\vec{j}_{am}|$) is: $\sqrt{j(j+1)}$ au, where j is the rotational quantum number. We can also calculate the j_{am} using the cross product of position ($\vec{q}^r$ or $q\vec{p}^r$) and momentum ($\vec{p}^r$ or $y\vec{p}^r$) vectors, as shown below for BC diatom.

$$\begin{aligned} |\vec{j}^x{}_{am}| &= q^r_2 p^r_3 - p^r_2 q^r_3 \\ |\vec{j}^y{}_{am}| &= p^r_1 q^r_3 - q^r_1 p^r_3 \\ |\vec{j}^z{}_{am}| &= q^r_1 p^r_2 - p^r_1 q^r_2 \end{aligned} \tag{2.68}$$

Using the Eq. 2.68 along with the $\sqrt{j(j+1)}$ form of the angular momentum, the rotational quantum number is obtained as $j = -\frac{1}{2} + \frac{\sqrt{1-4|\vec{j}_{am}|^2}}{2}$.

The vibrational quantum number of the diatom can be calculated using two different methods, one of which is based on the Dunham expansion [45, 46], and the other which uses the following formula [47].

$$v = -\frac{1}{2} + \frac{1}{\hbar\pi}\int_{r_-}^{r_+} \rho_r \mathrm{dr} \tag{2.69}$$

Where v is the vibrational quantum number, and ρ_r is the relative momentum of a diatom. The detailed mathematical formulation for solving the Eq. 2.69 is discussed in the Appendix I.

2.4.2.9 Calculating quantum states and binning in QCT

The quantum numbers calculated in the QCT dynamical approach, as shown above, are real numbers. However, in reality, these quantum states are discrete and are shown only by integer values. So, this conversion of real quantum numbers to integer values is a prerequisite to a more accurate picture of the system and calculation of state-to-sate observables, as will be described later in much detail. This conversion is performed by replacing, let's say, any real vibration quantum number (v) to the integer value $\bar{v}$ if: $(\bar{v} - 0.5) \le v < (\bar{v} + 0.5)$.

Now, since these quantum numbers calculated in the QCT are approximated to be integer values, there are two ways these trajectories can be treated. The treatment is related to the weight of the trajectories. Here, both approaches are discussed in detail.

1. **Histogram binning (HB)**

Histogram binning is about giving the equal weightage of one (1) to each trajectory irrespective of how far and close it is to the nearest integer value.

2. **Gaussian binning (GB)**

In Gaussian binning, trajectories are weighted with respect to their proximity with the nearest integer quantum number, which, in turn, is decided by the method described earlier. For the i^{th} trajectory, this weight is given based on the vibrational, rotational, or both quantum numbers as follows.

$$w_i^v = \frac{e^{-(v_i-\bar{v}_i)^2/\epsilon_v^2}}{\epsilon_v\sqrt{\pi}}, \qquad w_i^r = \frac{e^{-(j_i-\bar{j}_i)^2/\epsilon_r^2}}{\epsilon_r\sqrt{\pi}} \tag{2.70}$$

Here, w_i^r and w_i^v are rotational and vibrational Gaussian weights, respectively, $\bar{v}$ and $\bar{j}$ are the nearest integer vibrational and integer rotational quantum number to real quantum numbers for the product diatom. The constant parameter ϵ (ϵ_r for rotational weight and ϵ_v for vibrational weight) is related to the full width at half maxima (FWHM) of the Gaussian distribution function w_i. The expression for FWHM = 2 $\epsilon\sqrt{ln(2)}$. For a specific trajectory, the Gaussian distribution function is centered around value $\bar{v}_i$ or $\bar{j}_i$, and its width and value in vibrational and rotational binning are controlled by respective FWHM or ϵ.

2.4.2.10 Calculating reaction observables and Monte Carlo sampling

The aim of the dynamics calculations is to calculate the different reaction observables, such as reaction probability, ICS, DCS, rate constant, energy distribution, etc., at state-to-state or state-selected levels. Calculation of these observables requires the trajectories starting from all the initial collision parameters within the sphere of reactivity, which is the collision sphere beyond which reaction can't take place. This collision sphere is represented by the maximum value of the impact parameter, where the impact parameter is the distance between the centers of two colliding partners.

This requirement of sampling the full space by the trajectories is important since the observables are averaged over the whole reaction space. One such observable is average probability, which is an important reaction attribute and is further used to calculate other observables. This average probability for all the collisions below $b_{\max}$ is calculated as follows.

$$< P_r(E_{\text{col}}, v, j) > \propto \int_{b=0}^{b_{\max}} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \int_{\eta=0}^{2\pi} \int_{\epsilon=0}^{2\pi} P_r(b,\theta,\phi,\eta,\epsilon) b db \sin\theta d\theta d\phi d\eta d\epsilon \tag{2.71}$$

In the above expression of average reaction probability, all the collisions below $b_{\max}$ are mathematically expressed by integrating the probability function $P_r(b,\theta,\phi,\eta,\epsilon)$ over all the collision parameters. To numerically solve the integral, the values of the collision parameters are selected, a choice that depends on the numerical integration method implemented to solve Eq. 2.71. The Monte Carlo method is one such method. It approximates a multidimensional integral of the form: $Int = \int_0^1 \int_0^1 \int_0^1 \ldots \int_0^1 f(\beta) d\beta$ as follows [44].

$$I \approx \frac{1}{N} \sum_{i}^{N} f(\beta_i) \tag{2.72}$$

Here, the integral in Eq. 2.71 is converted to the limits of 0 to 1 and approximated as per Eq. 2.72. The parameters are modified as β_1, β_2, β_3, β_4, and β_5, and for each trajectory run, randomly selected in the range of 0 to 1. Since the range of collision parameters in actual form is $\theta = 0$ to π, $\phi = 0$ to 2π, $\eta = 0$ to 2π, $\epsilon = 0\ 2\pi$, and $b = 0$ to b_{max}, the modified parameters β_i, where i= 1 to 5, are converted to actual values of parameters as follows.

$$\theta = cos^{-1}(1 - 2\beta_1), \qquad \phi = 2\pi\beta_2, \qquad \eta = 2\pi\beta_3$$

$$\epsilon = 2\pi\beta_4, \qquad b = \sqrt{\beta_5}\, b_{\text{max}}$$

Under these sampling conditions, different reaction observables can be calculated. These observables can be calculated from a fixed initial rovibrational state of reactant diatom (v, j) to the specific product rovibrational state (v', j'), called state-to-state properties or from a fixed initial rovibrational state of reactant diatom (v, j) to all the product diatom states, termed as state-selected properties. State-to-state information is important for exploring reaction dynamics in finer detail. The mathematical formulas of a few different observables are as follows.

1. **Reaction probability**

   This is the probability of the product formation under the reaction condition. The state-to-state and total state-selected reaction probabilities are expressed as follows.

$$P_{vj}^{\text{GB}}(E_{\text{col}}) = \frac{\sum_i^{N_r} w_i}{N} \tag{2.73}$$

   Here, $P_{vj}^{E_{\text{GB}}}(E_{\text{col}})$ is the total probability. This gives us information about the trajectories starting from the initial (v, j) state of the diatom to all the possible rovibrational states of the product diatom. The parameter w_i is weight of the i^{th} trajectory. In the case of the HB, the $w_i = 1$ for all the trajectories, and in the case of GB, the weight is defined by Eq. 2.70. The form which the above probability gets for HB is as follows.

$$P_{vj}^{\text{HB}}(E_{\text{col}}) = \frac{N_r}{N} \tag{2.74}$$

Unit of the observable: unitless.

Also, $P^{\text{HB/GB}}_{vj}(E_{\text{col}}) = \sum_{v'j'} P^{\text{HB/GB}}_{vj\to v'j'}(E_{\text{col}})$, where $P^{\text{HB/GB}}_{vj\to v'j'}(E_{\text{col}})$ is state-to-state probability which is also expressed as follows.

$$P^{\text{GB}}_{vj\to v'j'}(E_{\text{col}}) = \frac{\sum_i^{N_r} w_i^{v'j'}}{N} \tag{2.75}$$

Here, the weight of only those trajectories that land into specific (v', j') are considered, and all the other trajectories are either not taken or have null weight.

2. **Integral cross-section (ICS)**

ICS for a reaction is the area around the target which, if accessed by the attacking species, results in the successful collision or, in other words, product formation. The mathematical expressions for state-to-state and state-selected ICSs are as follows.

$$\sigma^{E_{\text{col}}}_{vj\to v'j'} = \pi b^2_{max} P_{vj\to v'j'}(E_{\text{col}}) \qquad \sigma^{E_{\text{col}}}_{vj} = \sum_{v'j'} \sigma^{E_{\text{col}}}_{vj\to v'j'} \Rightarrow \pi b^2_{max} P_{vj}(E_{\text{col}}) \tag{2.76}$$

Unit of the observable: unit of $(b_{\text{max}})^2$.

The standard deviation in the ICS is calculated as follows [48].

$$\text{SD}(\sigma^{E_{\text{col}}}_{vj\to v'j'}) = \pi b^2_{\text{max}} P_{vj\to v'j'}(E_{\text{col}}) \sqrt{\frac{N_r^{v'j'} - N}{N_r^{v'j'} N}} \tag{2.77}$$

The given ICSs can be expressed in terms of HB ICSs and GB ICSs based on the type of probability used in the calculations.

3. **Rate constant**

The elementary reaction under investigation is an atom-diatom reaction, making it a second-order reaction.

$$k_{vj\to v'j'}(T) = \sqrt{\frac{8k_BT}{\pi\mu_{\text{A-BC}}}} \left(\frac{1}{k_BT}\right)^2 \int_0^\infty E_{\text{col}} \sigma^{E_{\text{col}}}_{vj\to v'j'} e^{-\frac{E_{\text{col}}}{k_bT}} dE_{\text{col}} \tag{2.78}$$

$$k_{vj}(T) = \sum_{v'j'} k_{vj\to v'j'}(T)$$

Unit of the observable: $\text{cm}^3\text{sec}^{-1}$ or $\text{cm}^3\text{sec}^{-1}\text{molecule}^{-1}$.

In the above equations, k_B is Boltzmann constant, $\mu_{\text{A-BC}} = \frac{m_A m_{BC}}{m_A + m_B + m_C}$ is the reduced mass of reactant atom-diatom pair, E_{col} is the collision energy. It can be seen that the limits of the integral go from 0 to ∞, but in computational settings, the above equation can be approximated as follows.

$$k_{vj\to v'j'}(T) = \sqrt{\frac{8k_BT}{\pi\mu}} \left(\frac{1}{k_BT}\right)^2 \sum_i \left(E^i_{\text{col}} \sigma^{E^i_{\text{col}}}_{vj\to v'j'} e^{-\frac{E^i_{\text{col}}}{k_bT}} dE_{\text{col}}\right) \tag{2.79}$$

In this expression, E^{i}_{col} is the collision energy at the i^{th} collision energy grid, dE_{col} is collision energy grid step size and all the other parameters are the same as in Eq. 2.78 but with respect to i^{th} grid point. It is clear from the equation that the rate constant at temperature T, has a contribution from the cross-sections at all the collision energies. Same as in the case of ICS, the rate constants can be calculated using the HB and GB method, and that depends on the type of cross section used in the Eq. 2.78.

4. **Differential cross section (DCS)**

The calculation of DCS starts from the calculation of scattering angle, that is, the angle between the relative velocities of 'product diatom with respect to atom' and 'line of approach of an atom during the collision.' The angle can be calculated as follows.

$$\vec{v}_f \cdot \vec{v}_i = |\vec{v}_f||\vec{v}_i|\cos\theta$$

$$\theta = cos^{-1}\left(\frac{\vec{v}_f \cdot \vec{v}_i}{|\vec{v}_f||\vec{v}_i|}\right)$$

Now, QCT approach takes the initial velocity only in the z direction, the formula can be written as follows.

$$\theta = cos^{-1}\left(\frac{\vec{v}^{z}_f \cdot \vec{v}^{z}_i}{|\vec{v}_f||\vec{v}_i|}\right)$$

Here, $\vec{v}^{z}_f = \frac{yp_z}{\mu}$, where yp_z is the z-coordinate of momentum and μ is the reduced mass of product diatom and atom system. The forward and backward scattering are the scattering angles (θ) of 0° and 180°, respectively. These angles are opposite to the experimental definition where 0° is backward scattering and 180° is forward.

Once the scattering angles are calculated for all the trajectories falling into a specific product channel, the DCSs can be calculated using two approaches: the Expansion of moments method [49] and the HB approach.

In the **expansion of moments** method, the DCS is expressed as follows.

$$\frac{d\sigma_{vj}(E_{\text{col}}, \theta)}{d\omega} = \frac{\sigma^{E_{\text{col}}}_{vj}}{2\pi}\left[\frac{1}{2} + \sum_{n=1}^{M} a_n P_n(\cos\theta)\right] \tag{2.80}$$

Here, a_n is given as follows.

$$a_n = \frac{2n+1}{2}\frac{1}{\sum_{i=1}^{N_r} w_i}\sum_{i}^{N_r} P_n(\cos\theta_i)$$

The Parameter $P_n(\cos\theta)$ is the Legendre polynomial of order n, θ is the scattering angle at which DCS is getting calculated, N_r is the number of reactive trajectories, and θ_i is the scattering angle of i^{th} trajectory. It can be seen from the above expressions of DCS and a_n that to calculate the DCS at an angle θ, information about the scattering angles at all the reactive trajectories is needed. The parameter M is the highest order of Legendre polynomial, which is to be converged to calculate the DCSs. It (parameter M) is an important parameter, and changing the value of M changes the behavior of the DCS. A high value of M will introduce oscillating behavior in DCSs. One can converge the parameter using two approaches. The first approach uses a statistical method known as the Kolmogorov–Smirnov test, which can be used when no other DCS data (such as quantum mechanical DCS or experimental data) is available. The second approach can only be used when DCS data from the other methods is available. In that case, one can vary the value of M, get the DCS, and compare to get the best value of M. The term a_n contains the weight of each reactive trajectory, and based on the binning approach, the DCS is either the Histogram or Gaussian binned DCS.

The variance in DCS at a scattering angle θ is calculated as follows.

$$\text{var}\left(\frac{d\sigma_{vj}(E_{\text{col}},\theta)}{d\omega}\right) = \left[\frac{d\sigma_{vj}(E_{\text{col}},\theta)}{d\omega}\right]^2 \left(\frac{N-N_r}{NN_r}\right) + \left(\frac{\sigma_{vj}^{E_{\text{col}}}}{2\pi}\right)^2 \text{var}(g(\cos\theta)) \tag{2.81}$$

Here, $g(\cos\theta) = \frac{1}{2} + \sum_{n=1}^{M} a_n P_n(\cos\theta)$ and $\text{var}(g(x))$ can be written as follows.

$$\text{var}[g(\cos\theta)] = \sum_{n=1}^{M}\left[\left(\frac{2n+1}{2}\right)^2 \langle P_n^2(\cos\theta)\rangle - a_n^2\right] P_n^2(\cos\theta)$$

In the above expression a_n has the same meaning as in Eq. 2.80 and $\langle P_n^2\rangle = \sum_{i=1}^{N_r} P_n^2(\cos\theta_i)/N_r$. The other method to calculate DCS uses the following formula.

$$\frac{d\sigma_{vj}(E_{\text{col}},\theta)}{d\omega} = \frac{\sigma_{vj}^{E_{\text{col}}}}{2\pi\sin\theta d\theta} P(\theta) \tag{2.82}$$

Here, $P(\theta) = \frac{N_r^\theta}{N_r}$, which is the probability of the trajectories scattering at angle θ. The Eqs. 2.80 and 2.82 can be used to calculate ICSs using the following formula.

$$\sigma_{v,j}(E_{col},\theta) = \int_{\pi=0}^{2\pi}\int_0^{\pi} \frac{d\sigma_{vj}(E_{\text{col}},\theta)}{d\omega} sin\theta d\theta d\phi = 2\pi \int_0^{\pi} \frac{d\sigma_{vj}(E_{\text{col}},\theta)}{d\omega} sin\theta d\theta \tag{2.83}$$

5. **Disposal of total energy in products**
The initial energy given to the colliding system remains the same as the reaction progresses. Upon the conversion of reactants into the products, this energy is redistributed in relative translation energy of product atom-diatom fragments and rovibrational energy

of product diatom. These different modes of energy are calculated as follows.

$$\begin{aligned}\langle f'_T\rangle &= \frac{\sum_i E_i^T}{N_r}\\ \langle f'_V\rangle &= \frac{\sum_i E_i^V}{N_r}\\ \langle f'_R\rangle &= \frac{\sum_i E_i^R}{N_r}\end{aligned} \tag{2.84}$$

Here, E_i^T is the relative translational energy and E_i^V and E_i^R are the vibrational and rotational energies of product diatom, respectively. The subscript i is the i^{th} reactive trajectory, and N_r is the total number of reactive trajectories.

2.4.2.11 Steps to perform the QCT calculation for a system

Let's consider that the reaction under the QCT study is:

$$\text{A} + \text{BC} \rightarrow \text{AB} + \text{C}.$$

The following steps are followed to perform the QCT calculations.

1. Choosing the input parameters for the reaction. The QCT code used to perform the dynamical calculations[1] reported in this thesis requires the following input parameters.

Table 2.3: Input parameters for the QCT calculations

Parameters	**Remarks**
Masses of the atoms (m_{A}, m_{B}, and m_{C})	Unit: atomic unit.
Equilibrium distance of the diatoms ($r_{\text{eq}}^{\text{AB}}$, $r_{\text{eq}}^{\text{BC}}$, and $r_{\text{eq}}^{\text{AC}}$)	Unit: Bohr.
Dissociation energy of the diatoms (D_e^{AB}, D_e^{BC}, and D_e^{AC})	Unit: Hartree.
reactant diatom distance at the outer and inner turning points	Unit: Bohr.
Morse parameters (α_{AB}, α_{BC}, and α_{AC})	Unit: Bohr^{-1}.

[1] Copy of the code can be accessed from `https://github.com/ajaymrawat`

Inner and outer turning points of reactant diatoms (Bohr)	These are the smallest and largest internuclear distances between the atoms when a diatom vibrates.
Initial distance between reactant atom and center of mass of diatom in z-axis (r_z)	This is the initial distance at the start of the trajectory run. Ideally, atom and diatom fragments should be kept at the asymptote at the start of the trajectory run so there is no interaction between them. During the calculations, this is done by varying diatomic distance from inner to outer turning points (of initial rovibrational level), calculating the potential at each point as a function of r_z, and finally looking for the asymptote. Unit: Bohr.
Distance between product atom and center of mass of diatom in z-axis (r_z^p)	This is the distance at which the product has formed and no longer can fall back to the interaction region. This is chosen in the same way as r_z. Since the products can attain any rovibrational level available in the energy range of collision, multiple calculations with different sets of inner and outer turning points can be used. This distance can be the same or different for all the three diatoms. In the case of the product being the same diatom as the reactant (elastic or inelastic collision), the distance is defined separately as a parameter named d_{nr}. Unit: Bohr.
d_{nr}	This is the interatomic distance of reactant diatom at the outer turning point of the higher rovibrational level. Unit: Bohr.

Total numbers of trajectories (n_{traj})	The value is converged by running QCT calculations for different n_{traj} at a specific collision energy and checking for convergence of reaction probabilities or ICSs.
Half of the vibrational time period of reactant diatom (v_{htp})	This is the time diatom takes to vibrate from inner to outer turning point, which is the half vibration. It is calculated by propagating the diatom from inner (outer) to outer (inner) turning points and calculating the time taken. In the QCT code, this is used for the randomization of the initial diatomic distance. Unit: atomic unit.
Maximum impact parameter (b_{max})	Impact parameter is the distance between the centers of colliding species in the perpendicular direction to the path of attacking species. The maximum impact parameter is the maximum value of the impact parameter beyond which collision between the two fragments doesn't result in product formation. Before the final calculations, b_{max} values are converged at each collision energy by running a small number of trajectories (20000 or 30000). Unit: Bohr.
Boundaries of interaction region (r_{int})	This is the sum of the three bond distances of the triatomic system at the start and end of the interaction region. This parameter calculates the collision time, the classical equivalent of a complex lifetime. The value is chosen by inspecting a set of trajectories for a point in trajectory run when the potential shows a sharp dip, indicating complex formation. Unit: Bohr.

2. As discussed above, convergence of the b_{max} requires running the calculations for small

number of trajectories. This needs clipping the PES with the QCT code. The PES reference should be the reactant A + BC asymptote. Each trajectory run should be checked for accuracy of propagation as described in section 2.4.2.3.

3. After b_{max} convergence, the final calculations can be performed for a larger number (n_{traj}) of trajectories.

4. The Final step is the calculation of reaction observables.

Chapter 3

Quantum and classical comparison and Isotopic effect in H + $LiH^+ \rightarrow H_2$ + Li^+ reaction

3.1 Introduction

Lithium is one of the most abundant elements present in the early universe after hydrogen (H), helium (He), and deuterium (D). In the beginning, these species were present in ionic form, and as the universe started cooling, the atoms and molecules formed by the recombination of ionic species. The Li atom led the formation of neutral LiH molecule and LiH^+ molecular ions. These molecular species formed through different ion-atom radiative associations, atom-atom radiative associations, and ion-atom radiative detachment processes [50–52]. These LiH and LiH^+ species are believed to be crucial in the evolution of early universe chemistry, and their seemingly simple reactions with H^+ and H exhibit exciting features. The molecular ion LiH^+, which is an important part of the Li reaction network, is formed by the ion-atom radiative association process between Li^+ and H [50]. These molecular ions further get destroyed by reacting with the abundant H atom through the following processes:

$$\mathrm{H} + \mathrm{LiH}^+ \rightarrow \mathrm{Li}^+ + \mathrm{H} + \mathrm{H}$$

$$\mathrm{H} + \mathrm{LiH}^+ \rightarrow \mathrm{H}_2 + \mathrm{Li}^+$$

The first process is termed CID, which becomes important once the collision energy is higher than ~ 0.112 eV, the binding energy of LiH^+, including its ZPE. The second reaction, H + $LiH^+ \rightarrow H_2$ + Li^+, is a depletion process, and the dynamics of this reaction is investigated in this work. The reaction has a large exoergicity of ≈ 4.36 eV (4.6 eV) with (without) zero point energy correction of reactant and product diatoms [53, 54]. An energetic representation of a few important configurations and diatomic states is shown in Fig. 3.1.

3.2 Potential energy surface involved in the reaction

The reaction takes place on the adiabatic singlet ground electronic state of the LiH_2^+ system, which is energetically well separated from the first excited electronic state with a large energy gap at all relevant nuclear geometries [53, 55]. In the past, various PESs are already being constructed for the LiH_2^+ system [54, 56–60]. However, the PES constructed by Mar-

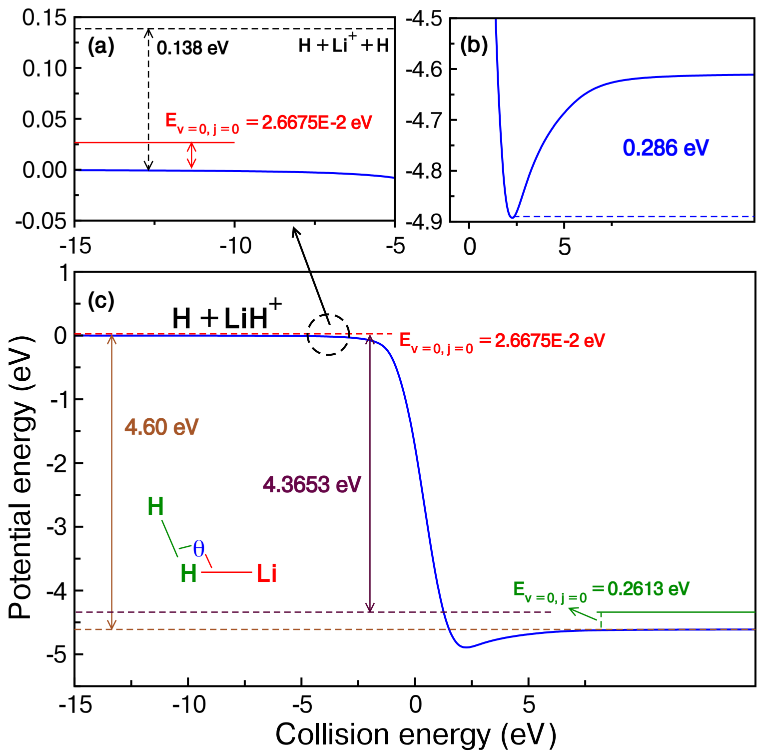


Figure 3.1: Energy diagram for the H + $LiH^+ \rightarrow H_2 + Li^+$ reaction. Panels (a) and (b) are the zoomed portion of reactant and product asymptotes, respectively.

tinazzo et al. [54] is used in the present work. This PES was initially constructed by fitting 11000 energy points and further improved with 600 MRCI energy points. The long-range part of the PES was modeled accurately by using appropriate two-body and three-body analytical functions. Recent PESs of LiH_2^+ reactive system include that reported by He et al.[59] and Dong et al. [60] However, these new PESs do not include the analytical long-range part, which is important for reactions involving ionic atoms and molecules. The PES of He et al. [59] uses a switching function to ensure the regularity in the potentials at large internuclear distances, but no long-range analytical function is used. The same use of the switching function is made in the work of Dong et al. [60] Despite the absence of the correction for basis set superposition error (BSSE), various later theoretical studies [57, 58, 61] show that the surface of Martinazzo et al. [54] is fairly accurate.

The LiH^+ depletion reaction, which is of interest in the current work, follows the minimum energy path (MEP) at the attacking angle ($\angle H-H-Li$) of 80° with no barrier. The MEPs at different attacking angles are shown below. It can be seen from the figure that the attack from the H side has a repulsive barrier. As the attacking angle increases and H approaches from the Li side, the approach becomes gradually attractive in nature. The global minimum

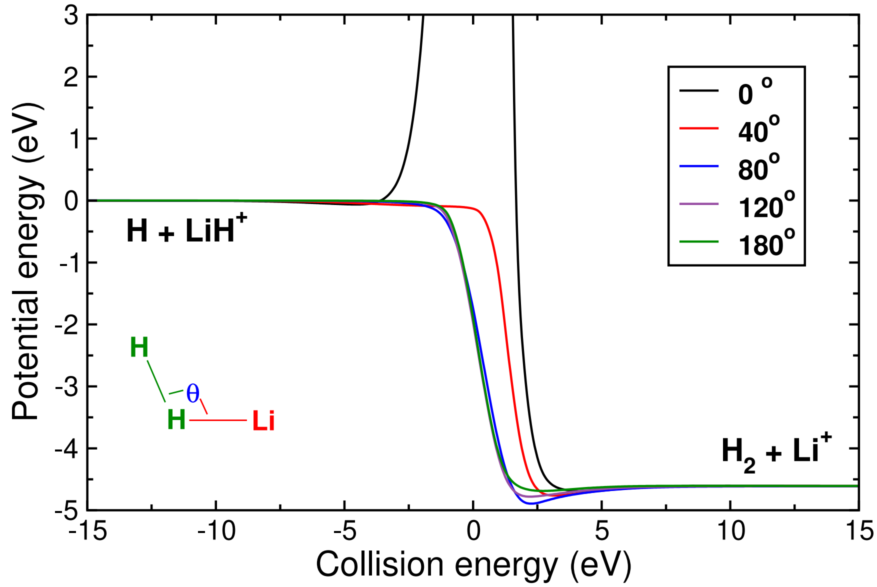


Figure 3.2: MEPs at different attacking angles (θ). The path with the deepest minima is at 80°.

(shallow) has the C_{2v} point group with energy of $\sim$ 0.286 eV with respect to the (H_2 + Li^+) product asymptote.

3.3 Previous studies and motivation

This reaction is important in the astrophysical evolution of Li chemistry and has been studied widely using QM [60, 62–68] and QCT [69–74] methods. Despite the considerable amount of work on this reaction system, only a few studies have been reported that are quantum mechanically exact and performed at the state-to-state level. Of these studies, only a few addressed the comparison of QM and QCT results and the isotopic effect on the dynamics of the reactions. The reaction, $H + LiH^+ \rightarrow H_2 + Li^+$, is studied by Dong et al. [60] by using QM-CC and QCT methods on a PES constructed by the same authors. [60] The QM-CC and QCT total ICSs and DCSs were calculated and compared. The QM-CC and QCT total ICSs didn't agree well with each other in lower and higher collision energy ranges. The DCSs were shown up to 0.35 eV of collision energy, and QCT and QM-CC DCSs showed reasonably good agreement. The most recent and rigorous state-to-state dynamical study for the same was performed by Sahoo et al. [65] using the PES of Martinazzo et al. [54]. The comparison of the QM-CC and previous QCT [69] results for the total ICSs in the collision energy range of 0.001 - 1.0 eV was done, and an excellent agreement was found. The QCT and QM results were also compared for other exoergic triatomic reactions. As far as other reaction systems are concerned, the comparison on $[NeHeH]^+$ system for Ne + $HeH^+ \rightarrow$

NeH^+ + He, a barrierless exoergic reaction, was done for total reaction probability, total ICSs, and rate constant and a good agreement was found [75]. Another barrierless exoergic reaction, H + LiH → H_2 + Li, was also studied by Wang et al. [76] in the collision energy range 0.02 - 0.56 eV. The QCT total reaction probability, total ICSs, and DCSs calculated in their study were compared with the different QM-CC results on the same LiH_2 PES. A good agreement was found between QCT and QM-CC results.

Regarding the studies concerning the isotopic effect in LiH_2^+ system, Li et al. [72] in 2010 studied the isotopic effect on the stereodynamics of the H/ D/ T + LiH^+ → HH/ HD/ HT + Li^+ and H + LiH^+/ LiD^+/ LiT^+ reactions. This study was done at a collision energy of 0.5 eV. A QCT method was employed to study the stereodynamics on the PES of Martinazzo et al. [54]. The DCSs were calculated at the initial state-selected level. The authors showed that in case of change in mass of the attacking atom, H/ D/ T + LiH^+, the DCS becomes backward dominated for H as an attacking atom and remains forward for D/ T as an attacking atom. But on changing the mass of the target diatom, H + LiH^+/ LiD^+/ LiT^+, the backward-dominated DCS remains unaffected. Later, Zhu et al. [67] also calculated the DCS for the H + LiH^+ → H_2 + Li^+ reaction. Their DCS at 0.5 eV was forward-biased, opposite to the backward-dominated DCS of Li et al. [72]. The same forward-dominated behavior of the DCS at 0.5 eV was also reported by Sahoo et al. [66]. These contradictions raise concerns about the studies on kinematic effect and stereodynamics reported by Li et al. [72] In this series of studies, Ya-Min Li et al. [77] also studied H/ D + LiH^+ → HH/ HD + Li^+ reaction using the QCT method to calculate the total ICS, product rotational alignment, and product vibrational distributions.

So, it is clear from the above discussion that despite several studies on the reaction system, a rigorous study on the comparison of QM-CC and QCT method and kinematic effect is still lacking. The work presented in this chapter aims to fill up this gap by presenting a rigorous comparison of various reaction attributes calculated by the QM-CC and QCT methods at the state-to-state level and by exploring the kinematic effect on them. For this purpose, the following reactions are considered in the following.

$$\mathrm{H} + \mathrm{LiH}^+\ (v = 0, j = 0) \rightarrow \mathrm{H_2}\ (v', j') + \mathrm{Li}^+ \qquad \text{(R1)}$$

$$\mathrm{D} + \mathrm{LiH}^+\ (v = 0, j = 0) \rightarrow \mathrm{HD}\ (v', j') + \mathrm{Li}^+ \qquad \text{(R2)}$$

$$\mathrm{H} + \mathrm{LiD}^+\ (v = 0, j = 0) \rightarrow \mathrm{HD}\ (v', j') + \mathrm{Li}^+ \qquad \text{(R3)}$$

$$\mathrm{D} + \mathrm{LiD}^+\ (v = 0, j = 0) \rightarrow \mathrm{D_2}\ (v', j') + \mathrm{Li}^+ \qquad \text{(R4)}$$

The first and foremost objective of performing a comparison of QM-CC and QCT methods for such an exoergic barrierless reaction is to obtain and understand any valid complementarity and agreement between the results obtained by the two methods both at initial state-specific and state-to-state levels. Such an attempt is essential to take advantage of the computationally less demanding QCT method in obtaining at least the thermal and state-to-state rate constants for astrochemical modeling if a good agreement between QM-CC and QCT results is achieved. In this work, the QM-CC results are presented only for reactions R1 and R2. These results are taken from two earlier studies of ours [65, 66]. The QCT calculations for all four reactions are performed, and the results are presented. The comparison between different QM-CC and QCT reaction observables like ICS, product vibrational level resolved ICS, product rotational level resolved ICS, DCS, and product energy disposal is reported. Further, these QM-CC and QCT comparisons are used to assert the applicability of the quasi-classical method to the above exoergic and barrierless reactions. The effect of isotopic substitution on product energy disposal and reaction mechanism is studied in terms of the observables mentioned above.

3.4 Computation details

As pointed out, the QM-CC results are presented only for R1 and R2 reactions, a detailed discussion of which can be found in Ref. [65]. The method used for these calculations is the same as discussed in section 2.4.1 of chapter 2. QCT methodology followed for the dynamical calculations of this chapter is the same as discussed in section 2.4.2 of Chapter 2. Hamilton's equations are solved using the fourth-order adaptive step size Runge-Kutta method [78] in the framework of Jacobi coordinates, and initial conditions as discussed in Chapter 2 are sampled using the sampling technique called Monte Carlo.

To start the propagation, the first step is to converge the input parameters, and the techniques used are the same as discussed in section 2.4.2 of Chapter 2. The input parameters converged for the four reactions are as follows.

Table 3.1: Input parameters for R1, R2, R3, and R4 reactions.

Parameters	**R1**	**R2**	**R3**	**R4**
Initial time step (h)	1.0	1.0	1.0	1.0
r_z (Bohr)	30.0	30.0	30.0	30.0
r_z^p (Bohr)	30.0	30.0	30.0	30.0
n_{traj}	500000	500000	500000	500000
d_{nr} (Bohr)	10.0	8.273	10.0	10.0
Inner turning point (Bohr)	3.6910	3.691	3.7399	3.7399
Outer turning point (Bohr)	4.83381	4.83381	4.71851	4.71851

Half vibrational time period (v_{htp}) (au)	4401	4401	4503	4503
Boundaries of interaction region (r_{int}) (Bohr)	17.0	17.0	17.0	17.0
$E_{v=0,j=0}$ of reactant diatom (Hartree)	9.8021E-4	9.8021E-4	7.4787E-4	7.4787E-4

Atomic mass(m_{H}/ m_{D}/ m_{Li}) (amu)	1.007/ 2.014/ 6.939
Equilibrium distance of the diatoms ($r_{\mathrm{eq}}^{\mathrm{H_2}}$/ $r_{\mathrm{eq}}^{\mathrm{LiH^+}}$/ $r_{\mathrm{eq}}^{\mathrm{LiD^+}}$/ $r_{\mathrm{eq}}^{\mathrm{HD}}$) (Bohr)	1.4455/ 4.1365/ 4.1365/ 1.4455
Dissociation energy of the diatoms ($D_{\mathrm{e}}^{\mathrm{H_2}}$/ $D_{\mathrm{e}}^{\mathrm{LiH^+}}$/ $D_{\mathrm{e}}^{\mathrm{LiD^+}}$/ $D_{\mathrm{e}}^{\mathrm{HD}}$) (Hartree)	0.1741/ 0.00509/ 0.00509/ 0.17413
Morse parameters ($\alpha^{\mathrm{H_2}}$/ $\alpha^{\mathrm{LiH^+}}$/ $\alpha^{\mathrm{LiD^+}}$/ α^{HD}) (Bohr^{-1})	1.08272/ 0.817606/ 0.817606/ 1.08272

Out of these input parameters, an important parameter is the maximum value of the impact parameter. It is the value of b beyond which collisions don't result in product formation, and thus, sampling them in the calculations produces erroneous results.

To converge the b_{max} value, a series of calculations are performed for different fixed b values, and the probability of the reaction is compared. A total of 20,000 trajectories are run in each of these calculations. The value of b at which the probability goes to zero is taken as the b_{max}. A representation plot of the process for a few collision energies is given in Fig. 3.3. It can be seen from the figure that as the b value increases, the reaction probability decreases and slowly approaches zero.

The converged b_{max} values for reactions R1, R2, R3, and R4 are given in Table 3.3.

Table 3.3: Maximum impact parameter (b_{max}) for R1, R2, R3, and R4 reactions.

Collision energy (eV)	**b_{max} (Bohr)**			
	R1	**R2**	**R3**	**R4**
0.005	15.3	15.5	15.1	15.1
0.01	13.9	14.0	13.4	13.4
0.02	12.7	12.8	12.2	12.2
0.03	12.2	12.2	11.6	11.7
0.04	11.8	11.9	11.3	11.3

0.05	11.6	11.6	11.0	11.0
0.06	11.4	11.4	10.8	10.8
0.07	11.2	11.2	10.7	10.7
0.08	11.1	11.1	10.6	10.6
0.09	11.0	11.0	10.5	10.5
0.10	10.9	11.0	10.4	10.4
0.12	10.8	10.8	10.2	10.2
0.14	10.7	10.7	10.1	10.1
0.16	10.6	10.6	10.0	10.0
0.18	10.6	10.6	10.0	10.0
0.20	10.5	10.6	9.9	9.9
0.25	10.4	10.4	9.7	9.8
0.30	10.3	10.4	9.6	9.6
0.35	10.1	10.2	9.5	9.5
0.40	10.0	10.3	9.4	9.4
0.45	9.8	10.1	9.3	9.2
0.50	9.6	9.8	9.2	8.2
0.55	8.3	9.8	9.1	8.0
0.60	8.1	9.8	9.0	7.9
0.65	8.1	9.6	8.9	7.6
0.70	7.9	9.4	8.8	7.4
0.75	7.8	9.3	8.7	7.4
0.80	7.8	9.1	8.5	7.3
0.85	7.7	8.6	8.4	7.2
0.90	7.6	8.7	8.5	7.1
0.95	7.5	6.5	8.5	7.1
1.00	7.5	6.5	8.2	6.9

As it can be seen from Table 3.1, trajectories were started at a distance of 30 Bohr between the attacking atom and reactant diatom and stopped after the formation of the product diatom occurs when the distance between the product diatom and atom reaches 30 Bohr. These two distances are chosen so there is no interaction between the atom and diatom pair on the reactants/ products side. Although one can be slightly relaxed when choosing these distances, the values shouldn't be so small that the trajectory run starts and ends in the interaction region. In the same way, a distance bigger than the required distance is not incorrect, but it will unnecessarily increase the propagation time. Also, excluding CID

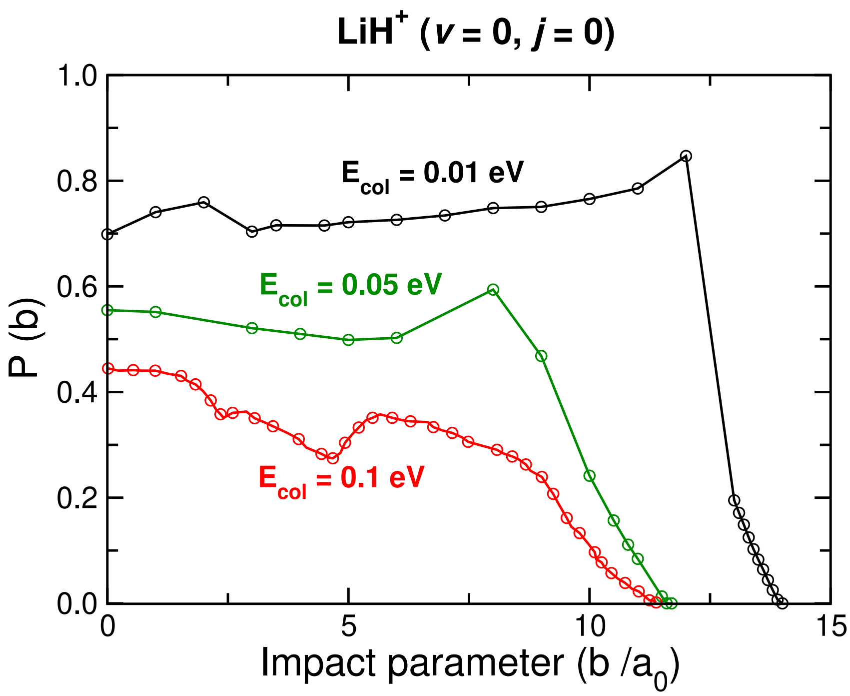


Figure 3.3: Plot showing the convergence of b_{max} for different collision energies.

trajectories is important during the propagation since the dissociation channel opens up at low collision energy. These CID trajectories are excluded from the reactive and non-reactive channels by comparing the ro-vibrational energies of the product diatoms at their respective asymptotes with the three-body dissociation energies. The procedure to calculate the ro-vibrational energies is discussed in section 2.4.2 of Chapter 2.

The final calculations of cross section and other observables were performed by running 5×10^5 trajectories. After each trajectory, product vibrational and rotational quantum levels are calculated using the procedure discussed in section 2.4.2.8 of Chapter 2. These vibrational and rotational quantum numbers calculated using the above procedure are generally real and should be converted into integer numbers. This conversion of real to integer values is carried out using the standard HB and GB techniques discussed in section 2.4.2.9 of Chapter 2. In the HB, the real quantum numbers of the trajectories are converted into the nearby integers. Each trajectory is given the same weight of 'one' irrespective of the difference between the nearby integer and the real quantum number value. In the case of GB, each trajectory is given some weight, which depends on the proximity of the calculated real-valued quantum number and a nearby integer value. The mathematical formulae used to calculate the QCT-HB and QCT-GB reaction observables are discussed in section 2.4.2.10 of Chapter 2.

3.5 Results and discussion

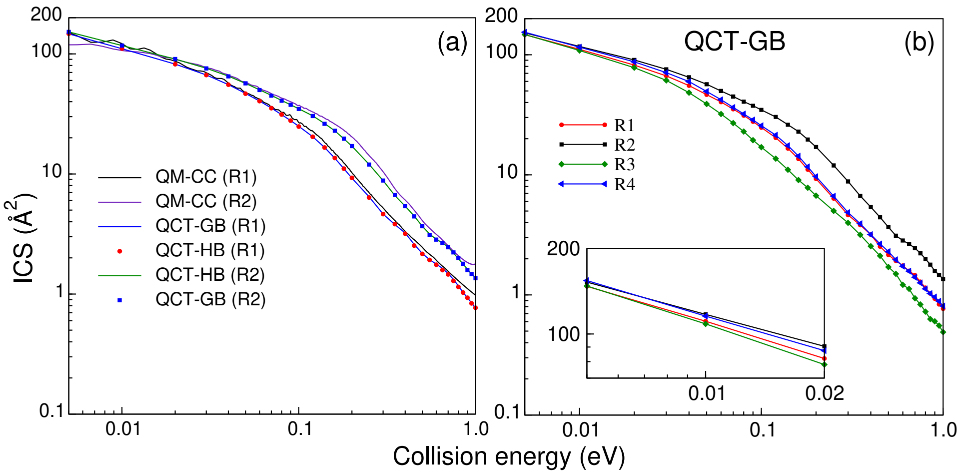


Figure 3.4: Initial state-selected QM-CC and QCT ICSs as a function of collision energy. QM-CC, QCT-HB, and QCT-GB ICSs for reactions R1 and R2 are shown in panel (a). Panel (b) shows QCT-GB ICSs for all four reactions. The QCT-GB ICSs in the collision energy range 0.005 - 0.02 eV are shown in the inset.

3.5.1 Total ICS

Total ICSs for the four reactions are shown on a logarithmic scale in Fig. 3.4. The QM-CC, QCT-HB, and QCT-GB ICSs for R1 and R2 are compared in panel (a), whereas panel (b) shows the QCT-GB ICSs for R1, R2, R3, and R4. It can be seen from panel (a) that both the QCT-HB and QCT-GB ICSs show excellent agreement throughout the collision energy range for R1 and R2. The QCT ICSs for R1 and R2 agree well with QM-CC ICS, but a few variations at the lower collision energies are seen for reaction R2. Both the QM-CC and QCT ICSs have no signature of resonance oscillations. Being a classical method, the latter can not capture quantum phenomena like resonance or interference. On the other hand, total QM-CC ICS is the summation of different state-to-state ICSs, which averages out the resonance structures. Also, resonance structures may vanish due to averaging over different partial waves, which is required to calculate the total ICS. QCT ICSs for all four reactions are shown in Panel (b). The standard deviations in the QCT ICSs for reactions R1, R2, R3, and R4 at selected collision energies are calculated using Eq. 2.77 and presented in Table 3.4.

All the reactions show the same pattern of ICSs as a function of collision energy, where the reactivity is high at the lower collision energy, and it decreases as the collision energy increases. Only the QCT-GB ICSs are shown because both the QCT-GB and QCT-HB ICSs agree well in the full collision energy range [cf., panel (a)]. The isotopic effect on the ICSs

Collision energy (eV)	**Standard deviation (Å^2)**			
	R1	**R2**	**R3**	**R4**
0.005	0.1322	0.1338	0.1246	0.1198
0.05	0.0081	0.0083	0.0072	0.0075
0.10	0.0062	0.0070	0.0051	0.0060
0.14	0.0052	0.0062	0.0041	0.0050
0.20	0.0040	0.0052	0.0032	0.0038
0.35	0.0026	0.0034	0.0022	0.0024
0.50	0.0018	0.0024	0.0015	0.0016
0.65	0.0013	0.0020	0.0013	0.0012
0.80	0.0011	0.0017	0.0009	0.0010
1.00	0.00086	0.00097	0.00074	0.00081

Table 3.4: Standard deviations in QCT ICSs for R1, R2, R3, and R4 reactions.

is evident from these plots, and various observations can be made. These observations can be explained a bit more clearly by categorizing the reaction pairs in two different ways: Type I and Type II. Type I is the reaction pair where both of the reactions of the pair have either the same attacking atom (R1 - R3 and R2 - R4) or the same target diatom (R1 - R2 and R3 - R4). Type II (R1 - R4) shows the reaction pair where the attacking atom and target diatom are different in both reactions. The observations are as follows.

1. The four reactions show a minor difference in the reactivity order at the lower end of the collision energy ($\approx$ 0.005 eV). The reactivity differences start appearing with increasing collision energy and follow the order R2 > R4 $\approx$ R1 > R3.
2. **Type I:** When the target diatom is the same (R1 - R2 and R3 - R4), a heavy attacking atom results in higher reactivity, whereas a lighter target diatom results in high reactivity when the attacking atom is the same, e.g., R1 - R3 and R2 - R4 pairs.
3. **Type II:** In the case of the pair of reactions R1 and R4, where both the attacking atom and target diatoms are different, the reactivity of both reactions is almost the same.
4. In both the cases discussed in Type I, the change in the ICSs between the reactions is plotted as a function of collision energy and shown in Fig. 3.5. The formula used to calculate the change in ICS between the reactions A and B at a collision energy of E_{col} is as follows.

$$\text{Change in the ICS (\%)} = \left(\frac{\sigma_A - \sigma_B}{\sigma_B}\right) \times 100 \tag{3.1}$$

It can be seen that the change in ICSs for the reactions in pairs (R1 - R2 or R3 - R4 or R1 - R3 or R2 - R4) follows the same qualitative pattern. In the lower range of collision energy, it increases almost linearly with collision energy, and then, with increasing collision energy, it becomes constant.

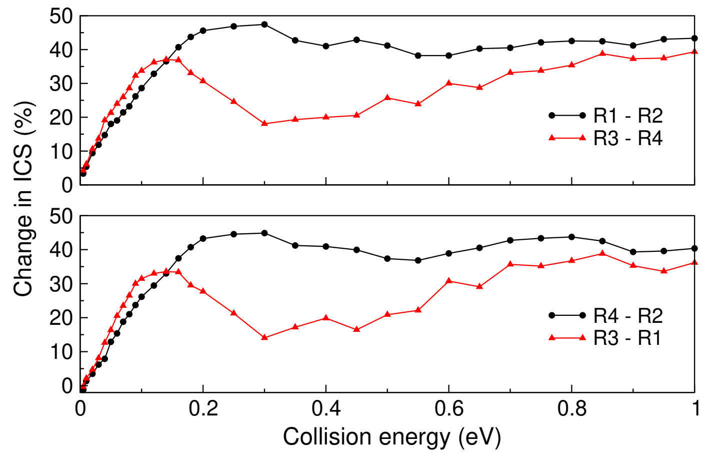


Figure 3.5: The change in QCT-GB ICS (%) for a pair of reactions as a function of collision energy.

3.5.2 Product vibrational level resolved ICS

Product vibrational level resolved QM-CC, QCT-HB, and QCT-GB ICSs are shown in Fig. 3.6 in terms of population distributions at three different collision energies 0.02 eV, 0.35 eV, and 0.65 eV. The first and second columns of Fig. 3.6 show the product vibrational level

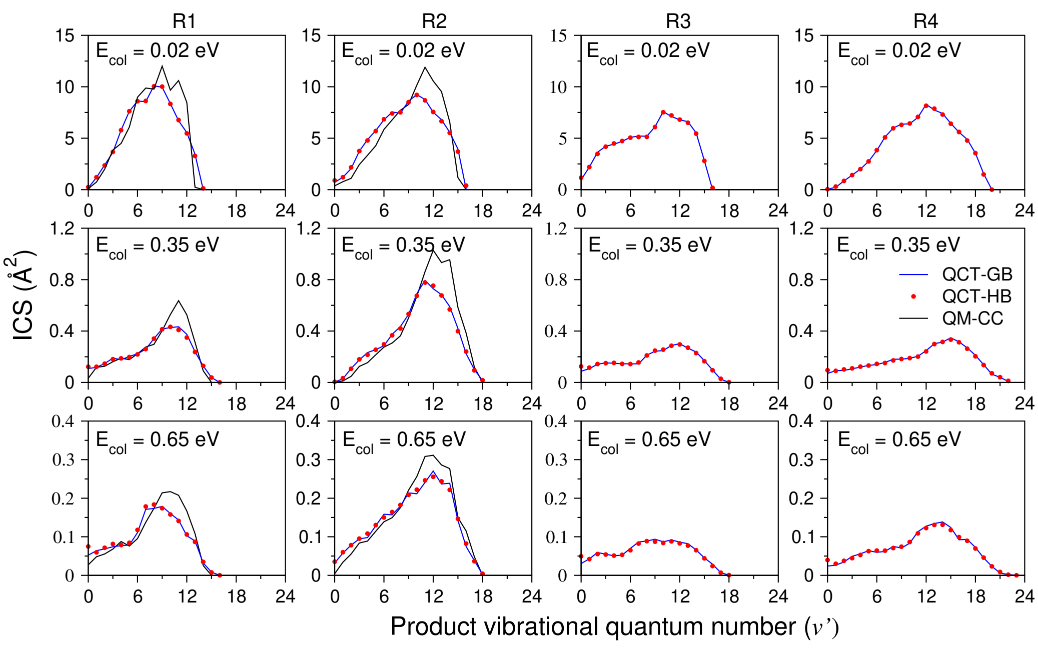


Figure 3.6: Product vibrational level distributions regarding ICSs at three different collision energies. QM-CC, QCT-HB, and QCT-GB ICSs are shown for R1 and R2 reactions, and only QCT-GB and QCT-HB ICSs are shown for reactions R3 and R4.

(v') resolved QM-CC, QCT-HB, and QCT-GB ICSs for R1 and R2, respectively. The third and fourth columns show only the QCT-HB and QCT-GB distribution for reactions R3 and R4, respectively. The QCT-HB and QCT-GB ICSs agree reasonably well with each other.

It can be seen that both the QM-CC and QCT ICSs are in good agreement and show the same inverse Boltzmann distribution where the products are vibrationally hot. The same behavior of the ICS distribution is also revealed by reactions R3 and R4. In the case of reactions R1 and R2, when the contribution of the v'-resolved cross section to the total ICS is compared, it is observed that the highest contribution comes from almost the same v' levels in both QM-CC and QCT methods. The effect of isotopes can be seen in the number of product vibrational levels accessed in different reactions, and this is because the product diatom is different in reactions R1 and R4 than in R2 or R3. In the case of R2 and R3, the product diatom is HD, whereas, for reactions R1 and R4, the product diatoms are H_2 and D_2, respectively. The energy difference between the successive vibrational levels of the product diatom follows the order H_2 > HD > D_2, which means that at given collision energy, the number of vibrational levels accessible for reactions follows the order R4 > R2 = R3 > R1. This is observed in the vibration level distributions in ICSs shown in Fig. 3.6. Another difference is the magnitude of ICS coming from different vibrational levels for different reactions. This difference mimics the trend of overall ICS as discussed in 3.5.1. In the case of reactions R3 and R4, as the collision energy increases, the product vibrational level resolved ICS distribution gets flat, and the contribution of different product vibrational levels in total ICS starts getting closer to each other. This change in product vibrational distribution is more prominent in R3.

3.5.3 Product rotational level resolved ICS

Product rotational level resolved ICSs are shown in Fig. 3.7 for all four reactions in terms of population distributions at 0.35 eV collision energy.

The QM-CC and QCT ICS distributions are presented for R1 and R2, whereas only QCT distributions are presented for R3 and R4. In the case of QCT, both the QCT-HB and QCT-GB distributions were calculated; however, only QCT-GB ICSs are presented because QCT ICS distributions calculated at the rotationally resolved level are the same for HB and GB methods. In Fig. 3.7, these rotational level resolved ICS distributions are shown as color-mapped plots where each block represents the numerical value of rotational level resolved ICS for a combination of product vibrational (v') and rotational (j') levels.

For all four reactions, the distribution of ICSs follows the same so-called diagonal pattern where vibrationally hot diatomic product states are rotationally cold and rotationally hot diatomic product states are vibrationally cold. The difference in the rotational distribution of R1, R2, R3, and R4 is the accessibility of different rotational and vibrational levels, as discussed previously, due to different product diatoms. These diatoms for reactions R1, R2,

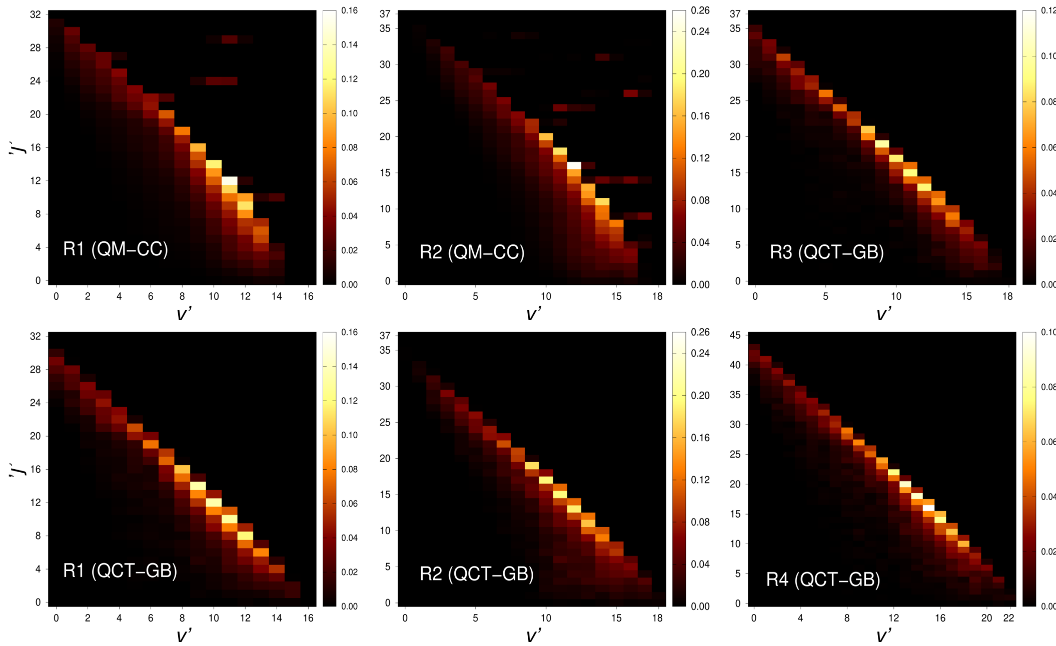


Figure 3.7: Product rotational level resolved ICSs of the four reactions as a function of v' (abscissa) and j' (ordinate) at a collision energy of 0.35 eV. The QM and QCT-GB ICS are shown for reactions R1 and R2, and only QCT-GB ICS is shown for reactions R3 and R4. The product diatoms for reactions R1, R2, R3, and R4 are H_2, HD, HD, and D_2, respectively.

R3, and R4 are H_2, HD, HD, and D_2, respectively. The above-discussed diagonal features of the plots, common for an exoergic reaction, [79] are discussed in detail in our earlier publication [65]. This diagonal feature in the j' resolved state-to-state ICSs can be seen more clearly in Fig. 3.8, where the QM-CC and QCT-GB ICSs for some specific v' of reactions R1 and R2 are plotted as a function of j'. The similarities of the QM-CC and QCT-GB ICSs at fixed vibrational levels can be observed from this figure, except that the highest contributing product rotational level in the QCT-GB ICS shifts to a slightly lower value than the QM-CC distribution. The same diagonal behavior is also observed in the rotational level resolved ICSs at 0.02 eV and 0.65 eV of collision energies, which are shown in Fig. 3.9 and 3.10. We note here that the product rotational level resolved ICSs for reactions R1 and R4 described above are obtained without the inclusion of nuclear spin statistics of the homonuclear product diatom. For a detailed explanation, the readers are referred to the last paragraph of section 3.6.

3.5.4 Product energy disposal mechanism

After calculating vibrationally and rotationally resolved state-to-state ICS and its distribution in the product vibrational and rotational levels, the energy disposed to different modes of products is also obtained. The average fractions of available energy entering into product

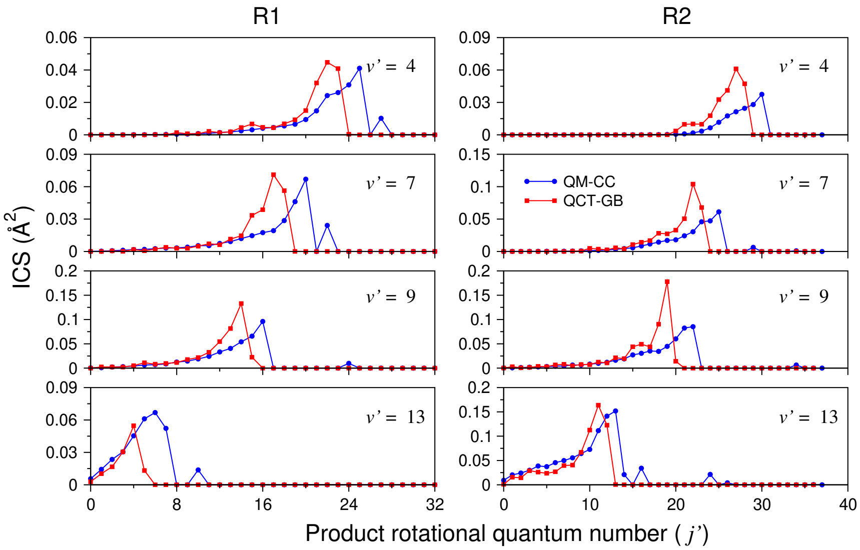


Figure 3.8: Initial state-selected QM-CC and QCT-GB ICSs for some selected product vibrational level (v') as a function of product rotational level (j') for the reactions R1 and R2 at 0.35 eV collision energy.

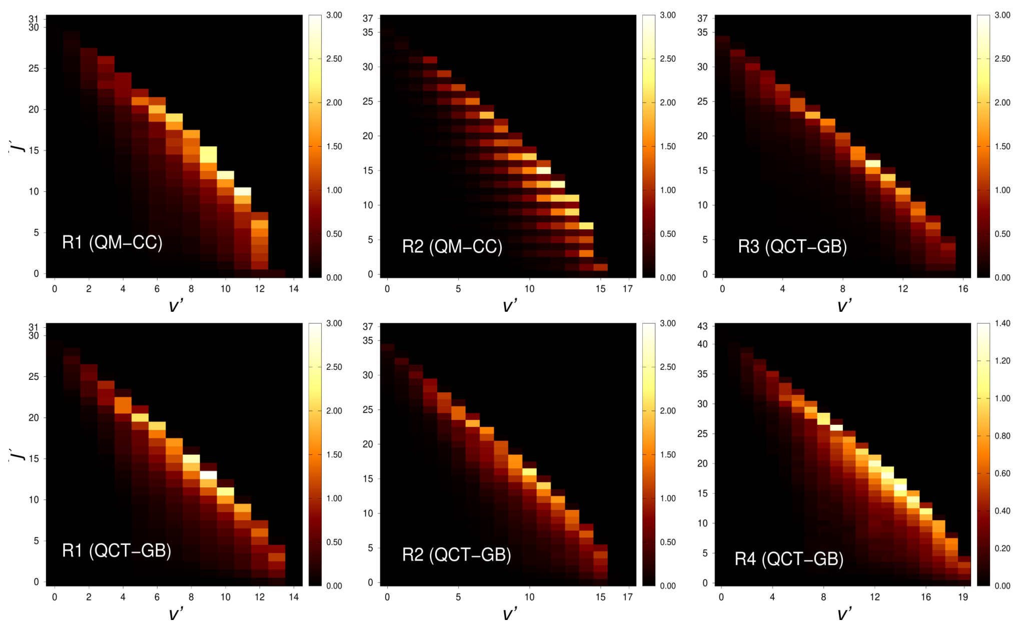


Figure 3.9: Product rotational level resolved ICSs of the four reactions as a function of v' (abscissa) and j' (ordinate) at a collision energy of 0.02 eV. The QM-CC and QCT-GB ICS are shown for reactions R1 and R2, and only QCT-GB ICS is shown for reactions R3 and R4. The product diatoms for reactions R1, R2, R3, and R4 are H_2, HD, HD, and D_2, respectively.

vibration ($\langle f'_V \rangle$), translation ($\langle f'_T \rangle$), and rotation ($\langle f'_R \rangle$) are shown in Fig. 3.11 as a function of collision energy. The QM average fraction ($\langle f' \rangle$) values are taken from an earlier QM-CC study of ours [65] and the equations used to calculate the fractions are given in the

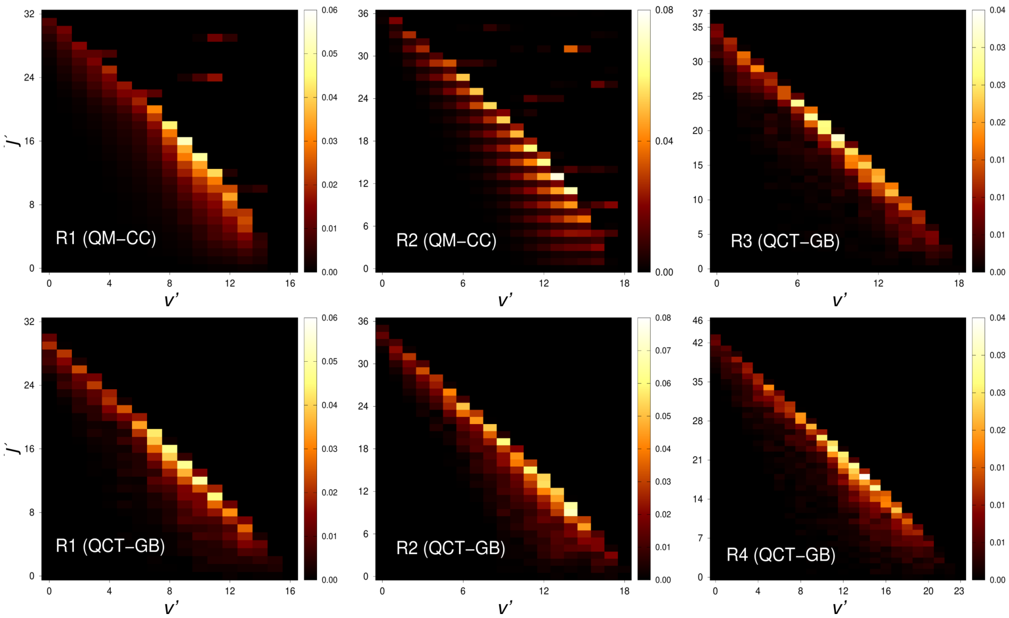


Figure 3.10: Same as Fig. 3.9 but at the collision energy of 0.65 eV.

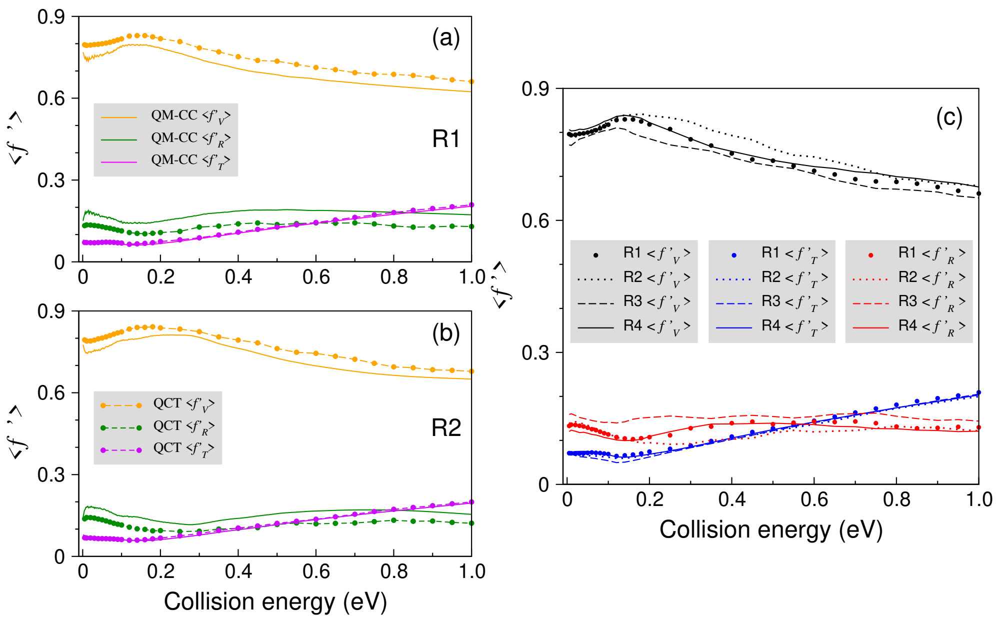


Figure 3.11: Average fraction of available energy disposed to the product vibration ($\langle f'_V \rangle$), rotation ($\langle f'_R \rangle$), and translation ($\langle f'_R \rangle$) as a function of collision energy. QM-CC and QCT energy disposal for R1 and R2 are shown in panels (a) and (b), respectively. In panel (c), only the QCT product energy disposal is compared for reactions R1, R2, R3, and R4.

same article. However, the QCT average energy fraction is calculated using the equations given in section 2.4.2.10 of Chapter 2. Panels (a) and (b) of Fig. 3.11 show the QCT and QM-CC energy disposal for R1 and R2, respectively. It can be seen from the figure that a significant amount of the available energy goes into the product vibration. This behavior of energy disposal is typical for exoergic reactions occurring on "attractive" PES [65, 79, 80]. Interestingly, the amount of energy going into product vibration decreases roughly after 0.15 eV, and with this, an increased amount of energy can be seen to flow to the product translation. Whereas the fraction of energy going into product rotation remains flat after collision energy of $\approx$ 0.5 eV. This particular feature in the behavior of energy disposal is discussed in detail in our earlier work [65] and won't be discussed here for brevity.

Comparing the ($\langle f' \rangle$) values calculated by the QCT method, it is observed that the QCT energy disposal values almost closely resemble that calculated by the QM-CC method. For reactions R1 and R2, both the QM-CC and QCT energy disposal to the product translation are the same, but the difference is seen in the energy disposal to vibrational and rotational degrees of freedom. The QCT $\langle f'_V \rangle$ are higher than the QM-CC $\langle f'_V \rangle$ values and QCT $\langle f'_R \rangle$ follows the opposite trend.

The QCT energy disposal for reactions R1, R2, R3, and R4 reactions are compared in Panel (c) of Fig. 3.11 to show the isotopic effect. It can be seen that the energy disposal mechanism for all four reactions is almost the same. This similarity in the energy disposal pattern signifies that the isotopic effect on the product energy disposal is quantitative but not qualitative. The trend of energy disposed of in product vibration is R2 > R4 $\approx$ R1 > R3, which is the same as the trend of total ICS for the reactions. For the reaction pairs R1 - R2 and R3 - R4, the product energy disposal in vibration is higher for the reaction with heavier attacking atoms. In contrast, for reaction pairs R1 - R3 and R2 - R4, product vibrational energy disposal is higher for the lighter target diatom. In the case of reaction pair R1 - R4, the energy flow to the product vibration is almost the same except at higher collision energy, where the product vibrational energy for reaction R4 is a little higher than that of R1. The above trend is followed in the whole collision energy range except for a bit of off-trend for reaction R1 at a lower and higher collision energy. The fraction of energy flowing into product translation is almost the same for all the reactions, and the trend observed in the case of energy disposal in product rotational mode is opposite to that of the vibrational mode.

3.5.5 Contribution of direct and indirect mechanisms in total ICS

The reaction R1, which has a shallow well in the product valley in the underlying PES, follows two different mechanisms depending on the collision energy as found in our earlier study [66]. It follows a mixed direct and indirect mechanism at lower collision energy, with a dominant indirect contribution. At higher collision energies, it mainly follows a direct stripping mechanism [65, 66]. This was confirmed by both QM partial wave contribution analysis and QCT method [66]. To understand the mechanism of other isotopic variants of the reaction, the average collision time of the four reactions is calculated. The collision time is the measure of the total time spent by the reactive system in the interaction region. To define the interaction region, a bunch of trajectories are followed to find the change of the potential as a function of the sum of three interatomic distances. The sum of distances at the point of the sudden drop of the potential at 17 a_0 is considered the boundary of the interaction region. A few of the such trajectories that were used to determine the start of interaction region at 17 a_0 for reaction R1 are shown in Fig. 3.12. The same exercise was performed for R1, R2, and R4 reactions. The collision time calculated for each trajectory

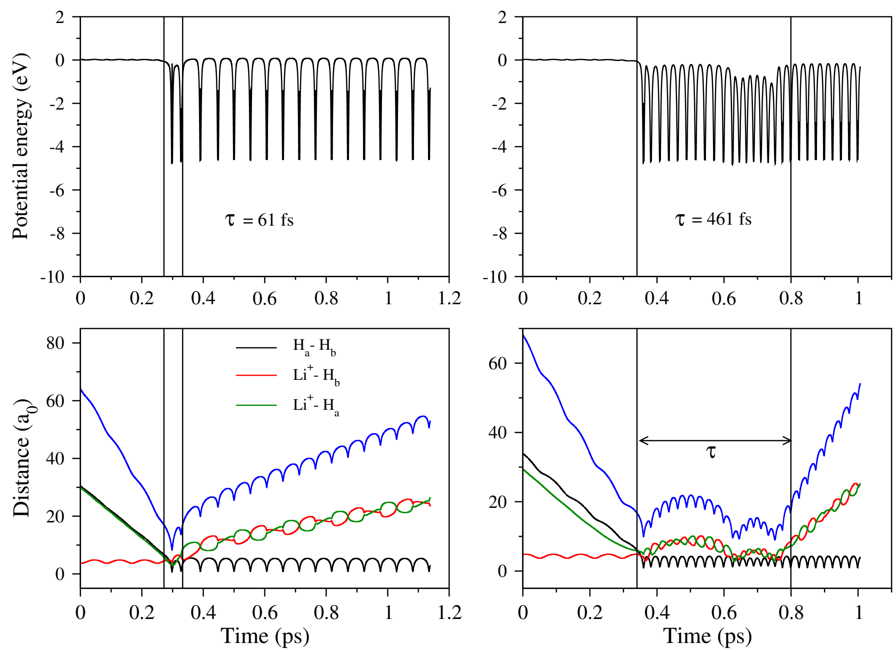


Figure 3.12: Tpical reactive trajectories of the H + LiH$^+$ ($v = 0, j = 0$) $\longrightarrow$ H_2 + Li^+ reaction. The solid blue curve corresponds to the sum of all three internuclear distances ($R_{H_aH_b}$ + $R_{Li^+H_b}$ + $R_{Li^+H_b}$). The two vertical lines represent the first and last instances of time at which the geometrical criterion that the sum of all three internuclear distances is 17 a_0 is satisfied. The total time spent by the trajectory inside these two vertical lines is regarded as the collision time.

at specific collision energy is later used to calculate the average collision time for a reaction as follows.

$$\tau_{avg} = \frac{1}{N_r}\sum_{i=1}^{N_r}\tau_i \tag{3.2}$$

Here, τ_{avg} is the average collision time, N_r is a number of reactive trajectories, and τ_i is the collision time of i^{th} trajectory. The detailed procedure to calculate the collision time and average collision time is discussed in an earlier publication [66]. The plot of average

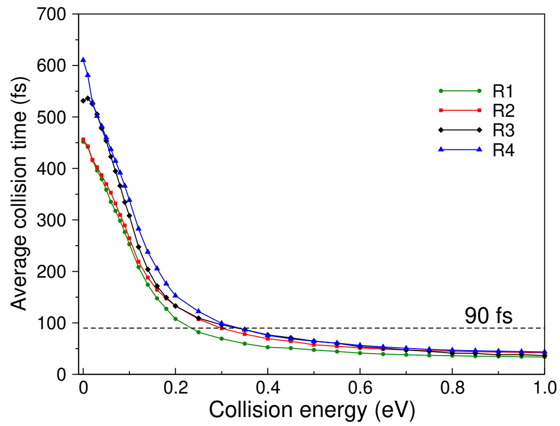


Figure 3.13: The average collision time of R1, R2, R3, and R4 reactions as a function of collision energy. The horizontal dashed line is a 90 fs line (see text for details).

collision time as a function of collision energy is shown in Fig. 3.13, and a line for threshold time of 90 fs is also shown. This threshold time is the boundary where the direct and indirect trajectories are separated. The trajectory is considered to be indirect when the collision time is higher than the threshold value. If the collision time is lower than the threshold time, it is considered as a direct trajectory. It can be seen from Fig. 3.13 that all four reactions have an average collision time higher than the threshold time in the lower collision energy range and a lower value of average collision time at the higher collision energy range. It indicates a significant number of direct trajectories at higher and indirect trajectories at lower collision energy ranges. The same can be observed clearly in Fig. 3.14, where the contribution of direct and indirect trajectories to the total ICS is shown as a function of collision energy. Fig. 3.14 confirms that a significant part of the total reactivity for all four reactions comes

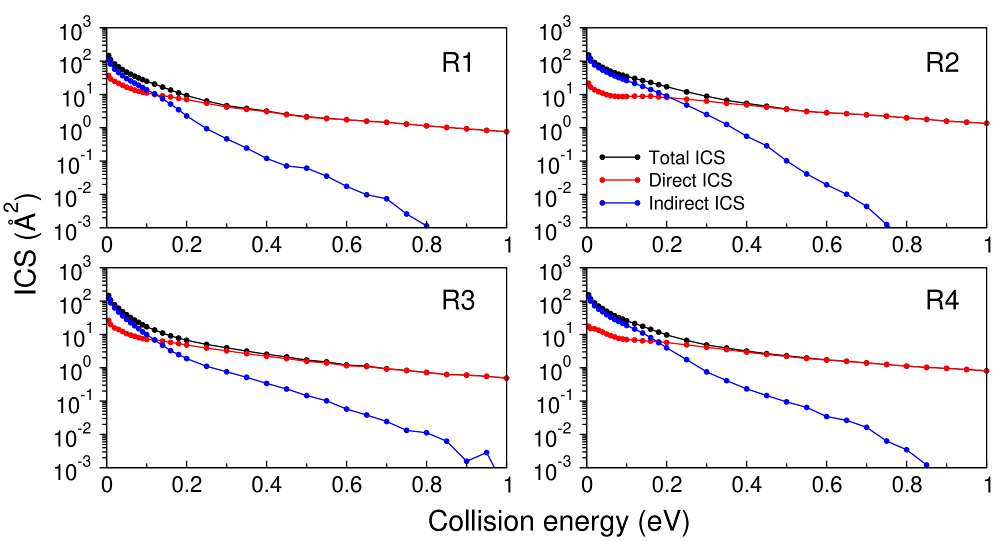


Figure 3.14: Initial state-specific d total ICS with contribution from direct and indirect mechanisms for all four reactions as a function of collision energy.

from the indirect trajectories at lower collision energies. As the collision energy increases, direct trajectories start making a significant contribution to the total reactivity.

The effect of isotopic substitution can be seen in Fig. 3.15, where the fractions of direct and indirect trajectories are shown as a function of collision energy. It can be seen that at

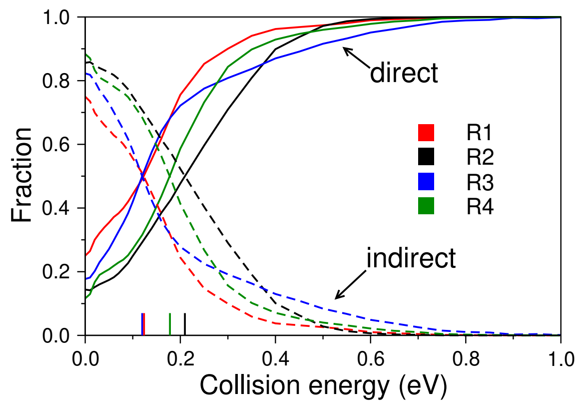


Figure 3.15: Fractions of direct and indirect trajectories for reactions R1, R2, R3, and R4 as a function of collision energy. The fraction represents the ratio of the number of a specific type of trajectory divided by the total number of reactive trajectories. The vertical colored lines at the lower end of the collision energies represent the points where the mechanism shifts predominantly from direct to indirect or vice-versa.

a higher collision energy range, all four reactions follow a direct mechanism with a negligible contribution from the indirect mechanism. The contribution is uneven for different mass combinations at the lower collision energy range. At very low collision energy, the reactions R2 and R4 have the highest and comparable contribution to the indirect mechanism. Reactions R3 and R1 have the lowest contribution from the indirect mechanism. The shift from predominantly indirect to direct mechanism occurs at collision energies centered around 0.1 eV for R1 and R3 and around 0.2 eV for reactions R2 and R4.

3.5.6 DCS

The DCSs for the reactions R1, R2, R3, and R4 are plotted in Fig. 3.16 as a function of center-of-mass scattering angle (θ) for three different collision energies, 0.02 eV, 0.35 eV, and 0.65 eV. The first and second columns of the figure show the comparison of QM-CC (dashed line) and QCT (solid line) DCSs for reactions R1 and R2, respectively, at three different collision energies mentioned above. It can be seen that for both the reactions R1 and R2, QCT DCSs closely follow the QM-CC DCSs except at $\theta = 0\,^\circ$ and 180°. The same QM-CC and QCT DCSs for reaction R1 and R2 at a collision energy of 0.02 eV are shown along with the standard deviation (in QCT DCSs) in Fig. 3.17. These standard deviations

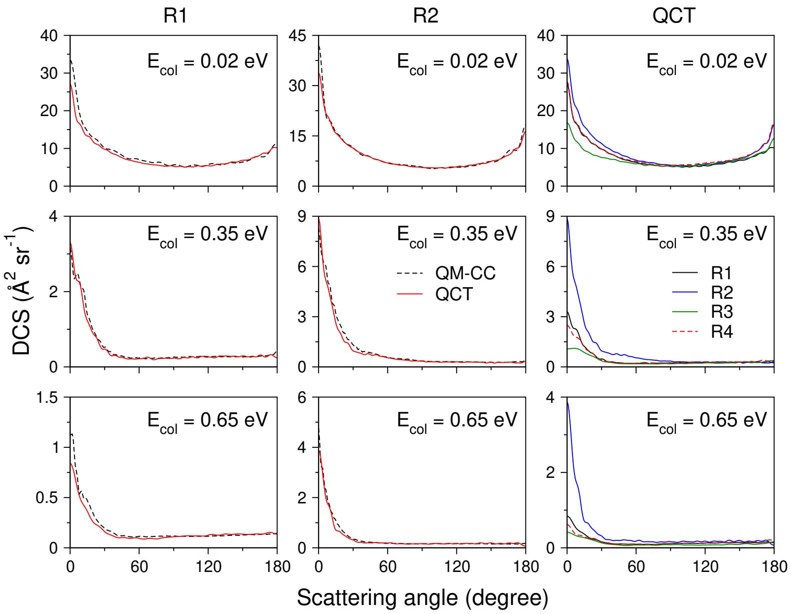


Figure 3.16: Initial state-selected total DCSs of the four reactions as a function of center-of-mass scattering angle (θ) at E_{col} = 0.02, 0.35, and 0.65 eV. The first and second columns show the comparison between the QM-CC (dashed line) and QCT (solid line) DCS for reactions R1 and R2. The QCT DCSs of all four reactions are compared in the panels in the third column.

are calculated using Eq. 2.81.

To observe the kinematic effect on the DCSs, the QCT DCSs for R1, R2, R3 and R4 reactions are shown in the third column of Fig. 3.16. At first glance, it is clear that all the reactions show the same qualitative behavior of product scattering: a dominant forward scattering at all the collision energies and slight backward scattering, which decreases at 0.35 eV and 0.65 eV. Studying this trend alongside the fraction of direct and indirect trajectories (cf., Fig. 3.15) at different collision energies and the contribution of these different types of trajectories in the DCS (cf., Fig. 3.18) reveals an interesting observation: direct trajectories are dominantly forward scattered and indirect trajectories are more or less symmetrically scattered. This can explain the trend in DCSs with increasing collision energy. The indirect trajectories, prominent at the lower collision energy along with the dominant forward scattered direct trajectories, result in the slight backward peak, which disappears at higher collision energies because the direct trajectories are more prominent there.

Another feature of the comparison plot of DCSs (third column of Fig. 3.16) is the quantitative order of DCSs. It can be seen that in the forward scattered region, the quantitative order of the DCSs is R2 > R4 $\approx$ R1 > R3, which is the same as the order of ICSs for the four reactions. This correlation can be understood by looking at Eq. 2.80, where the terms

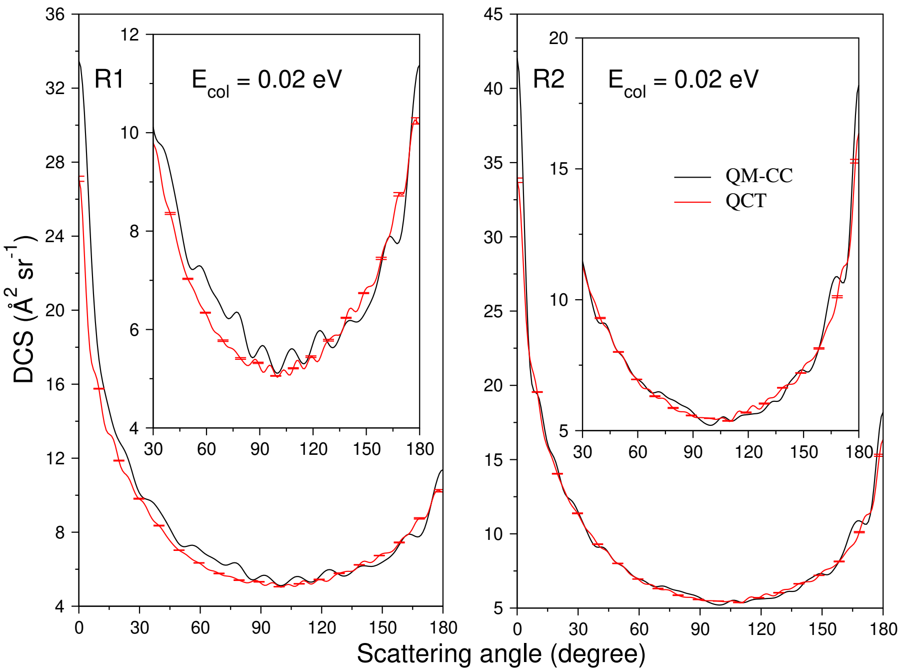


Figure 3.17: Initial state-selected QM-CC and QCT DCSs for R1 and R2 at 0.02 eV of collision energy. Standard deviations in QCT DCSs at selected scattering angles (θ) are also shown. DCSs from 30 ° to 180 ° scattering angles for both of the reactions are shown in the inset.

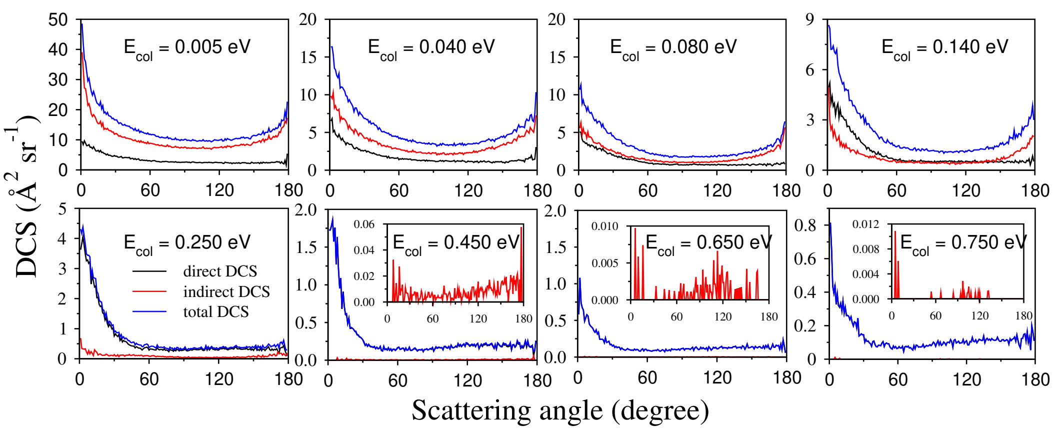


Figure 3.18: DCSs calculated using the histogram binned (HB) method as a function of scattering angle at different collision energies for reaction R1. The DCSs calculated for the direct and indirect trajectories are shown along with the total DCSs. Indirect DCSs at collision energies of 0.45 eV, 0.65 eV, and 0.75 eV are very low and thus enlarged in the inset.

inside the bracket contribute mostly to the qualitative nature (shape) of the DCSs and the first term ($\sigma_{vj,GB}^{E_{col}}$) controls the quantitative nature.

The same isotopic effect was also studied by Li et al. for reactions R1, R2, and R3 [72]. These calculations were only performed at a collision energy of 0.5 eV using the QCT

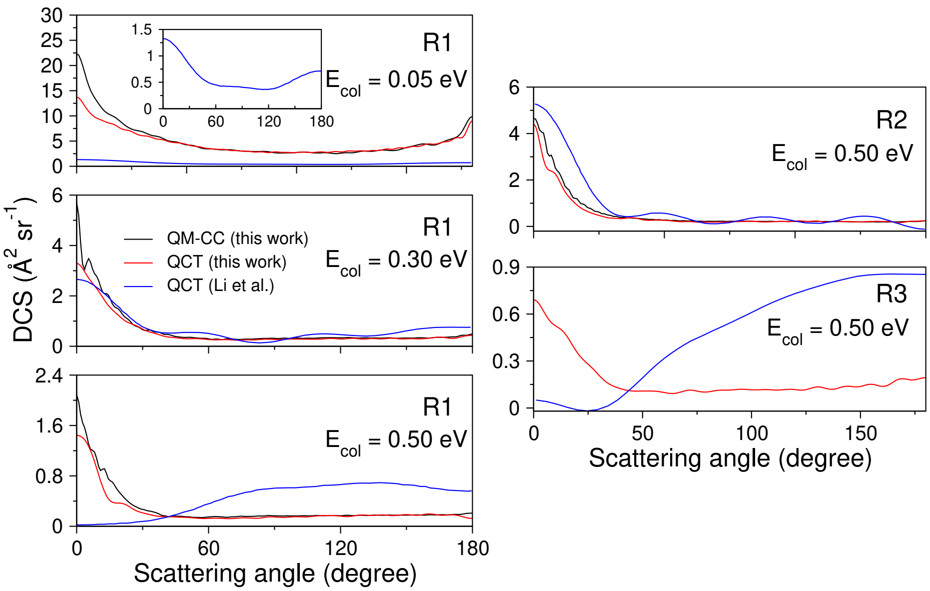


Figure 3.19: Initial state-selected total DCSs as a function of center-of-mass scattering angle (θ) for reactions R1, R2, and R3 compared with the available literature results at different collision energies. The QM-CC and QCT DCS calculated in this work are compared with the QCT DCS reported by Li et al. [72]. The QCT DCS reported by Li et al. for reaction R1 at 0.05 eV is magnified for clarity and shown in the inset.

method. In another work by Li. et al. [70], the DCSs for R1 were calculated at a few other collision energies. The QCT DCSs for the reactions R1, R2, and R3 calculated in this work are compared with the QCT DCSs of Li et al. [70, 72] and shown in Fig. 3.19. The comparison for the reaction R1 is made at 0.05, 0.30, and 0.50 eV; for reactions R2 and R3, the comparison is made at 0.50 eV collision energy. In the case of reactions R1 and R2, both the QCT DCSs of this work and that of Li et al., [70, 72] are compared with the QM-CC DCSs of the present work. At 0.05 eV collision energy QCT DCSs of Li et al. [70, 72] and the present results for reaction R1 have the same qualitative behavior, however the agreement between the QM-CC and QCT DCSs of the present work is relatively quite better. The same features can be observed at 0.30 eV collision energy. However, the two QCT DCSs for reaction R1 at a higher collision energy of 0.50 eV are drastically different. It can be seen that the QM-CC and QCT DCSs of the present work agree well with each other and have prominent forward scattering behavior. The QCT DCS of Li et al., [72], however, have a backward bias. The same difference is also seen for reaction R3, where the QCT DCS of the present work is forward-biased. However, in the case of reaction R2, all three DCSs are qualitatively the same except for the fact that the QCT DCS of Li et al. [72] seems to be overestimated in the forward scattering region. Overall, it can be seen that the current QCT and QM-CC DCSs are in good agreement with each other, ensuring the accuracy of current calculations. This explains the difference between the QCT DCS of Li et al. and the results of the present investigation, indicating a possible discrepancy in the calculations of Li et al. [72].

3.5.7 Reaction probability and initial state-specific rate constants

The reaction R1 and its isotopic variants, R2, R3, and R4, are exoergic. The MEP for the reactions is barrierless and exoergic for all the attacking angles ($\angle H_AH_BLi$), except at attacking angles $< 7°$, as shown in Fig. S1 of Ref. 65. The total reaction probability for total angular momentum quantum number $J = 0$ obtained by the QM-CC, QCT-HB, and QCT-GB methods for R1 and R2 is shown in Fig. 3.20.

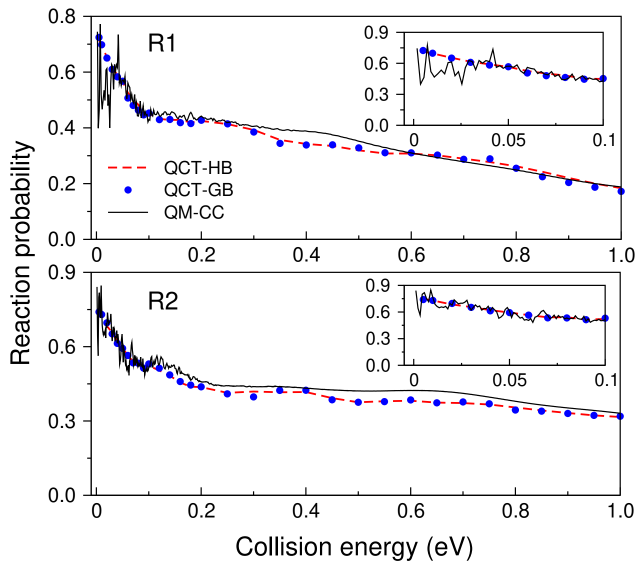


Figure 3.20: Initial state-selected total reaction probabilities for $J = 0$ of reaction R1 and R2 as a function of collision energy calculated by the QM-CC, QCT-HB, and QCT-GB methods. The total reaction probability in the lower collision energy range is shown in the inset for both reactions.

It can be seen from the figure that the probabilities of both reactions exhibit sharp resonance oscillations at lower collision energy, and the contour of the probabilities decreases on average with increasing collision energy. The oscillations are due to the formation of quasi-bound complexes supported by the shallow well. Moreover, these total reaction probabilities show no energy threshold, signifying the barrierless nature of the reactions. In the case of R1 and R2, when the total reaction probability is calculated using the QCT method, it is found that both the QCT-HB and QCT-GB methods tend to reproduce the overall energy dependence of the reaction probabilities obtained by the QM-CC methods. The resonance oscillations found at the lower end of the collision energies in QM-CC reaction probabilities are absent in classical (QCT) reaction probabilities. The QCT total reaction probabilities also follow the QM-CC behavior at the higher J value (cf. Fig. 3.21).

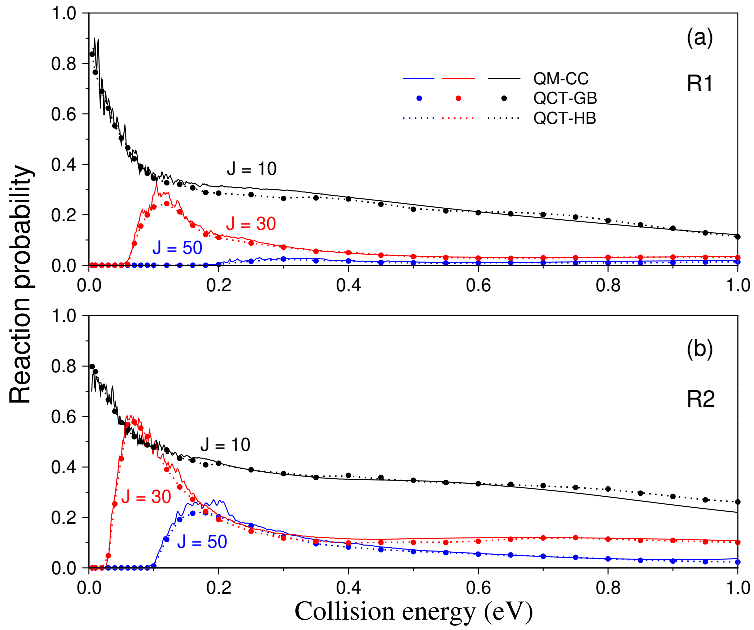


Figure 3.21: Initial state-selected QM-CC, QCT-HB, and QCT-GB total reaction probabilities at J = 10, 30, and 50 for reaction R1 and R2 as a function of collision energy.

Initial reactant diatom state-specific rate constants are shown in Figs. 3.22 and 3.26. Fig. 3.22 shows the QM-CC and QCT state-specific ($v = 0, j = 0$) rate constants for reactions R1 (panel a) and R2 (panel b) at a temperature range from 60 to 10,000 K. It is to be noted that the minimum energies at which the dynamical calculations are performed for QM-CC and QCT methods are 10^{-3} eV and 5×10^{-3} eV respectively. As it can be seen from Eq. 2.79, to calculate the rate constant at any temperature, the ICS values at all the energies (0 to ∞) are required. So, the information at this minimum energy grid value becomes crucial. The contribution of ICS values at lower energies is prominent in calculating the rate constants at lower temperatures. In the current study, because of the limitation in the minimum energy value at which the data is available, the minimum temperature at which rate constants are calculated is taken as 60 K (which corresponds to ~ 0.005 eV) in the present calculation. A detailed discussion on the contribution of collision energies from different ranges can be found in section 4.3.3 of Chapter 4.

It can be seen from the two panels that both the QM-CC and QCT rate constants follow the same trend for reactions R1 and R2; however, the comparison between QM-CC and QCT rate constants in the lower temperature range is better for reaction R1 than R2. Looking closely, it can be seen that the rate constants for R1 and R2 initially increase with increasing temperature and then show negative temperature dependence. In case of reaction R1, the

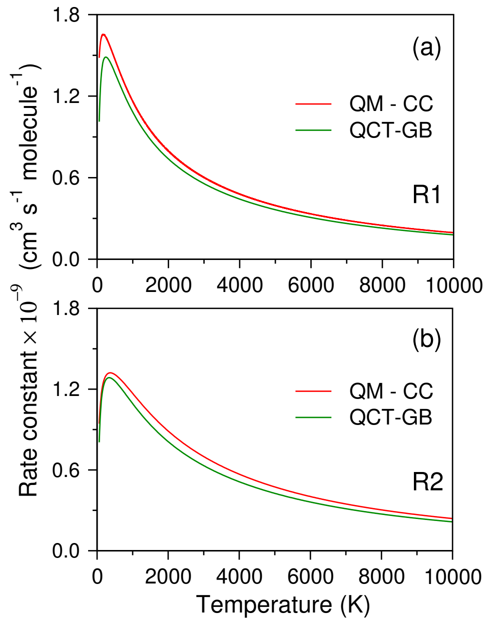


Figure 3.22: Initial state-selected ($v = 0, j = 0$) rate constants for reaction R1 and R2 as a function of temperature calculated by the QM-CC and QCT-GB methods.

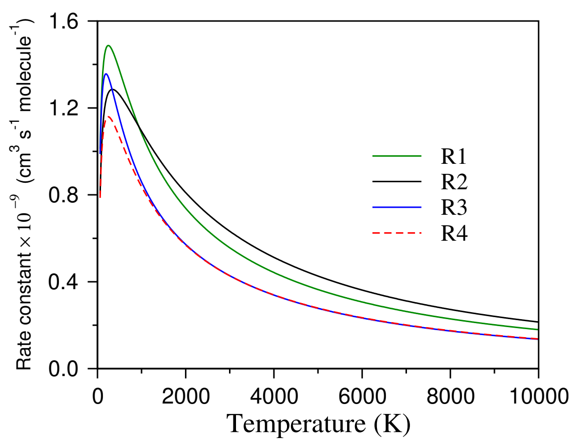


Figure 3.23: QCT-GB initial state-selected ($v = 0, j = 0$) rate constants for reactions R1, R2, R3, and R4 as a function of temperature.

QM-CC rate constant reaches to the maximum value of 1.65×10^{-9} cm^3 sec^{-1} molecule^{-1} at 174 K and QCT rate constant reaches to maximum value of 1.48×10^{-9} cm^3 sec^{-1} molecule^{-1} at 250 K. In the case of reaction R2, as seen in panel (b) of Fig. 3.22, the maximum rate constants for QM-CC and QCT methods are observed at 370 K and 340 K, respectively. This nature of rate constant as a function of temperature is also found for other exoergic barrierless reactions [81–84]. A different study [64] on reaction R1 in the ultracold region also showed the same behavior of rate constant in the low-temperature range. It is also to be pointed out that the reactant diatoms have dissociation energy (D_0) of ≈ 0.112 eV which means that all the possible ro-vibrational levels of the reactant diatom are open in the temperature range considered in the present study. Thus, in order to obtain the thermal QM-CC and QCT rate constants, an extensive number of calculations considering all the reactant ro-vibrational levels are required, which, however, is certainly beyond the present scope and is planned for future work.

3.6 Effect of isotopic substitution

Since the QCT results for the reaction R1 and R2 are identical to the QM-CC results, it establishes the reliability of the method for the type of reaction being studied in the current work. This justifies the use of only QCT method to study the dynamics and the effect of the isotopic substitution for R3 and R4. It is known that the PES of a reactive system doesn't depend on the mass of the participating nuclei, but the latter can govern the dynamics of the reaction. So, the isotopic effects can be understood more clearly by investigating the effect of nuclear masses on the PES. The mass-weighted coordinates represent this effect, and the such PESs are called mass-weighted PESs [85, 86] The physical effect of mass (and mass-weighted coordinates) appears in the reaction dynamics as to what extent the reactant and product valleys are skewed to each other in the mass-weighted PES. This also results in the variation of the MEPs for different isotopic variants of the concerned reaction. The mass-weighted PESs of all four reactions are calculated and are shown in Figs. 3.24 and 3.25.

The mass-weighted coordinates are obtained from the internuclear distances, as discussed in Appendix J. The skewness between the reactant and product valleys can be measured in terms of an angle called the skew angle (β) of the reaction. The skew angle can also be written in a slightly different form called mass-factor ($\cos^2\beta$). The skew angles for the four reactions are calculated by using the formula given in Eq. J.5 of Appendix J and are listed in Table 3.5 [86]. It is important to note that the higher the skew angle, the lower the skewness. The order of skew angles for the reactions is: R3 > R4 > R1 > R2.

Table 3.5: Skew angles in the four isotopic variant reactions

Reagent pair	**Skew angle (β) (degree)**
H + LiH^+ (R1)	48.6
D + LiH^+ (R2)	40.3
H + LiD^+ (R3)	59.4
D + LiD^+ (R4)	51.5

It is observed that the order of the ICSs for the reactions is: R2 > R4 $\approx$ R1 > R3. In the case of Type I, the reaction pairs with the same attacking atom but a different target diatom (R1 - R3 and R2 - R4); the reaction with the lighter diatom has high reactivity. On the other

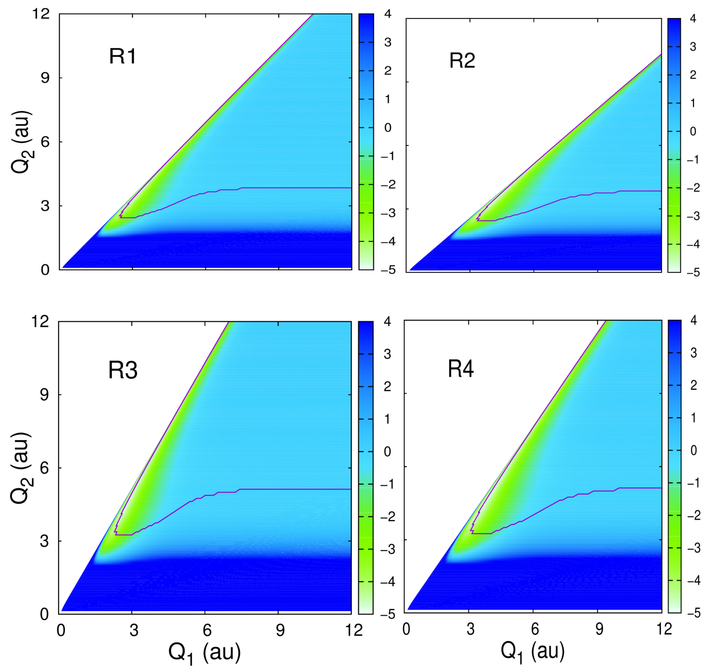


Figure 3.24: Contour map of the mass-weighted potential energy surfaces and MEPs (shown as a violet-colored line) at an attacking angle ($\angle H_AH_BLi$) of 80°, showing the effect of mass on the PES and MEP for R1, R2, R3, and R4. The skew angles for R1, R2, R3, and R4 are 48.6°, 40.3°, 59.4°, and 51.5°, respectively.

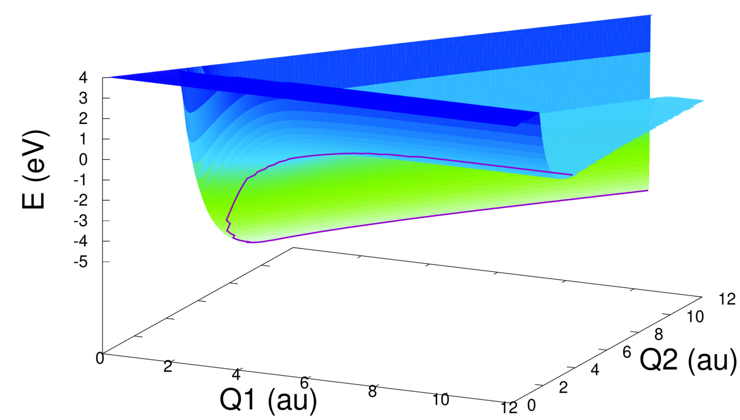


Figure 3.25: 3D view of the mass-weighted PES contour map for reaction R1 as shown in Fig. 3.24 .

hand, the reaction pair with the same target diatom but a different heavier attacking atom (R1 - R2 and R3 - R4), the reactivity is high. It can be seen that this reactivity order is the opposite of the skew angle order. Now, when the order of reactivity is compared for the Type II reaction pair (R1 - R4), it is found that the reactivity is R4 $\approx$ R1. This similarity in reactivity of the reactions R1 and R4 can be understood when the skew angles are compared. The skew angles for R1 and R4 are 48.6° and 51.5°, respectively, which are very close and result in almost similar reactivity. This trend of the isotopic effect on the reactivity of the reactions and its relation with the skew angle observed in the current work is not unique to only the current (LiH_2^+) barrierless exoergic system. The same trend in ICSs was also observed in the LiH_2 system by He et al [87]. Irrespective of the barrierless exoergic nature of the reactions, the same trend is also observed for various other reactions [88–90].

The skewness of mass-weighted PES is well known to be related to the product energy disposal of reactions with a barrier according to Polanyi's rule [86] (See Refs. 91–94 for related work on sudden vector projection model). In the present case of an exoergic-barrierless reaction, the trend observed in the reactivity of two different types of reaction pairs is also observed in the energy disposal to the product vibration. It is observed that reactions with sharply turning (low skew angle) reactants and product valleys result in highly vibrating product diatom. The reactions where the skew angles are close to each other have an almost equal amount of energy disposal in product vibration. Interestingly, this observation is in line with exoergic reactions having a barrier along the reaction path. Hence, it seems that the role of the barrier in determining the effect of mass on the product energy disposal of exoergic reactions is redundant. Comparing the product vibrational energy disposal for reactions R2 and R3, it is observed that more energy is disposed to product vibration in the case of R2 than R3. The two reactions have the same product diatom (HD) but different isotopic combinations of reactant species. This means that for reactions R2 and R3, the number of product vibrational and rotational levels are the same. This suggests that the product energy disposal is affected solely by the skewness of the mass-weighted PES of the system. The isotopic effect at any specific collision energy is seen only in the energy disposal to the vibrational and rotational degrees of freedom. The energy disposal to the product translation remains the same for all reactions. This is because the underlying PES is attractive for all isotopic combinations, and the energy given to the reactant translation predominantly flows to product vibration.

As there is a significant isotopic effect on the observables like ICS and product energy disposal, the effect on the observables like DCS also follows the same trend. In the case of DCS, the isotopic effect, largely seen in the qualitative nature of the angular distributions, can also be correlated with the order of skew angles. Here, similar to ICS and product energy disposal, DCSs for the reactions R1 and R4 are very close. This trend, along with the trend in the magnitude of DCSs in the forward scattering region (which was discussed earlier in section 3.5.6), is opposite to the order of the skew angles, R3 > R4 > R1 > R2.

The effect of isotopes on the initial state-specific rate constants can be seen in Fig. 3.26. The rate constant shown is calculated using Eq. 2.79 and shows the substantial effect of

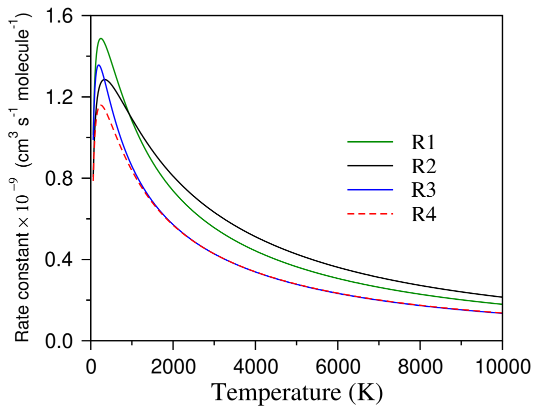


Figure 3.26: QCT-GB initial state-selected (v = 0, j = 0) rate constants for reactions R1, R2, R3, and R4 as a function of temperature.

isotopic substitution. The order of the rate constants for the reactions is not straightforward and changes as the temperature increases. Interestingly, putting the reduced mass equals to 'one' in Eq. 2.79 reveals a pattern, as shown in Fig. 3.27. Now, the order of rate constant,

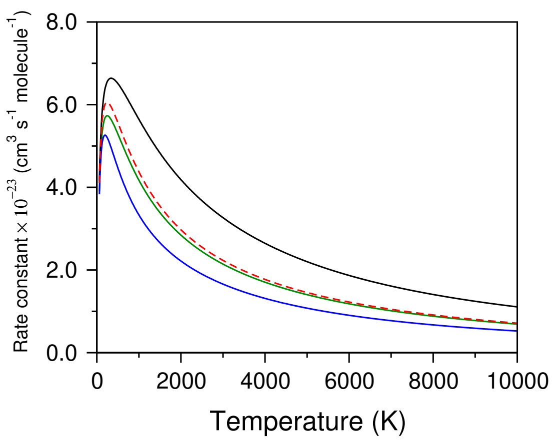


Figure 3.27: QCT-GB initial state-selected (v = 0, j = 0) rate constants calculated without the reduced mass (see text for details) for reactions R1, R2, R3, and R4 as a function of temperature. The line types are the same as mentioned in the Fig. 3.26.

R2 > R4 ≈ R1 > R3, has the same relationship with skew angles as observed for ICSs. Here, It is important to note that in the case of a product diatom having two identical nuclei, such as H_2 and D_2 in the present case, the effect of nuclear spin statistics must be incorporated to obtain physically meaningful cross sections [95]. It was found that for H_2 product, a post-antisymmetrization factor of 3/2 (1/2) corresponding to odd (even) rotational level should be multiplied to the cross section obtained from the QM scattering calculations (cf. Eq. 2.54) [95]. This generally gives a saw-toothed type product rotational distribution, [95] which is also obtained in the present case of reaction R1. However, no significant difference was found in the total ICS, total DCS, initial state-specific rate constant, and product energy disposal for the results obtained without and with the inclusion of the nuclear spin factor. Although there are obvious differences in the product rotational level resolved ICSs, the overall qualitative behavior of the distribution does not change and hence does not affect the outcome of the present study. For the sake of comparison, the product rotational level resolved state-to-state ICSs with and without the nuclear spin factor are shown in Fig. 3.28 in terms of color-mapped plot, same as in Fig. 3.7. As can be seen from the figure, the qualitative feature of the distribution does not change, unaltering the present conclusions.

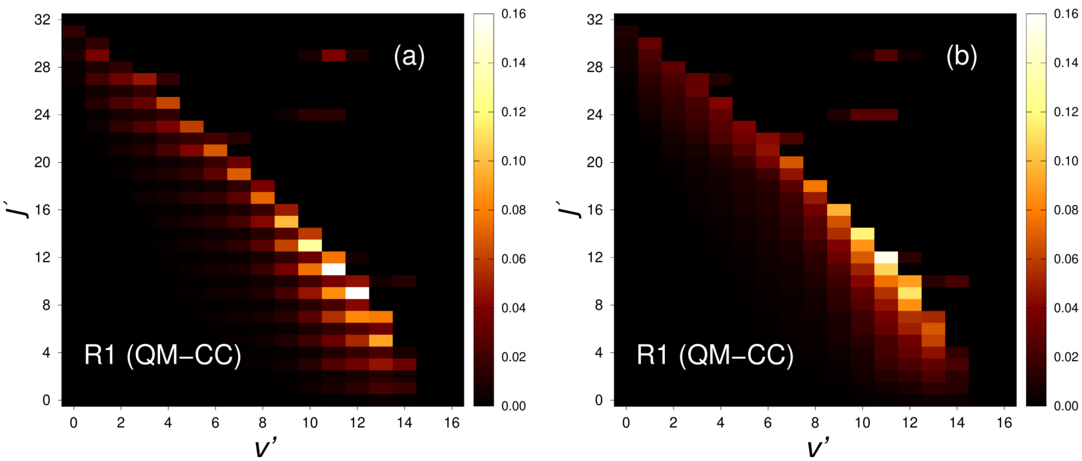


Figure 3.28: Product rotational level resolved QM-CC ICSs with (panel a) and without (panel b) nuclear spin factor for reaction R1 as a function of v' (abscissa) and j' (ordinate) at a collision energy of 0.35 eV.

3.7 Roaming trajectories in reaction R1

As discussed in section 3.5.5, the total reaction has contributions from the direct and indirect trajectories. The direct trajectories are short-lived, and the indirect trajectories are long-lived. Both the tightly bound and loosely bound complexes during the course of the reaction can contribute to the indirect trajectories. Loosely bound complexes are generally formed when the system has low recoil energy to form the product. Formation of such complexes is common in ion-molecule reactions dominated by the long-range potential and can lead to the roaming mechanism. The roaming mechanism is where the trajectories form the product through pathways other than the minimum energy path. These pathways are energetically high, and atoms access the region very close to the asymptote before returning to the interaction region and forming the product [96–98]. The mechanism was first reported for the formaldehyde [99] and later demonstrated in many other reactions [96, 99–104] through the classical and quantum studies.

Out of many roaming trajectories obtained in the reaction H + LiH^+ $\rightarrow$ H_2 + Li^+ in the current investigation, a few representative trajectories are shown in Fig. 3.29. It can be seen

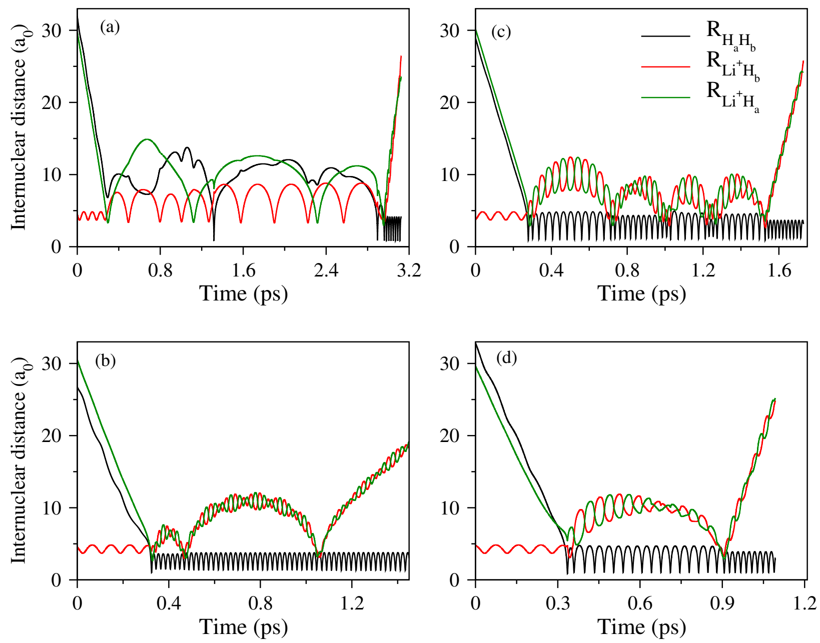


Figure 3.29: Four representative roaming reactive trajectories for the reaction R1 shown as the evolution of the three internuclear distances in time. The roaming of the attacking H atom and the recoiling Li^+ ion are shown in panels (a) and (b–d), respectively.

from the panel (a) of Fig. 3.29 that the attacking H_a atom and target H_b atom from the LiH_b^+ roam around the Li^+ ion before leaving as the product diatom H_2. However, in the case of the panel (b-d) of the same figure, the Li^+ ion roams after forming product diatom H_2. A snapshot of a trajectory shown in panel (a) at different times of propagation is also shown in Fig. 3.30 (refer to the Supplementary files of Ref. [66] for the animation of the trajectory).

3.8 Chapter summary

The present chapter discusses the results of dynamical investigations carried out for the four different isotopic variants of the $H + LiH^+ \rightarrow H_2 + Li^+$ reaction. The work presented focuses on two different themes: the first is the comparison between the QM and QCT dynamics for these exoergic-barrierless reactions, and the second is to study the isotopic effect on different dynamical attributes, such as total ICSs, product vibrational and rotational level resolved ICSs, product energy disposal, DCSs, etc. The reaction mechanisms for different reaction variants are also compared. Calculations for reactions R1 and R2 are carried out using the QCT and QM-CC methods, and reactions R3 and R4 are only studied using the QCT method. The comparison of the QM-CC and QCT methods is carried out using the results of R1 and R2 reactions only. A study on the effect of isotopic substitution is done for

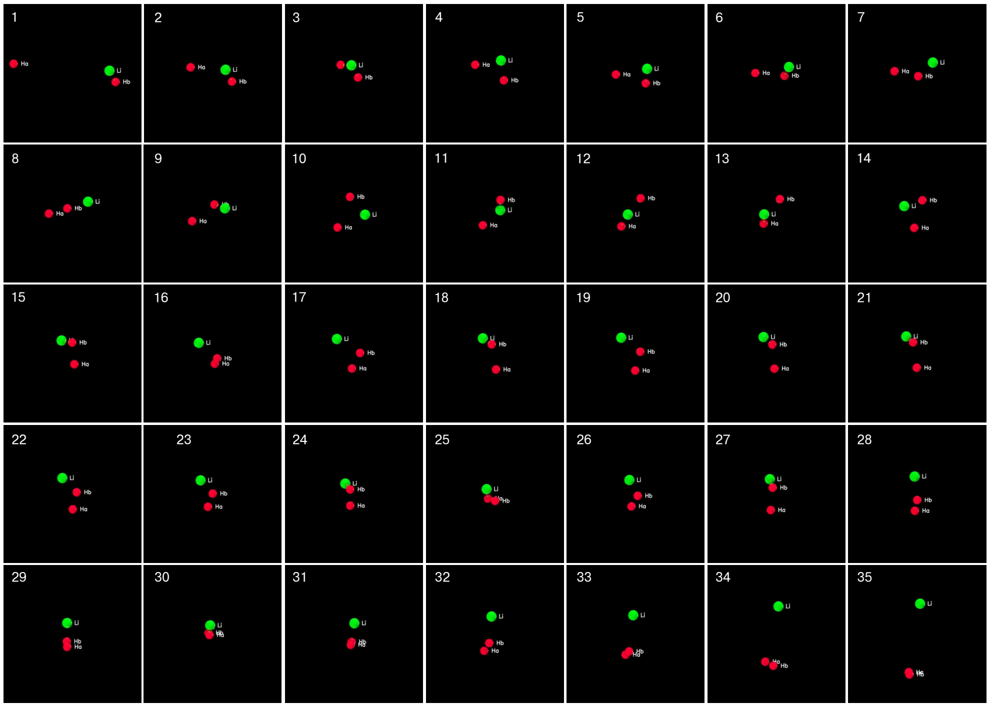


Figure 3.30: Roaming of attacking H_a atom before forming the product H_2 or H_aH_b.

all four reactions using QCT dynamics results. A very good agreement between the QM and QCT total ICS, DCS, and initial state-specific rate constant is found in the case of reactions R1 and R2. This comparison is also found to be valid at the state-to-state level, where the QCT and QM product vibrational and rotational level resolved ICSs have the same behavior.

Generally, the agreement between the QM-CC and QCT results depends on the quantum effects, such as quantum tunneling and quantum resonance in the reaction. For the reactions involving heavier atoms or reactions having a barrierless exoergic nature, these effects are not prominent, and thus, a better agreement between the QM-CC and QCT results is expected, a result that is also observed in other studies [75, 76, 105]. In the case of the current reaction system, not only is it barrierless in nature, but the energy well present in the reaction path is very shallow (≈0.286 eV below the product asymptote). Due to the shallow nature of the well, the quantum mechanical resonance oscillations are not too strong and almost disappear with partial wave summation, facilitating a good agreement between the QM-CC and QCT results for the reactions in the whole collision energy range. Despite this, a few minor disagreements are seen, which are found to be quantitative in nature. A significant isotopic effect is found in the overall reactivity and the product angular distributions of the four reactions. It is found that the mass combination of the reactants changes the nature of the underlying PES by altering the skew angle between the reactant and product valley and changes the dynamical observables and the reaction mechanism.

While the present work is restricted to the initial ro-vibrational ground level ($v = 0$, $j = 0$) of the reactant diatom, dynamical studies with adequate ro-vibrational excitation are necessary at thermal energies to understand the primordial chemistry of LiH^+. Owing to the good agreement between the QM-CC and QCT results, the present study surely provides a solid ground to implement the QCT method instead of using computationally expensive quantum dynamical methods to study the present reactions with ro-vibrationally excited reactants in the context of the astrochemical abundance of the fragments involved.

Chapter 4

Fitting ground electronic state potential energy surface of $HeLiH^+$ system and dynamics of He + $LiH^+ \rightarrow LiHe^+$ + H reaction

4.1 Introduction

Just after the Big Bang, the universe was simply a hot plasma of matter and radiation. The formation of the first elements, mainly nuclei of H, D, ^{3}He, and ^{7}Li, took place in the initial three minutes after the explosion in the process called primordial nucleosynthesis or Big Bang nucleosynthesis (BBN) [106]. Initially, radiation and matter were coupled, and the universe was opaque. It was not until ≈ 374 thousand years or redshift (z) ≈ 40000, when the temperature got reasonably cold (≈ 3000 K), decoupling radiation and matter and resulting in a transparent universe [107]. It was also when nuclei started combining with the electrons (e^-), forming ions and subsequently neutral atoms in order of their ionization potential.

After the formation of atomic ions and neutral atoms, different molecular species were formed. HeH^+ was the first molecular species of the universe, and the formation of H_2^+, H_2, HD, and HD^+ followed. Li and Li^+, which were present in the environment, reacted with the other atoms, ions, and electrons to form LiH, LiH^+, $LiHe^+$ molecular species. LiH^+ is an essential species of Li reaction network that takes part in various reactions involving radiation, electrons, and atomic or molecular species [108–110]. Out of this reaction, the reactive and inelastic collision of LiH^+ with other atomic species, such as H, has been the subject of different studies [65, 66, 111–115]. However, its collision with He has not been explored much. LiH^+ can react with He in the following way.

R1: $\quad He + LiH^+ \rightarrow LiHe^+ + H$

The reverse reaction of R1, shown below as R2, is also a possible reaction that contributes to Li chemistry.

R2: $\quad H + LiHe^+ \rightarrow LiH^+ + He$

The other possibility is collision-induced dissociation (CID).

R3: He + $LiH^+ \rightarrow Li^+$ + H + He

R4: H + $LiHe^+ \rightarrow Li^+$ + He + H

4.1.1 Possible reactions of ground electronic state PES and their energetics

These reactions R1, R2, and CID process (R3 and R4) take place on the electronic ground state of $HeLiH^+$ surface. Reaction R1 has a small endothermicity of ~0.0322 eV (~0.0574 eV) with (without) zero-point energy (ZPE) of the LiH^+ reagent diatom. Moreover, the reaction (R1) becomes exothermic by ~0.01 eV upon considering the reagent diatom LiH^+ to v = 1, j = 0 level without containing ZPE of $LiHe^+$. However, the reaction remains endothermic by ~ 0.0052 eV when ZPE of $LiHe^+$ is also considered. Reverse reaction of R1, R2 is exothermic by ~ 0.0728 eV (~ 0.0574 eV) with (without) ZPE of $LiHe^+$ diatom. In the case of R1 and R2 reactions, CID channel (reactions R3 and R4, respectively) opens up at the collision energies of 0.1072 eV and 0.0596 eV, respectively, for v = 0, j = 0 asymptote of reagent diatoms. Such low energy for CID also makes it an important candidate for the Li chemical network. The energetics of these reactions is shown in Fig. 4.1. In all of these

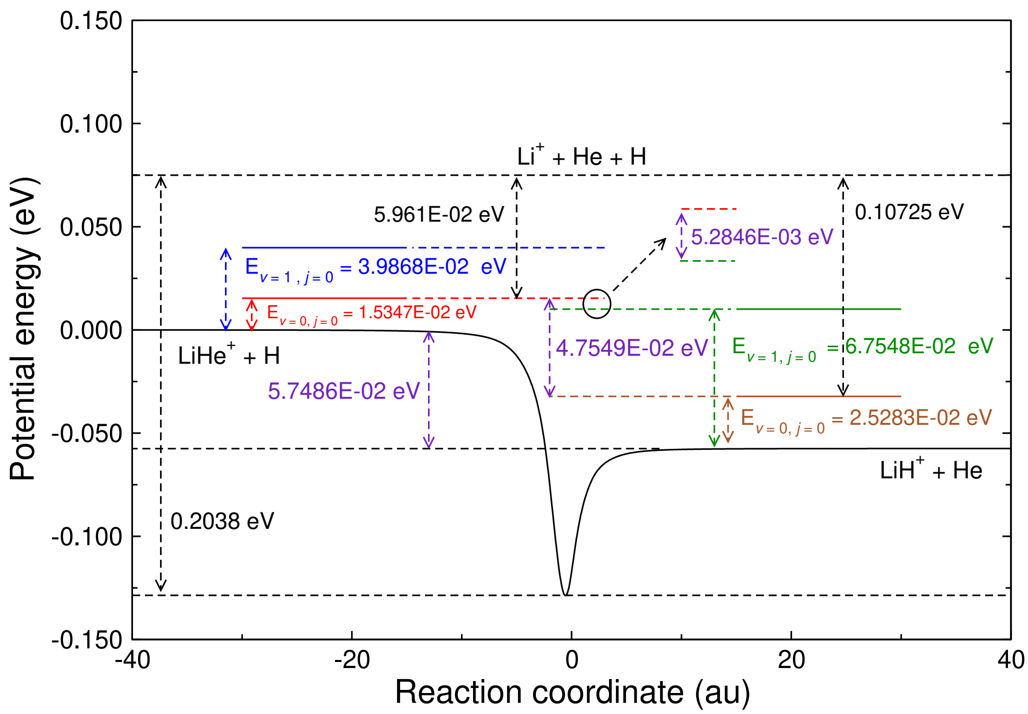


Figure 4.1: Energetics of R1, R2, R3, and R4 reactions.

cases, the global minima of the $HeLiH^+$ system is at the - 0.2038 eV below the three-body dissociation asymptote.

4.1.2 Previous studies and motivation of the present work

To study the above reactions theoretically, PES is the central instrument. In the case of current $HeLiH^+$ system, till date, the only available electronic ground potential energy surface (PES) was constructed by Wernli et al.[116] The total potential of the system was represented as many body expansion (MBE) and *ab initio* data points were calculated at the multireference configuration interaction (MRCI) level of theory. The two-body and three-body parts in the expansion were fitted using the Aguado and Paniagua [117] functional form. The basis set employed for He, H, and Li atoms were aug-cc-PVQZ, $(13s6p3d1f)/[8s6p3d1f]$ and $(15s10p6d1f)/[12s10p6d3f]$, respectively. [118–120] The core orbitals of H and Li were contracted with 6 and 4 primitives, respectively. The surface has been used subsequently in different investigations closely related to the adiabatic ground electronic state of $HeLiH^+$.[121, 122] Among these, Tacconi et al. [121] studied the formation and destruction of LiH^+ and $LiHe^+$ diatoms at various initial ro-vibrational states using the quantum reactive scattering approach with negative imaginary potential. These dynamics calculations were performed using the coupled state (CS) approximation, where the coupling between different z projections along the body fixed z axis are neglected. Integral cross sections (ICSs) and temperature dependent rate constants were the major findings of the work. Later, these results were used to assess the reaction network and abundance of $LiHe^+$ in the early universe.[122] This study on the abundance showed that in the z value of 10 to 20, $LiHe^+$ has a very low fractional abundance that is not sufficient enough to be detected by the available instruments.

The ICSs for the formation of LiH^+ (reaction R2) presented in the work of Taconi et al.[121] increase at the higher collision energy. This behavior for an exoergic reaction R2, prima facie, appears to defy the universal nature of ICSs with increasing collision energy for such reactions. As for the destruction of LiH^+ (reaction R1), studied in their work, a closer look raises the same question about the non-appearance of the effect of dissociation channel in the ICSs or any other observables. These concerns point to some possible errors in the underlying PES or the dynamics calculations themselves. Since the results of these dynamics calculations were used for other subsequent work of importance, revisiting the system appears to be of significant importance.

With this, we take up the problem and start from the first step: construction of global PES. Since the affordability of performing electronic structure calculations at a much higher level of theory opens up the possibility of constructing a more accurate surface and studying the dynamics in greater details. This detailed investigation can settle the above concerns and will further improve the present knowledge and thus the importance of $LiHe^+$ and LiH^+

diatoms in Li reaction network.

The present study primarily focuses on the construction of an adiabatic electronic ground state PES of the $HeLiH^+$ system by fitting the FCI *ab initio* data points. The PES is fitted using a combination of machine learning and Aguado Paniagua function.[117] The fitted PES is used to study the dynamics of R1 (He + LiH^+ ($v = 0, j = 0$) $\rightarrow$ $LiHe^+$ (v', j') + H) reaction using a quantum time-dependent and quasi-classical trajectory (QCT) approach. Few results are also presented for reaction R3, that is the CID channel of reaction R1.

4.2 Discussion on fitting

The choice of the chemical fragments of the system to generate *ab initio* data and fit with the analytical form depends on the representation of the total potential. In the current study, many body expansion (MBE) of Sorbie and Murrell [4], which expresses the total potential for a triatomic system as a sum of one, two, and three-body terms, is used to express the total potential of the system. The expression for the total potential (V_{total}) in MBE is

$$V_{\text{total}}(R_{\text{AB}}, R_{\text{BC}}, R_{\text{AC}}) = \sum_{i=\text{A,B,C}} V_1^i + \sum_{i=\text{AB, BC, AC}} V_2^{\text{i}}(R_i) + V_3(R_{\text{AB}}, R_{\text{BC}}, R_{\text{AC}}). \qquad (4.1)$$

In Eq. 4.1, V_1^i , $\text{V}_2^i(R_i)$, and $V_3(R_{\text{AB}}, R_{\text{BC}}, R_{\text{AC}})$ are one, two, and three-body terms, respectively. The interatomic distance among A, B, and C atoms are represented as R_{AB}, R_{BC}, and R_{AC}.

4.2.1 Generating *ab initio* data points

The single point *ab initio* energies are calculated at several configurations using the full configuration-interaction (FCI) method in Molpro[123]. These FCI energies extrapolated to the complete basis set (CBS) limit are used to construct the electronic ground state PES of the $HeLiH^+$ system. The $HeLiH^+$ system has 5 electrons; among them, two electrons, which are in the lowest molecular orbital, are considered as the core electrons in the FCI energy calculations. To calculate the FCI/CBS energies at each nuclear configuration, the FCI calculations were performed using aug-cc-pVnZ (n = 3, 4) basis sets and then extrapolated by the following two-point extrapolation formula.[124, 125]

$$E_{\text{CBS}} = E_X + \frac{E_X - E_{X\text{-}1}}{\left[1 - (X{+}k)^{-1}\right]^{-3} - 1} \qquad (4.2)$$

Here, E_{X-1} and E_X are the energies obtained with basis sets of the cardinality X-1 and X, respectively, and k has been interpreted as an angular momentum offset number, which depends on the correlation level employed. For the FCI level of calculations, the value of k is -1 [124, 125].

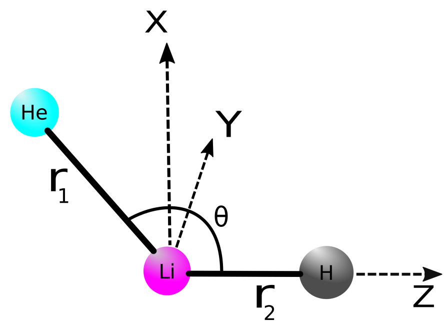


Figure 4.2: Internal coordinates used in $HeLiH^+$ *ab initio* data point generation and fitting.

The coordinate system used for calculating the *ab initio* data points is shown in Fig. 4.2. The Li-H moiety of $HeLiH^+$ complex is placed along the z-axis. The distances from He to Li, Li to H, and He to H atom are defined as $\vec{r}_1$, $\vec{r}_2$ and $\vec{r}_3$, respectively. The angle between the $\vec{r}_1$ and $\vec{r}_2$ ($\angle$He-Li-H) is defined as θ, which now onwards be referred as attacking angle. The *ab initio* energies are calculated within ($\vec{r}_1$, $\vec{r}_2$, θ) space. According to the coordinate system, linear He-Li-H^+/Li-H-He^+ configuration corresponds to θ = 180.0°/0°. The insertion of the He atom in between Li-H^+ is also considered in the present potential energy calculation by the inclusion of points along $\vec{r}_1$ and $\vec{r}_3$ (where θ = 0°) within linear Li-He-H^+ configuration. The details of the chosen three-dimensional space is specified in Table 4.1.

Table 4.1: Range of Coordinates and details of the grids along $\vec{r}_1$, $\vec{r}_2$, $\vec{r}_3$ and θ

θ**(deg.)**	$\vec{r}_1(a_0)$	$\vec{r}_2(a_0)$	**grid length**(a_0) [d$\vec{r}_1$, d$\vec{r}_2$, d$\vec{r}_3$]
	2.2 to 11.0	2.0 to 10.8	0.4
180.0 (linear He-Li-H), 165.0,	12.0 to 40.0	12.0 to 40.0	2.0
150.0, 120.0, 90.0, 60.0	50.0 to 150.0	50.0 to 150.0	10.0
	1.0 to 11.0	0.8 to 10.8	0.4
45.0,30.0,15.0,5.0	12.0 to 40.0	12.0 to 40.0	2.0
	50.0 to 150.0	50.0 to 150.0	10.0
	$\vec{r}_2(a_0)$	$\vec{r}_3(a_0)$	
	1.4 to 10.2	1.4 to 9.0	0.4
0.0 (Linear Li-H-He)	12.0 to 40.0	12.0 to 40.0	2.0
	50.0 to 150.0	50.0 to 150.0	10.0
	$\vec{r}_1(a_0)$	$\vec{r}_3(a_0)$	
	1.4 to 10.2	1.4 to 11.0	0.4
0.0 (linear Li-He-H)*	12.0 to 40.0	12.0 to 40.0	2.0
	50.0 to 150.0	50.0 to 150.0	10.0

*This configuration indicates the insertion of the He atom into the Li-H^+.

4.2.2 Fitting of *ab initio* data points

4.2.2.1 Fitting of two body terms

The *ab initio* data points for HeH, LiH^+, and $LiHe^+$ diatoms are fitted with analytical functional forms. Aguado Paniagua functional form [117] is used to fit the HeH diatomic *ab initio* data points, and Neural network (NN) functional form is used to fit the data points of LiH^+ and $LiHe^+$ diatoms. The Aguado function, that is only used for HeH diatomic potential, possess the mathematical form,

$$V_{\mathrm{HeH}}(R_{\mathrm{HeH}}) = \frac{c_0 e^{-\alpha_{\mathrm{HeH}} R_{\mathrm{HeH}}}}{R_{\mathrm{HeH}}} + \sum_{i=1}^{N} c_i \rho_{\mathrm{HeH}}^{i}, \tag{4.3}$$

where, $\rho_{\mathrm{HeH}} = R_{\mathrm{HeH}} e^{-\beta_{\mathrm{HeH}}^{(2)} R_{\mathrm{HeH}}}$. The coefficients c_i's (i = 0 to N) are the linear parameters with a precondition of $c_0 > 0$ and α_{HeH} and $\beta_{\mathrm{HeH}}^{(2)}$ are two non-linear parameters. The value of N is optimized for the diatomic system. The parameters c_i, α_{HeH} and $\beta_{\mathrm{HeH}}^{(2)}$ are optimized using Levenberg Marquardt (LM) optimization algorithm. The first term in Eq. 4.3 is short-range, and the second is a long-range contribution in total potential.

The feed-forward neural network (FNN) that is used to fit the data points for LiH^+ and $LiHe^+$ diatoms has a total of four layers (one input layer + two hidden layers + one output layer). The mathematical form of the neural network function is

$$V = b^{(3)} + \sum_{k=1}^{N_k} w_{\mathrm{k}}^{(3)} f^{(2)} \left[b_{\mathrm{k}}^{(2)} + \sum_{j=1}^{N_j} w_{jk}^{(2)} f^{(1)} \left(b_j^{(1)} + w_{ij}^{(1)} G_i \right) \right]. \tag{4.4}$$

Here, $b_j^{(1)}$, $b_k^{(2)}$, and $b^{(3)}$ are the biases associated with the neurons of the first hidden layer, second hidden layer, and output layer, respectively. The $w_{ij}^{(1)}$, $w_{jk}^{(2)}$, and $w_k^{(3)}$ are the weights associated with corresponding neurons of the input-first hidden layers, first-second hidden layers, and second hidden-output layers, respectively. The functions $f^{(1)}$ and $f^{(2)}$ are the activation functions, and V is the fitted potential, which is the output of the final layer (output layer). The network used in the present study is built and trained with the Keras API of the TensorFlow [126], and the LM algorithm is used as an optimization algorithm. The issue of overfitting is the major bottleneck in NN training. In the case of diatoms where the number of input variables is only one (interatomic distance), it is relatively easy to detect the overfitting visually, but as the number of input parameters increases, the detection of overfitting becomes challenging. Certain techniques can be used to overcome this. This present investigation implements the technique of regularisation of biases, weights, and activity (output of layers) of hidden layers to tackle overfitting. The early stopping procedure has also been implemented as an additional measure to reduce overfitting. The parameters used in the NN fitting for both diatoms are given in Table 4.2.

	$LiHe^+$	LiH^+
Neurons per layer	1+45+45+1	1+40+40+1
Learning rate	0.1	0.1
Batch size	32	32

Table 4.2: Hyperparameters for the fitting of neural network

4.2.2.2 Fitting of three-body term

The three-body term in the MBE form of the total potential is fitted using the triatomic Aguado Paniagua function, which is expressed as [117]

$$V^{(3)}_{\mathrm{ABC}} = \sum_{i\,j\,k\,=\,1}^{M} d_{\mathrm{ijk}} \rho^{i}_{\mathrm{AB}} \rho^{j}_{\mathrm{BC}} \rho^{k}_{\mathrm{AC}} \ . \tag{4.5}$$

Here, d_{ijk} are linear parameters and $\rho_{\mathrm{AB}} = R_{\mathrm{AB}} e^{-\beta^{(3)}_{\mathrm{AB}} R_{\mathrm{AB}}}$. The ρ_{BC} and ρ_{AC} possess similar formulae. The parameters i, j, and k follow the two constrains: $i + j + k \neq i \neq j \neq k$ and $i + j + k \leq \mathrm{M}$. The performance of the fitting or optimization of the parameters is measured by the root mean square error (RMSE) value, which is calculated as follows.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} \left(V^{\mathrm{fit}}_{i} - V^{\mathrm{ab\ initio}}_{i}\right)^2}{N}} \ , \tag{4.6}$$

where, $V^{\mathrm{ab\ initio}}_{i}$ and V^{fit}_{i} are the *ab initio* and corresponding fitted energies, respectively. N represents the total number of data points considered in the fitting. An accurately fitted surface must possess a low RMSE value, but vice-versa is not necessarily true. Thus, RMSE shouldn't be taken blindly as a signature of an accurately fitted surface.

4.2.2.3 Long range part of the potential

In the case of fitted functions, the fitting is generally good in the range of input data but may deviate in the extrapolating region. The NN method used here works exceptionally well in fitting and interpolating the data in the range of training inputs but fails miserably in extrapolation or beyond the training space, as can be seen for one and two-dimensional fitting shown in Figs. 4.7 and 4.3, respectively.

It can be seen from Fig. 4.3 that the NN function reproduces the potentials accurately in the region sampled by *ab initio* data points used in the training, but beyond the range, extrapolated energies are not accurate. The same can be observed from the comparison curve shown for the diatom (Fig. 4.7).

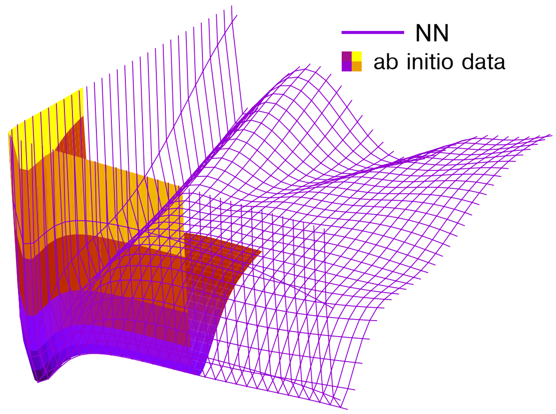


Figure 4.3: NN fitted surface with the *ab initio* energy points used in the fitting. The NN surface is extrapolated beyond the range of *ab initio* energy points.

Contrary to the NN, the AP function doesn't show the behavior as dramatic when extrapolating, but it has the functional bias of approaching zero as interatomic distance increases. To overcome these shortcomings, analytical long-range function (LR) is used at higher interatomic distances. The final potential uses the fitted potential function in the short range and the LR function in the long-range as follows.

$$V_2(R) = (1 - f(R))V_2^{\text{fit}}(R) + f(R)V_{\text{LR}}(R) \tag{4.7}$$

Here, $V_2^{\text{fit}}(R)$ is the analytical potential function fitted with the *ab initio* data points and $V_{\text{LR}}(R)$ is an analytical form of the long-range potential. The parameter $f(R)$ is the switching function that smoothly connects the $V_2^{\text{fit}}(R)$ and $V_{\text{LR}}(R)$. Out of the various possibilities, the following switching function is used in the present investigation [127].

$$f(R) = \begin{cases} 0, & R \geq R_0 + 8 \\ 0.5\sin\left[\frac{\pi}{16}(R - R_0)\right] + 0.5, & R_0 - 8 < R < R_0 + 8 \\ 1, & R \geq R_0 + 8 \end{cases} \tag{4.8}$$

Here, R_0 is the point where the switching between the *ab initio* fitted and analytical long-range potential takes place. The different combinations of parameters R_0 and the angle ($\frac{\pi}{16}$ in the current fit) control the smoothness of the switching. These parameters are crucial for the accuracy of fitting and should be optimized carefully because a rather careless choice of these parameters can introduce artificial features, e.g., energy wells or barriers, in the PES. The R_0 values for $LiHe^+$, LiH^+, and HeH are 25 a_0, 20 a_0, and 15 a_0, respectively. The LR formula for the three diatoms is as follows [128, 129].

$$\begin{gathered} V_{\text{LR}}^{\text{LiHe}^+}(R) = V_{\text{LR}}^{\text{LiH}^+}(R) = -\frac{\alpha_{\text{d}}q^2}{2R^4} - \frac{\alpha_{\text{q}}q^2}{2R^6} - \frac{\alpha_{\text{o}}q^2}{2R^8} - \frac{\beta_{\text{ddq}}q^3}{6R^7} - \frac{\gamma_{\text{d}}q^3}{24R^8}\,, \\ V_{\text{LR}}^{\text{HeH}}(R) = -\frac{C_6}{R^6} - \frac{C_8}{R^8} - \frac{C_{10}}{R^{10}} \end{gathered} \tag{4.9}$$

HeH			LiHe+	LiH+
		α_d	1.384	4.5
C_6	5.642	α_q	2.275	15.0
C_8	83.67	α_o	10.620	131.25
C_{10}	1743.08	α_ddq	20.41	159.75
		γ_d	37.56	1333.125

Table 4.3: Parameters in the analytical long-range formula

Here, α_d is dipole polarizability, α_q is quadrupole polarizability, α_o is octapole polarizability, β_ddq is first hyperpolarizability, and γ_d is second hyperpolarizability. The parameters C_6, C_8, and C_{10} are dispersion coefficients. It is clear from the long-range formulae that in the case of HeH, where both the atoms of the pair are neutral, dispersion energy is a contributing factor, whereas, in the case of LiH^+ and $LiHe^+$, the maximum contribution comes from dipole induced interactions. The values of these parameters are taken from the literature and shown in Table 4.3 [128, 129].

4.2.3 Computational details

The dynamical investigations are performed using both the quantum and quasi-classical methods. Reaction R1 is studied using both the QM and QCT methods, and only the QCT method is used for R3. A discussion of different aspects of these methods is given below.

4.2.3.1 QM method

Full Coriolis coupled QM calculations are performed using the real wave packet methodology coded in the DIFFREALWAVE [21, 130, 131], and a detailed discussion is presented in section 2.4.1 of Chapter 2. The integral cross section, reaction probability, and rate constants are calculated using the formulas discussed in section 2.4.1.5 in Chapter 2. The converged input parameters used in the DIFFREALWAVE code are as follows.

Table 4.4: Converged QM input parameters for reaction R1.

Parameters	**Converged value**
Total grid points for the product Jacobi, $N_{R_c}/N_{r_c}/N_{\gamma_c}$	139/ 139/ 55
Minimum and maximum grid length along R_c (R_c^{min}/R_c^{max}) (a_0)	0.5/22.0
Minimum and maximum grid length along r_c (r_c^{min}/r_c^{max}) (a_0)	0.5/19.5
Location of the dividing surface in the product channel, R'_d (a_0)	13.5
Cut-off potential (V_cut) (Hartree)	0.50

Starting point of absorption along R_c and r_c (R_c^{abs}/r_c^{abs} (a$_0$)	15.0/15.0
Absorption strength along R_c and r_c (C_{abs}/c_{abs}	8.0/3.0
Center of the initial WP in reagent Jacobi coordinates (R_0) (a$_0$)	11.5
Initial translational energy (E_{trans}) (eV)	0.28
Width of the initial WP (α)	10.0
Smoothness of the initial WP (β)	0.80
Total vibrational levels of the product (LiH^+) diatom	6
Number of rotational levels of the product (LiH^+) diatom	18
Number of Chebyshev iterations (n_{steps})	55000
Total propagation time (fs)	6032.16
Range of total angular momentum (J) in dynamics	J = 0 - 55

Input parameters are converged one by one. Several calculations are performed for different parameter values under convergence while keeping all the other parameters constant, and the procedure is repeated for all the parameters. At the time of the convergence of the first parameter, all the parameters except the one under convergence are given some starting values. As the parameters converge, the starting values are replaced with the converged values. The total reaction probability is used to check the convergence of the parameters. The parameter is considered converged when changing its value doesn't change the total reaction probability.

Notes on input parameters

Some of the input parameters used in the DIFFEREALWAVE code, as shown in table 4.4, are correlated with each other, and the values of these parameters can't be changed freely. The relation between different parameters is as follows.

1. $R_0 < R_c^{abs}$. It means that initially, the WP shouldn't be located in the absorption region.
2. $R_c^{abs} < R_c^{max}$ and $r_c^{abs} < r_c^{max}$. It means that the absorption should start well before the maximum grid value along R_c and r_c.
3. $r'_d < R_c^{abs}$, which means that the analysis of the WP should be done before the absorption function is activated.

There are other parameters used in the DIFFREALWAVE code that are correlated with each other. However, those parameters are specific to the working of the DIFFREALWAVE algorithm, and a detailed discussion of those can be found in the work of Hankel et al. [21].

A few important parameters are α, β, and E_{trans}. Out of the three, α and E_{trans} decide the spread of the WP in the collision energy grid. The range of energy covered by the WP can be calculated as follows.

$$E_{\pm} = \frac{(k_0 \pm \alpha)^2}{2\mu} \tag{4.10}$$

Here, initial angular momentum, k_0, is equals to $\sqrt{2\mu_{\text{A-BC}}E_{\text{trans}}}$. E_+ and E_- are the maximum and minimum energy of the collision energy range WP can access. The parameter β may not control the width, but it controls the smoothness of the WP, ensuring that the WP maintains a constant amplitude in a collision energy range. This overall effect can be seen clearly in Fig. 4.4, where the norm of the WP is given at different translation energies.

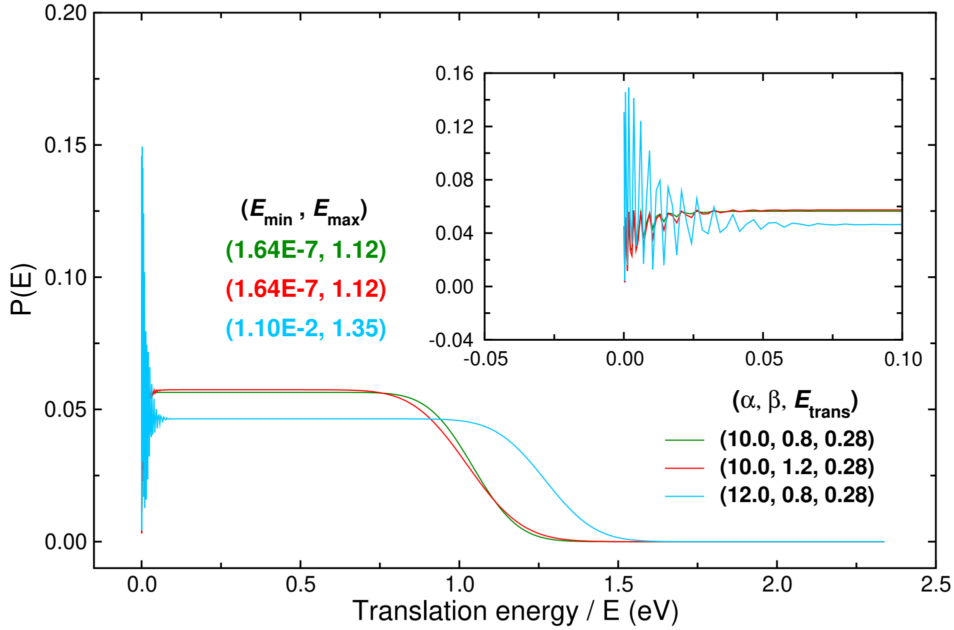


Figure 4.4: Probability of wave packet along translation energy grid for different combinations of α, β, and E_{trans}. The minimum and maximum energies of collision energy range at different parameter combinations are given in corresponding colors.

It can be seen from the figure that different combinations of three parameters give different WP spreads with varying smoothness. Thus, these parameters are converged in such a way that a constant amplitude WP covers the energy range of interest in the investigation.

The parameter n_{step} is the total Chebyshev iterations required in the complete propagation of the wave packet. It gives the WP propagation time. It is converged for the dynamics in such a way that the WP gets enough time to transform into the product. A short n_{step} value can result in the incomplete transformation of the WP into the product because some of the long-living complexes may not get enough time to relax into products. However, a higher than required value will unnecessarily increase the computational cost without affecting the outcome of the reaction. The n_{step} can be transformed into the propagation time (t) using the following relation.

$$t = \frac{a_s \hbar n_{\text{step}}}{\sqrt{1 - E_s^2}} \tag{4.11}$$

Here, scaled energy $E_s = a_s E + b_s$. The parameters E is the initial translation energy given to the wave packet (E_{trans}), a_s and b_s are the scaling parameters used for scaling the Hamiltonian used in the DIFFREALWAVE. The scaling parameters $a_s = 2/\Delta E$, and $b_s = -1 - a_s E_{min}$. $\Delta E = E_{max} - E_{min}$, where E_{max} and E_{min} are the maximum and minimum boundaries of Hamiltonian.

4.2.3.2 QCT method

As for the QCT calculations, the chapter follows the same methodology as followed in Chapter 3 and is discussed in detail in section 2.4.2 of Chapter 2. The converged input parameters for the reaction are as follows.

Table 4.5: QCT input parameters for R1 reaction.

Parameters	**R1**
Initial time step (h)	1.0
r_{z} (Bohr)	30.0
r_z^p (Bohr)	32.0
n_{traj} in final calculations	500000
d_{nr} (Bohr)	25.0
Inner turning point of LiH^+ (Bohr)	3.72781
Outer turning point of LiH^+ (Bohr)	4.90569
Half vibrational time period of LiH^+ (ν_{htp}) (au)	1768
Boundaries of interaction region (r_{int}) (Bohr)	17.0
$E_{v=0,j=0}$ of LiH^+ (eV)	2.5276563075E-2
Atomic mass(m_{He}/ m_{Li}/ m_{H}) (amu)	4.00150609431/ 7.01435769687/ 1.00727645232
Equilibrium distance of the diatoms ($r_{\text{eq}}^{\text{LiHe}^+}$/ $r_{\text{eq}}^{\text{LiH}^+}$/ $r_{\text{eq}}^{\text{HeH}}$) (Bohr)	3.6339267853d0/ 4.1884376875d0/ 6.6837367473d0
Dissociation energy of the diatoms ($D_{\text{e}}^{\text{LiHe}^+}$/ $D_{\text{e}}^{\text{LiH}^+}$/ $D_{\text{e}}^{\text{HeH}}$) (Hartree)	2.754888299740748E-3/ 4.871482473865499E-3/ 2.070375889575279E-005

Morse parameters (α^{LiHe^+}/ α^{LiH^+}/ α^{HeH}) (Bohr^{-1})	0.7716d0/ 0.904839d0	1.10856d0/

Note: The symbol of the parameters have the same value as discussed in section 2.4.2 of Chapter 2.

The converged values of maximum impact parameters (b_{max}), which are shown in Table 4.7 are calculated by running 20000 trajectories. Once the b_{max} values are converged, a total of 500,000 trajectories are run for the final calculations.

Table 4.7: Maximum impact parameter (b_{max}) for R1 reaction.

Collision energy (eV)	b_{max} (Bohr)	Collision energy (eV)	b_{max} (Bohr)
0.01	-	0.25	4.52
0.02	-	0.30	4.12
0.03	-	0.35	2.76
0.04	4.24	0.40	2.68
0.05	5.07	0.45	2.46
0.06	5.55	0.50	2.19
0.07	5.73	0.55	2.29
0.08	5.72	0.60	2.19
0.09	5.59	0.65	2.22
0.10	5.50	0.70	2.06
0.105	5.49	0.75	2.07
0.11	5.40	0.80	1.79
0.12	5.39	0.85	1.86
0.13	5.31	0.90	1.79
0.14	5.31	0.95	1.82
0.15	5.42	1.00	1.76
0.20	5.22		

The CID channel (reaction R3) was also studied using the QCT method. The converged input parameters for the CID calculations were kept the same as given in Table 4.5. However, the b_{max} values are converged by running the same 20000 numbers of trajectories, which are shown in Table 4.8.

A total of 500,000 trajectories are run to perform the final calculations. The QCT-HB approach is used to calculate the number of trajectories going into the CID channel. The detailed procedure to calculate these trajectories is discussed in section 2.4.2.7 of Chapter 2.

Table 4.8: Maximum impact parameter (b_{max}) for CID channel (R3).

Collision energy (eV)	b_{max} (Bohr)	Collision energy (eV)	b_{max} (Bohr)
0.75	-	0.20	6.29
0.08	-	0.30	6.71
0.09	-	0.40	6.82
0.10	-	0.50	7.02
0.11	4.36	0.60	7.01
0.12	5.34	0.70	7.16
0.14	5.77	0.80	7.22
0.16	5.97	0.90	7.16
0.18	6.15	1.00	7.33

4.2.4 Convergence of J_{max} values

As mentioned, the J_{max}=55 is taken as a converged value. When calculating cross sections, the S-matrix elements are required for all J collisions possible within the collision energy range of the investigation. Now, as the value of J increases, the centrifugal barrier of the reaction increases (cf. Fig. 5.2), pushing the threshold energy of the reaction further in the higher collision energy range. As the threshold energy increases with increasing J value, the

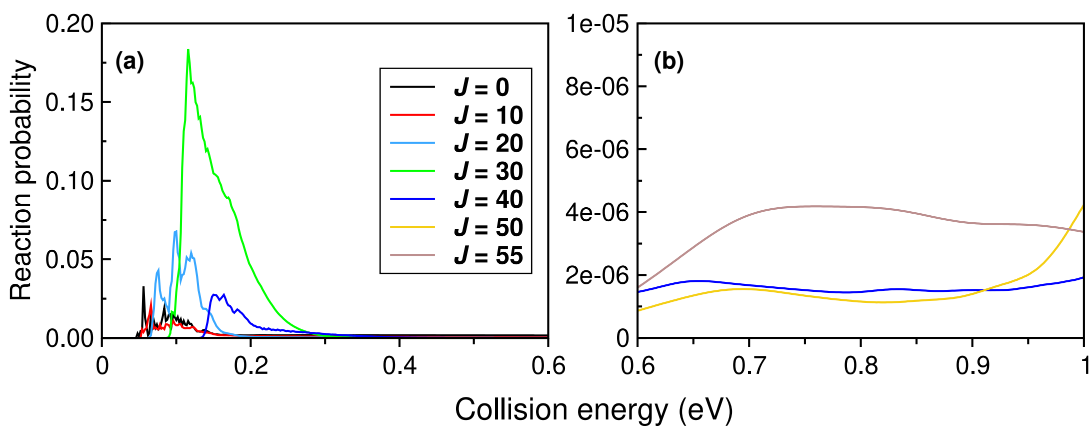


Figure 4.5: Convergence of J_{max} value. Reaction probabilities for different J values are shown as a function of collision energies. Panel (a) shows the probabilities from 0.0 eV to 0.6 eV, and the probabilities for the same J collisions from 0.6 eV to 1.0 eV are shown in panel (b).

appearance of non-zero probability also shifts to higher collision energies. This means that within a specific collision energy range, only a certain number of J (up to J_{max}) collisions will contribute to the total reaction probability and the rest of the collision with $J > J_{max}$ will be irrelevant. Thus, a converged J_{max} value can be obtained by plotting the total reaction probability plot for different J as a function of collision energy as done in Fig. 4.5 for the current reaction. The figure shows that the probability of J=55 is the order of 10^{-6} eV but not exactly equal to zero. This can be considered negligible, converging the value of $J_{\max}$ to 55.

Note: Sometimes when converging the value of $J_{\max}$, the probability may not reach to 0. However, if the probabilities are sufficiently small after some J values, convergence can be considered.

4.3 Results and Discussion

4.3.1 Fitting of PES

The NN method was used to fit data points of the $LiHe^+$ and LiH^+ diatoms, whereas the HeH diatom was fitted using the AP function (cf. sec. 4.2.2.1). The conventional method using the AP function was also used to fit $LiHe^+$ and LiH^+, but the quality of fit was inferior to NN, especially in the interaction region of the potential curve as shown in Fig. 4.6.

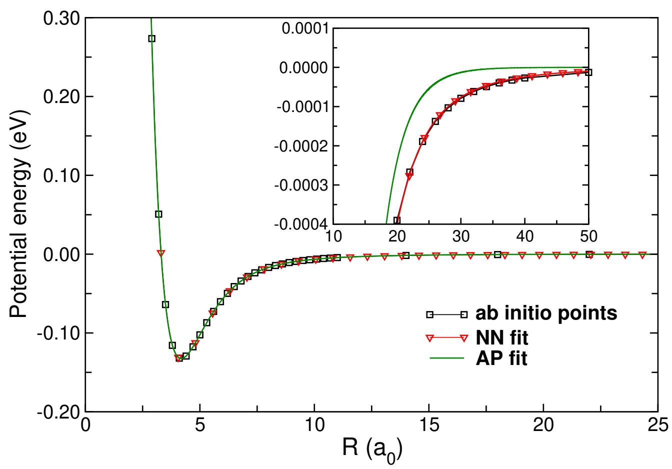


Figure 4.6: NN and AP functional fit for the LiH^+ diatom shown with the *ab initio* data used for fitting. A specific region of the same plot is zoomed in the inset.

However, in the case of HeH, the NN approach did not work well. A major difficulty was experienced in the neck region (between the minimum potential region and asymptotic region) of the potential curve, where there was an uncompromisable qualitative mismatch between NN fit and the actual behavior of HeH potential as shown in Fig. 4.7. Since the three-body potential part depends on the quality of the two-body fits in MBE, the accuracy in the two-body fittings is essential to the accurate three-body and overall fit. Considering this, the fitting of LiH^+ and $LiHe^+$ was performed using NN, and the AP function was used for the HeH fit.

A total of 121, 91, and 126 *ab initio* energy data points were used for the fitting of $LiHe^+$, LiH^+, and HeH curves, respectively. In the case of both $LiHe^+$ and LiH^+, a ratio of 85% and

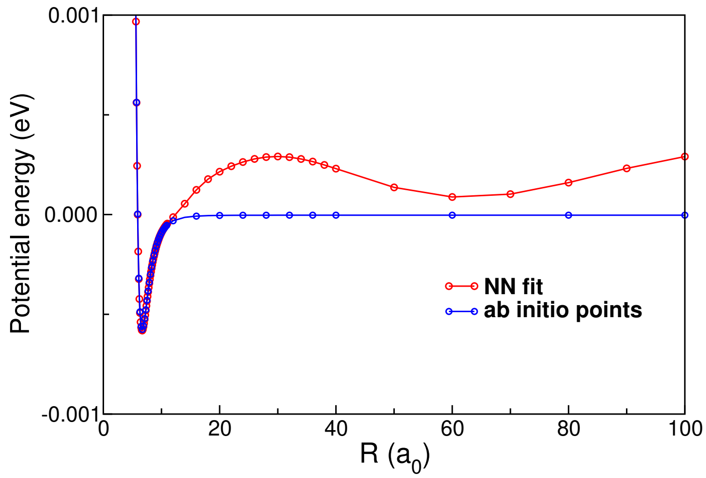


Figure 4.7: Oscillations in NN fit shown with the *ab initio* data points used in the fitting.

15% is used for training and validation sets, respectively. The fitting RMSE for the HeH diatom is 9.61E-06 eV. In the case of $LiHe^+$ and LiH^+, the RMSE of the fitting is 2.15E-05 eV (6.93E-06 eV) and 1.00E-04 eV (8.16E-04 eV), respectively, for training (validation) data sets. As for the test set in the case of NN fit, separate points that are not used in the training or validation process are calculated using the same *ab initio* level of the method as the training data and compared with the fitted values. These data points show an excellent fit, as shown in the V_{fit} vs. $V_{\mathrm{ab\ initio}}$ curve in Fig. 4.8. The convergence of RMSE for the training and validation set shows the good fit of the data, and further, it can be verified from the learning curves shown in Fig. 4.9. These learning curves are important to check for overfitting, which is a crucial bottleneck in any ML fitting procedure. It can be seen from the learning curves that the RMSE curves for both the training and validation data sets converge into each other as training progresses.

The quality of the diatomic fit, as already evident from the low RMSE values, can easily be seen in Fig. 4.10. Fig. 4.10 (a) shows the fit in complete interatomic space, whereas individual fittings in the short, intermediate, and long ranges of potential are shown in panels (b), (c), and (d) of Fig. 4.10, respectively, for better comparison. The fitting reproduces the nature of diatomic potential well in short, intermediate, and long ranges. The small van-der walls well of just 5.89E-04 eV in the case of HeH is also fitted successfully. Further, the quality of the fit can also be seen from the *ab initio* vs fitted data plot in Fig. 4.11. In this plot, all the data points fall along the diagonal, which suggests the fit is of good quality. The equilibrium distance (R_{eq}) and minimum energy (E_{eq}) obtained from current fitting, Wernli et al.[116] surface, and *ab initio* optimization of the $LiHe^+$ and LiH^+ diatoms are compared in Table 4.9. The comparison is shown only for $LiHe^+$ and LiH^+ diatoms because no *ab initio* optimized geometry was obtained for HeH. It can be observed from the table

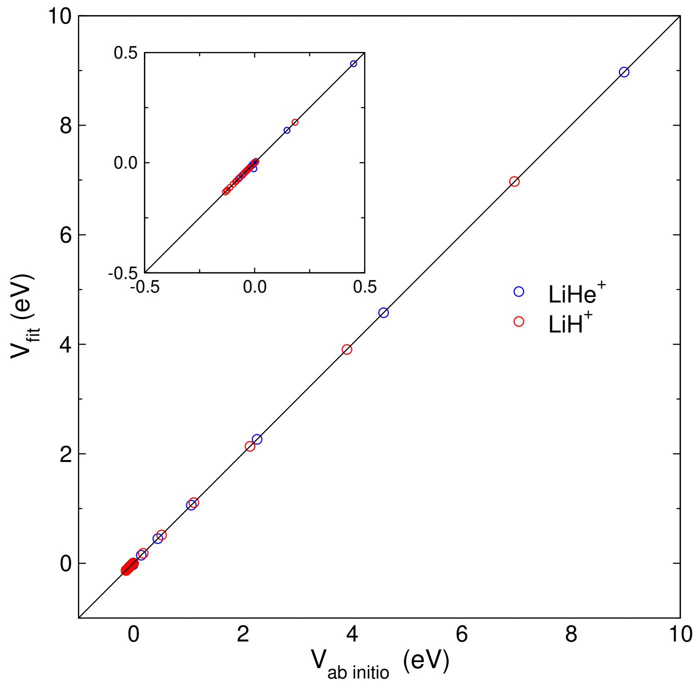


Figure 4.8: A plot between the $E_{ab\text{-}initio}$ and E_{fit} for the test data set in the NN fitting of $LiHe^+$ and $LiHe^+$ diatoms. The same data points from the lower energy region (< 0.5 eV) are shown in the inset.

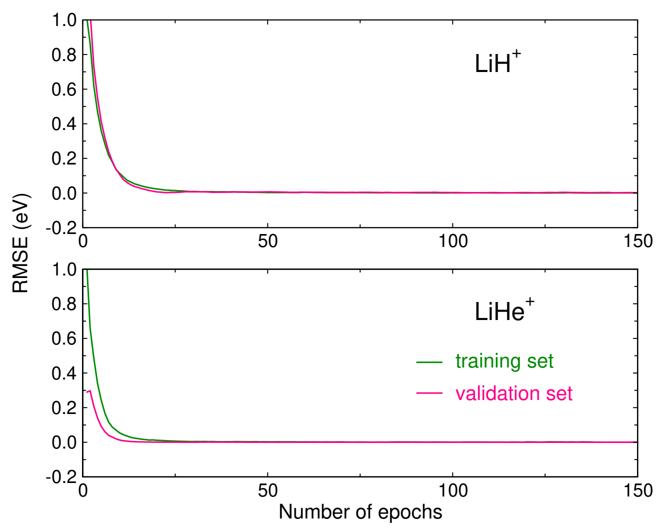


Figure 4.9: The learning curves from the NN training of LiH^+ and $LiHe^+$ diatoms. The training was a little longer, but only 150 epochs are shown.

that both the fitted and optimized parameters are close to each other but a little higher than the values obtained from Wernli et al. [116] surface.

It is understood that the total potential contributes to two-body and three-body interactions. After fitting the two-body (diatomic) interaction, the process of fitting the three-body in-

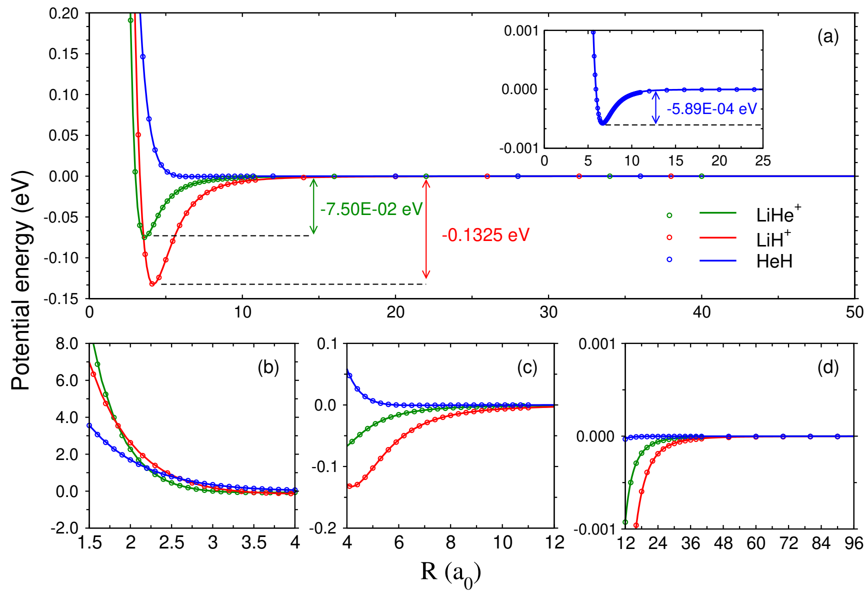


Figure 4.10: The *ab initio* data points (circular symbols) for the three diatoms, $LiHe^+$, LiH^+, and HeH, along with the fitted potential curves (solid lines). The shallow van-der Walls potential well for HeH is shown in the inset of panel (a). Panels (b), (c), and (d) show the fitting at the short, intermediate, and high interatomic distances separately. The color coding, as defined in panel (a), is the same for all the panels.

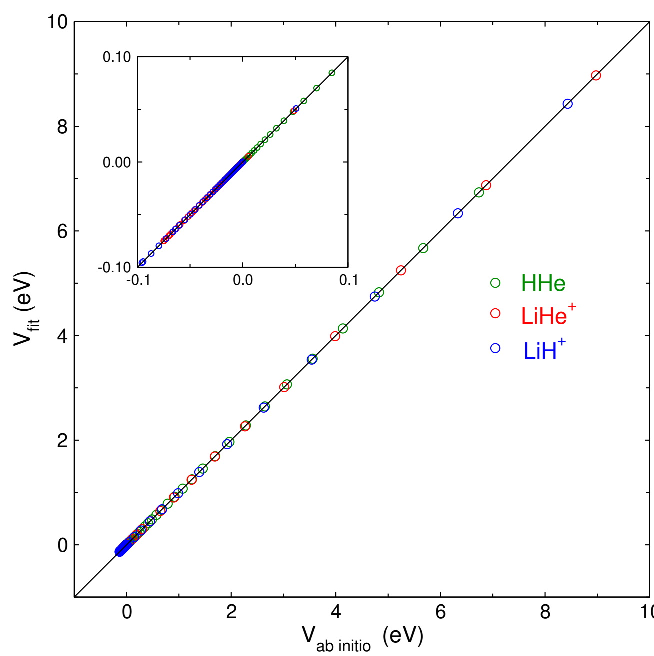


Figure 4.11: A plot of $E_{ab\ initio}$ vs E_{fit} for $LiHe^+$, $LiHe^+$, and HeH. The same data points from the lower energy region (< 0.10 eV) are shown in the inset.

teraction starts by generating the data points to be used in fitting. These data points are generated by subtracting the two-body interactions from the total potential at the grid spec-

	R_{eq} (a_0)	E_{eq} (eV)
$LiHe^+$		
This work	3.638	-7.509E-2
Wernli et al.[116]	3.586	-7.965E-2
ab initio data	3.641	
LiH^+		
This work	4.183	-0.1325
Wernli et al.[116]	4.142	-0.1380
ab initio data	4.186	

Table 4.9: Comparison of equilibrium distance and energy at the minima for the LiH^+ and $LiHe^+$ diatoms

ified in Table 4.1. Initial fit was performed with 29039 *ab initio* data points of energy < 10 eV. This initial fit was used to identify and remove the data points which have fitting errors ($|E_{ab\ initio}$-$E_{fit}|$) > 0.04 eV. Finally, 29011 *ab initio* data points with energy < 10 eV and fitting error < 0.04 eV were taken as input data sets to fit the three-body potential. The best fit was obtained for the order (M) = 13 with the RMSE of 1.63E-3 eV (13.14 cm^{-1}). The overall RMSE of the fitted surface that is obtained when the potential of two-body and three-body are added together is 1.76E-3 (14.21 cm^{-1}). This total RMSE is for the energy space up to 10 eV, and RMSE improves to 9.28E-4 eV (7.48 cm^{-1}) when energy space only up to 2 eV is considered. This information about RMSE in different energy regions is important because not all reaction events will venture into the coordinate space of higher energy. To give a little more idea about the overall fitting error or accuracy in different energy spaces for the current PES, RMSE values for two-body, three-body, and overall $HeLiH^+$ system fitting in different energy regimes are provided in Table 4.10.

Table 4.10: RMSE (eV) for diatoms in different energy regions

	$LiHe^+$	**LiH^+**	**HeH**	**$HeLiH^+$**
< 18 eV	1.195E-04	-	-	-
< 10 eV	1.20E-04	-	-	1.76E-003
< 8 eV	1.19E-04	3.2E-04	-	1.68E-003
< 6 eV	1.17E-04	4.3E-05	9.6E-06	1.60E-003
< 4 eV	1.14E-04	3.5E-05	9.7E-06	1.33E-003
< 2 eV	1.04E-04	3.4E-05	9.8E-06	9.28E-004

To check the quality of the fit for three-body, a plot showing the fitting error ($E_{ab\ initio}$-E_{fit}) with respect to *ab initio* energies at each grid point is shown in Fig. 4.12. The figure also shows the information regarding the attacking angle for each data point in a different color. It can be seen that most of the points fall closer to the horizontal 0.0 eV line (zero line), which is the signature of a good fit. However, a few points deviate from the zero line and indicate a relatively large error. It is clear from Fig. 4.12 and Fig. 4.13 that most of the fitted points with the large fitting error belong to the region of low attacking angles (~ 0°) and high energy (≥ 2.0 eV), a consequence of which can be seen in decreasing RMSE with lowering of energy space from 10 eV to 2 eV as discussed in the previous paragraph.

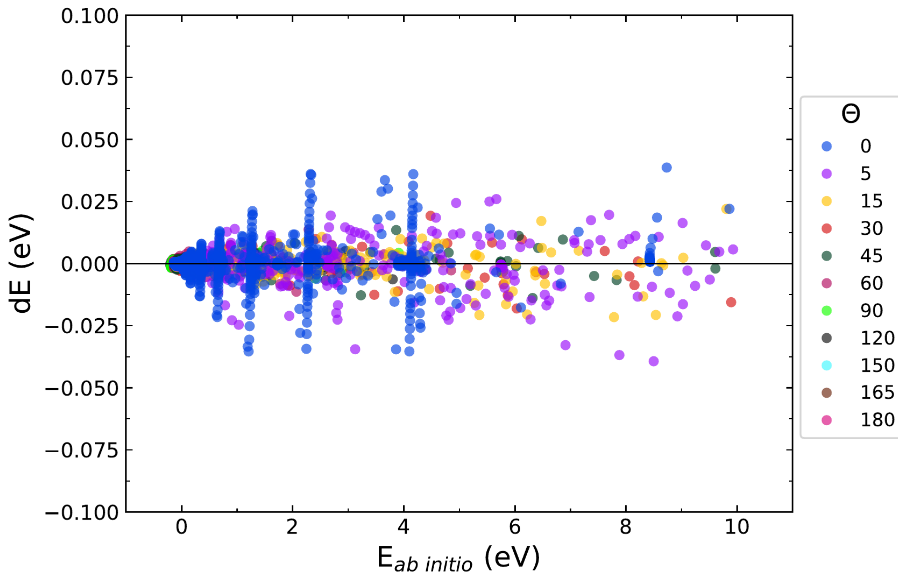


Figure 4.12: Fitting error (dE = $E_{ab\ initio}$-E_{fit}) vs $E_{ab\ initio}$ plot for overall fitting of $HeLiH^+$ system. The data points for different attacking angles (θ = ∠He-Li^+-H) are colored differently.

Further, the accuracy of the overall fitting is verified by comparing the *ab initio* and fitted potential points at the attacking angle of 15°, 30°, 60° and 180° in Fig. 4.14. In all of these potential cuts, the $LiHe^+$ bond distance is fixed at 3.40 a_0, and potential is plotted against LiH^+ bond distance. The figure has three panels, among which panel (a) shows the fit in overall coordinate space, whereas panels (b) and (c) show the fitting in short and long ranges, respectively. It can be seen from the plots that the *ab initio* points are in good agreement with the fitted potential cuts in short as well as long ranges, especially in the case of 15° attacking angle where the barrier around 3.0 a_0 is also reproduced perfectly. A few more potential cuts at different configurations are shown in Fig. 4.15. Fig. 4.15 also shows a good comparison at attacking angles of 95°, the configuration which was not used in the fitting, and thus confirms the accuracy of fitted potential.

4.3.2 Features of Fitted PES

The features of the PES can be understood better in the context of two asymptotes, LiH^+ + He and $LiHe^+$ + H, of the reaction R1 or R2. In the case of both of these asymptotes, the

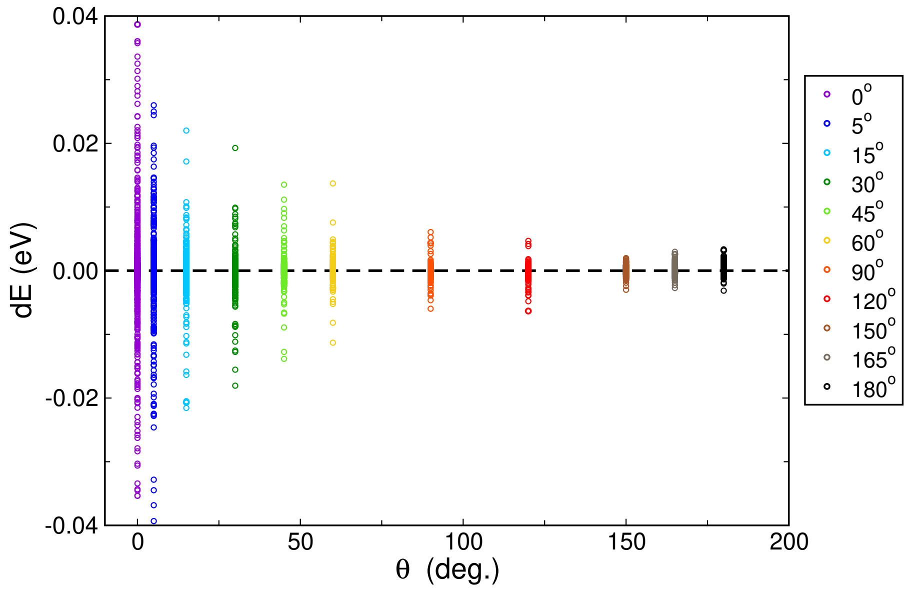


Figure 4.13: Fitting error (dE = $E_{ab\ initio}$-E_{fit}) vs attacking angle (θ) plot for overall fitting of $HeLiH^+$ system.

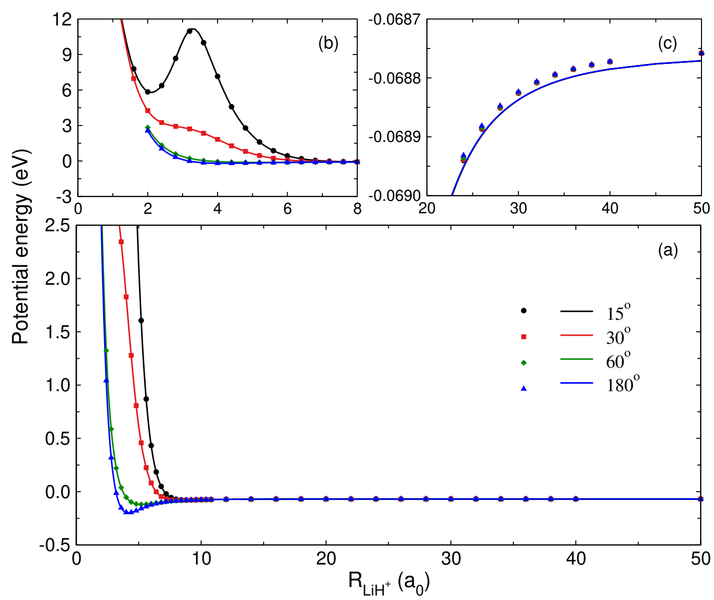


Figure 4.14: Comparison of *ab initio* data points (symbols) with the fitted potential (solid lines) at different attacking angles (θ) for $LiHe^+$ bond distance fixed at 3.40 a_0 in panel (a). The fitting comparison at the long-range and short-range region is zoomed in panels (b) and (c).

attack of the He(H) atom is favored from the Li side rather than the opposite H(He) side of the diatom. This can be seen clearly in the potential plots of He around LiH^+ and H around $LiHe^+$, respectively, in panel (a) and (b) of Fig. 4.16. In both of these panels, the diatomic distance is fixed at 4.17 a_0 for $LiHe^+$ and 3.62 a_0 for LiH^+ and attack from Li side results

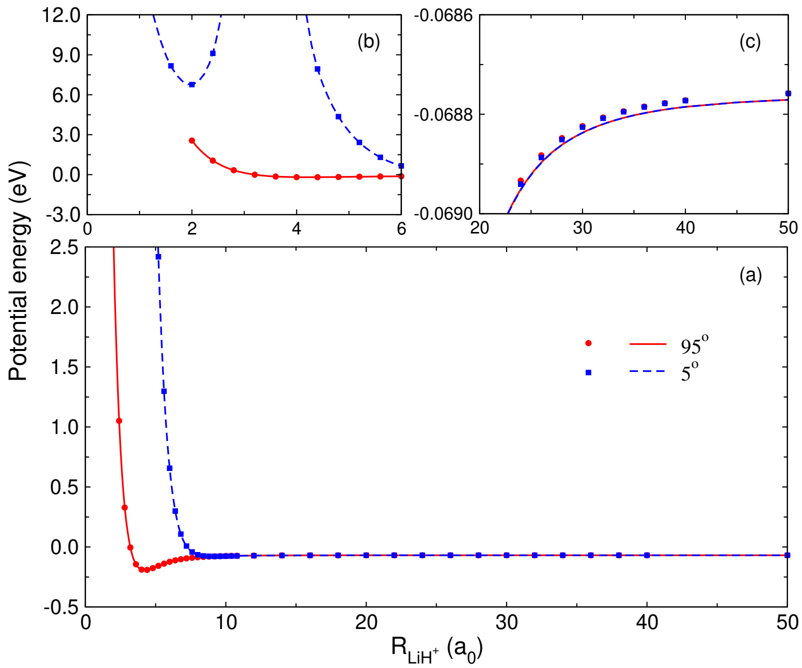


Figure 4.15: Comparison of ab-initio data points (symbols) with the fitted potential (solid lines) at different attacking angles (θ) for $LiHe^+$ bond distance fixed at 3.40 a_0 in panel (a). The fitting comparison at the long-range and short-range region is zoomed in panels (b) and (c).

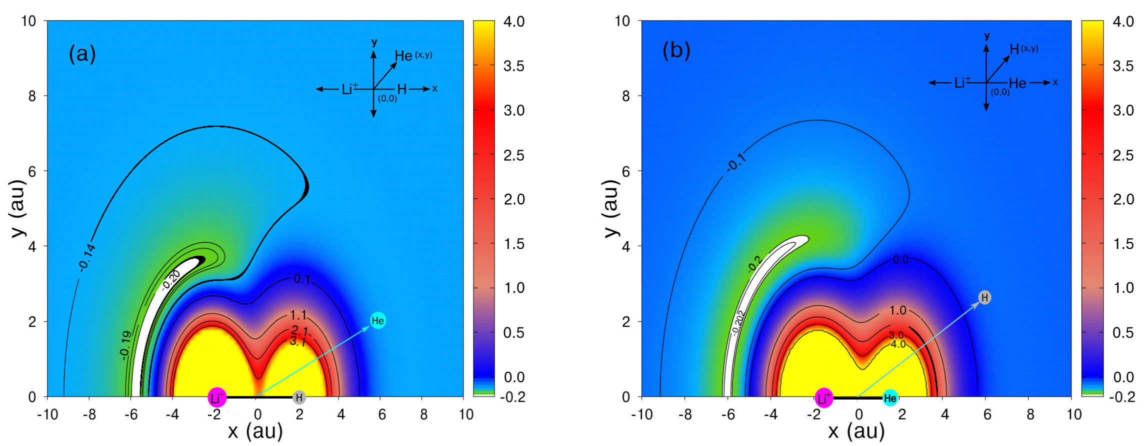


Figure 4.16: Contour plots of $HeLiH^+$ with $LiHe^+$ bond distance fixed at 4.17 a_0 in panel (a), LiH^+ bond distance fixed at 3.62 a_0 in panel (b) and origin at the center of diatom with fixed bond distance in both the cases.

in an energy minimum, whereas the attack from the opposite side has a repulsive barrier. The preference given to the Li side by H or He while attacking $LiHe^+$ or LiH^+ diatom, respectively, can also be observed from the MEP curves as shown in Fig. 4.17. It is clear from the MEPs that a higher energy barrier is faced when an atom attacks from the opposite

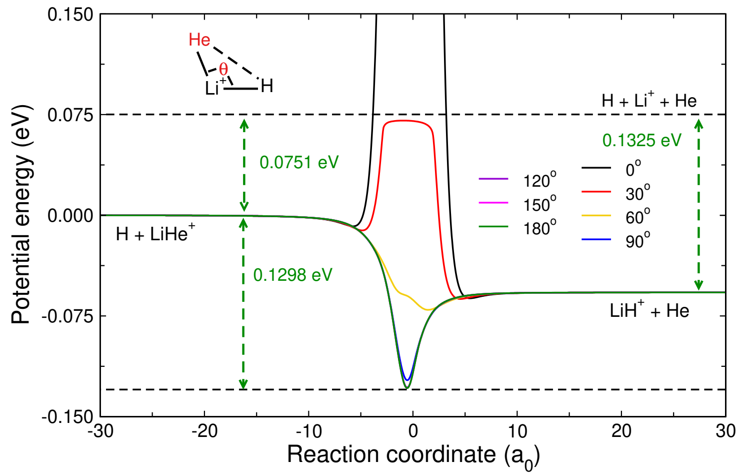


Figure 4.17: Minimum energy paths (MEPs) at different attacking angles θ.

side than Li (close to 0°), and as the angle of attack starts approaching 180°, the MEP starts getting attractive. Interestingly, after 90°, the MEPs are very close to each other, and after 120° no significant difference is observed, the trend which can be seen more clearly in Fig. 4.18. This similarity in the MEPs is evidence of almost similar potential surfaces after 90°,

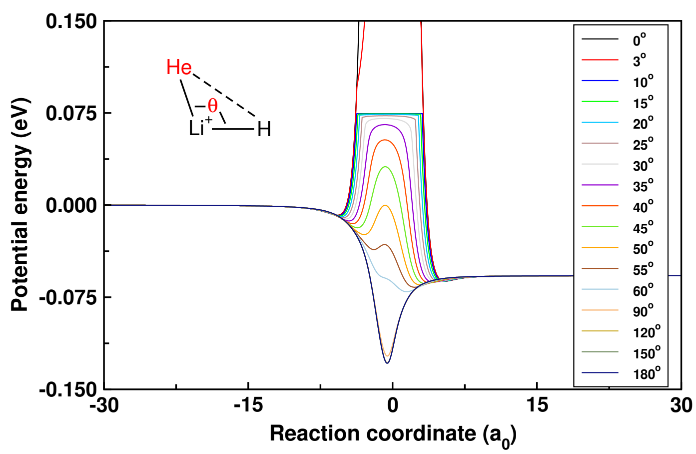


Figure 4.18: Minimum energy paths showing the changing nature of the potential surface.

and the same can be seen in Fig. 4.19. The potential surfaces are shown at the attacking angles of 5°, 45°, 80°, and 180°, and as it was evident from the MEP plots, the potential surfaces have a barrier at the attacking angle close to 0° and as the attacking angle increases, surfaces become barrierless.

Although the potential surfaces become evidently similar as the attacking angle increases, the global minima are only obtained at the linear configuration when the attacking angle is 180°. The same is also obtained from the optimized calculations, and the fitted surface reproduces the parameters accurately, which are tabulated along with the global minima

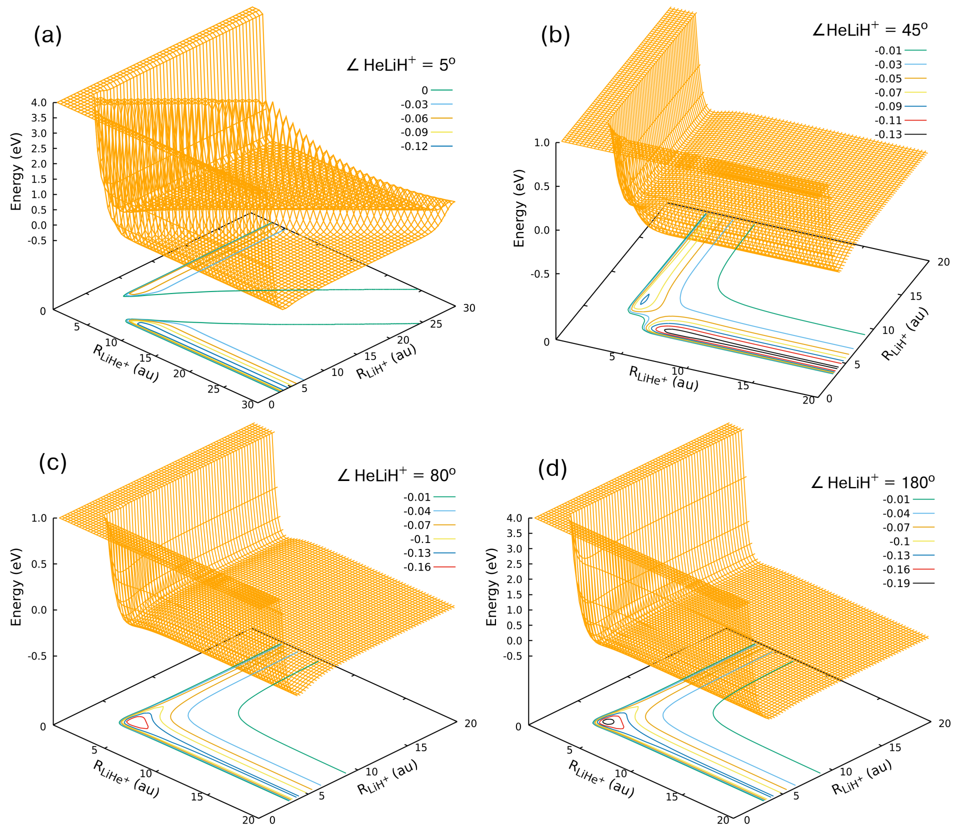


Figure 4.19: Potential energy surfaces with contour plots at the different attacking angles (θ) of 5°, 45°, 80°, and 180°.

parameters from Wernli et al. [116] surface in Table 4.11.

Table 4.11: Parameters of global minima with respect to three-body dissociation

Parameters of global minima	**This work**	***ab initio* optimization**	**Wernli et al. surface[116]**
R_{LiHe^+} (a_0)	3.636	3.650	3.590
R_{LiH^+} (a_0)	4.192	4.191	4.157
R_{HeH} (a_0)	7.829	7.841	7.747
$\angle HeLiH^+$	180°	180°	180°
E (eV)	-0.2038	-0.2034	-0.2152

4.3.3 Reaction dynamics

The reaction dynamics calculations for reaction R1 were performed using the QM and QCT methods discussed in section 4.2.3. Both the QCT and QM methods are used to calculate the reaction observables. In the case of QM-CC method, several calculations are performed to converge the input parameters to get the converged results up to the collision energy of 1 eV and J_{max} of 55.

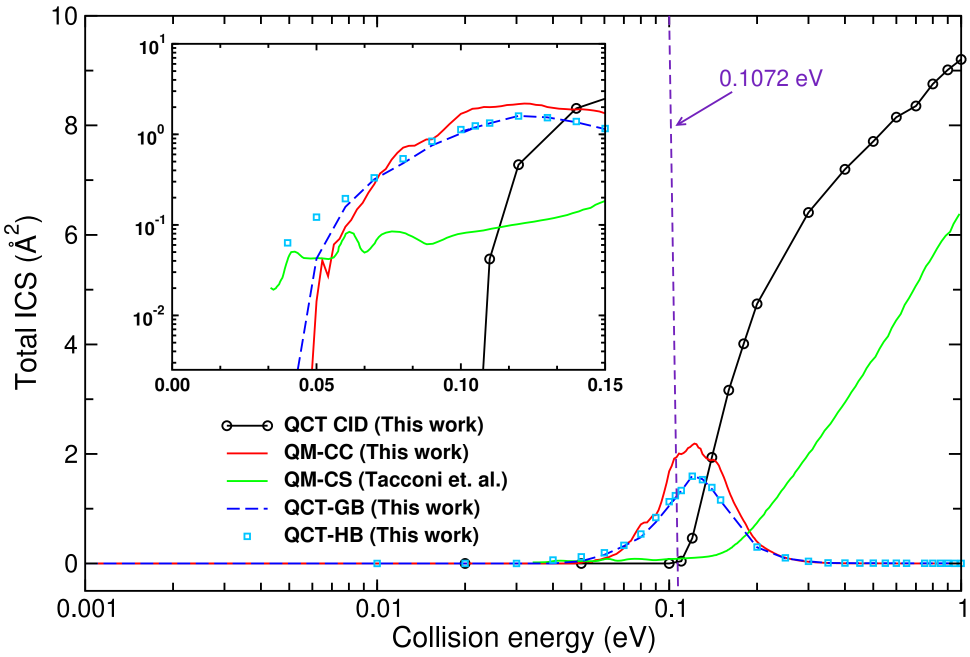


Figure 4.20: Total ICS for reaction R1 as a function of collision energy. The QCT ICS of current work is compared with the QM-CS ICS of Tacconi et al.[121] The black line shows the collision energy at which the dissociation channel opens up. Both the axes are logarithmic, but the plot with linear axes is also shown in the inset.

The total ICSs for reaction R1 calculated using QCT-HB, QCT-GB, and QM-CC methods are shown and compared with the total ICS calculated in the work of Tacconi et al.[121] in Fig. 4.20. It is clear from the plot that the results of the current study vary from the results of Tacconi et al.[121]. Although the trend of ICS is the same as in the case of Tacconi et al. in the lower collision energy range (< 0.12 eV), the magnitudes are one order larger in the ICSs of current work. The trend is the opposite in collision energies > 0.12 eV, where the ICSs of the present study decrease and that of Tacconi et al. increase. However, this point of reversal of the ICS trend is not random. The three-body dissociation channels of the system for LiH^+ (v=0, j=0) state of reagent diatom opens up at ~ 0.107 eV of collision energy. This point (~ 0.12 eV) is close to the collision energy point at which the dissociation channel opens up, making three-body dissociative products another possible reaction channel. This expectation is found valid in the QCT calculations where the number of the trajectories going into the dissociation channel starts increasing and dominating other reaction channels after the opening of the three-body dissociation channel. The domination of the CID channel after the point of dissociation can be seen as the CID cross section shown in the same Fig. 4.20. It can be clearly seen from the figure that the CID cross sections start appearing after the opening of the dissociation channel at ~ 0.107 eV and soon become the prominent reaction channel. Moreover, this reversal of the ICS trend after the opening of the dissociation channel is not new and has also been observed in a few other theoretical and experimental studies [132–134].

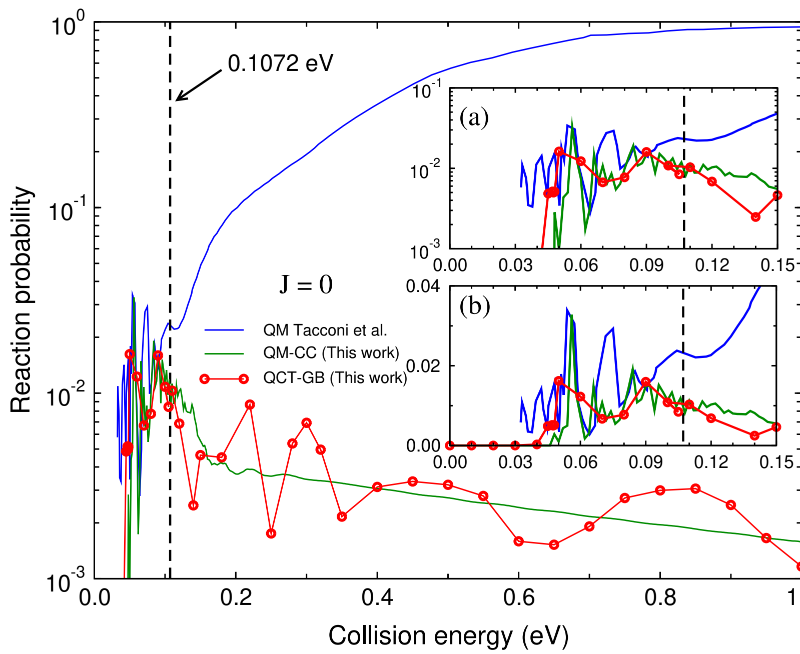


Figure 4.21: Comparison plot of QM (Tacconi et al.[121]) reaction probability with QCT-GB and QM-CC reaction probabilities of this work for reaction R1 (He + LiH^+(v = 0, j = 0) → $LiHe^+$ + H) at total angular momentum (J) = 0 (impact parameter b = 0) as a function of collision energy. Reaction probabilities at the lower collision energy range are shown in the inset, taking the y-axis in the logarithmic scale in panel (a) and the normal scale in panel (b). The black dashed line is the collision energy at which the dissociation channel opens up. The y and x axes are logarithmic and linear, respectively.

The same trend reversal near the collision energy at which the dissociation channel opens up is also observed for the reaction probabilities for J = 0, 10, 20, and 30 collisions. The reaction probability for J = 0 is shown in Fig. 4.21, and that of J = 10, 20, and 30 is shown in Fig. 4.22. Fig. 4.21 also shows the comparison of the QM total reaction probability of Tacconi et al.[121] with QCT-HB, QCT-GB, and QM-CC total reaction probabilities of present calculations. The reaction probabilities of Tacconi et al. keep increasing collision energy, but the QCT-GB results initially increase up to ~ 0.09 eV and then show the reversal of behavior. The same pattern of reversal is also followed by the QM-CC reaction probability of current work, which closely follows QCT-GB probability, as can be seen from panels (a) and (b) of Fig. 4.21. Also, both the QM-CC and QCT-GB reaction probabilities have a threshold very close to each other, which, within the numerical error range, matches with

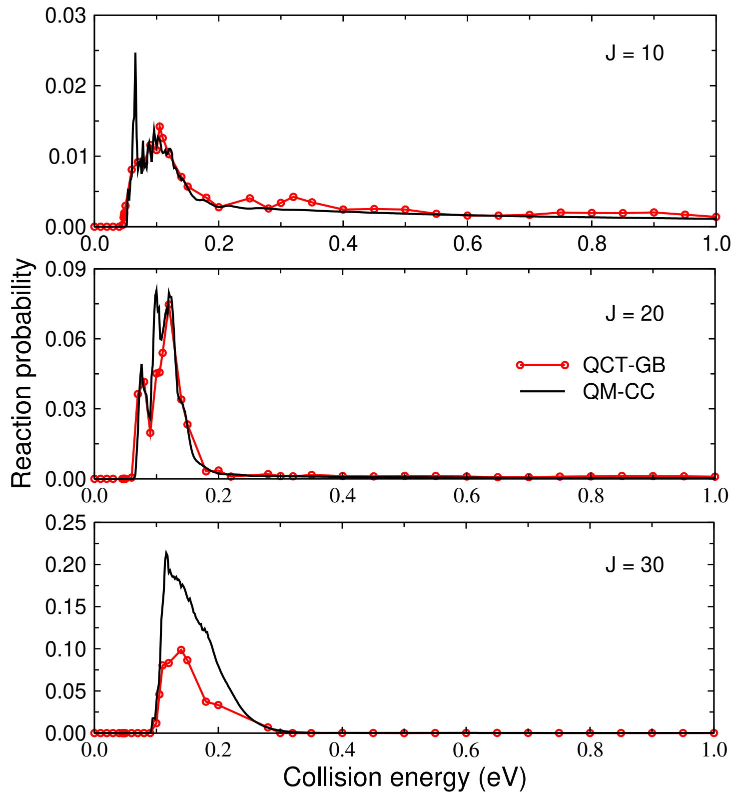


Figure 4.22: Comparison plot of QM-CC and QCT-GB reaction probabilities for reaction R1 (He + LiH^+(v = 0, j = 0) → $LiHe^+$ + H) at different total angular momentum values as a function of collision energy.

the endothermicity (~ 0.0475 eV) for reaction R1 when the zero-point energies of both the reagent (LiH^+) and product ($LiHe^+$) diatoms are considered.

This similarity, instead of an exact match between the energy threshold appearing in QM-CC and QCT total reaction probabilities, provides an insight into the QCT method in case of endothermic reactions. The QCT method, being a classical method, can produce product states with real ro-vibrational quantum numbers, which is otherwise not possible in QM methods. This otherwise benign error is a significant source of divergence between QM and QCT results in the energy region below the ZPE. The possibility of the existence of these below ZPE states in endothermic reactions is the reason that threshold energy appears a little earlier in QCT than the actual threshold appearing in QM calculations. In fact, it was found true when restrictions were gradually introduced on trajectories in QCT calculations, and the actual energy threshold was reproduced, as shown in Fig. 4.23. The Figure shows the total reaction probability at J = 0 for QCT-HB, QCT-GB, and QCT-GB+ZPER. In the QCT-HB approach, each trajectory is treated equally, whereas a weight factor is introduced on the trajectories based on the closeness of their ro-vibrational states to the closest integer value in the QCT-GB approach. Finally, an additional restriction of removing all the trajectories, which result in product states below ZPE, is introduced along with the QCT-GB approach in QCT-ZPE+GB. As pointed out, it can be seen that the energy threshold in J = 0 reaction probability appears earlier in QCT-HB than QCT-GB, and, finally, QCT-GB+ZPER has the same threshold as QM-CC. This threshold is encountered at 0.04754 eV of energy when v = 0, j = 0 level energy of LiH^+ and $LiHe^+$ diatoms is considered.

The effect of these different thresholds appearing in QCT-HB, QCT-GB, QCT-ZPE+GB, and QM-CC methods can also be observed in the QCT and QM-CC ICSs calculated in the present work (cf. inset of Fig. 4.20). It can be seen that all three ICSs follow the same trend, except in the energy range < 0.07 eV, where the QCT-HB ICSs start varying from the QCT-GB and QM-CC cross sections. As the collision energy is close to the endothermicity values (~ 0.0475 eV), the trajectories going below the ZPE start increasing in numbers. QCT-HB method, as already pointed out, accepts these trajectories with equal (unit) weight, resulting in increased ICSs than QM-CC ICSs as observed in Fig. 4.20. Since QCT-GB can mitigate this unnatural behaviour of trajectories by weighting the trajectories, QCT-GB ICS values are much closer to the QM-CC values. This feature appearing in ICSs due to the leaking of trajectories below ZPE when the QCT-HB method is used is not unique and was also reported earlier with an interesting consequence where quantum mechanically endothermic reaction behaves as exothermic.[135]

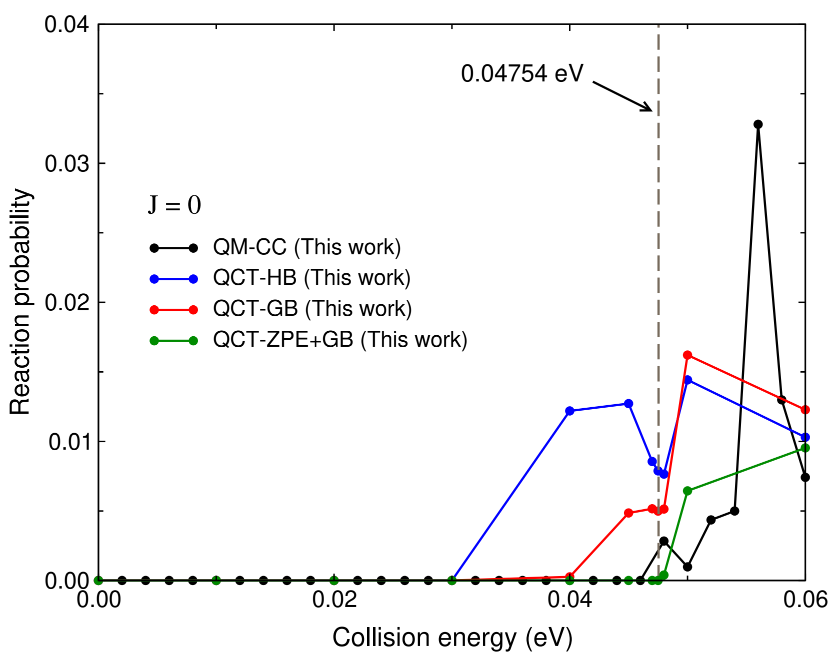


Figure 4.23: Comparison plot of J = 0 reaction probability for QM-CC, QCT-HB, QCT-GB, and QCT-ZPE+GB methods for reaction R1 (He + LiH^+(v = 0, j = 0) → $LiHe^+$ + H) near threshold energy (0.04754 eV).

The state-specific rate constant values are also calculated for reaction R1 using both the QM-CC and QCT methods (QCT-HB and QCT-GB) and compared with the rate constants of Tacconi et al.[121] in Fig. 4.24. In the work of Tacconi et al. [121], rate constants are

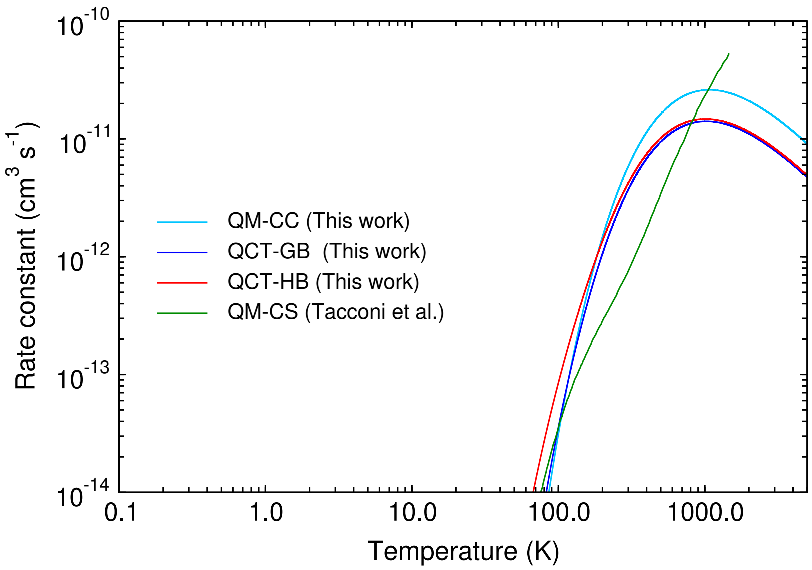


Figure 4.24: Comparison plot of reaction rate constants from Tacconi et al. [121] and this work for reaction R1 (He + LiH$^+$ ($v = 0$ $j = 0$) $\rightarrow$ LiHe$^+$ + H) at different temperatures. Both the axes are logarithmic.

calculated up to 1000 K, whereas in the current study, rate constants are shown up to 5000 K. Since the cross sections at the higher collision energies are essentially very low and reaction being endoergic has no cross sections below the endoergic energy of ~ 0.0475 eV, the cross section data up to 1.0 eV is enough to get the converged state-specific rate constants even well below 100 K and above 5000 K of temperatures. This can be understood more clearly as follows.

Theoretically, the rate constant at any temperature depends on the cross-section and Boltzmann distribution at all the energies that can be understood by looking at the rate constant equation given below.

$$k_{vj}(T) = \sqrt{\frac{8k_BT}{\pi\mu}}\left(\frac{1}{k_BT}\right)^2 \sum_i \left(E^i_{\mathrm{col}}\sigma_{vj}^{E^i_{\mathrm{col}}} e^{-\frac{E^i_{\mathrm{col}}}{k_bT}} dE_{\mathrm{col}}\right) \tag{4.12}$$

$$k_{vj}(T) = \sum_i \left[\sqrt{\frac{8k_BT}{\pi\mu}}\left(\frac{1}{k_BT}\right)^2 \left(E^i_{\mathrm{col}}\sigma_{vj}^{E^i_{\mathrm{col}}} e^{-\frac{E^i_{\mathrm{col}}}{k_bT}} dE_{\mathrm{col}}\right)\right]$$

$$k_{vj}(T) = \sum_i k_{v,j}^{E^i_{col}}(T) \tag{4.13}$$

The contribution of each energy in the rate constant at a specific temperature can be quantified. As a general trend, while calculating the rate constants, the contributions from the lower collision energies are major at low temperatures, and contributions from the higher collision energies are major at high temperatures. So, to calculate the rate constants at the lower temperatures, cross section data at the low energies should be available, and cross section data at the high temperature is needed to calculate the rate constants at high temperatures. In the present case, the cross sections have two features:

1. Reaction is endoergic in nature for the initial reactant state ($v = 0, j = 0$), meaning that cross sections below the endoergicity are essentially zero.
2. The cross sections at the higher collision energies are very close to zero.

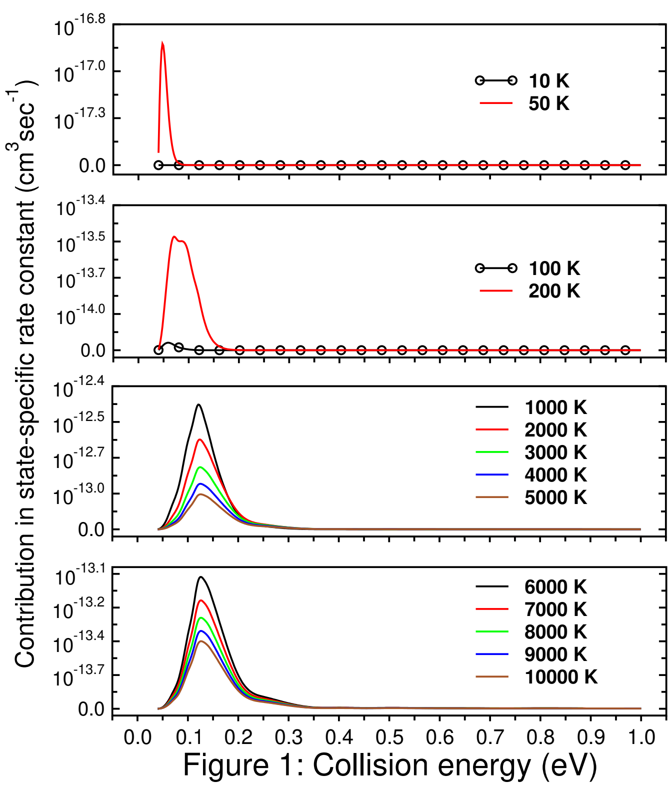


Figure 4.25: Contribution of each collision energy in the total rate constant value at different temperatures.

These two features have an impact on the convergence of the $k(T)$ when it comes to the respective range of collision energy where the cross section data is required. Fig. 4.25 shows the contribution of the collision energies in total rate constants at different temperatures calculated using Eq. 4.13. It is clear that the contribution of the energies below ~ 0.04 eV is zero, which is because of the reaction's endoergic character. The contribution of the higher collision energies is also very low. Also, compared to the peak contribution coming from the energy range 0.01 - 0.3 eV, the contribution of the lower and higher end of the collision energies is negligibly small, as seen in Fig. 4.26. Since data at the collision energies from where the major contribution is expected is available, it is confident to say that the $k(T)$ converges well within the temperature range of 100 - 5000 K.

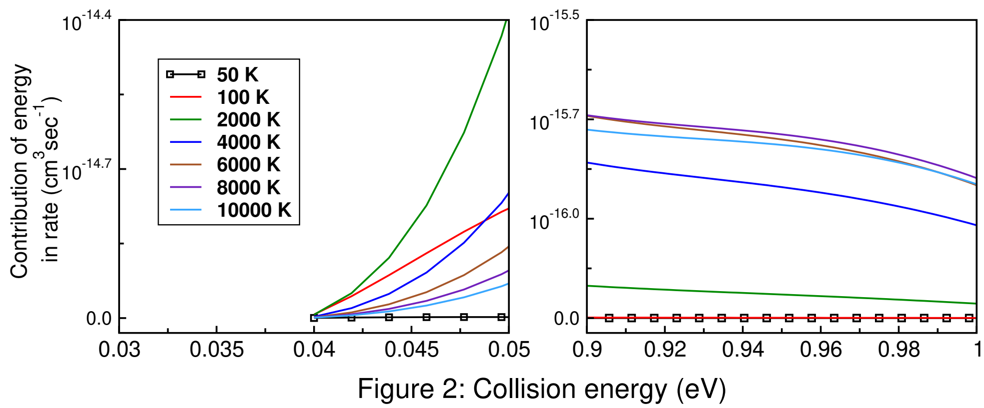


Figure 4.26: Contribution of each collision energy in the total rate constant value at different temperatures. Energies at lower and higher ends are zoomed.

Even if the rate constants are calculated using the collision energies in different ranges, as shown in Fig. 4.27, it is observed that the rate constants don't vary much. It can be seen from Fig. 4.27 that the rate constants change significantly when the energies above 0.3 eV are discarded; otherwise, the rate constants are more or less the same. It is to be noted that the cross-section data is only available up to 2.0 eV of collision energy and, thus, extrapolation is used to calculate the cross sections up to 2.0 eV, which is further used in rate constant calculation.

Coming to Fig. 4.24, it is clear that all the rate constants are almost of the equivalent order, except as the temperature increases, the rate constant of Tacconi et al.[121] increases, whereas QM-CC and QCT rate constants calculated in current work decreases after a maximum around ~ 1065 K. This decrease in the rate constants is the consequence of the behavior of ICS or opening of the three body dissociation channel at the higher collision energy, an effect that doesn't seems to be included in the calculations of Tacconi et al. [121]. A hint that can be realized from the temperature (~ 1065 K) corresponding to a maximum of the rate constants of current work, which is very close to the temperature (~ 1250 K) that corresponds to the opening of the CID channel.

The behavior of the rate constants from the two studies can also be connected with the nature of cross sections as a function of collision energy. The cross sections or reactivity of the reaction (R1) in the current work increase at the lower range of collision energy; thus,

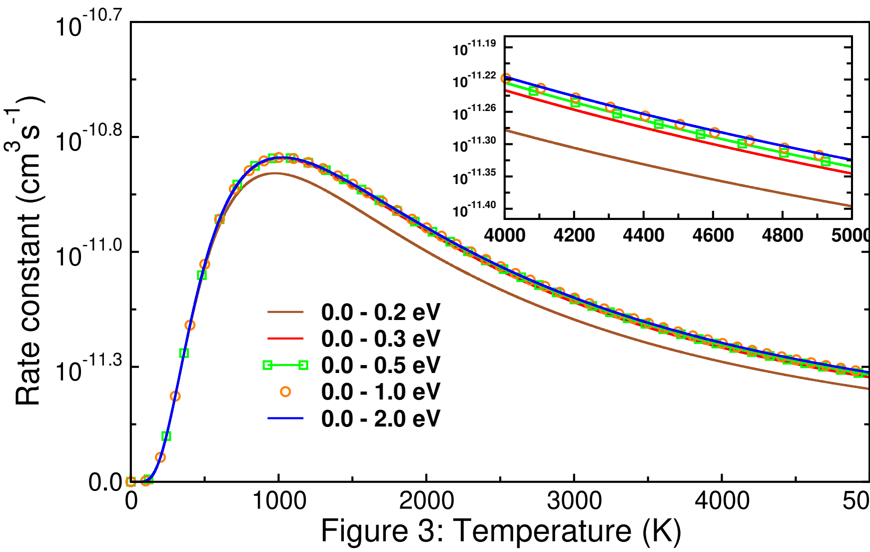


Figure 4.27: Rate constants calculated using collision energies in different ranges.

rate constants show the initial increment at lower temperatures, but as the cross sections start falling down after the opening of three body dissociation channel, rate constants start decreasing too. Since the cross sections of Tacconi et al.[121] keep on increasing with the collision energy, their rate constants also increase with temperature without any fall.

4.4 Chapter summary

The current work discusses the construction of ground electronic state PES of the $HeLiH^+$ system, which is fitted using NN and AP functions. The MBE approach was used to represent the total potential into one-body, two-body, and three-body potential parts, which were fitted independently and later added to give the total fitted potential at a fixed nuclear configuration. $LiHe^+$ and LiH^+ diatoms were fitted using the NN function, and HeH and the three-body parts of the potential were fitted using the AP function. The analytical long range was used with the diatomic parts and stitched with the fitted analytical function using a switching function.

Many attempts were made to get the best fit of the surface, clearly implying that the fitting process is not very straightforward and needs to be fine-tuned to get the expected results. Also, different fitting methods have their bottlenecks and must be dealt with accordingly. In the case of the ML methods, such as NN, used in the present work, the accuracy of the fit is a delicate balance between the number of points used in training data and the hyperparameters of the network. Neither an unnecessary deep network nor the vast amount of training data alone can result in a well-trained network. Even different NN models with the same hyperparameters but different starting points, controlled by the initialized values of weights and biases, will drastically affect the learning outcomes. In the current work, while fitting the NN model, an additional difficulty of the appearance of artificial wells and barriers of shallow energy order (10^{-5} eV) was faced. Such features were not apparent when visualizing the potential curves and surface in low energy resolution (zoomed-out plots) or simply aiming for a small RMSE of the fitting and could only be realized after carefully examining the fitted curve.

Once the fitted surface with desired accuracy was obtained, it was used to perform the preliminary dynamics calculations using both the QCT and QM-CC methods and total ICSs, total reaction probability, and total rate constant of endoergic reaction R1 (He + LiH^+(v = 0, j = 0) → $LiHe^+$ + H) were presented and compared with the existing results of Tacconi et al. [121]. Both the QM-CC and QCT results of the current work follow each other with reasonable similarity and differ from the results of Tacconi et al. [121]. Also, these results appear more in sync with the effect of the presence of a three-body dissociation channel. These primary dynamical results are interesting because of the presence and effect of a three-body dissociation channel on the overall dynamics of the reactions and also in the sense that the new cross section, rate constant trends calculated in the current study can significantly affect the abundance calculations of LiH^+ and $LiHe^+$ species in the interstellar medium.

Chapter 5

State-to-state dynamics of isotopic exchange reaction $^{18}O + {}^{16}O^{16}O\ (v = 0, j = 1) \rightarrow {}^{18}O^{16}O + {}^{16}O$

5.1 Introduction

During the night of September 3rd and 4th 1980, a balloon-borne mass spectrometer recorded the distribution of heavy ozone [$^{50}O_3$ ($^{18}O^{16}O^{16}O$)], isotopologues of ozone, in the stratosphere with the descent of the balloon flight. The results showed the unusually enhanced amount of heavy $^{50}O_3$, an enrichment of heavy ^{18}O isotope in the atmospheric ozone. The isotopic ratio $^{50}O_3/^{48}O_3$ in the atmosphere was found greater than the ratio $^{50}O_3/^{48}O_3$ of the standard and a maximum of 40% enrichment was reported at the altitude of 32 km, which was decreasing at the altitude higher or lower than this [136].

Later, in laboratory studies, the equal enrichment of $^{49}O_3$ and $^{50}O_3$ was reported, confirming that the fractionation of isotopes is mass-independent [137]. Subsequent measurements also confirmed the enrichment of heavier isotopes of atmospheric O_3 [138–141]; however, the enrichment ratio and its variation with altitude varied from study to study. This became an interesting problem, and several attempts were made to explain the mass-independent fractionation (MIF) of ozone, but no satisfactory answer has yet been found. However, it is largely believed that the MIF found is a consequence of the isotopic effect on the ozone formation reaction. The ozone formation in the atmosphere is a three-body recombination process: $O + O_2 + M \rightarrow O_3 + M^*$. The process is shown to take place via two different mechanisms: the energy transfer mechanism called the Lindemann mechanism and the radical complex mechanism known as the Chaperon mechanism [142–144]. The Lindemann mechanism is a two-step process.

$$\text{Step 1:} \qquad O + O_2 \rightleftarrows O_3^* \tag{5.1}$$

$$\text{Step 2:} \quad O_3^* + M \longrightarrow O_3 + M^* \tag{5.2}$$

In the first step, atomic and molecular oxygen collide to form the metastable ozone (O_3^*). Later, intermediate metastable O_3^* forms the stable O_3 by transferring the excess energy to the third body M, which can be an atmospheric atom or molecule, such as nitrogen, argon, etc. The isotopic effect observed in the rate of recombination reaction is due to the lifetime of the metastable ozone $^{a+b+c}O_3^*$.

However, step 2 is not the only pathway through which metastable ozone can transform. It can also break down into atomic and molecular oxygen, where the attacking atom and molecule on the reactant side exchange the oxygen atom: $O_3^* \longrightarrow O + O_2$. The overall reaction is called the isotopic exchange reaction, competing with the second step (eq. 5.2) of the Lindemann mechanism. The reaction is as follows.

$$^{a}O + {}^{b}O^{c}O \longrightarrow O_3^* \longrightarrow {}^{b}O + {}^{a}O^{c}O \tag{5.3}$$

Here, a, b, and c are the isotopic masses of oxygen atoms. These three masses can be the same or different based on the isotopes involved. Reaction 5.3 is either thermoneutral, exothermic, or endothermic, determined by the isotopic combination involved. Interestingly, the isotopic effect observed in the ozone formation reaction is also found in this reaction, making some combinations of isotopes react faster than others. The experimental thermal rate constants for the different isotopic variants of the reaction 5.3 show a negative temperature dependence [145].

5.2 Summary of Potential energy surfaces

The construction of O_3 PES was subjected to extensive studies over the years. The overview presented in this section will shed light on the importance and sophistication of electronic structure calculations and their impact on the fine topographical features of the surface. The discussion will also highlight how these fine topographic features of the underlying PES affect the outcomes of the chemical processes occurring on the PES.

The first global potential energy surface (PES) was constructed by Siebert et al. [146] in 2001. The *ab initio* data points were calculated using the internally contracted multireference interaction method with Davidson correction (MRCI+Q), and the complete active space self-consistent field / CASSCF(12,9) was used as the reference function. The basis set employed for the oxygen atom was cc-pVQZ. Other than the accurate depiction of the potential well region, one of the noteworthy features of the PES was the small barrier of height 50 cm^{-1} close to the $O + O_2$ asymptote. The dissociation energy obtained from the fitted PES was ~0.1 eV (~900 cm^{-1}) lesser than the experimental value of 1.052 eV [146]. However, it was reported in the same article and subsequent study by the same group in 2003 [147] that using the larger basis set (quintet zeta), the barrier above the asymptote transformed to a barrier of height ~114 cm^{-1} below the $O + O_2$ asymptote and dissociation energy error was dropped to ~0.04 eV (~322 cm^{-1}). This modified surface [147] was obtained by augmenting the original surface of Siebert et al. (Ref. [146]) by adding an analytical correction term, which was calculated only around the barrier, not globally. This correction was the difference between the original surface and energy values calculated at the higher basis set and extrapolated to the complete basis set (CBS) limit along the minimum energy path (MEP). New modified PES supported a van der Walls well, separated from the global minima by the submerged barrier. In the literature, the original and modified Siebert et al. PESs are known as SSB and mSSB surfaces, respectively.

Since it is shown that the SSB PES has certain drawbacks, which were corrected in mSSB PES, another attempt by Babikov et al. [148] was made to correct the surface in 2003. The correction term added to their potential was global. The PES also contains the submerged barrier and the dissociation energy calculated from the fitted PES matches the experimental value [148]. Until now, all the PESs have either the barrier or submerged reef near the O + O_2 asymptote, although the experimental results suggested that the reaction is barrierless. On top of it artificially removing the barrier from the SSB surface in the work of Siebert et al. produced reaction rate constants much closer to the experimental values with negative temperature dependence also found in the experimental rate constants for $^{16}O + {}^{36}O_2 \longrightarrow {}^{34}O_2 + {}^{18}O$ isotopic exchange reaction [147]. These discrepancies in the results prompted several others to perform the ab-inito calculations, where the same submerged reef structure was obtained [149, 150].

In this series, another attempt was made to construct a PES by Holka et al. [151] in 2010, where the correct features were incorporated via fitting of *ab initio* data, not by applying correction functions. Their surface also found the submerged barrier reported in other potential surfaces, but the discrepancies related to the mismatch between the experimental dissociation energy of O + O_2 asymptote and experimental data hinting towards the barrierless reaction remained unexplained. It was only in 2011 that Dawes et al. [152] could finally obtain the reef-free potential near the O + O_2 asymptote via *ab initio* calculations. Data points are extrapolated to the CBS limit using the energies calculated with aug-cc-pVTZ and aug-cc-pVQZ basis sets and MRCI with Davidson correction (MRCI+Q) method, and calculations used two lowest full-valence 13-state dynamical state-averaged CASSCF (DW-SA-CASSCF) reference wavefunctions. However, only the 2-D surface was constructed, and the Quasi-statistical model (QSM) studies based on the PES gave the rate constants for $^{16}O + {}^{32}O_2 \rightarrow {}^{32}O_2 + {}^{16}O$ reaction which were in much better agreement with the experiment. Also, the negative temperature dependence in rate constants was reported based on *ab initio*-based PES for the first time. Later, in 2013, Dawes et al. [153] constructed the full dimensional PES for O_3. Their calculations use explicitly correlated RCI (MRCI-F12) method with VQZ-F12 basis sets without CBS extrapolation. The data points also included the long-range model of Lepper et al. [154] and correction for spin-orbital (SO) coupling. The rate constants based on the 3-D QSM calculations were reported for $^{16}O + ^{32}O_2 \rightarrow ^{32}O_2 + ^{16}O$ reaction, which show the same negative temperature dependence as shown in their earlier study [152]. The dissociation energy obtained from fitted PES (9253 cm^{-1}) is slightly higher than the experimental energy (9219±10 cm^{-1}), and vibrational levels for ^{16}O and ^{18}O are reproduced with reasonable accuracy.

However, it was found later that the reef structure reappears in configurations other than the MEP [155], which makes the Dawes surface not completely free from the submerged barrier. Before this surface, two other PES were reported in the literature. The first PES was reported by Babikov et al. [156] in 2013. The *ab initio* energy points were calculated at icMR-CISD+Q level with CASSCF(12,9) active space. The data points for fitting PES were extrapolated to the CBS limit using energies at aug-cc-pVQZ and aug-cc-pv5Z basis sets and fitting reports the reef structure globally. The second potential surface has the same reef-free feature as Ref. [152] and a much better agreement regarding theoretical and experimental energies of vibrational levels of O_3 was reported by Tyuterev et al. [157] in 2013 before the 2013 PES of Dawes. Two versions of the PESs were reported; one surface has the reef structure, and the other is free from it (abbreviated as NR_PES in the original article). The energy points were calculated using the ic-MRCI/AV5Z method. In the reef-containing version of the PES, data points calculated using the AV5Z and CBS extrapolation were used around the well region and beyond, respectively. In the NR_PES, the finding related to the use of several electronic states from the work of Dawes et al. [152] was included in the ic-MRCI method. This NR_PES reproduced the vibrational-rotational bands with < 1 cm^{-1} RMS deviation between the experimental and calculated values. Another surface and the one used in the current work was constructed by Varga et al. [158] in 2017. Their surface used the extended multi-state complete active space second-order perturbation theory (XMS-CASPT2) to calculate the *ab initio* energies with minimally augmented cc-pVTZ (maug-cc-pVTZ) basis set. The reference states used for the *ab initio* method were SA-CASSCF(12,9) calculations.

5.3 Previous studies and motivation of current work

The isotopic variant of the exchange reaction studied in this work is

$$^{18}O + ^{16}O^{16}O(v = 0, j = 1) \longrightarrow ^{18}O^{16}O + ^{16}O. \tag{5.4}$$

The reaction can be termed an 8+66 variant of the isotopic exchange reaction. The reaction is thermoneutral without considering the ZPEs of the reactant and product diatoms and exothermic when the ZPEs are considered. This makes the 8+66 variant of the reaction exothermic barrierless. This reaction has been the subject of several studies in the past. A brief discussion of these studies is presented in this section, which highlights the journey to reproduce the correct experimental findings theoretically and shows the evolution of the understanding of the 8+66 isotopic exchange reaction and its current state.

DLLJG PES [153] was the first potential surface to provide the reef-free potential structure. After the construction of the DLLJG PES [153], several investigations using this PES were carried out by different groups on the 8+66 variant of O + O_2 isotopic exchange reaction. Some of these groups were Dawes, Guo, co-workers, and Honvault, Mahapatra, and co-workers.

In 2014, Guo et al. [159] reported the quantum scattering calculations for 6+66, 6+88, and 8+66 variants of the isotopic exchange reaction using the DLLJG [153] PES. The initial rovibrational state of the reactant diatom considered in the dynamic calculations of the above reactions is $(v, j) = (0, 0)$. However, in the case of the 8+66 reaction, $(0, 5)$, $(0, 9)$, and $(0, 21)$ states were also considered, which were further used for interpolating results for other initial states of the reactant diatoms. Here, the negative temperature dependence of the thermal rate constant is reported for the first time via rigorous quantum dynamical calculations. This dependence couldn't be obtained in the quantum or other method-based calculations using PES reported before DLLJG (except Tyuterev et al. surface [157]). QSM calculations performed by Guo et al. [152, 153] did find such temperature dependence of thermal rate coefficients. However, the exchange reaction being non-statistical [160–162], the QSM study lacks the desired rigor.

The same DLLJG surface was used in investigating the state-to-state dynamics of the 8+66 variant of isotopic exchange reaction in the collision energy range up to 0.35 eV by Xie et al. [163] in 2015. The time-dependent quantum wave packet method was used to propagate the wave function by taking the hypothetical $j = 0$ initial state of the reactant diatom. State-to-state ICSs and DCSs are reported and compared with experimental findings. DCSs were reported to be forward-biased in the collision energy range under consideration. A detailed discussion of DCSs is also reported in the study. Subsequently, Guo et al. [164] also reported a quantum dynamical study of 8+66 and 6+88 reactions on the DLLJG potential surface using the time-dependent methodology. The initial states reported in the study are $j = 0, 1, 3, 5, 9$ and 21. For $J > 30$, probabilities were calculated using J-shifting rule. The ICS (up to 0.2 eV collision energy), initial state-selected rate constants, and thermal averaged rate constants were reported for 8+66 and 6+88 reactions. The thermal averaged rate constants show a negative temperature dependence, but the magnitude is much smaller than the experimental data reported in Rajagopala et al. [165]. A comparison of results from the DLLJG surface is made with the results from the mSSB surface, and the thermal rate constants are in contrast to each other, where the mSSB thermal rate constant shows the positive temperature dependence. The two cross sections are also in contrast to each other. The results used by Guo et al. in Ref. [164] for comparison was published at the same time (in 2015) by Rajagopala et al. [165]. Rajagopala et al. [165] reported a quantum dynamical investigation for the 8+66 and 6+88 isotopic exchange reactions using the DLLJG surface [153] for the initial diatom states of $j = 1, 3, 5, 7, 9, 11,$ and 13. The initial state-specific ICSs were reported for the two isotopic variants of the reactions. These ICSs are further used to calculate the state-selected and thermal averaged rate constants for the reactions. Calculated thermal averaged rate constants correctly exhibit the negative temperature dependence. Still, the magnitude was 2 to 4 times shorter than the experimental values, and cross sections are rich in resonance structure in the collision energy below 0.1 eV.

The first match between the experimental and theoretical investigations was reported in 2017 by Guillon et al. [155]. The thermal rate constant results for 8+66 isotopic exchange reaction were reported by them, and the PES used in the work was constructed by Tyuterev et al. [157]. The dynamical study used the TI method, and ICSs, thermal rate constants, etc., observables were calculated and compared with the results of Rajagopala et al. [165].

Coming to the statistical and non-statistical nature of the reaction, the lifetime of the collision complex plays a crucial role. If the lifetime of collision complexes is significantly long, the energy distribution among the different degrees of freedom takes place. In this case, products go to all the possible channels equally, losing any memory of the initial collision conditions. However, when the lifetime of the complex is low, the complex breaks down before the energy distribution and product states show the effect of initial collision conditions. Among many other observables, the signature of statistical and non-statistical nature can be observed as symmetrical and non-symmetrical DCSs, respectively. In the case of the 8+66 variant of the isotopic exchange reaction, the non-statistical nature of the reaction was reported. A crossed-molecular beam experiment at 0.31 eV (7.3 kcal/mol) reported the forward scattered DCS, confirming the non-statistical character of the reaction [160]. In 2009, Zhang et al. [161] reported the non-statistical nature of isotopic exchange reaction. The exact quantum mechanical was used to study the 6+66, 8+66, and 6+88 variants of the reaction on the Babikov et al. [148] PES in the collision energy range of 0.0 - 0.1 eV. A predominant forward scattering was observed for all the variants of the exchange reaction, and results for the 8+66 isotopic variant are consistent with the experimental DCSs reported by Wyngarden et al. [160]. In 2014, the forward scattered DCS was reported in yet another crossed-molecular beam experiment by Wyngarden et al. [162] at 0.2472 eV (5.7 kcal/mol). The experiment confirmed the non-statistical nature of the 8+66 exchange reaction.

The current investigation presented in this chapter uses the TDQM method to investigate the same 8+66 isotopic exchange reaction using the Varga et al. [158] PES. Different reaction attributes, such as reaction probability and ICSs, are compared with the previous results. A short discussion of the state-to-state level observables is also presented.

5.4 Features of O_3 PES

Ozone has two types of minima: one is called ring-structure with the same three bond distances and D_{3h} symmetry, and the second is of C_{2v} symmetry. In the Varga et al. PES [158] the global minima (C_{2v} symmetry) is found at 1.16 eV energy below the O + O_2 asymptote. The bond distances are 1.2683 Å, 1.2683 Å, and 2.1894 Å, and the angle between the same bond distances is 119.4°.

5.5 Features of 8+66 isotopic exchange reaction

The reaction under study is

$$^{18}\mathrm{O} + {}^{16}\mathrm{O}^{16}\mathrm{O}(v = 0, j = 1) \longrightarrow {}^{18}\mathrm{O}^{16}\mathrm{O} + {}^{16}\mathrm{O}.$$

The energies of the initial rovibrational level ($v = 0, j = 1$) of reagent diatom ($^{32}O_2$) and ZPE ($v = 0, j = 0$) of product diatom ($^{34}O_2$) from their corresponding minima are 0.097350 eV and 0.094237 eV, respectively. When energies of the initial rovibrational state of reactant and ZPE of product diatoms are not considered, the reaction is thermoneutral, but the reaction becomes exothermic by ~0.003113 eV otherwise. A few other important features of the reaction are as follows.

1. Reaction is non-statistical in nature.
2. The rate coefficients show the negative temperature dependence.

The minimum energy path, which shows the energy of the reaction, is depicted in Fig. 5.1. Varga et al. [158] PES is used in the calculation of MEP shown in the figure.

Fig. 5.1 also shows the absence of the much controversial submerged reef structure, a prominent feature of almost every PES constructed before the Dawes surface [153]. It can be seen that the reef structure is not present on both the reactant and product asymptotes, where a shoulder appears instead. A deep well of energy ~ 1.158 eV from the reactant asymptote is also shown in the MEP of the reaction.

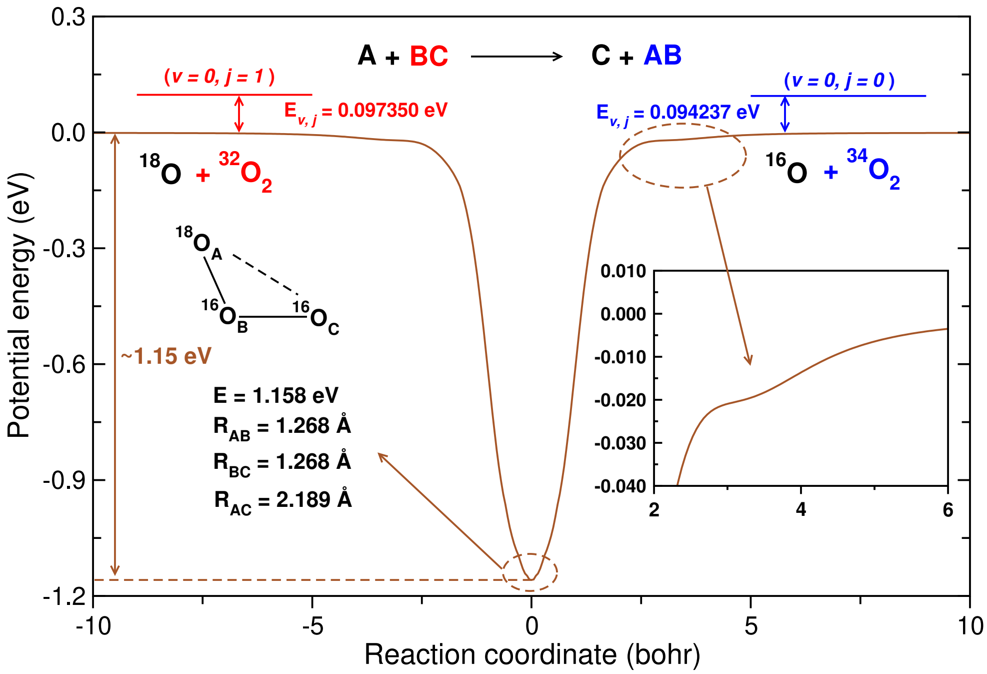
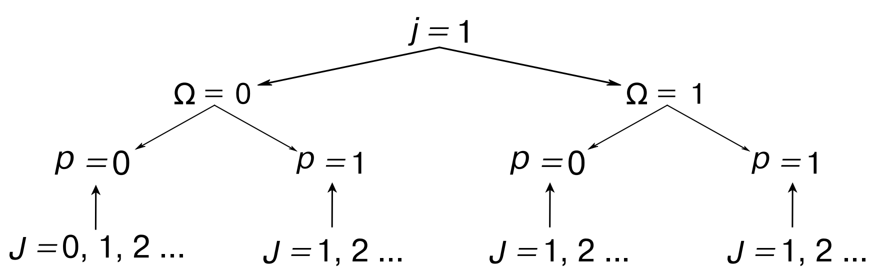


Figure 5.1: Minimum energy path calculated at the angle ($\angle O_A$-O_BO_C) of global minima, 119.4°. As shown in the inset, no reef structure is present on the reactant or product asymptote.

5.6 Computational details

The TDQM method, which is discussed in section 2.4.1 of Chapter 2, is used in the current work. Since the calculations are performed for ($v = 0, j = 1$) the initial state of the reactant diatom, a total of four different sets of calculations are carried out for (v, j, Ω, p) values of (0, 1, 0, 0), (0, 1, 0, 1), (0, 1, 1, 0), and (0, 1, 1, 1) for different values of total angular momentum (J). A cartoon diagram of four sets of calculations for various J values is shown below.



The dynamical investigations are carried out in the collision energy range of 0.0 - 0.1 eV, and the maximum value of J ($J_{\max}$) is 80 for all four sets. However, calculations are converged in the collision energy range of 0.004 - 0.1 eV only because the combination of E_{trans} and α values gives a constant amplitude WP only in this collision range (cf. section 4.2.3.1 of Chapter 4). The input parameters used in the TDQM method are converged by running several calculations for the initial reactant diatomic state of (v = 0, j = 1, Ω = 0, p = 0) at J = 0. The converged parameters are as given in Table 5.1.

Table 5.1: Converged QM input parameters.

Parameters	**Converged value**
Total grid points along the product Jacobi, $N_{R_c}/N_{r_c}/N_{\gamma_c}$	383/ 255/ 160
Minimum and maximum grid length along R_c (R_c^{min}/R_c^{max}) (a$_0$)	0.1/27.0
Minimum and maximum grid length along r_c (r_c^{min}/r_c^{max}) (a$_0$)	0.1/19.0
Location of the dividing surface in the product channel, $R_d^{'}$ (a$_0$)	19.0
Cut-off potential (V_{cut}) (Hartree)	0.22
Starting point of absorption along R_c and r_c (R_c^{abs}/r_c^{abs} (a$_0$)	23.5/16.0
Absorption strength along R_c and r_c (C_{abs}/c_{abs}	0.3/0.3
Center of the initial WP in reagent Jacobi coordinates (R_0) (a$_0$)	12.0
Initial translational energy (E_{trans}) (eV)	0.26
Width of the initial WP (α)	17.5
Smoothness of the initial WP (β)	0.80
Number of vibrational levels of the product ($^{34}O_2$) diatom	1
Number of rotational levels of the product ($^{34}O_2$) diatom	27
Number of Chebyshev iterations (n_{steps})	80000
Total propagation time (fs)	6941.88
Range of total angular momentum (J) in dynamics	J = 0 - 80

These converged parameters are used for all four sets at all the J values. However, as the J value increases, a few modifications are made in the parameters, such as the number of iteration steps (n_{step}), center of the initial wave packet (R_0), etc. The modifications are required because increasing J changes the effective potential of the reaction system, as shown in Fig. 5.2. It can be seen from the figure that the increment in the effective potential pushes the asymptote further in the higher grid value of the distance between the reactant atom and diatom, increasing the value of the center of the initial wave packet. This change in R_0 is to be followed by other parameters, such as the point of initiation of the absorption region in r grid (r_c^{abs}), the maximum value of r grid (r_c^{max}).

Increasing the value of J also affects the number of Chebyshev iteration steps (n_{step}). This is because prolonged wave packet (WP)propagation is required to converge the initial parameters at a lower collision energy range. In contrast, convergence at a higher collision energy range can be obtained by short-time propagation. Since increasing J shifts the energy threshold of the reaction due to the centrifugal barrier, probabilities are only possible

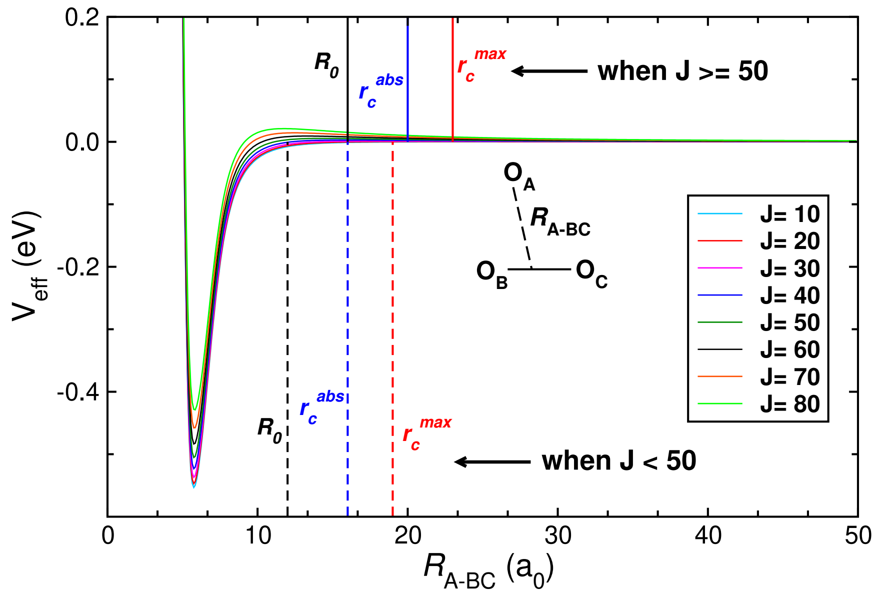


Figure 5.2: Effective potential energy (V_{eff}) for different J values. The locations of R_0, r_c^{abs}, and r_c^{max} parameters for J < 0 and J ≥ 50 are shown as dashed and solid horizontal lines, respectively.

at the higher collision energies, requiring a short propagation time to converge. This is why the n_{step} values can be decreased for higher J calculations. The modifications in J and other parameters are listed in Table 5.2.

Table 5.2: Change in the converged parameters with increasing J.

Parameters	Old values	New values
Center of the initial WP in reagent Jacobi coordinates (R_0) (a_0)	12.0	16.0
Starting point of absorption along r_c [r_c^{abs}] (a_0)	16.0	20.0
Maximum grid length along r_c (r_c^{max}) (a_0)	19.0	23.0
Number of Chebyshev iterations (n_{steps})	80000	75000
Absorption strength along R_c and r_c (C_{abs}/c_{abs})	0.3/ 0.3	50.0/ 50.0
Total propagation time (fs)	6941.88	6508.01

Also, the number of processors of the number of the product helicity quantum levels considered in the calculations is not equal to min(J_{max}, njab) but a truncated value of Ω'_{max} ($N_{\Omega'}$) is used. To get the truncated values for the number of Ω' at different J: first $N_{\Omega'}$ values for different J, such as 10, 20, 30, 40, 50, 60, 70, and 80 were obtained, and then the $N_{\Omega'}$ values for the rest of the J are extrapolated (XMGRACE plotting package was used in the current work). The values of truncated $N_{\Omega'}$ at different J are as given in Table 5.3.

Table 5.3: Truncated numbers of Ω'.

Total angular momentum (J)	$N_{\Omega'}$
21/ 22/ 23/ 24/ 25/ 26/ 27/ 28/ 29/ 30	19/ 20/ 20/ 21/ 22/ 22/ 23/ 24/ 25/ 25
31/ 32/ 33/ 34/ 35/ 36/ 37/ 38/ 39/ 40	26/ 27/ 27/ 28/ 28/ 29/ 30/ 30/ 31/ 31
41/ 42/ 43/ 44/ 45/ 46/ 47/ 48/ 49/ 50	32/ 33/ 34/ 34/ 35/ 36/ 36/ 37/ 38/ 38
51/ 52/ 53/ 54/ 55/ 56/ 57/ 58/ 59/ 60	39/ 40/ 41/ 42/ 43/ 44/ 45/ 46/ 47/ 47
61/ 62/ 63/ 64/ 65/ 66/ 67/ 68/ 69/ 70	48/ 48/ 48/ 49/ 49/ 49/ 49/ 49/ 49/ 48
71/ 72/ 73/ 74/ 75/ 76/ 77/ 78/ 79/ 80	48/ 48/ 48/ 48/ 47/ 47/ 47/ 47/ 47/ 46

A few parameters related to the reaction system are given in Table 5.4.

Table 5.4: Parameters related to the O_3 reaction system.

Parameters	Values
Masses of the oxygen isotopes (m_A/ m_B/ m_C) (amu)	17.99916/ 15.99943/ 15.99943
Rovibrational energy of $^{32}O_2$ ($E_{v=0,j=1}$) (eV)	0.097350
Rovibrational energy of $^{34}O_2$ ($E_{v=0,j=1}$) (eV)	0.094237

Convergence of $J_{\mathbf{max}}$ values

As mentioned, the $J_{\max}$ value of 80 is taken for all four initial reactant diatomic states corresponding to four different sets of calculations, which can be seen from Fig. 5.3. This figure shows the total reaction probabilities for different J collisions in the collision energy range of 0.0 eV - 0.1 eV. It is clear from the figure that the $J \geq 80$ collisions have no probability below 0.1 eV, meaning that calculations up to J = 80 are sufficient for studying the dynamics up to the collision energy of 0.1 eV.

Time-dependent quantum calculations are expensive!!
In this work, a total of 317 calculations, excluding the calculations done for converging the input parameters, were performed for different combinations of (J, v, j, Ω, p), and each calculation, running on a parallel computing algorithm, took an average of 26 days. This translates to an enormous ~23 years of computing time used for all the calculations.

5.7 Results and Discussion

It is already clear from the above discussions that the best-known surfaces for the O_3 system are constructed by Dawes et al. [153], and Tyuterev et al. [157]. These two surfaces are used

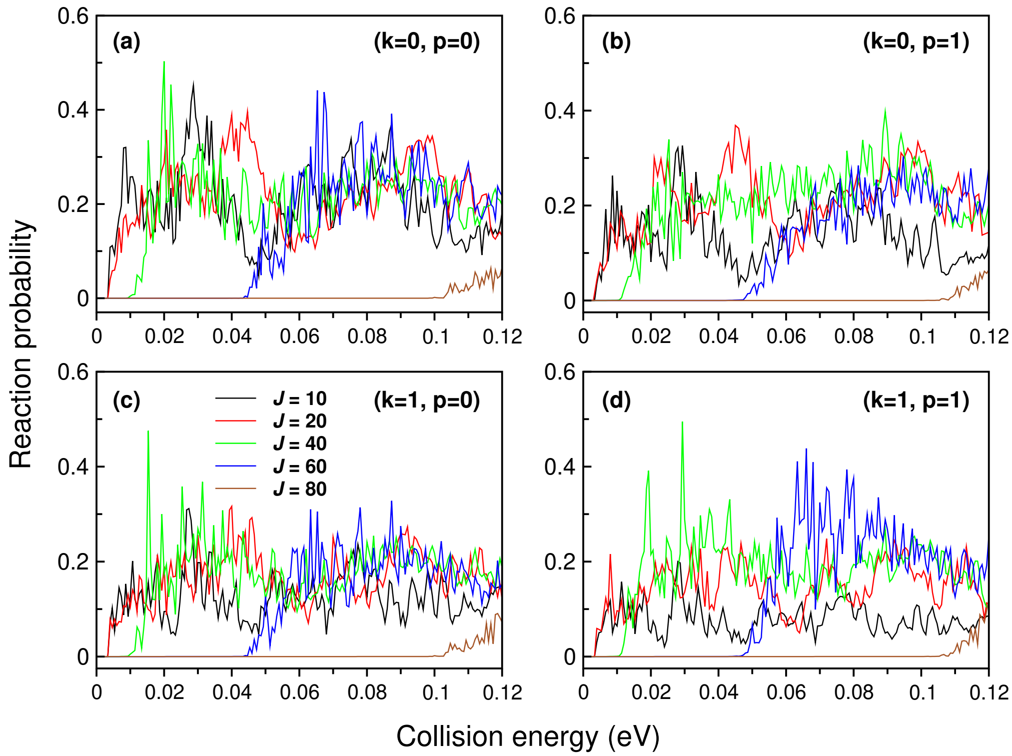


Figure 5.3: Total reaction probabilities for collision with different J. Panel a, b, c, and d show the collision for different initial states of reactant diatom as specified in the corresponding panels.

in investigating the O + O_2 isotopic exchange reaction from various different perspectives. In the current investigations, different reaction attributes have been calculated and presented below.

5.7.1 Reaction probability of J = 0

The total reaction probabilities shown in Fig. 5.3 for ($J = 0, v = 0, j = 1, k = 0/1, p = 0, 1$) initial reactant diatomic states are summed up using Eqs. 2.49 - 2.51 to get the total reaction probabilities for ($J = 0, v = 0, j = 1$) initial reactant diatomic state as shown in Fig. 5.4.

The probability values start from the collision energy of 0.004 eV because, as already discussed, the calculations are not converged at energies $\lesssim 0.004$ eV. The same total probabilities at different J collisions are shown in Fig. 5.5.

The following features are observed from the two figures.

1. The probability plot for $J = 0$ has no visible trend with respect to the collision energy. However, with increasing J collisions, a clear trend of decreasing probability emerges with increasing collision energy, which is the typical trend for exoergic barrierless reac-

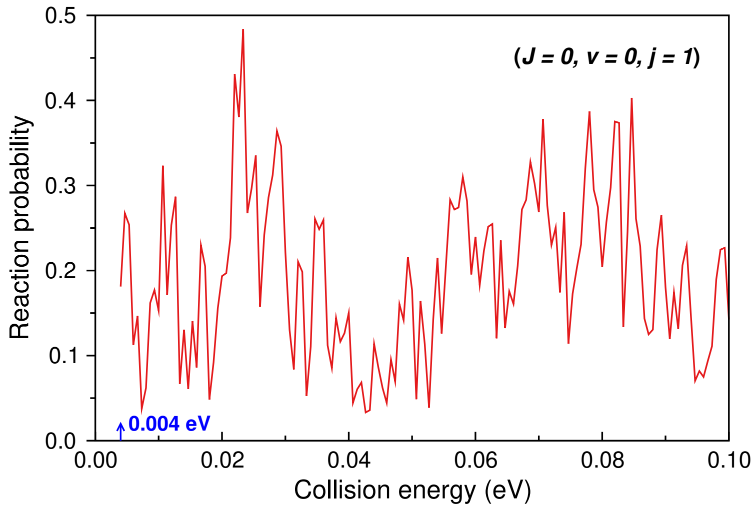


Figure 5.4: Reaction probability for total angular momentum $J = 0$. The probabilities start from 0.004 eV because the dynamics calculations are converged in the collision energy range of 0.004 - 0.1 eV.

tions.

2. There is no threshold for $J = 0$ collisions. However, as the J values increase, the collision threshold increases. This is because the threshold appearing in an exoergic barrierless reaction is totally because of the centrifugal barrier, which, for $J = 0$ collisions, has zero value and shows an increase for collision with increasing J.

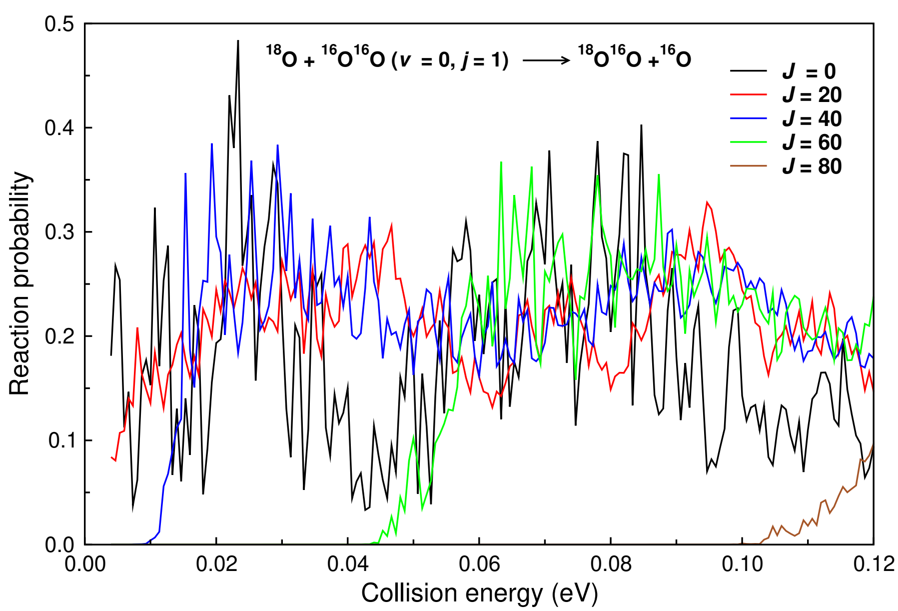


Figure 5.5: Reaction probability for total angular momentum $J > 0$.

3. Rich structures (fluctuations) called resonances are seen for all the J values. These reso-

nances, which appear in the full range of collision energies considered, are the signature of metastable complexes forming because of the deep well present in the potential surface.

5.7.2 Integral cross section (ICS)

The total ICS for the reaction is shown in Fig. 5.6. The ICSs decrease with collision energies, a typical feature of exoergic barrierless reactions. The cross sections obtained in our calculations are compared with the results of Guillon et al. [155], and Rajagopala et al. [165], and it can be seen that the three cross sections are close to each other. All three cross sections show the same resonance structures as in reaction probabilities. However, these features are more prominent in the results of current calculations and the work of Rajagopala et al. [165]. These resonances are quite interesting since such features usually get destroyed in highly averaged observables like ICS.

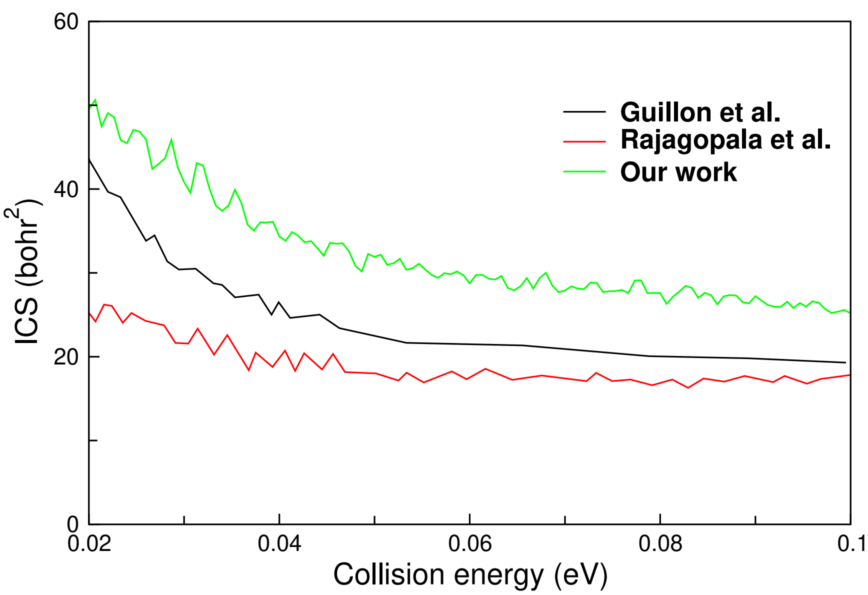


Figure 5.6: ICSs as a function of collision energy. The results of the current work are compared with the results from the earlier investigations of Guillon et al. [155], and Rajagopala et al. [165].

5.7.3 State-to-state ICS

The ICSs from the current work are also resolved further at state-to-state levels. However, before discussing state-to-state ICSs, the energetics of the rotational and vibrational states of product diatom is discussed. The energies of different vibrational levels of the product diatom $^{34}O_2$ are as follows.

It can be seen from the table that in the collision energy range considered, which is 0.0 -

Table 5.5: Vibrational level energies for the product diatom ($^{34}O_2$).

v' level	$E_{v'}$ (eV)
0	0.94237E-01
1	0.28287
2	0.46851
3	0.65101

0.1 eV, only the first vibrational level of the product diatom is open, which can also be seen from the converged value of the total product vibrational level presented in Table 5.1. In this collision energy range, the total energy for the system with reactant diatom in (v = 0, j = 1) state ($E_{v=0,j=1}$ = 0.097350 eV) is in the range of 0.097350 - 0.197350 eV.

To show the possible number of rotational levels open in the energy range of calculations, energies of different rotational levels at the first vibrational level of the product diatom (v' = 0, j') are as follows.

Table 5.6: Rotational level energies for the product diatom ($^{34}O_2$) at first vibrational level (v' = 0, j').

(v'=0, j') levels	$E_{v'=0,j'}$ (eV)
(0, 0)	0.94237E-01
(0, 1)	0.94574E-01
(0, 2)	0.95247E-01
(0, 3)	0.96257E-01
(0, 4)	0.97603E-01
(0, 5)	0.99286E-01
(0, 6)	0.10130
(0, 7)	0.10366
(0, 8)	0.10635
(0, 9)	0.10938
(0, 10)	0.11274
(0, 11)	0.11644
(0, 12)	0.12048
(0, 13)	0.12485
(0, 14)	0.12956
(0, 15)	0.13460
(0, 16)	0.13997
(0, 17)	0.14569

(0, 18)	0.15173
(0, 19)	0.15811
(0, 20)	0.16483
(0, 21)	0.17188
(0, 22)	0.17926
(0, 23)	0.18697
(0, 24)	0.19502
(0, 25)	0.20341
(0, 26)	0.21212
(0, 27)	0.22117
(0, 28)	0.23054
(0, 29)	0.24025
(0, 30)	0.25029
(0, 31)	0.26067
(0, 32)	0.27137
(0, 33)	0.28240
(0, 34)	0.29376
(0, 35)	0.30545
(0, 36)	0.31747
(0, 37)	0.32981
(0, 38)	0.34249
(0, 39)	0.35549
(0, 40)	0.36882

According to the energies shown in Table 5.6, 25 vibrational levels are open in the collision energy range (total energy range) of 0.0 - 0.1 eV (0.097350 - 0.197350 eV). The contour diagram for product rotational level resolved ICSs at product vibrational quantum number $(v') = 0$ is shown in Fig. 5.7 as a function of collision energy and product rotational quantum number (j').

The white curve superimposed on the ICSs shows the j' quantum level contributing the maximum to the ICS (j'_{max}) at a specific collision energy. It can be seen that j'_{max} value increases as the collision energy increases. In other words, the product rotational level resolved ICS becomes rotationally hot as the collision energy increases. The same trend is seen distinctly in Fig. 5.8 when the product rotational level resolved ICSs are plotted with respect to j' for a few selected collision energies.

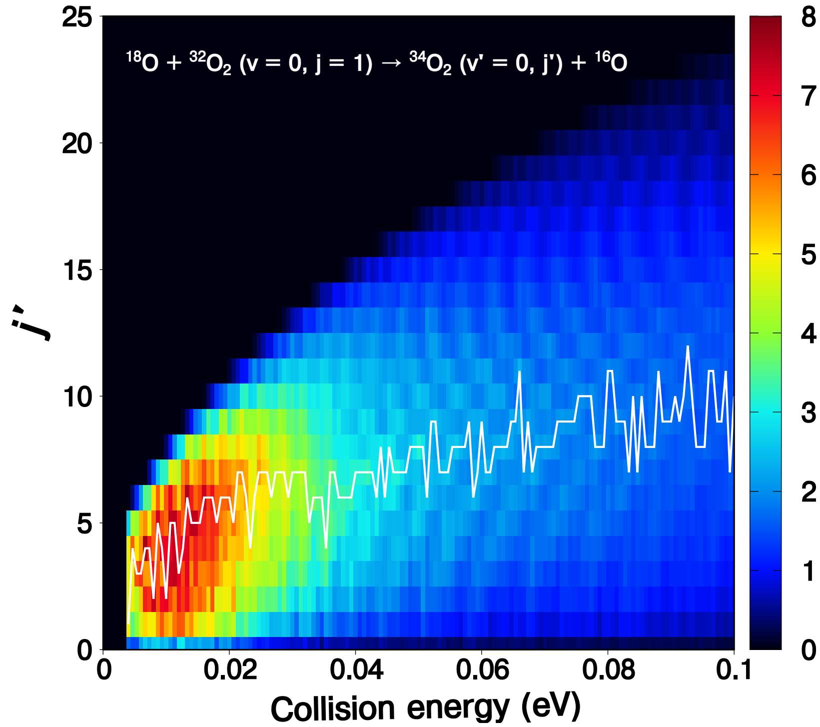


Figure 5.7: Rotational resolved state-to-state ICSs with respect to the collision energy and product rotational quantum number (j').

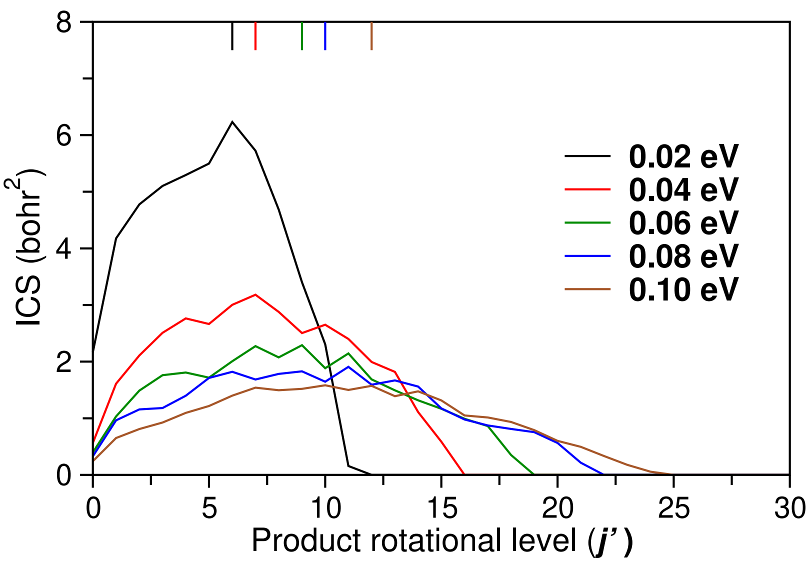


Figure 5.8: Product rotational level resolved state-to-state ICSs with respect to the product rotational levels (j') at different collision energies.

5.8 Chapter summary

The chapter presents the quantum dynamical investigation for isotopic exchange reaction: $^{16}O + ^{32}O_2$ ($v = 0, j = 1$) $\rightarrow$ $^{34}O_2 + ^{18}O$. This particular isotopic combination is popularly called the 8+66 variant of the reaction. The PES used for the purpose was constructed by

Varga et al. [158] in 2017. The PES is free from the artificial submerged reef structure, which plagued many earlier surfaces. The work uses the TDQM approach implemented in the DIFFREALWAVE code [21]. Total reaction probability for different J collisions are calculated. The probabilities show the resonance features for all the J collisions when plotted with respect to the collision energy. These features are because of the deep potential well present in the reaction path. Interestingly, resonance features are also present in the ICS, although the ICS is an averaged quantity. When the ICSs are resolved rotationally, the distribution turns rotationally hot as the collision energy increases.

Overall, these resonance features are interesting features and warrant a more detailed investigation to uncover their genesis. Since the results of the current work do not converge in low collision energies, the ICSs are shown only for the collision energies > 0.02 eV. This limitation means that the rate constants can not be reported without errors. However, these limitations will not affect the results presented in the present chapter. Rectifying these limitations is the purpose of future work, which will focus more on the mechanistic aspect of the reaction.

Chapter 6

Summary and Outlook

The work reported in this thesis covers the concepts of construction of PES and dynamics of chemical reactions involving atom-diatom collisions. The construction of PES is important because the fitted surface efficiently calculates the potential energies at different nuclear configurations spanned in reaction space, eliminating the need to perform computationally expensive ab initio calculations. The accuracy of these PESs is paramount since electronic potential controls the motion of the colliding partners, affecting the overall accuracy of the reaction dynamics studies and reaction attributes. The reaction dynamics studies are important to understand the reactions at the atomic level and calculate different reaction attributes, such as integral cross section and rate constants. These reaction attributes further provide the atomic-level understanding of a reaction. This thesis reports the reaction dynamics studies of three different reactions. Two of them are the collisions of Li-bearing molecular ion LiH^+ with H and He atoms, respectively, and the third is the isotopic exchange reaction in ^{18}O + $^{32}O_2$ collision. In addition, a detailed discussion on the construction of a ground electronic state of the $HeLiH^+$ is presented.

The first reaction containing LiH^+ molecular ion is a destruction channel of LiH^+ by collision with H atom: H + $LiH^+ \rightarrow H_2 + Li^+$. This is a barrierless exoergic reaction with exoergicity of $\sim$ 4.60 eV. The study explores two themes: first is the comparison of QM and QCT results, and second is the investigation of the isotopic effect on different reaction attributes, such as total reaction probability, ICS, rate constants, etc. Four isotopic variants of the reaction, H / D + LiH^+/LiD^+ $\rightarrow$ H_2 / HD / D_2 + Li^+, are used for this purpose. Out of these four reactions, H / D + LiH^+ $\rightarrow$ H_2 / HD + Li^+, which are studied using both the QM and QCT theoretical approaches, are used in QM and QCT comparison, and all the four reactions are used for investigating the isotopic effect. Only the QCT method is used for studying H / D + LiD^+ $\rightarrow$ HD / D_2 reactions. The QCT method used in the investigation uses both the GB and HB approaches. The QM results compared with QCT results are taken from our previous work [65, 66]. In all four reactions, the initial reactant diatomic state is (v = 0, j = 0). The QM and QCT results converge for all the scalar and vector reaction attributes. The QM and QCT total ICSs follow each other at all the collision energies, and no resonance features are observed in either of the approaches. The same level of convergence between the two methods is also found even at the state-to-state level for vibration-resolved and rotational-resolved ICSs. In addition, DCSs from the two methods closely follow each other except at the poles, where slight quantitative differences are found. This convergence in the QM and QCT results is crucial because, owing to the excellent agreement between the two methods, the computationally cheaper QCT method can replace computationally expensive QM calculations. This replacement can save hours of computational time in cases where a large number of QM calculations are a must for meaningful conclusions.

As for the isotopic effect, a significant effect of isotopes on reaction attributes is observed. These observations can be correlated with the effect of mass on the PES. This PES is called the mass-weighted PES, and the effect of mass is seen as the skewed reactant and product channels on this type of potential surface. In the case of the reactions having the same skew angles, reaction attributes follow the same qualitative and quantitative patterns. The QCT DCSs for the four isotopic variants of reactions are quantitatively different at the poles, and this difference, again, correlates with the order of skew angles. As for the mechanism followed by the reactions, all the isotopic variants follow the indirect mechanism with a higher average collision time in the lower collision energy range, whereas the direct mechanism with trajectories with shorter collision time becomes prominent with increasing collision energy. The presence of the roaming mechanism is also reported for the reaction R1.

As for the construction of PES, a potential fit for the ground electronic state of the $HeLiH^+$ system is reported. The total potential of the system is represented using the MBE approach. The fitting uses an ML-based neural network algorithm and conventional fitting, which uses the AP functional form. A grid was prepared using the two interatomic distances and the angle between them. These grid points were used to accurately represent the complete reaction space in the input data. The density of these grid points was higher in the interaction region than in the asymptote, where a sparse grid was used. An iterative process of fitting and adding points in the region of high error was used to finally get the best fit with a total RMSE of 14.21 cm^{-1}. The energy space of the data points in the best-fit surface was < 10 eV. Before this fit, the data points in the energy region higher than > 10 eV were removed because such energetically higher spaces are unlikely to be accessed in the energy space investigated. In the final surface, the potential in the short range is calculated using the fitted functional form, and the potential in the long-range is calculated using the analytical expression. The potentials at the short and long ranges are connected using the smooth switching function.

Once the PES is constructed, the quantum and classical dynamics of the endothermic He + $LiH^+ \rightarrow LiHe^+$ + H reaction was investigated for the initial reactant diatom state of ($v = 0$, $j = 0$). The dynamics of the CID channel for the He + LiH^+ ($v = 0, j = 0$) collisions, He +

$LiH^+ \rightarrow Li^+ + H + He$, is also reported, but it is only studied using the classical method. The time-dependent wave packet approach was used as a quantum method, whereas Hamilton's equation-based QCT approach was used as a classical method. The CID channel is found to be of importance when studying the dynamics of the reaction because of the low binding energy of LiH^+ which opens the dissociation at a low collision energy of ~ 0.10725 eV. The effect of this is observed in all the reaction attributes, such as reaction probability, cross section, rate constants. The cross sections of the reaction increase with the collision energy at first, which is the same as the usual trend found in endothermic reactions, but later, it drops rapidly to almost zero as the CID pathway becomes accessible around ~ 0.107 eV. However, the ICS of the CID reaction increases with increasing collision energy after the appearance of the CID channel.

The isotopic exchange reaction $^{18}O + {}^{32}O_2 \rightarrow {}^{34}O_2 + {}^{16}O$ is investigated for the ($v = 0, j = 1$) initial state of reactant diatom in the collision energy range of 0.0 eV - 0.1 eV. Since this reaction competes with the ozone formation reaction in the atmosphere, studying this reaction is important to understand the MIF observed in the ozone, which enriches the heavy ozone. Since the beginning, scientific investigations on this reaction have focused on resolving the conflict between the contradicting experimental and theoretical rate constants, and it was only in 2017 when Guillon et al. [155] reported the matching experimental and theoretical rate constants. Despite these many studies, a detailed mechanistic investigation of the reaction is still not reported. The work reported in this thesis explores the reaction from a mechanistic point of view. Since the reaction possesses a major bottleneck in the computational cost of dynamics calculations, quantum dynamics is performed only for the 8+66 isotopic variant of the reaction, and corresponding results are reported. The calculations are performed using the DIFFREALWAVE code, which uses the time-dependent wave packet method and includes the Coriolis coupling. The ICSs obtained from the current investigation are compared with the previous results, and all the results were found to be close to each other. Further, product rotational level resolved ICSs shows that the rotational quantum number with the maximum contribution at a fixed collision energy increases with collision energy. Interesting resonance features resulting from metastable states forming at the deep well present in the potential surface are observed in the total reaction probability and ICSs.

The work reported in this thesis is just a stepping stone and paves the way for further studies. The roaming trajectories observed in the $H + LiH^+ \rightarrow H_2 + Li^+$ reaction can be further explored to shed light on this relatively recent mechanism of product formation. The fitted $HeLiH^+$ potential surface also facilitates the reaction involving the collision between H and $LiHe^+$: H + $LiHe^+$ $\rightarrow$ LiH^+ + H. This reaction is a possible destruction and formation channel of $LiHe^+$ and LiH^+, respectively. This reaction, along with the LiH^+ destruction reported in the current thesis, can further be used to calculate the thermal rate constants of astrochemical importance. Regarding the isotopic exchange reaction of the 8+66 isotopic variant, interesting resonance structures appearing in the reaction probability and ICSs require a detailed analysis. Moreover, the quantum dynamics calculations reported in this thesis use only a single electronic state. This approach is called the adiabatic approximation, and its accuracy depends on the interactions between the electronic states. The approximated results are accurate if the electronic states are energetically far from each other. However, when the electronic states are energetically close, adiabatic calculations lack accuracy, and the inclusion of non-adiabatic effects is a must. A non-adiabatic approach to the dynamics will also be used in the future.

Appendix A

Time dependent and time independent Schrödinger equations

The time-dependent form of the Schrödinger equation (SE) is written as

$$i\hbar\frac{\partial\Psi(r_e, R_N, t)}{\partial t} = \hat{H}\Psi(r_e, R_N, t), \tag{A.1}$$

where r_e and R_N are the electronic and nuclear position coordinates, respectively, $\hat{H}$ is Hamiltonian operator, and $\psi(r_e, R_N, t)$ is the wave function at time t and given position coordinates. The Eq. A.1 is a first-order initial value problem, meaning that at a given initial condition, the equation has a unique solution.

Eq. A.1 can have a variety of solutions. Let's consider that a specific set of solutions exists where the time and position variables can be separated as

$$\Psi(r_e, R_N, t) = F(t)\psi(r_e, R_N). \tag{A.2}$$

Putting Eq. A.2 in A.1 and solving it further gives the following form.

$$i\hbar\frac{\partial F(t)\psi(r_e, R_N)}{\partial t} = \hat{H}F(t)\psi(r_e, R_N)$$

$$i\hbar\psi(r_e, R_N)\frac{dF(t)}{dt} = \hat{H}F(t)\psi(r_e, R_N)$$

$$i\hbar\frac{dF(t)}{F(t)dt} = \frac{\hat{H}\psi(r_e, R_N)}{\psi(r_e, R_N)} \tag{A.3}$$

It is clear from Eq. A.3 that the LHS and RHS sides are functions of different variables. These two sides are equal, meaning that these sides should be equal to a constant term, such as E. Later, it was realized that this constant E is equivalent to the energy.

$$i\hbar\frac{dF(t)}{F(t)dt} = \frac{\hat{H}\psi(r_e, R_N)}{\psi(r_e, R_N)} = E$$

Writing the above equation separately.

$$i\hbar\frac{dF(t)}{dt} = EF(t) \tag{A.4}$$

$$\hat{H}\psi_E(r_e, R_N) = E\psi_E(r_e, R_N) \tag{A.5}$$

Solving Eq. A.4 further gives us

$$F(t) = F(t_0)\exp\left(\frac{-iEt}{\hbar}\right), \tag{A.6}$$

where $F(t_0)$ is the time-dependent part of the wave function at time t_0. The Eq. A.5 is called the time-independent SE. The overall solution of the Eq. A.1 is written as

$$\psi(r_e, R_N, t) = F(t_0)\psi_E(r_e, R_N)\exp\left(\frac{-iEt}{\hbar}\right). \tag{A.7}$$

It is clear from Eq. A.7 that the time appears as a phase factor in the solution, which means that the probability of the state (at specific position coordinates) is a constant of time. These types of states are called stationary states. Now, the initial state of the overall wave function can be obtained by putting $t = 0$ in Eq. A.7.

$$\psi(r_e, R_N, t = 0) = F(t_0)\psi_E(r_e, R_N) \tag{A.8}$$

It can be seen from Eq. A.8 that for Eq. A.1 to have the stationary states as the solutions, the initial states should be an energy eigenstate $\psi_E(r_e, R_N)$. However, it is to be noted that not all the solutions of Eq. A.1 are of the above type, and for the other types of solutions, the probability is not a constant of time. In such cases, a more involved process is required to solve the time evolution of the system.

Appendix B

Matrix form of time-independent Schrödinger equation

The time-independent Schrödinger equation (TISE) for a molecular system is as follows.

$$\hat{H}\psi(r_e, R_N) = E\psi(r_e, R_N) \tag{B.1}$$

The total wave function $\psi(r_e, R_N)$ is written as a linear combination of electronic wave functions.

$$\psi(r_e, R_N) = \sum_i \Phi_i(r_e; R_N)\mathcal{X}_i(R_N)$$

Putting this expression is Eq. B.1 gives us the following equation.

$$\sum_i (\hat{T}_\mathrm{N} + \hat{H}_\mathrm{e} + \hat{V}_\mathrm{NN})\Phi_i(r_e; R_N)\mathcal{X}_i(R_N) = \sum_i E\Phi_i(r_e; R_N)\mathcal{X}_i(R_N)$$

Now, to solve this equation for the j^{th} electronic state, left multiply the equation with $\Phi_j^*(r_e; R_N)$ and integrate in electronic space as follows.

$$\sum_i \langle \Phi_j^*(r_e; R_N)|(\hat{T}_\mathrm{N} + \hat{H}_\mathrm{e} + \hat{V}_\mathrm{NN})\mathcal{X}_i(R_N)|\Phi_i(r_e; R_N)\rangle = \sum_i E\, \langle \Phi_j^*(r_e; R_N)|\Phi_i(r_e; R_N)\rangle\, \mathcal{X}_i(R_N)$$

$$\sum_i \langle \Phi_j|(\hat{H}_\mathrm{e} + \hat{V}_\mathrm{NN})|\Phi_i(r_e; R_N)\rangle\, \mathcal{X}_i(R_N) + \sum_i \langle \Phi_j^*(r_e; R_N)|\hat{T}_\mathrm{N}\mathcal{X}_i(R_N)|\Phi_i(r_e; R_N)\rangle = E\mathcal{X}_j(R)$$

$$V_j\mathcal{X}_j(R_N) + \sum_i \langle \Phi_j^*(r_e; R_N)|\hat{T}_\mathrm{N}\mathcal{X}_i(R_N)|\Phi_i(r_e; R_N)\rangle = E\mathcal{X}_j(R_N) \tag{B.2}$$

Solving the term containing $\hat{H}_\mathrm{N}$,

$$\sum_i \langle \Phi_j^*(r_e; R_N)|\hat{T}_\mathrm{N}\mathcal{X}_i(R_N)|\Phi_i(r_e; R_N)\rangle = -\sum_i \sum_\alpha \frac{1}{2m_\alpha} \langle \Phi_j^*(r_e; R_N)|\nabla_\alpha^2 \cdot \mathcal{X}_i(R_N)|\Phi_i(r_e; R_N)\rangle$$

Here, $\nabla_\alpha^2 = \frac{\partial^2}{\partial \mathrm{x}_\alpha^2} + \frac{\partial^2}{\partial \mathrm{y}_\alpha^2} + \frac{\partial^2}{\partial \mathrm{z}_\alpha^2}$. Operating the double derivative gives us the following form.

$$
\begin{aligned}
= -\sum_\alpha \frac{1}{2m_\alpha}\Bigg[\sum_i \Big[&\langle \Phi_j^*(r_e;R_N)|\Phi_i(r_e;R_N)\rangle\, \nabla_\alpha^2 \mathcal{X}_i(R_N) + \\
&\langle \Phi_j^*(r_e;R_N)|\nabla_\alpha^2|\Phi_i(r_e;R_N)\rangle\, \mathcal{X}_i(R_N) + \\
&2\,\langle \Phi_j^*(r_e;R_N)|\nabla_\alpha|\Phi_i(r_e;R_N)\rangle\, \nabla_\alpha \mathcal{X}_i(R_N)\Big]\Bigg]
\end{aligned}
$$

$$
\begin{aligned}
&= -\sum_\alpha \frac{1}{2m_\alpha}\Bigg[\nabla_\alpha^2 \mathcal{X}_j(R_N) + \sum_i \Big[\langle \Phi_j^*(r_e;R_N)|\nabla_\alpha^2|\Phi_i(r_e;R_N)\rangle + \\
&\quad 2\,\langle \Phi_j^*(r_e;R_N)|\nabla_\alpha|\Phi_i(r_e;R_N)\rangle\, \nabla_\alpha\Big]\mathcal{X}_i(R_N)\Bigg] \\
&= -\sum_\alpha \frac{1}{2m_\alpha}\Bigg[\nabla_\alpha^2 \mathcal{X}_j(R_N) + \sum_i \big[(G_{ji} + 2F_{ji}\nabla_\alpha)\mathcal{X}_i(R_N)\big]\Bigg] \\
&= -\sum_\alpha \frac{1}{2m_\alpha}\nabla_\alpha^2 \mathcal{X}_j(R_N) - \sum_\alpha \frac{1}{2m_\alpha}\sum_i \big[(G_{ji} + 2F_{ji}\nabla_\alpha)\mathcal{X}_i(R_N)\big] \\
&= \hat{T}_{\mathrm{N}}\mathcal{X}_j(R_N) - \sum_\alpha \frac{1}{2m_\alpha}\sum_i \big[(G_{ji} + 2F_{ji}\nabla_\alpha)\mathcal{X}_i(R_N)\big] \\
&= \hat{T}_{\mathrm{N}}\mathcal{X}_j(R_N) - \sum_i \big[\Lambda_{ji}\mathcal{X}_i(R_N)\big]
\end{aligned}
$$

Thus, the final equation can be written as follows.

$$
\hat{T}_{\mathrm{N}}\mathcal{X}_j(R_N) + V_j\mathcal{X}_j(R) - \sum_i \Lambda_{ji}\mathcal{X}_i(R_N) = 0 \tag{B.3}
$$

Appendix C

Regression

Regression is a statistical method of finding a mathematical relationship between the dependent and independent variables. In the context of potential energy surface (PES) construction, it is to find a functional relation between the molecular coordinates and corresponding potential energies, which, respectively, are independent and dependent variables. Regression has two types, linear and nonlinear, which depend on the mathematical representation of the function.

To understand it, take an example of a one-dimensional problem, which has only one independent and one dependent variable. The data points are: (x_1,y_1), (x_2,y_2), (x_3,y_3), ...(x_n,y_n) and mathematical relation between x and y is $f(x;\, a_0,\, a_1,\, ...\, a_n)$, where a_i's (i = 1, 2, ... n) are the coefficients. The form of this function $f(x;\, a_0,\, a_1,\, ...\, a_n)$ dictates if the regression is linear or nonlinear.

In **linear regression**, the function $f(x;\, a_0,\, a_1,\, ...\, a_n)$ can be written as a linear combination of the parameters (a_i's). However, the independent variable x may have a linear or nonlinear form. The mathematical expression is as follows.

$$f(x; a_0, a_1, ...a_n) = \sum_{i=0}^{n} a_i \phi_i(x) \tag{C.1}$$

Here, $\phi_i(x)$ can be linear, parabolic, or any other function.

In the same way, when the function $f(x;\, a_0,\, a_1,\, ...\, a_n)$ can not be expressed as a linear combination of the parameters (a_i's), such as Eq. C.1, it is called **nonlinear regression**. A few examples of such functional forms are the Michaelis–Menten equation in enzyme kinetics ($v = (v_{max}[c])/(k_m + [c])$, where v_{max} and k_m are the parameters) and functional form of a feed-forward neural network.

Once the choice of the function is made, the best mathematical form can be obtained by optimizing the values of parameters (a_i's) to fit the function with the data best. There are various such optimizing algorithms, and in all of these, the process of best fit is minimizing the loss function, which is a measure of the error between the actual and fitted outputs or dependent variables. There are several loss functions, such as root mean square error (RMSE), mean square error (MSE), square error, etc. Among these loss functions, one such function, RMSE, is expressed as follows.

$$\text{RMSE} = \sqrt{\sum_{i}^{n} \left(y_i^p - y_i\right)^2 / n} \tag{C.2}$$

Here, y_i's and y_i^p are the actual and predicted values of the dependent variables, respectively. The total number of data points is n.

The approach to minimize the loss function is different in linear and nonlinear regression problems. Whereas linear regression-based optimization schemes use analytical equations for minimizing the loss function, nonlinear regression-based schemes use an iterative process to reach the point of minima. More about the working process of these two types of approaches can be understood from **linear least square fit** and **gradient descent** methods.

Appendix D

Neural network: mathematical expression

Let's take a neural network composed of the input layer, output layer, and two hidden layers. Each layer in the above network has i, j, k, and l numbers of neurons, respectively. Each neuron gets its value by passing the weighted sum of the values of neurons from the previous layer and the bias associated with the neuron in question. The values of all the neurons give the final output, a mathematical expression of which is discussed below.

The value of the neuron in the first hidden layer depends on the values of neurons in the input layer. The value of the j^{th} neuron in the first hidden layer can be written as follows.

$$= f^{h1}\left(b_j^{h1} + \sum_i w_{ij}^{h1} x_i\right)$$

Here, b_j^{h1} is the bias value, w_{ij}^{h1} is the weight associated with the j^{th} neuron of hidden layer and i^{th} neuron of the input layer, x^i is the value of the i^{th} neuron in the input layer, and f^{h1} is the activation function of the first hidden layer. The output of the k^{th} neuron in the second hidden layer is as follows.

$$= f^{h2}\left(b_k^{h2} + \sum_j w_{jk}^{h2} f^{h1}\left(b_j^{h1} + \sum_i w_{ij}^{h1} x_i\right)\right)$$

Here, b_k^{h2} is the bias value of the k^{th} neuron of the second hidden layer, w_{jk}^{h2} is the weight associated with the k^{th} neuron of second hidden layer and j^{th} neuron of the first hidden layer, and f^{h2} is the activation function of the second hidden layer. In the same way, the final output of the network is written as below.

$$\text{Output} = f^{o}\left(b_l^{o} + \sum_k w_{kl}^{o} f^{h2}\left(b_k^{h2} + \sum_j w_{jk}^{h2} f^{h1}\left(b_j^{h1} + \sum_i w_{ij}^{h1} x_i\right)\right)\right)$$

Here, b_l^{o} is the bias value of the l^{th} neuron of the output layer, w_{kl}^{o} is the weight associated with the l^{th} neuron of output layer and k^{th} neuron of the second hidden layer, and f^{o} is the activation function of the output layer.

Appendix E

Wavepacket propagation in reagent Jacobi coordinates

As discussed in section 2.4.1.1 of chapter 2, the bottleneck in choosing coordinates for wave propagation is that the reactants and products are best described by two different types of coordinates: reagent Jacobi and product Jacobi, respectively.

If only the state-selected reaction observables, such as total reaction probability, total integral cross-section, etc., are required, wavepacket propagation only in the reagent Jacobi coordinates is sufficient. The procedure to perform such calculations is discussed in the work of Stephen K. Gray et al. [166] for the reactions where two reactive channels are the same ($O + H_2$) is as follows.

1. Propagate the wave packet in reagent Jacobi and analyze the products of elastic and inelastic collisions at the analysis line R_∞. This will give us the state-to-state probability for the elastic and inelastic collisions: $P^{nr}(E; v, j \rightarrow v', j')$, where nr is non-reactive.
2. Now, since the sum of probabilities of individual product channels is equal to one, the probability for reactive collision is as follows.

$$P^r(E; v, j) = 1 - P^{nr}(E; v, j)$$

   where r shows the reactive collision.

However, the probabilities or other observables at state-to-state levels can't be calculated because $P^{nr}(E; v, j \rightarrow v', j') + P^r(E; v, j \rightarrow v', j') \neq 1$, meaning that the at the state-to-state level total probabilities are not one.

Appendix F

Split Operator Method

Split operator (SO) is a method to approximate the time evolution operator ($e^{-i\hat{H}\Delta t/\hbar}$) by splitting it into kinetic ($\hat{K}$) and potential ($\hat{V}$) energy parts. The splitting means that the effective time evolution operator acting on the wave function is: $e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$ not $e^{-\hat{H}\Delta t/\hbar}$. Since $\hat{K}$ and $\hat{V}$ are non-commutative, splitting the time evolution operator as $\hat{T}$ and $\hat{V}$ introduces the error. This error is of second order and can be evaluated as below.

$$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar} + \text{Error} \tag{F.1}$$

Expanding the LHS and RHS in terms of the Tylor series.

LHS:

$$e^{-i\hat{H}\Delta t/\hbar} = 1 - \frac{i(\hat{T}+\hat{V})\Delta t}{\hbar} - \frac{(\hat{T}+\hat{V})^2\Delta t^2}{2\hbar^2} + i\frac{(\hat{T}+\hat{V})^3\Delta t^3}{6\hbar^3} + \ldots \tag{F.2}$$

RHS:

$$e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar} = \left(1 - \frac{i\hat{T}\Delta t}{\hbar} - \frac{\hat{T}^2\Delta t^2}{2\hbar^2} + \frac{i\hat{T}^3\Delta t^3}{6\hbar^3} + \ldots\right)\left(1 - \frac{i\hat{V}\Delta t}{\hbar} - \frac{\hat{V}^2\Delta t^2}{2\hbar^2} + \frac{i\hat{V}^3\Delta t^3}{6\hbar^3} + \ldots\right)$$

The error is:

$$\text{Error} = e^{-i\hat{H}\Delta t/\hbar} - e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$$

$$= \left(1 - \frac{i(\hat{T}+\hat{V})\Delta T}{\hbar} - \frac{(\hat{T}+\hat{V})^2\Delta t^2}{2\hbar^2} + \frac{i(\hat{T}+\hat{V})^3\Delta t^3}{6\hbar^3}\right) - \left(1 - \frac{i\hat{T}\Delta t}{\hbar} - \frac{\hat{T}^2\Delta t^2}{2\hbar^2} + \frac{i\hat{T}^3\Delta t^3}{6\hbar^3}\right)\left(1 - \frac{i\hat{V}\Delta t}{\hbar} - \frac{\hat{V}^2\Delta t^2}{2\hbar^2} + \frac{i\hat{V}^3\Delta t^3}{6\hbar^3}\right)$$

$$\begin{aligned} = &\left(\frac{\hat{T}\hat{V}}{2\hbar^2} + \frac{\hat{V}\hat{T}}{2\hbar^2} + \frac{\hat{T}\hat{V}}{\hbar}\right)\Delta t^2 \\ &- \left(\frac{i\hat{T}\hat{V}^2}{2\hbar^2} + \frac{i\hat{T}^2\hat{V}}{6\hbar^3} + \frac{i\hat{T}\hat{V}\hat{T}}{6\hbar^3} + \frac{i\hat{V}\hat{T}\hat{V}}{6\hbar^3} + \frac{i\hat{V}^2\hat{T}}{6\hbar^3} + \frac{i\hat{T}\hat{V}^2}{2\hbar^3} + \frac{i\hat{T}^2\hat{V}}{2\hbar^3}\right)\Delta t^3 \\ &- \left(\frac{\hat{T}\hat{V}^3}{6\hbar^4} + \frac{\hat{T}^2\hbar V^2}{4\hbar^4} + \frac{i\hat{T}^3\hat{V}}{6\hbar^4}\right)\Delta t^4 + \left(\frac{i\hat{T}^2\hat{V}^3}{12\hbar^5} + \frac{i\hat{T}^3\hat{V}^2}{12\hbar^5}\right)\Delta t^5 + \frac{\hat{T}^3\hat{V}^3}{36\hbar^6}\Delta t^6 \end{aligned} \tag{F.3}$$

It is clear from Eq. F.3 that the error contains the terms with Δt^2, Δt^3, Δt^4, and higher order. Thus error is termed as an error of second-order. However, the error can be reduced by splitting the $\hat{T}$ and $\hat{V}$ parts of the effective time-evolution operator.

F.1 Potential or kinetic referenced split operator method

It is already stated that the second-order SO is written as $e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$. The error terms in this contains terms of Δt^2, Δt^3, and higher orders. This error can be reduced further in two different ways by splitting the kinetic or potential energies symmetrically as follows.

$$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{V}\Delta t/2\hbar}e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/2\hbar} + \text{Error} \tag{F.4}$$

$$e^{-i\hat{H}\Delta t/\hbar} = e^{-i\hat{T}\Delta t/2\hbar}e^{-i\hat{V}\Delta t/\hbar}e^{-i\hat{T}\Delta t/2\hbar} + \text{Error} \tag{F.5}$$

The splitting in Eqs. F.4 and F.5 are known as the 'kinetic energy referenced' and 'potential energy referenced' approaches, respectively. The error in these two approaches can be calculated as in the case of second-order splitting. Here, the error for 'kinetic energy referenced' splitting will be calculated.

The LHS of Eq. F.4 is the same as Eq. F.2. When the extension up to the second order of the Taylor series is taken for the terms in Eq. F.4, the two sides are written as follows.

RHS

$$\begin{aligned}
e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar} = &\\
&\left(1 - \frac{i\hat{V}\Delta t}{2\hbar} - \frac{\hat{V}^2\Delta t^2}{8\hbar^2}\right)\left(1 - \frac{i\hat{T}\Delta t}{\hbar} - \frac{\hat{T}^2\Delta t^2}{2\hbar^2}\right)\left(1 - \frac{i\hat{V}\Delta t}{2\hbar} - \frac{\hat{V}^2\Delta t^2}{8\hbar^2}\right)\\
= 1 &- \frac{i\hat{V}\Delta t}{2\hbar} - \frac{\hat{V}^2\Delta t^2}{8\hbar} - \frac{i\hat{T}\Delta T}{\hbar} - \frac{\hat{T}\hat{V}\Delta t^2}{2\hbar^2} + \frac{i\hat{T}\hat{V}^2\Delta t^3}{8\hbar^3} - \frac{\hat{T}^2\Delta t^2}{2\hbar^2} + \frac{i\hat{T}^2\hat{V}\Delta t^3}{4\hbar^3} + \frac{\hat{T}^2\hat{V}^2\Delta t^4}{16\hbar^4}\\
&- \frac{i\hat{V}\Delta t}{2\hbar} - \frac{\hat{V}^2\Delta t^2}{4\hbar^2} + \frac{i\hat{V}^3\Delta t^3}{16\hbar^3} - \frac{\hat{V}\hat{T}\Delta t^2}{2\hbar^2} + \frac{i\hat{V}\hat{T}\hat{V}\Delta t^3}{4\hbar^3} + \frac{\hat{V}\hat{T}\hat{V}^2\Delta t^4}{16\hbar^4} + \frac{i\hat{V}\hat{T}^2\Delta t^3}{4\hbar^3} + \frac{\hat{V}\hat{T}^2\hat{V}\Delta t^4}{8\hbar^4}\\
&- \frac{\hat{V}\hat{T}^2\hat{V}^2\Delta t^5}{32\hbar^5} - \frac{\hat{V}^2\Delta t^2}{8\hbar^2} + \frac{i\hat{V}^3\Delta t^3}{16\hbar^3} + \frac{\hat{V}^4\Delta t^4}{64\hbar^4} + \frac{\hat{V}^2\hat{T}\Delta t^3}{8\hbar^3} + \frac{\hat{V}^2\hat{T}\hat{V}\Delta t^4}{16\hbar^4} - \frac{i\hat{V}^2\hat{T}\hat{V}^2\Delta t^5}{64\hbar^5}\\
&+ \frac{\hat{V}^2\hat{T}^2\Delta t^4}{16\hbar^4} - \frac{i\hat{V}^2\hat{T}^2\hat{V}\Delta t^5}{32\hbar^5} - \frac{\hat{V}^2\hat{T}^2\Delta t^6}{128\hbar^6}
\end{aligned}$$

LHS

$$e^{-i\hat{H}\Delta t/\hbar} = 1 - \frac{i\hat{T}\Delta t}{\hbar} - \frac{i\hat{V}\Delta t}{\hbar} - \frac{\hat{T}^2\Delta t^2}{2\hbar^2} - \frac{\hat{V}^2\Delta t^2}{2\hbar^2} - \frac{\hat{T}\hat{V}\Delta t^2}{2\hbar^2} - \frac{\hat{V}\hat{T}\Delta t^2}{2\hbar^2}$$

The error term is:

$$\text{Error} = e^{-i\hat{H}\Delta t/\hbar} - e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}e^{-i\hat{T}\Delta t/\hbar}.$$

$$
\begin{aligned}
= &-\left(\frac{i\hat{T}\hat{V}^2}{8\hbar^3}+\frac{i\hat{T}^2\hat{V}}{4\hbar^3}+\frac{i\hat{V}\hat{T}\hat{V}}{4\hbar^3}+\frac{i\hat{V}\hat{T}^2}{4\hbar^3}+\frac{i\hat{V}^3}{16\hbar^3}+\frac{i\hat{V}^3}{16\hbar^3}+\frac{\hat{V}^2\hat{T}}{8\hbar^3}\right)\Delta t^3 \\
&-\left(\frac{\hat{T}^2\hat{V}^2}{16\hbar^4}+\frac{\hat{V}\hat{T}\hat{V}^2}{16\hbar^4}+\frac{\hat{V}\hat{T}^2\hat{V}}{8\hbar^4}+\frac{\hat{V}^4}{64\hbar^4}+\frac{\hat{V}^2\hat{T}\hat{V}}{16\hbar^4}+\frac{\hat{V}^2\hat{T}^2}{16\hbar^4}\right)\Delta t^4 \\
&+\left(\frac{i\hat{V}^2\hat{T}\hat{V}^2}{64\hbar^5}+\frac{i\hat{V}^2\hat{T}^2\hat{V}}{32\hbar^5}+\frac{\hat{V}\hat{T}^2\hat{V}^2}{32\hbar^5}\right)\Delta t^5+\frac{\hat{V}^2\hat{T}^2\Delta t^6}{128\hbar^6}
\end{aligned} \tag{F.6}
$$

The Eq. F.6 contains the error terms of third, fourth, and higher order of Δt but not second order, thus decreasing the total error.

Further splitting of the time evolution operator gives rise to the split operator method of higher order. These high-order methods have low error, but the downside is increased computational overhead.

Appendix G

Chebyshev Polynomials

Chebyshev polynomials are the solution of the Chebyshev differential equation. The Chebyshev differential equation is expressed as

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + n^2y = 0, \tag{G.1}$$

where, $n = 0, 1, 2, 3...$, is the order of the polynomial. The solutions of the above differential equations, Chebyshev polynomial of order n, are:

$$T_n(x) = \frac{(-2)^n(n!)}{(2n)!}\sqrt{1-x^2}\frac{d^n}{dx^n}(1-x^2)^{n-\frac{1}{2}}, \qquad x \in [-1,1].$$

If we put $x = cos\theta$, Eq. G.1 can also be expressed as:

$$\frac{d^2y}{d\theta} + n^2y = 0, \tag{G.2}$$

where the solutions of this transformed form of the Chebyshev differential equation are:

$$T_n(\theta) = cos(n\theta).$$

So, the $T_n(x)$ can also be expressed as

$$T_n(x) = cos(ncos^{-1}x).$$

Examples are,

$$T_0(x) = 1, \;\; T_1(x) = x, \;\; T_2(x) = 2x^2 - 1$$

The above solutions can be expressed for the complex Chebyshev polynomial as:

$$Q_n(\omega) = i^nT_n(-i\omega), \qquad \omega \in [-i, i] \tag{G.3}$$

G.1 Recursion relation

Chebyshev polynomials of different order are related to each other through a recursion formula.

1. Real Chebyshev polynomials

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \tag{G.4}$$

2. Complex Chebyshev polynomials

   Putting $x = -i\omega$ in Eq. G.4 and multiplying both sides with i^{n+1} gives:

$$Q_{n+1}(\omega) = 2\omega Q_n(\omega) + Q_{n-1}(\omega) \tag{G.5}$$

Any arbitrary function $f(x)$ ∀ x ∈[-1,1] is expanded using the Chebyshev polynomial as:

$$f(ax) = \sum_{n=0}^{\infty} A_n(a) T_n(x), \tag{G.6}$$

where, $A_n(a)$ is coefficient of $T_n(x)$ Chebyshev polynomial. These polynomials of n^{th} order are orthogonal to each other, which is shown as follows.

$$\int_0^{\pi} T_n(\theta) T_m(\theta) d\theta = \delta_{nm} \begin{cases} 0; n \neq m \\ n = m \begin{cases} \pi; n = m = 0 \\ \frac{\pi}{2}; n = m \neq 0 \end{cases} \end{cases} \tag{G.7}$$

or

$$\int_{-1}^{1} \frac{T_n(x) T_m(x)}{\sqrt{1-x^2}} dx = \delta_{nm} \begin{cases} 0; n \neq m \\ n = m \begin{cases} \pi; n = m = 0 \\ \frac{\pi}{2}; n = m \neq 0 \end{cases} \end{cases} \tag{G.8}$$

The coefficients $A_n(a)$ can be calculated as:

$$A_n = \int_{-1}^{1} \frac{f(ax) T_n(x)}{\sqrt{1-x^2}} dx. \tag{G.9}$$

$$\int_{-1}^{1} \frac{f(x) T_n(x)}{\sqrt{1-x^2}} dx = \int_{-1}^{1} \frac{\sum_{k=0}^{n} A_k T_k(x) T_n(n)}{\sqrt{1-x^2}} dx = \sum_{k=0}^{n} A_k \left[\int_{-1}^{1} \frac{T_k(x) T_n(x)}{\sqrt{1-x^2}} \right]$$

At $k = n$,

$$\int_{-1}^{1} \frac{f(x) T_n(x)}{\sqrt{1-x^2}} dx = A_n \left(\frac{\pi}{2 - \delta_{n0}} \right).$$

$$A_n = \left(\frac{2 - \delta_{k0}}{\pi} \right) \left[\int_{-1}^{1} \frac{f(x) T_n(x)}{\sqrt{1-x^2}} dx \right] \tag{G.10}$$

Finally, the Eq. G.6 is written as:

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{2 - \delta_{n0}}{\pi} \right) \left[\int_{-1}^{1} \frac{f(x) T_n(x)}{\sqrt{1-x^2}} dx \right] T_n(x) \tag{G.11}$$

G.2 Expansion of time evolution operator

The time evolution operator for a time step t is written as:

$$f(\hat{H}) = exp\left(\frac{-i\hat{H}t}{\hbar}\right).$$

The operator can be expressed as,

$$f(\hat{H}) = \sum_{n=0}^{\infty} A_n T_n(\hat{H}).$$

Putting the values of A_n from Eq. G.10 gives the following expansion.

$$f(H) = \sum_{n=0}^{\infty}\left(\frac{2-\delta_{n0}}{\pi}\right)\left[\int_{-1}^{1}\frac{exp\left(\frac{-i\hat{H}t}{\hbar}\right)T_n(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H}\right]T_n(\hat{H}) \tag{G.12}$$

Solving the integral inside the square bracket:

$$\int_{-1}^{1}\frac{exp\left(\frac{-i\hat{H}t}{\hbar}\right)T_n(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H} = \int_{-1}^{1}\frac{\left\{\cos\left(\frac{\hat{H}t}{\hbar}\right) - i\sin\left(\frac{\hat{H}t}{\hbar}\right)\right\}T_n(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H}$$

$$= \left[\int_{-1}^{1}\frac{\cos\left(\frac{\hat{H}t}{\hbar}\right)T_n(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H} - i\int_{-1}^{1}\frac{\sin\left(\frac{\hat{H}t}{\hbar}\right)T_n}{\sqrt{1-\hat{H}^2}}(\hat{H})d\hat{H}\right] \tag{G.13}$$

Using the following property of the integrals to solve the equation further:

$$\int_{-a}^{a} f(x)dx = \begin{cases} 0, & when\ f(-x) = -f(x) \\ 2\int_{0}^{a} f(x)dx, & when\ f(-x) = f(x). \end{cases}$$

Now, *sin(x)* is odd function because *sin(-x) = -sin(x)* and *cos(x)* is even function because *cos(-x) = cos(x)*. Also, Chebyshev polynomial $T_n(x)$ is odd $\forall\ n \in$ odd and is even $\forall\ n \in$ even.

Putting the integral from Eq. G.13 to Eq. G.12, expanding the summation over n, and separating the even and odd terms of Chebyshev polynomial and solving for the *cosin* and *sin* parts individually.

The solution of the *cosin* part is as follows.

$$\sum_{n=0}^{\infty}\left(\frac{2-\delta_{n0}}{\pi}\right)T_n(\hat{H})\int_{-1}^{1}\frac{\cos\left(\frac{\hat{H}t}{\hbar}\right)T_n(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H} =$$

$$\sum_{n=0}^{\infty}\left(\frac{2-\delta_{n0}}{\pi}T_n(\hat{H})\right)\left[\int_{-1}^{1}\frac{\cos\left(\frac{\hat{H}t}{\hbar}\right)T_{2n}(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H} + \int_{-1}^{1}\frac{\cos\left(\frac{\hat{H}t}{\hbar}\right)T_{2n+1}(\hat{H})}{\sqrt{1-\hat{H}^2}}d\hat{H}\right]$$

Now, $\because \int_{-a}^{a} f(\hat{H})dx = 0 \; \forall \; n \in$ odd. So,

$$\int_{-1}^{1} \frac{\cos\left(\frac{\hat{H}t}{\hbar}\right) T_{2n+1}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} = 0.$$

Thus,

$$\sum_{n=0}^{\infty} \left(\frac{2-\delta_{n0}}{\pi} T_n(\hat{H})\right) \int_{-1}^{1} \frac{\cos\left(\frac{\hat{H}t}{\hbar}\right) T_n(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} = \sum_{n=0}^{\infty} \left(\frac{2-\delta_{2n0}}{\pi}\right) T_{2n}(\hat{H}) \left[\int_{-1}^{1} \frac{\cos\left(\frac{\hat{H}t}{\hbar}\right) T_{2n}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H}\right]. \quad \text{(G.14)}$$

The *sin* part is solved the same way as follows.

$$\sum_{n=0}^{\infty} \left(\frac{2-\delta_{n0}}{\pi}\right) T_n(\hat{H}) \int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_n(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} = \sum_{n=0}^{\infty} \left(\frac{2-\delta_{n0}}{\pi}\right) T_n(\hat{H}) \left[\int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_{2n}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} + \int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_{2n+1}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H}\right]$$

Now, $\because \int_{-a}^{a} f(x)dx = 0 \; \forall \; n \in$ odd. So,

$$\int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_n(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} = 0.$$

Thus,

$$\sum_{n=0}^{\infty} \left(\frac{2-\delta_{n0}}{\pi}\right) T_n(\hat{H}) \int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_n(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} = \sum_{n=0}^{\infty} \left(\frac{2-\delta_{(2n+1)0}}{\pi}\right) T_{2n+1}(\hat{H}) \left[\int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_{2n+1}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H}\right] \quad \text{(G.15)}$$

We can write the final equation as,

$$f(\hat{H}) = \sum_{n=0}^{\infty} \left[\left(\frac{2-\delta_{2n0}}{\pi}\right) T_{2n}(\hat{H}) \int_{-1}^{1} \frac{\cos\left(\frac{\hat{H}t}{\hbar}\right) T_{2n}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H} - i \left(\frac{2-\delta_{(2n+1)0}}{\pi}\right) T_{2n+1}(\hat{H}) \int_{-1}^{1} \frac{sin\left(\frac{\hat{H}t}{\hbar}\right) T_{2n+1}(\hat{H})}{\sqrt{1-\hat{H}^2}} d\hat{H}\right] \quad \text{(G.16)}$$

Now, using the properties:

$$\int_{-1}^{1} \frac{sin\Big(ax\Big)T_{2n+1}(x)}{\sqrt{1-x^2}}dx = (-1)^n \pi J_{2n+1}(a),$$

$$\int_{-1}^{1} \frac{cos\,(ax)\,T_{2n}(x)}{\sqrt{1-x^2}}dx = (-1)^n \pi J_{2n}(a),$$

where, $J_n(a)$ are the ..... and $a > 0$. So,

$$f(\hat{H}) = \sum_{n=0}^{\infty}\left[\left(\frac{2-\delta_{2n0}}{\pi}\right)T_{2n}(\hat{H})(-1^n)\pi J_{2n}\left(\frac{t}{\hbar}\right) - i\left(\frac{2}{\pi}\right)T_{2n+1}(\hat{H})(-1^n)\pi J_{2n+1}\left(\frac{t}{\hbar}\right)\right] \quad \text{(G.17)}$$

Rewriting the above equation using the properties of odd and even functions, such that $T_{2n}(-\hat{H}) = T_{2n}(-\hat{H})$ and $-T_{2n+1}(-\hat{H}) = T_{2n+1}(-\hat{H})$.

$$f(\hat{H}) = \sum_{n=0}^{\infty}\left[(2-\delta_{2n0})\,T_{2n}(-\hat{H})(-1^n)J_{2n}\left(\frac{t}{\hbar}\right) + i2T_{2n+1}(-\hat{H})(-1^n)J_{2n+1}\left(\frac{t}{\hbar}\right)\right]$$

Thus, the expansion of the time evolution operator is:

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = \sum_{0}^{\infty} i^n(2-\delta_{n0})T_n(-\hat{H})J_n\left(\frac{t}{\hbar}\right). \quad \text{(G.18)}$$

From Eq. G.3: $Q_n(\omega) = i^n T_n(-i\omega)$. Putting $\omega = -i\hat{H}$ gives: $Q_n(-i\hat{H}) = i^n T_n(-\hat{H})$. Putting this in Eq. G.18 gives the complex Chebyshev polynomial form of expansion as,

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = \sum_{0}^{\infty}(2-\delta_{n0})Q_n(-i\hat{H})J_n\left(\frac{t}{\hbar}\right). \quad \text{(G.19)}$$

Now, as a final step, the operator H is converted into a scaled form: $\hat{H}_s$. This is done because the requirement of expanding any function $f(x)$ in Chebyshev polynomial form is that $x \in [-1, 1]$. The scaling is done as defined as:

$$\hat{H}_s = a_s\hat{H} + b_s,$$

where, $a_s = 2/\Delta E$, $b_s = -1 - a_s E_{min}$, and $\Delta E = E_{max} - E_{min}$. Now, $\hat{H} = \frac{\hat{H}-b_s}{a_s}$. Using these, Eqs. G.18 and G.19 are written as,

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = exp\left(\frac{-i(E_{max}+E_{min})t}{2\hbar}\right)\sum_{0}^{\infty} i^n(2-\delta_{n0})T_n(-\hat{H}_s)J_n\left(\frac{t\Delta E}{2\hbar}\right), \quad \text{(G.20)}$$

$$exp\left(\frac{-i\hat{H}t}{\hbar}\right) = exp\left(\frac{-i(E_{max}+E_{min})t}{2\hbar}\right)\sum_{0}^{\infty}(2-\delta_{n0})Q_n(-i\hat{H}_s)J_n\left(\frac{t\Delta E}{2\hbar}\right). \quad \text{(G.21)}$$

G.3 Expansion of $\sqrt{1-\hat{H}_s^2}$

The operator $\sqrt{1-\hat{H}_s^2}$ is used in the modified form of the real wave packet method. This function is expanded using Chebyshev polynomial as:

$$\sqrt{1-\hat{H}_s^2} = \sum_{n=0} A_n T_n(\hat{H}_s). \tag{G.22}$$

Calculating coefficient A_n using Eq. G.10 as,

$$A_n = \left(\frac{2-\delta_{n0}}{\pi}\right)\int_{-1}^{1} \frac{\sqrt{1-\hat{H}_s^2}\, T_n(\hat{H}_s)}{\sqrt{1-\hat{H}_s^2}} d\hat{H}_s = \left(\frac{2-\delta_{n0}}{\pi}\right)\int_{-1}^{1} T_n(\hat{H}_s)d\hat{H}_s \tag{G.23}$$

Putting the value of A_n from Eq. G.23 to Eq. G.22 and expanding and separating the Chebyshev polynomials based on the odd and even value of n.

$$\sqrt{1-\hat{H}_s^2} = \sum_{n=0} T_n(\hat{H}_s)\left(\frac{2-\delta_{n0}}{\pi}\right)\int_{-1}^{1} T_n(\hat{H}_s)d\hat{H}_s$$

$$= \sum_{n=0} T_{2n}(\hat{H}_s)\left(\frac{2-\delta_{2n0}}{\pi}\right)\int_{-1}^{1} T_{2n}(\hat{H}_s)d\hat{H}_s + \sum_{n=0} T_{2n+1}(\hat{H}_s)\left(\frac{2-\delta_{(2n+1)0}}{\pi}\right) \int_{-1}^{1} T_{2n+1}(\hat{H}_s)d\hat{H}_s \tag{G.24}$$

It is known that $T_n(x)$ is odd $\forall\, x \in$ odd and even $\forall\, x \in$ even. So, putting $\int_{-1}^{1} T_{2n+1}(\hat{H}_s)d\hat{H}_s = 0$, the equation becomes,

$$= \sum_{n=0} T_{2n}(\hat{H}_s)\left(\frac{2-\delta_{2n0}}{\pi}\right)\int_{-1}^{1} T_{2n}(\hat{H}_s)d\hat{H}_s \tag{G.25}$$

Using $T_n(x) = cos(ncos^{-1}x)$ and writing,

$$T_{2n}(\hat{H}_s) = cos(2ncos^{-1}\hat{H}_s) = 2cos^2(ncos^{-1}\hat{H}_s) - 1 \Rightarrow 2T_n(\hat{H}_s)^2 - 1.$$

Put the value of $T_{2n}(\hat{H}_s)$ in Eq. G.24.

$$= \sum_{n=0} T_{2n}(\hat{H}_s)\left(\frac{2-\delta_{2n0}}{\pi}\right)\int_{-1}^{1} (2T_n(\hat{H}_s)^2 - 1)d\hat{H}_s \tag{G.26}$$

Integral in the Eq. G.26 can be solved easily, and the equation is written as,

$$= \sum_{n=0} T_{2n}(\hat{H}_s)\left(\frac{2-\delta_{2n0}}{\pi}\right)\left(\frac{-2}{4n^2-1}\right)$$

$$= \frac{2}{\pi}T_0(\hat{H}_s) + \sum_{n=1} T_{2n}(\hat{H}_s)\left(\frac{2}{\pi}\right)\left(\frac{-2}{4n^2-1}\right)$$

$$\sqrt{1-\hat{H}_s^2} = \frac{2}{\pi}\left(1 - \sum_{n=1}\frac{2T_{2n}(\hat{H}_s)}{4n^2-1}\right) \tag{G.27}$$

The Eq. G.27 is the expansion of $\sqrt{1-\hat{H}_s^2}$ in Chebyshev polynomials and is solved in the modified real wave packet method.

Appendix H

Hamiltonian for A + BC collision

For the A + BC collision, the total Hamiltonian can be written as follows.

$$\hat{H} = \hat{T}_{\mathrm{A}} + \hat{T}_{\mathrm{B}} + \hat{T}_{\mathrm{C}} + \hat{V}_{\mathrm{elec}}$$

The expression in terms of Jacobi coordinates is written as:

$$\hat{H} = -\frac{\hbar^2}{2\mu_{\mathrm{r}}}\nabla^2_{\mathrm{r}} - \frac{\hbar^2}{2\mu_{\mathrm{R}}}\nabla^2_{\mathrm{R}} - \frac{\hbar^2}{2M}\nabla^2_{\mathrm{cm}} + \hat{V}_{\mathrm{elec}}$$

Here, the first term shows the relative kinetic energy (KE) of the diatomic part, the second part is the relative KE of the atom from the center of mass of the diatom, and the third part is the KE of the center of mass of the system. The total mass of the system (M) can be written as the sum of three atomic masses. Since the dynamics is not affected by the motion of the system as a whole, the KE part of the center of the mass of the whole system can be neglected, making the final Hamiltonian as follows.

$$\hat{H} = -\frac{\hbar^2}{2\mu_{\mathrm{r}}}\nabla^2_{\mathrm{r}} - \frac{\hbar^2}{2\mu_{\mathrm{R}}}\nabla^2_{\mathrm{R}} + \hat{V}_{\mathrm{elec}} \tag{H.1}$$

The term ∇_{a}, where a can be replaced with r and R is called the Laplacian operator and is written as:

$$\nabla^2_{\mathrm{a}} = \frac{\partial^2}{\partial x^2_{\mathrm{a}}} + \frac{\partial^2}{\partial y^2_{\mathrm{a}}} + \frac{\partial^2}{\partial z^2_{\mathrm{a}}}$$

The operator ∇ for r coordinate has the following form in the spherical coordinates.

$$\nabla^2_r = \frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r} + \frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{r^2\sin^2(\theta)}\frac{\partial}{\partial\phi^2} = \frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r} - \frac{\hat{j}^2}{\hbar^2 r^2}$$

The above expression can also be written as follows.

$$\nabla^2_r = \frac{1}{r}\frac{\partial^2}{\partial r^2}r - \frac{\hat{j}^2}{\hbar^2 r^2}$$

The two forms can be shown as equivalent by solving $\frac{1}{r}\frac{\partial^2}{\partial r^2}r$ and $\frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r}$ separately. Now, putting the expressions of ∇^2_r and ∇^2_R in Eq. H.1, the following form of the Hamiltonian can be obtained.

$$\hat{H} = -\frac{\hbar^2}{2\mu_{\mathrm{r}}}\left(\frac{1}{r}\frac{\partial^2}{\partial r^2}r\right) + \frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2} - \frac{\hbar^2}{2\mu_{\mathrm{R}}}\left(\frac{1}{R}\frac{\partial^2}{\partial R^2}R\right) + \frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2} + \hat{V}_{\mathrm{elec}}$$

This form of the Hamiltonian can be transformed into a different form, and the procedure for that in the case of time-independent Schrödinger equation (SE) is as follows.

$$\hat{H}\psi = E\psi$$

$$-\frac{\hbar^2}{2\mu_{\mathrm{r}}}\frac{1}{r}\left(\frac{\partial^2}{\partial r^2}r\right)\psi - \frac{\hbar^2}{2\mu_R}\frac{1}{R}\left(\frac{\partial^2}{\partial R^2}R\right)\psi + \frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2}\psi + \frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2}\psi + \hat{V}_{\mathrm{elec}}\psi = E\psi$$

Multiplying and dividing the LHS of the above equation by $\frac{Rr}{Rr}$.

$$\frac{Rr}{Rr}\left[-\frac{\hbar^2}{2\mu_{\mathrm{r}}}\frac{1}{r}\left(\frac{\partial^2}{\partial r^2}r\right)\psi - \frac{\hbar^2}{2\mu_R}\frac{1}{R}\left(\frac{\partial^2}{\partial R^2}R\right)\psi + \frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2}\psi + \frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2}\psi + \hat{V}_{\mathrm{elec}}\psi\right] = E\psi$$

Rearranging the above equation.

$$\frac{1}{Rr}\left[-\frac{\hbar^2}{2\mu_{\mathrm{r}}}R\left(\frac{\partial^2}{\partial r^2}r\right)\psi - \frac{\hbar^2}{2\mu_R}r\left(\frac{\partial^2}{\partial R^2}R\right)\psi + Rr\frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2}\psi + Rr\frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2}\psi + Rr\hat{V}_{\mathrm{elec}}\psi\right] = E\psi$$

$$-\frac{\hbar^2}{2\mu_{\mathrm{r}}}\frac{\partial^2}{\partial r^2}\psi' - \frac{\hbar^2}{2\mu_R}\frac{\partial^2}{\partial R^2}\psi' + \frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2}\psi' + \frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2}\psi' + \hat{V}_{\mathrm{elec}}\psi' = E\psi'$$

This form of the equation is further used in dynamics [30, 167]. The Hamiltonian is written as follows.

$$-\frac{\hbar^2}{2\mu_R}\left(\frac{\partial^2}{\partial R^2}\right) + \frac{\hat{j}^2}{2\mu_{\mathrm{r}}r^2} - \frac{\hbar^2}{2\mu_r}\left(\frac{\partial^2}{\partial r^2}\right) + \frac{\hat{L}^2}{2\mu_{\mathrm{R}}R^2}\psi + \hat{V}_{\mathrm{elec}} \tag{H.2}$$

Here, the Hamiltonian has two angular momentum. One is the rotational angular momentum of the diatom ($\hat{j}$) around the diatomic center of mass, and another one is the orbital angular momentum ($\hat{L}$) of the attacking atom around the center of mass of the diatom. These two angular momentum belong to two different parts of the triatomic system and are represented by two different vector fields. A modified form of the Hamiltonian can be written by introducing the total angular momentum as follows.

$$\vec{J} = \vec{j} + \vec{L}$$

$$\vec{L} = \vec{J} - \vec{j}$$

$$\vec{L}\cdot\vec{L} = (\vec{J} - \vec{j})\cdot(\vec{J} - \vec{j}) = J^2 - j^2 - (\vec{J}\cdot\vec{j} + \vec{j}\cdot\vec{J})$$

The term $J.j + j.J$ can be calculated as follows.

$$[\vec{J}, \vec{j}] = [(\vec{L} + \vec{j}), \vec{j}] = [\vec{L}, \vec{j}] + [\vec{j}, \vec{j}]$$

Since $\vec{L}$ and $\vec{j}$ belong to two different vector spaces that evolve independently of each other thus, these operators will be commutative with each other and also with themselves. So,

$$[\vec{J}, \vec{j}] = 0$$

means,

$$\vec{J} \cdot \vec{j} = \vec{j} \cdot \vec{J}$$

Thus,

$$\vec{L}^2 = \vec{J}^2 + \vec{j}^2 - 2(\vec{J} \cdot \vec{j}) \tag{H.3}$$

Putting Eq. H.3 in H.2 gives us the following form of the Hamiltonian.

$$\hat{H} = -\frac{\hbar^2}{2\mu_{\mathrm{r}}}\frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2\mu_{\mathrm{R}}}\frac{\partial^2}{\partial R^2} + \left(\frac{1}{2I_{\mathrm{r}}} + \frac{1}{2I_{\mathrm{R}}}\right)\hat{j}^2 + \frac{\left(\hat{J}^2 - 2(\vec{J} \cdot \vec{j})\right)}{2I_{\mathrm{R}}} + \hat{V}_{\mathrm{elec}} \tag{H.4}$$

The expression $2(\vec{J} \cdot \vec{j})$ is written as follows.

$$\vec{J} = \vec{J}_x\hat{i} + \vec{J}_y\hat{j} + \vec{J}_z\hat{k}$$
$$\vec{j} = \vec{j}_x\hat{\vec{i}} + \vec{j}_y\hat{\vec{j}} + \vec{j}_z\hat{k}$$
$$\vec{J} \cdot \vec{j} = \vec{J}_x \cdot \vec{j}_x + \vec{J}_y \cdot \vec{j}_y + \vec{J}_z.\vec{j}_z$$

We know that raising and lowering operators are:

$$\vec{J}_{\pm} = \vec{J}_x \pm i\vec{J}_y$$

$$\vec{j}_{\pm} = \vec{j}_x \pm i\vec{j}_y$$

$$\vec{J}_x = \frac{\vec{J}_+ + \vec{J}_-}{2}; \qquad \vec{j}_x = \frac{\vec{j}_+ + \vec{j}_-}{2}$$
$$\vec{J}_y = \frac{\vec{J}_+ - \vec{J}_-}{2i}; \qquad \vec{j}_y = \frac{\vec{j}_+ - \vec{j}_-}{2i}$$

Calculating $\vec{J}_x \cdot \vec{j}_x + \vec{J}_y \cdot \vec{j}_y$:

$$\vec{J}_x \cdot \vec{j}_x + \vec{J}_y \cdot \vec{j}_y = \left(\frac{\vec{J}_+ + \vec{J}_-}{2}\right)\left(\frac{\vec{j}_+ + \vec{j}_-}{2}\right) + \left(\frac{\vec{J}_+ - \vec{J}_-}{2i}\right)\left(\frac{\vec{j}_+ - \vec{j}_-}{2i}\right)$$

$$\Rightarrow \frac{1}{4}\left[\vec{J}_+\vec{j}_+ + \vec{J}_+\vec{j}_- + \vec{J}_-\vec{j}_+ + \vec{J}_-\vec{j}_- - \vec{J}_+\vec{j}_+ + \vec{J}_+\vec{j}_- + \vec{J}_-\vec{j}_+ - \vec{J}_-\vec{j}_-\right]$$

$$\vec{J}_x \cdot \vec{j}_x + \vec{J}_y \cdot \vec{j}_y = \frac{1}{2}\left[\vec{J}_+\vec{j}_- + \vec{J}_-\vec{j}_+\right]$$

$$\vec{J} \cdot \vec{j} = \frac{1}{2}\left(\vec{J}_+\vec{j}_- + \vec{J}_-\vec{j}_+ + 2\vec{J}_z.\vec{j}_z\right)$$

Putting the above expression in Eq. H.4:

$$\hat{H} = -\frac{\hbar^2}{2\mu_\mathrm{r}}\frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2\mu_\mathrm{R}}\frac{\partial^2}{\partial R^2} + \left(\frac{1}{2I_\mathrm{r}} + \frac{1}{2I_\mathrm{R}}\right)\hat{j}^2 + \frac{\left(\hat{J}^2 - 2\hat{J}_z\hat{j}_z\right)}{2I_\mathrm{R}} + \hat{V}_\mathrm{elec} - \frac{\left(\hat{J}_+\hat{j}_- + \hat{J}_-\hat{j}_+\right)}{2I_\mathrm{R}} \quad \text{(H.5)}$$

Eq. H.5 is the form of the equation used in the DIFFREALWAVE code as shown in Eq. 2.37 of Chapter 2.

Appendix I

Vibrational quantum number in QCT method

For a diatom, the following classical equation gives us the vibrational quantum number [47].

$$v = -\frac{1}{2} + \frac{1}{\hbar\pi}\int_{r_-}^{r_+} \rho_r \mathrm{dr} \tag{I.1}$$

Where, v is the vibrational quantum number and ρ_r is: $\rho_r = \sqrt{2\mu}(a + b\zeta + c\zeta^2)^{1/2}$. All the parameters are as follows.

$$a = E_{vj} - D - Aj_{am}^2, \qquad b = 2D - Bj_{am}^2, \qquad c = -(D + Cj_{am}^2)$$

$$A = \frac{1}{2\mu r_{eq}^2}\left[1 - \frac{3}{\alpha r_{eq}}\left(1 - \frac{1}{\alpha r_{eq}}\right)\right], \quad B = \frac{2}{\mu\alpha r_{eq}^3}\left(1 - \frac{3}{2\alpha r_{eq}}\right)$$

$$C = \frac{1}{2\mu\alpha r_{eq}^3}\left(1 - \frac{3}{\alpha r_{eq}}\right)$$

$$\zeta = exp(-\alpha(r - r_{eq}))$$

Here, D is the dissociation energy, j_{am} is the rotational angular momentum, μ is the reduced mass of diatom, r_{eq} is the equilibrium interatomic distance, and α is Morse parameter for the diatom.

This integral in the above form is solved in the range of the inner turning point (r_-) to the outer turning point (r_+). These limits are transformed to -1 and +1 and solved using the following properties of the Chebyshev quadrature.

$$\int_{-1}^{+1} \frac{f(y)}{\sqrt{1-y^2}}\mathrm{dy} = \sum_{i=1}^{k} w_i f(y_i)$$

Here, $w_i = \pi/k$ and k is the total number of points. Take the Eq. I and follow the following procedure.

$$v = -\frac{1}{2} + \frac{1}{\hbar\pi}\int_{r_-}^{r_+} \sqrt{2\mu}\left(a + b\zeta + c\zeta^2\right)^{\frac{1}{2}} dr$$

$$= -\frac{1}{2} + \frac{1}{\hbar\pi}\int_{r_-}^{r_+} \sqrt{2\mu}\left(a + be^{-\alpha(r-r_{eq})} + ce^{-2\alpha(r-r_{eq})}\right)^{\frac{1}{2}} dr$$

Step 1: Transforming the limits and rewriting the integral.

Take $y = \frac{2r-r_+-r_-}{r_+-r_-}$ and $dy = \frac{2}{r_+-r_-}dr \Rightarrow dr = \frac{r_+-r_-}{2}dy$

$$\rho_r = \int_{-1}^{+1} \sqrt{2\mu}\left(a + b\zeta + c\zeta^2\right)^{\frac{1}{2}} \frac{(r_+ - r_-)}{2} dy$$

Here, $\zeta = exp\left(-\alpha\left(\frac{y(r_+-r_-)}{2} + \frac{(r_++r_-)}{2} - r_{eq}\right)\right)$

Step 2: Multiply and divide above form of ρ_r with $\sqrt{1-y^2}$.

$$\rho_r = \int_{-1}^{+1} \sqrt{2\mu}\left(a + b\zeta + c\zeta^2\right)^{\frac{1}{2}} \frac{(r_+ - r_-)}{2}\left(\frac{\sqrt{1-y^2}}{\sqrt{1-y^2}}\right) dy$$

$$= \int_{-1}^{+1} \left(\sqrt{2\mu}\left(a + b\zeta + c\zeta^2\right)^{\frac{1}{2}} \frac{(r_+ - r_-)}{2}\sqrt{1-y^2}\right)/\sqrt{1-y^2}\, dy$$

$$= \frac{(r_+ - r_-)}{2}\sqrt{2\mu}\int_{-1}^{+1} \left(\left(a + b\zeta + c\zeta^2\right)^{\frac{1}{2}} \sqrt{1-y^2}\right)/\sqrt{1-y^2}\, dy$$

$$= \frac{(r_+ - r_-)}{2}\sqrt{2\mu}\int_{-1}^{+1} \frac{f(y)}{\sqrt{1-y^2}} dy$$

Step 3: Writing the transformed expression of v.

$$v = -\frac{1}{2} + \frac{1}{\pi\hbar}\frac{(r_+ - r_-)}{2}\sqrt{2\mu}\int_{-1}^{+1} \frac{f(y)}{\sqrt{1-y^2}} dy$$

$$v = -\frac{1}{2} + \frac{1}{\pi\hbar}\frac{(r_+ - r_-)}{2}\sqrt{2\mu}\sum_{i=1}^{k} w_i f(y_i)$$

Step 4: Converting the parameters from the above form to the atomic unit.

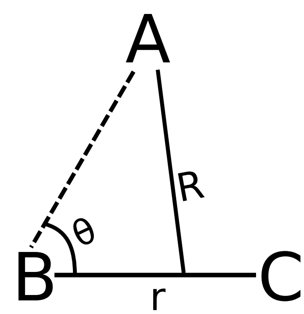
$$v = -\frac{1}{2} + \frac{1}{\pi}\frac{(r_+ - r_-)}{2}\sqrt{2\mu}\quad\sum_{i=1}^{k} w_i f(y_i)$$

The above expression can be implemented in the code to get the vibrational quantum number (v) of the diatom.

Appendix J

Mass-weighted Coordinates

Mass-weighted PES [85, 86] uses the mass-weighted coordinates Q_1 and Q_2, as shown in Fig. 3.24. For a reaction A + BC → AB + C, these coordinates are expressed in terms of internuclear distances as follows.



$$Q_1 = \sqrt{\mu_{A-BC}}R \tag{J.1}$$

$$Q_2 = \sqrt{\mu_{BC}}r \tag{J.2}$$

Here, R is the distance between the center of mass of diatom BC to atom A. μ_{BC} and μ_{A-BC} are reduced masses of diatom BC and reaction system A + BC. For atoms A, B, and C (having masses m_A, m_B, and m_C, respectively), the above-given reduced masses are calculated as follows.

$$\mu_{A-BC} = \frac{m_A(m_B + m_C)}{(m_A + m_B + m_C)} \tag{J.3}$$

$$\mu_{BC} = \frac{(m_B m_C)}{(m_B + m_C)} \tag{J.4}$$

The skew angles of the four reactions are calculated using the following equation, [86]

$$\beta = \cos^{-1}\left(\sqrt{\frac{m_A m_C}{(m_B + m_C)(m_A + m_B)}}\right) . \tag{J.5}$$

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List of publications

Chapter 3:

1. Jayakrushna Sahoo, **Ajay Mohan Singh Rawat**, and S. Mahapatra. Theoretical study of the energy disposal mechanism and the state-resolved quantum dynamics of the H + $LiH^+ \rightarrow H_2 + Li^+$ reaction. J. Phys. Chem. A **125**, 3387 (2021).

2. Jayakrushna Sahoo, **Ajay Mohan Singh Rawat**, and S. Mahapatra. Quantum interference in the mechanism of $H + LiH^+ \rightarrow H_2 + Li^+$ reaction dynamics. Phys. Chem. Chem. Phys. **23**, 27327 (2021).

3. **Ajay Mohan Singh Rawat**[†], Jayakrushna Sahoo[†], and S. Mahapatra. A combined quantum mechanical and quasi-classical state-to-state dynamical study to understand the isotopic effect in $H/D + LiH^+ /LiD^+ \rightarrow H_2/HD/D_2 + Li^+$ reactions. J. Phys. Chem. A **127**, 51, 10733 (2023).

Chapter 4:

4. **Ajay Mohan Singh Rawat**, Mohammed Alamgir, Sugata Goswami, and Susanta Mahapatra. A new ground electronic state potential energy surface of $HeLiH^+$: analytical representation and investigation of the dynamics of $He + LiH^+$ (v=0, j=0) $\rightarrow LiHe^+ + H$ reaction. J. Chem. Phys. **161**, 124308 (2024).

Chapter 5:

5. **Ajay Mohan Singh Rawat**, Jayakrushna Sahoo, and Susanta Mahapatra. State-to-state dynamics of isotopic exchange reaction $^{18}O + {}^{16}O^{16}O$ (v=0, j=1) $\rightarrow {}^{18}O^{16}O + {}^{16}O$. (*manuscript under preparation*).

Posters and Oral presentations in conferences

1. Poster presentation in 'DAE Symposium on Current Trends in Theoretical Chemistry (CTTC-2020)'. Virtual conference. September 2021 at BARC, Mumbai, India.
2. Poster presentation in '17th Theoretical Chemistry Symposium (TCS-2021)'. Virtual conference. December 2021 at IISER Kolkata, India.
3. Oral presentation in 'Chemfest-2023, annual in-house symposium'. February 2023 at School of Chemistry, University of Hyderabad, India.
4. Poster presentation in 'Structure and Dynamics: Spectroscopy and Scattering (SDSS 2023)'. October 2023 at IACS Kolkata, India.
5. Poster presentation in '18th Theoretical Chemistry Symposium (TCS-2023)'. November 2023 at IIT Madras, India.